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Strategic Management of Technological Innovation

SEVENTH EDITION



Melissa A. Schilling

Mc Graw Hill

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Seventh Edition

Melissa A. Schilling
New York University

Mc Graw Hill



STRATEGIC MANAGEMENT OF TECHNOLOGICAL INNOVATION

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About the Author

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Melissa Schilling is the John Herzog family professor of management and organizations at New York University's Stern School of Business. Professor Schilling teaches courses in strategic management, corporate strategy and technology, and innovation management. Before joining NYU, she was an Assistant Professor at Boston University (1997–2001), and has also served as a Visiting Professor at INSEAD and the Bren School of Environmental Science & Management at the University of California at Santa Barbara. She has also taught strategy and innovation courses at Siemens Corporation, IBM, the Kauffman Foundation Entrepreneurship Fellows program, Sogang University in Korea, and the Alta Scuola Polytecnica, a joint institution of Politecnico di Milano and Politecnico di Torino.

Professor Schilling's research focuses on technological innovation and knowledge creation. She has studied how technology shocks influence collaboration activity and innovation outcomes, how firms fight technology standards battles, manage platform ecosystems, and utilize collaboration, protection, and timing of entry strategies. She also studies how product designs and organizational structures migrate toward or away from modularity. Her most recent work focuses on knowledge creation, including how breadth of knowledge and search influences insight and learning, and how the structure of knowledge networks influences their overall capacity for knowledge creation. Her research in innovation and strategy has appeared in the leading academic journals such as *Academy of Management Journal, Academy of Management Review, Management Science, Organization Science, Strategic Management Journal,* and *Journal of Economics and Management Strategy* and *Research Policy.* She also sits on the editorial review boards of *Academy of Management Journal, Academy of Management Discoveries, Organization Science, Strategy Science,* and *Strategic Organization.* She is the author of *Quirky: The Remarkable Story of the Traits, Foibles, and Genius of Breakthrough Innovators Who Changed the World,* and she is coauthor of *Strategic Management: An Integrated Approach.* Professor Schilling won the Organization Science and Management Science Best Paper prize in 2007, an NSF CAREER award in 2003, and Boston University's Broderick Prize for research in 2000.

Preface

Innovation is a beautiful thing. It is a force with both aesthetic and pragmatic appeal: It unleashes our creative spirit, opening our minds to hitherto undreamed of possibilities, while accelerating economic growth and providing advances in such crucial human endeavors as medicine, agriculture, and education. For industrial organizations, the primary engines of innovation in the Western world, innovation provides both exceptional opportunities and steep challenges. While innovation is a powerful means of competitive differentiation, enabling firms to penetrate new markets and achieve higher margins, it is also a competitive race that must be run with speed, skill, and precision. It is not enough for a firm to be innovative–to be successful it must innovate better than its competitors.

As scholars and managers have raced to better understand innovation, a wide range of work on the topic has emerged and flourished in disciplines such as strategic management, organization theory, economics, marketing, engineering, and sociology. This work has generated many insights about how innovation affects the competitive dynamics of markets, how firms can strategically manage innovation, and how firms can implement their innovation strategies to maximize their likelihood of success. A great benefit of the dispersion of this literature across such diverse domains of study is that many innovation topics have been examined from different angles. However, this diversity also can pose integration challenges to both instructors and students. This book seeks to integrate this wide body of work into a single coherent strategic framework, attempting to provide coverage that is rigorous, inclusive, and accessible.

Organization of the Book

The subject of innovation management is approached here as a strategic process. The outline of the book is designed to mirror the strategic management process used in most strategy textbooks, progressing from assessing the competitive dynamics of the situation, to strategy formulation, and then to strategy implementation. The first part of the book covers the foundations and implications of the dynamics of innovation, helping managers and future managers better interpret their technological environments and identify meaningful trends. The second part of the book begins the process of crafting the firm's strategic direction and formulating its innovation strategy, including project selection, collaboration strategies, and strategies for protecting the firm's property rights. The third part of the book covers the process of implementing innovation, including the implications of organization structure on innovation, the management of new product development processes, the construction and management of new product development teams, and crafting the firm's deployment strategy. While the book emphasizes practical applications and examples, it also provides systematic coverage of the existing research and footnotes to guide further reading.

Complete Coverage for Both Business and Engineering Students

This book is designed to be a primary text for courses in the strategic management of innovation and new product development. Such courses are frequently taught in both

business and engineering programs; thus, this book has been written with the needs of business and engineering students in mind. For example, Chapter Six (Defining the Organization's Strategic Direction) provides basic strategic analysis tools with which business students may already be familiar, but which may be unfamiliar to engineering students. Similarly, some of the material in Chapter Eleven (Managing the New Product Development Process) on computer-aided design or quality function deployment may be review material for information system students or engineering students, while being new to management students. Though the chapters are designed to have an intuitive order to them, they are also designed to be self-standing so instructors can pick and choose from them "buffet style" if they prefer.

New for the Seventh Edition

This seventh edition of the text has been comprehensively revised to ensure that the frameworks and tools are rigorous and comprehensive, the examples are fresh and exciting, and the figures and cases represent the most current information available. Some changes of particular note include:

Six New Short Cases

Netflix and the Battle of the Streaming Services. The new opening case for Chapter Four is about a battle unfolding for dominance in movie and television streaming. Though the case focuses on Netflix, it also details the moves made by competitors such as Amazon Prime Video, Disney, Hulu, and HBO. The case reveals the very interesting synergies Netflix has reaped in being both a content developer and a distributor, and it highlights the tradeoffs content developers make in choosing to have their content exclusive to a particular streaming service.

Failure to Launch at Uber Elevate. The opening case for Chapter Five in the sixth edition was about UberAIR, Uber's plan for launching an air taxi service; the opening case for Chapter Five for the seventh edition is about Uber's withdrawal of plans to launch its own air taxi service and the other companies that are still moving forward. This case highlights the range of challenges in launching something as new as air taxi service. While battery life and flight time are still considered areas that need improvement, the primary challenges to this market are now regulatory and infrastructure oriented: Where will the eVTOLs land? Who will regulate air traffic and how? Will the eVTOLs be too noisy? Will the eVTOLs be manned by pilots or autonomous? It is pretty easy to conclude from the case that Uber probably tried to enter this market too early, but it remains unclear whether the remaining players (who are almost all manufacturing startups dedicated wholly to producing eVTOLs) will fare better.

Zeta Energy and The "Holy Grail" of Batteries. Chapter Eight now opens with a case about Zeta Energy, a young battery technology startup that is in the process of developing a lithium metal sulfur battery. The technology is impressive and the potential markets are huge and diverse (e.g., electric vehicles, grid storage, consumer devices, and drones), but Zeta faces a dilemma of how to reach the stage of commercialization. Battery development is expensive and risky; Zeta has had problems raising enough funding to build the kind of facility it needs to produce the batteries at scale. The case highlights the various partnering strategies Zeta is considering, setting up a nice opportunity for students to analyze the pros and cons of types of collaboration agreements and types of partners.

The Patent Battle Over CRISPR Cas-9 Gene Editing. The new opening case for Chapter Nine is on what has been described as one of the most important patent battles in the last 50 years. CRISPR Cas-9 is a breakthrough technology that enables live animals (including humans) to be gene edited—potentially enabling us to eliminate and/or treat a wide range of diseases. Even more exciting is the fact that the technology itself is relatively inexpensive and simple, prompting a flood of students, researchers, and manufacturers to enthusiastically begin using it. The ownership of the intellectual property rights, however, are contested between a group at Berkeley and a group at MIT. The way each group's patents were filed, concomitant with the change of patent law, collectively created one of the most interesting—and high-stakes—battles patent lawyers have seen in decades.

How Apple Organizes for Innovation. Chapter Ten now opens with a case that describes how Apple is organized. The case tells the story of when Steve Jobs returned to Apple and dramatically reorganized the firm, yielding a big firm that has a structure that is much more commonly seen in small firms. The case provides detail on why Jobs felt the structure was appropriate, what its tradeoffs are, notably highlighting how much power the structure gives to its top leader. While this was probably a very desirable feature for Jobs, the case raises the question of whether or not the same structure makes sense for Apple under Tim Cook and whether it would make sense for different kinds of firms.

Magna International's Carbon Fiber "Lightweighting" Project. The opening case for Chapter Twelve describes in detail how Magna International, a Tier 1 automotive supplier, developed a scalable manufacturing method for carbon fiber auto parts in response to BMW's announcement of its intentions to build cars with the new material. With details and quotes from Tom Pilette, the VP of Product and Process Development of Magna who led the project, we learn about how the team was assembled and managed, how the team culture evolved, how team members were compensated, and more. BMW ends up deciding to make carbon fiber composites in house rather than buying from a supplier, but Magna's efforts transform it into an award-winning world leader in carbon fiber composite manufacturing.

Cases, Data, and Examples from around the World

Careful attention has been paid to ensure that the text is global in its scope. The opening cases and examples feature companies from China, India, Israel, Japan, The Netherlands, Kenya, the United States, and more. Wherever possible, statistics used in the text are based on worldwide data.

More Comprehensive Coverage and Focus on Current Innovation Trends

In response to reviewer suggestions, the new edition now provides an extensive discussion of the use of "Big Data" in guiding innovation, the strengths and weaknesses of grand prizes (like the XPRIZE) in generating innovation, characteristics of breakthrough innovators, the role of organization culture in innovation, a detailed example of Failure Modes and Effects Analysis that helps students set up their own FMEA spreadsheet, and more. The suggested readings for each chapter have also been updated to identify some of the more recent publications that have gained widespread attention in the topic area of each chapter. Despite these additions, great effort has also been put into ensuring the book remains concise—a feature that has proven popular with both instructors and students.

Supplements

The teaching package for *Strategic Management of Technological Innovation* is available online from Connect at connect.mheducation.com and includes:

- An instructor's manual with suggested class outlines, responses to discussion questions, and more.
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- Jordan Cunningham, Eastern Washington University



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Chapter **One**

Introduction

THE IMPORTANCE OF TECHNOLOGICAL INNOVATION

technological innovation The act of introducing a new device, method, or material for application to commercial or practical objectives.

In many industries, **technological innovation** is now the most important driver of competitive success. Firms in a wide range of industries rely on products developed within the past five years for almost one-third (or more) of their sales and profits. For example, at Johnson & Johnson, products developed within the last five years account for over 30 percent of sales, and sales from products developed within the past five years at 3M have hit as high as 45 percent in recent years.

The increasing importance of innovation is due in part to the globalization of markets. Foreign competition has put pressure on firms to continuously innovate in order to produce differentiated products and services. Introducing new products helps firms protect their margins, while investing in process innovation helps firms lower their costs. Advances in information technology also have played a role in speeding the pace of innovation. Computer-aided design and computer-aided manufacturing have made it easier and faster for firms to design and produce new products, while flexible manufacturing technologies have made shorter production runs economical and have reduced the importance of production economies of scale.[1] These technologies help firms develop and produce more product variants that closely meet the needs of narrowly defined customer groups, thus achieving differentiation from competitors. For example, in 2021, Toyota offered dozens of different passenger vehicle lines under the Toyota brand (e.g., Camry, Prius, Highlander, Yaris, Land Cruiser, and Tundra). Within each of the vehicle lines, Toyota also offered several different models (e.g., Camry L, Camry LE, Camry SE, and Camry Hybrid SE) with different features and at different price points. In total, Toyota offered over 200 car models ranging in price from $16,605 (for the Yaris sedan) to $85,665 (for the Land Cruiser), and seating anywhere from three passengers (e.g., Tacoma Regular Cab truck) to eight passengers (Sienna Minivan). On top of this, Toyota also produced a range of luxury vehicles under its Lexus brand. Similarly, in 2021, Samsung produced more than 43 unique smartphones, from the Galaxy A01 priced at roughly $100 to the Galaxy Fold priced at roughly $2000. Companies can use broad portfolios of product models to help ensure they can penetrate almost every conceivable market niche. While producing multiple product variations used to be expensive and time-consuming,

flexible manufacturing technologies now enable firms to seamlessly transition from producing one product model to the next, adjusting production schedules with real-time information on demand. Firms further reduce production costs by using common components in many of the models.

As firms such as Toyota, Samsung, and others adopt these new technologies and increase their pace of innovation, they raise the bar for competitors, triggering an industry-wide shift to shortened development cycles and more rapid new product introductions. The net results are greater market segmentation and rapid product obsolescence.[2] Product life cycles (the time between a product's introduction and its withdrawal from the market or replacement by a next-generation product) have become as short as 4 to 12 months for software, 12 to 24 months for computer hardware and consumer electronics, and 18 to 36 months for large home appliances.[3] This spurs firms to focus increasingly on innovation as a strategic imperative—a firm that does not innovate quickly finds its margins diminishing as its products become obsolete.

THE IMPACT OF TECHNOLOGICAL INNOVATION ON SOCIETY

If the push for innovation has raised the competitive bar for industries, arguably making success just that much more complicated for organizations, its net effect on society is more clearly positive. Innovation enables a wider range of goods and services to be delivered to people worldwide. It has made the production of food and other necessities more efficient, yielded medical treatments that improve health conditions, and enabled people to travel to and communicate with almost every part of the world. To get a real sense of the magnitude of the effect of technological innovation on society, look at Figure 1.1, which shows a timeline of some of the most important technological innovations developed over the last 200 years. Imagine how different life would be without these innovations!

gross domestic product (GDP) The total annual output of an economy as measured by its final purchase price.

The aggregate impact of technological innovation can be observed by looking at **gross domestic product (GDP)**. The gross domestic product of an economy is its total annual output, measured by final purchase price. Figure 1.2 shows the average GDP per capita (i.e., GDP divided by the population) for the world from 1980 to 2019. The figures have been converted into U.S. dollars and adjusted for inflation. As shown in the figure, the average world GDP per capita has risen pretty steadily since 1980. In a series of studies of economic growth conducted at the National Bureau of Economic Research, economists showed that the historic rate of economic growth in GDP could not be accounted for entirely by growth in labor and capital inputs. Economist Robert Merton Solow argued that this unaccounted-for residual growth represented technological change: Technological innovation increased the amount of output achievable from a given quantity of labor and capital. This explanation was not immediately accepted; many researchers attempted to explain the residual away in terms of measurement error, inaccurate price deflation, or labor improvement.

FIGURE 1.1
Timeline of Some of the Most Important Technological Innovations in the Last 200 Years

1800	-	1800—Electric battery
	-	1804—Steam locomotive
	-	1807—Internal combustion engine
	-	1809—Telegraph
	-	1817—Bicycle
1820	-	1821—Dynamo
	-	1824—Braille writing system
	-	1828—Hot blast furnace
	-	1831—Electric generator
	-	1836—Five-shot revolver
1840	-	1841—Bunsen battery (voltaic cell)
	-	1842—Sulfuric ether-based anesthesia
	-	1846—Hydraulic crane
	-	1850—Petroleum refining
	-	1856—Aniline dyes
1860	-	1862—Gatling gun
	-	1867—Typewriter
	-	1876—Telephone
	-	1877—Phonograph
	-	1878—Incandescent lightbulb
1880	-	1885—Light steel skyscrapers
	-	1886—Internal combustion automobile
	-	1887—Pneumatic tire
	-	1892—Electric stove
	-	1895—X-ray machine
1900	-	1902—Air conditioner (electric)
	-	1903—Wright biplane
	-	1906—Electric vacuum cleaner
	-	1910—Electric washing machine
	-	1914—Rocket
1920	-	1921—Insulin (extracted)
	-	1927—Television
	-	1928—Penicillin
	-	1936—First programmable computer
	-	1939—Atom fission
1940	-	1942—Aqua lung
	-	1943—Nuclear reactor
	-	1947—Transistor
	-	1957—Satellite
	-	1958—Integrated circuit
1960	-	1967—Portable handheld calculator
	-	1969—ARPANET (precursor to Internet)
	-	1971—Microprocessor
	-	1973—Mobile (portable cellular) phone
	-	1976—Supercomputer
1980	-	1981—Space shuttle (reusable)
	-	1987—Disposable contact lenses
	-	1989—High-definition television
	-	1990—World Wide Web protocol
	-	1996—Wireless Internet
2000	-	2002—CRISPR Cas9 gene editing
	-	2003—Map of human genome
	-	2008—Blockchain
	-	2010—Synthetic life form
	-	2017—SpaceX reusable rocket

But in each case the additional variables were unable to eliminate this residual growth component. A consensus gradually emerged that the residual did in fact capture technological change. Solow received a Nobel Prize for his work in 1981, and the residual became known as the Solow Residual.[4] While GDP has its shortcomings as a measure of standard of living, it does relate very directly to the amount of goods consumers can purchase. Thus, to the extent that goods improve quality of life, we can ascribe some beneficial impact of technological innovation.

Sometimes technological innovation results in negative **externalities**. Production technologies may create pollution that is harmful to the surrounding communities; agricultural and fishing technologies can result in erosion, elimination of natural habitats, and depletion of ocean stocks; medical technologies can result in unanticipated consequences such as antibiotic-resistant strains of bacteria or moral dilemmas regarding the use of genetic modification. However, technology is, in its purest essence, knowledge—knowledge to solve our problems and pursue our goals.[5] Technological innovation is thus the creation of new knowledge that is applied to practical problems. Sometimes this knowledge is applied to problems hastily, without full consideration of the consequences and alternatives, but overall it will probably serve us better to have more knowledge than less.

externalities
Costs (or benefits) that are borne (or reaped) by individuals other than those responsible for creating them. Thus, if a business emits pollutants in a community, it imposes a negative externality on the community members; if a business builds a park in a community, it creates a positive externality for community members.

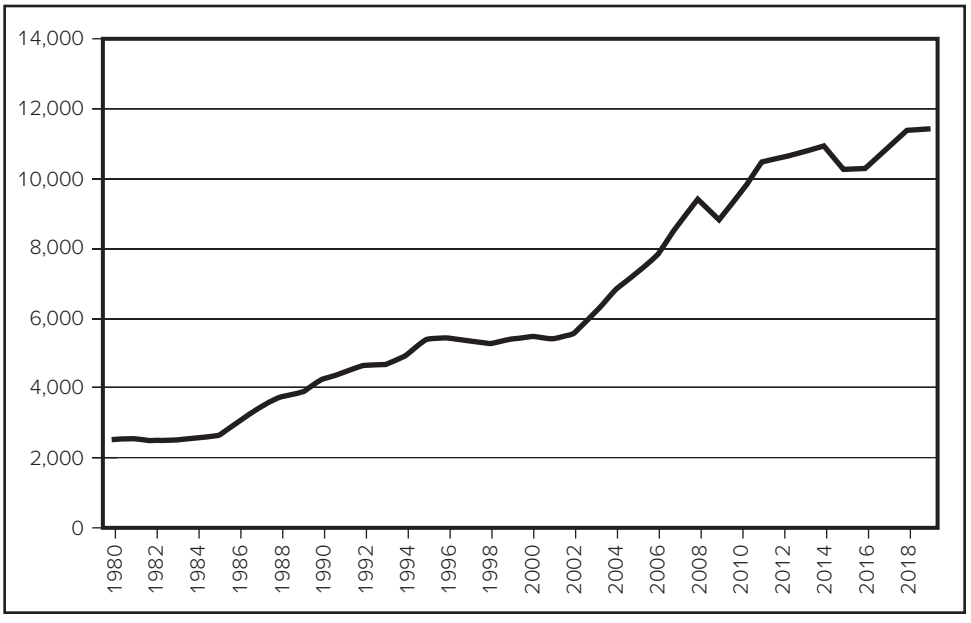


FIGURE 1.2
World Gross Domestic Product per Capita, 1980–2019 (in real 2019 U.S. dollars)

Source: "World GDP Per Capita 1960–2021," Macrotrends, https://www.macrotrends.net/countries/WLD/world/gdp-per-capita.

INNOVATION BY INDUSTRY: THE IMPORTANCE OF STRATEGY

As will be shown in Chapter Two, the majority of effort and money invested in technological innovation comes from industrial firms. However, in the frenetic race to innovate, many firms charge headlong into new product development without clear strategies or well-developed processes for choosing and managing projects. Such firms often initiate more projects than they can effectively support, choose projects that are a poor fit with the firm's resources and objectives, and suffer long development cycles and high project failure rates as a consequence (see the accompanying Research Brief for a recent study of the length of new product development cycles). While innovation is popularly depicted as a freewheeling process that is unconstrained by rules and plans, study after study has revealed that successful innovators have clearly defined innovation strategies and management processes.[6]

The Innovation Funnel

Most innovative ideas do not become successful new products. Many studies suggest that only one out of several thousand ideas results in a successful new product: Many projects do not result in technically feasible products and, of those that do, many fail to earn a commercial return. According to a 2012 study by the Product Development and Management Association, only about one in nine projects that are initiated is successful, and of those that make it to the point of being launched to the market, only about half earn a profit.[7] Furthermore, many ideas are sifted through and abandoned before a project is even formally initiated. According to one study that combined data from

Research Brief How Long Does New Product Development Take?[a]

In a large-scale survey administered by the Product Development and Management Association (PDMA), in 2012, researchers examined the length of time it took firms to develop a new product from initial concept to market introduction. The study divided new product development projects into categories representing their degree of innovativeness: "radical" projects, "more innovative" projects, and "incremental" projects. On average, *incremental* projects took only 33 weeks from concept to market introduction. *More innovative* projects took significantly longer, clocking in at 57 weeks. The development of *radical* products or technologies took the longest, averaging 82 weeks. The study also found that on average, for *more innovative* and *radical* projects, firms reported significantly shorter cycle times than those reported in the previous PDMA surveys conducted in 1995 and 2004.

[a] Adapted from Stephen K. Markham and Hyunjung Lee, "Product Development and Management Association's 2012 Comparative Performance Assessment Study," *Journal of Product Innovation Management* 30, no. 3 (2013): 408–29.

prior studies of innovation success rates with data on patents, venture capital funding, and surveys, it takes about 3000 raw ideas to produce one significantly new and successful commercial product.[8] The pharmaceutical industry demonstrates this well—only one out of every 5000 compounds makes it to the pharmacist's shelf, and only one-third of those will be successful enough to recoup their R&D costs.[9] Furthermore, most studies indicate that it costs at least $1.4 billion and a decade of research to bring a new Food and Drug Administration (FDA)–approved pharmaceutical product to market![10] The innovation process is thus often conceived of as a funnel, with many potential new product ideas going in the wide end, but very few making it through the development process (see Figure 1.3).

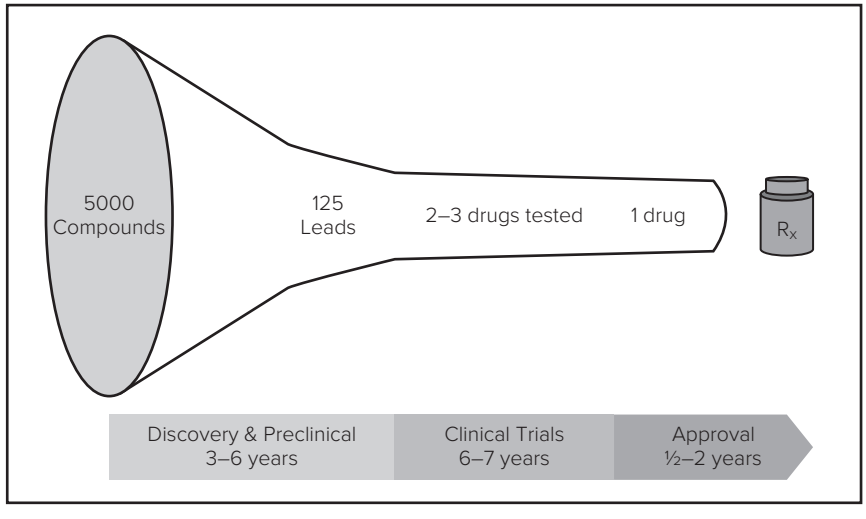


FIGURE 1.3
The New Product Development Funnel in Pharmaceuticals

The Strategic Management of Technological Innovation

Improving a firm's innovation success rate requires a well-crafted strategy. A firm's innovation projects should align with its resources and objectives, leveraging its core competencies and helping it achieve its strategic intent. A firm's organizational structure and control systems should encourage the generation of innovative ideas while also ensuring efficient implementation. A firm's new product development process should maximize the likelihood of projects being both technically and commercially successful. To achieve these things, a firm needs (*a*) an in-depth understanding of the dynamics of innovation, (*b*) a well-crafted innovation strategy, and (*c*) well-designed processes for implementing the innovation strategy. We will cover each of these in turn (see Figure 1.4).

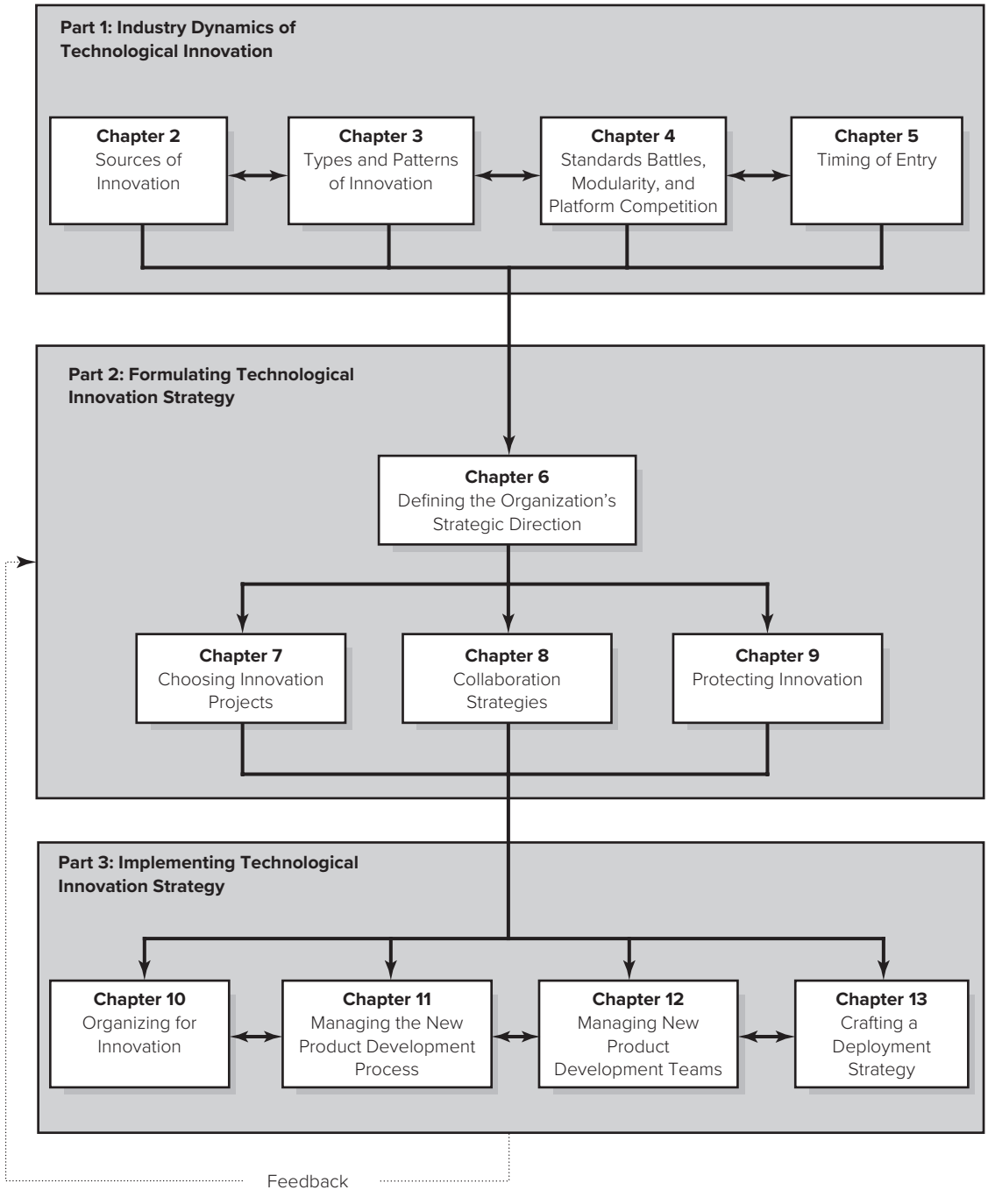
In Part One, we will cover the foundations of technological innovation, gaining an in-depth understanding of how and why innovation occurs in an industry, and why some innovations rise to dominate others. First, we will look at the sources of innovation in Chapter Two. We will address questions such as: Where do great ideas come from? How can firms harness the power of individual creativity? What role do customers, government organizations, universities, and alliance networks play in creating innovation? In this chapter, we will first explore the role of creativity in the generation of novel and useful ideas. We then look at various sources of innovation, including the role of individual inventors, firms, publicly sponsored research, and collaborative networks.

In Chapter Three, we will review models of types of innovation (such as radical versus incremental and architectural versus modular) and patterns of innovation (including s-curves of technology performance and diffusion, and technology cycles). We will address questions such as: Why are some innovations much harder to create and implement than others? Why do innovations often diffuse slowly even when they appear to offer a great advantage? What factors influence the rate at which a technology tends to improve over time? Familiarity with these types and patterns of innovation will help us distinguish how one project is different from another and the underlying factors that shape the project's likelihood of technical or commercial success.

In Chapter Four, we will turn to the particularly interesting dynamics that emerge in industries characterized by network externalities and other sources of increasing returns that can lead to standards battles and winner-take-all markets. We will address questions such as: Why do some industries choose a single dominant standard rather than enabling multiple standards to coexist? What makes one technological innovation rise to dominate all others, even when other seemingly superior technologies are offered? How can a firm avoid being locked out? Is there anything a firm can do to influence the likelihood of its technology becoming the dominant design? When are platform ecosystems likely to displace other forms of competition in an industry?

In Chapter Five, we will discuss the impact of entry timing, including first-mover advantages, first-mover *dis*advantages, and the factors that will determine the firm's optimal entry strategy. This chapter will address such questions as: What are the advantages and disadvantages of being first to market, early but not first, and late? What determines the optimal timing of entry for a new innovation? This chapter reveals a number of consistent patterns in how timing of entry impacts innovation success, and it

FIGURE 1.4
The Strategic Management of Technological Innovation



outlines what factors will influence a firm's optimal timing of entry, thus beginning the transition from understanding the dynamics of technological innovation to formulating technology strategy.

In Part Two, we will turn to formulating technological innovation strategy. Chapter Six reviews the basic strategic analysis tools managers can use to assess the firm's current position and define its strategic direction for the future. This chapter will address such questions as: What are the firm's sources of sustainable competitive advantage? Where in the firm's value chain do its strengths and weaknesses lie? What are the firm's core competencies, and how should it leverage and build upon them? What is the firm's strategic intent—that is, where does the firm want to be 10 years from now? Only after the firm has thoroughly appraised where it is currently can it formulate a coherent technological innovation strategy for the future.

In Chapter Seven, we will examine a variety of methods of choosing innovation projects. These include quantitative methods such as discounted cash flow and options valuation techniques, qualitative methods such as screening questions and balancing the research and development portfolio, as well as methods that combine qualitative and quantitative approaches such as conjoint analysis and data envelopment analysis. Each of these methods has its advantages and disadvantages, leading many firms to use a multiple-method approach to choosing innovation projects. This chapter also includes some of the sources of funding an innovative startup might use to finance their projects.

In Chapter Eight, we will examine collaboration strategies for innovation. This chapter addresses questions such as: Should the firm partner on a particular project or go solo? How does the firm decide which activities to do in-house and which to access through collaborative arrangements? If the firm chooses to work with a partner, how should the partnership be structured? How does the firm choose and monitor partners? We will begin by looking at the reasons a firm might choose to go solo versus working with a partner. We then will look at the pros and cons of various partnering methods, including joint ventures, alliances, licensing, outsourcing, and participating in collaborative research organizations. The chapter also reviews the factors that should influence partner selection and monitoring.

In Chapter Nine, we will address the options the firm has for appropriating the returns to its innovation efforts. We will look at the mechanics of patents, copyright, trademarks, and trade secrets. We will also address such questions as: Are there ever times when it would benefit the firm to not protect its technological innovation so vigorously? How does a firm decide between a wholly proprietary, wholly open, or partially open strategy for protecting its innovation? When will open strategies have advantages over wholly proprietary strategies? This chapter examines the range of protection options available to the firm, and the complex series of trade-offs a firm must consider in its protection strategy.

In Part Three, we will turn to implementing the technological innovation strategy. This begins in Chapter Ten with an examination of how the organization's size and structure influence its overall rate of innovativeness. The chapter addresses such questions as: Do bigger firms outperform smaller firms at innovation? How do formalization, standardization, and centralization impact the likelihood of generating innovative ideas and the organization's ability to implement those ideas quickly and efficiently? Is it possible to achieve creativity and flexibility at the same time as efficiency and

reliability? How does the firm's culture influence its innovation? How do multinational firms decide where to perform their development activities? How do multinational firms coordinate their development activities toward a common goal when the activities occur in multiple countries? This chapter examines how organizations can balance the benefits and trade-offs of flexibility, economies of scale, standardization, centralization, and tapping local market knowledge.

In Chapter Eleven, we will review a series of "best practices" that have been identified in managing the new product development process. This includes such questions as: Should new product development processes be performed sequentially or in parallel? What are the advantages and disadvantages of using project champions? What are the benefits and risks of involving customers and/or suppliers in the development process? What tools can the firm use to improve the effectiveness and efficiency of its new product development processes? How does the firm assess whether its new product development process is successful? This chapter provides an extensive review of methods that have been developed to improve the management of new product development projects and to measure their performance.

Chapter Twelve builds on the previous chapter by illuminating how team composition and structure will influence project outcomes. This chapter addresses questions such as: How big should teams be? What are the advantages and disadvantages of choosing highly diverse team members? Do teams need to be colocated? When should teams be full time and/or permanent? What type of team leader and management practices should be used for the team? This chapter provides detailed guidelines for constructing new product development teams that are matched to the type of new product development project under way.

Finally, in Chapter Thirteen, we will look at innovation deployment strategies. This chapter will address such questions as: How do we accelerate the adoption of the technological innovation? How do we decide whether to use licensing or OEM agreements? Does it make more sense to use penetration pricing or a market-skimming price? When should we sell direct versus using intermediaries? What strategies can the firm use to encourage distributors and complementary goods providers to support the innovation? What are the advantages and disadvantages of major marketing methods? This chapter complements traditional marketing, distribution, and pricing courses by looking at how a deployment strategy can be crafted that especially targets the needs of a new technological innovation.

Summary of Chapter

1. Technological innovation is now often the single most important competitive driver in many industries. Many firms receive more than one-third of their sales and profits from products developed within the past five years.
2. The increasing importance of innovation has been driven largely by the globalization of markets and the advent of advanced technologies that enable more rapid product design and allow shorter production runs to be economically feasible.
3. Technological innovation has a number of important effects on society, including fostering increased GDP, enabling greater communication and mobility, and improving medical treatments.

4. Technological innovation may also pose some negative externalities, including pollution, resource depletion, and other unintended consequences of technological change.
5. While government plays a significant role in innovation, industry provides the majority of R&D funds that are ultimately applied to technological innovation.
6. Successful innovation requires an in-depth understanding of the dynamics of innovation, a well-crafted innovation strategy, and well-developed processes for implementing the innovation strategy.

Discussion Questions

1. Why is innovation so important for firms to compete in many industries?
2. What are some advantages and disadvantages of technological innovation?
3. Why do you think so many innovation projects fail to generate an economic return?

Suggested Further Reading

Classics

Edwin Mansfield, "Contributions of R and D to Economic Growth in the United States," *Science* CLXXV (1972): 477–86.

Joseph Schumpeter, *The Theory of Economic Development* (1911; English translation, Cambridge, MA: Harvard University Press, 1936).

Kenneth J. Arrow, "Economic welfare and the allocation of resources for inventions," in *The Rate and Direction of Inventive Activity: Economic and Social Factors*, ed. R. Nelson (Princeton, NJ: Princeton University Press, 1962), 609–25.

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Frank R. Lichtenberg, "Pharmaceutical Innovation and Longevity Growth in 30 Developing and High-Income Countries, 2000–2009," *Health Policy and Technology* 3 (2014): 36–58.

Melissa A. Schilling, "Towards Dynamic Efficiency: Innovation and Its Implications for Antitrust," *Antitrust Bulletin* 60, no. 3 (2015): 191–207.

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Endnotes

1. James P. Womack, Daniel T. Jones, and Daniel Roos, *The Machine That Changed the World* (New York: Rawson Associates, 1990).
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5. Herbert A. Simon, "Technology and Environment," *Management Science* 19 (1973), 1110–21.
6. Shona Brown and Kathleen Eisenhardt, "The Art of Continuous Change: Linking Complexity Theory and Time-Paced Evolution in Relentlessly Shifting Organizations," *Administrative Science Quarterly* 42 (1997): 1–35; K. Clark and T. Fujimoto, *Product Development Performance* (Boston: Harvard Business School Press, 1991); Robert Cooper, "Third Generation New Product Processes," *Journal of Product Innovation Management* 11 (1994): 3–14; D. Doughery, "Reimagining the Differentiation and Integration of Work for Sustained Product Innovation," *Organization Science* 12 (2001): 612–31; and Melissa A. Schilling and Charles William Leslie Hill, "Managing the New Product Development Process: Strategic Imperatives," *Academy of Management Executive* 12, no. 3 (1998): 67–81.
7. Stephen K. Markham and Hyunjung Lee, "Product Development and Management Association's 2012 comparative performance assessment study," *Journal of Product Innovation Management* 30, no. 3 (2013): 408–29.
8. Greg Stevens and James Burley, "3,000 Raw Ideas Equals 1 Commercial Success!" *Research Technology Management* 40, no. 3 (1997): 16–27.
9. Standard & Poor's Industry Surveys, Pharmaceutical Industry, 2008.
10. Joseph A. DiMasi, Henry G. Grabowski, and Ronald W. Hansen, "Innovation in the Pharmaceutical Industry: New Estimates of R&D Costs," *Journal of Health Economics* 47 (May 2016): 20–33.

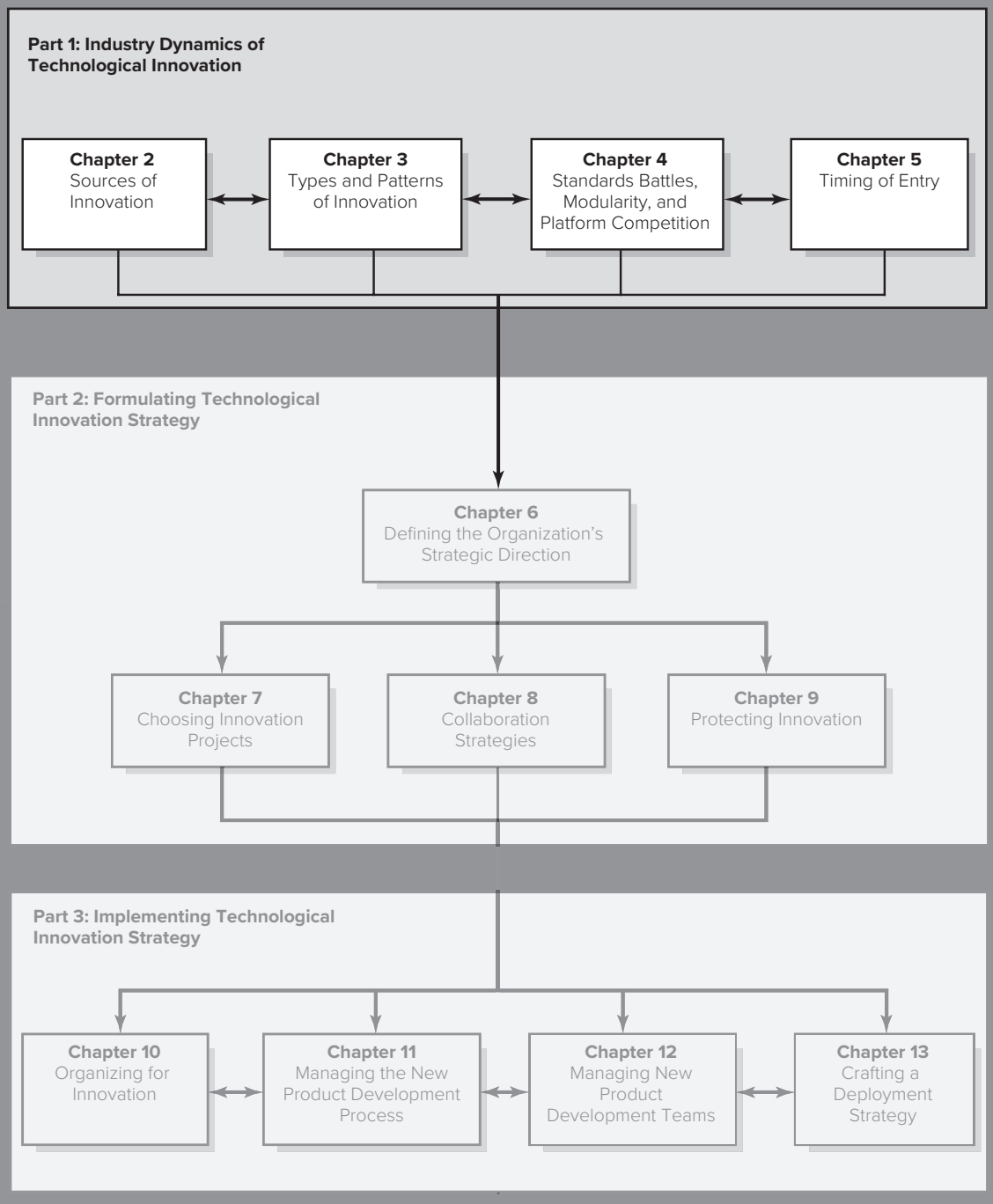
Part **One**

Industry Dynamics of Technological Innovation

In this section, we will explore the industry dynamics of technological innovation, including:

- The sources from which innovation arises, including the roles of individuals, organizations, government institutions, and networks.
- The types of innovations and common industry patterns of technological evolution and diffusion.
- The factors that determine whether industries experience pressure to select a dominant design, and what drives which technologies to dominate others.
- The effects of timing of entry, and how firms can identify (and manage) their entry options.

This section will lay the foundation that we will build upon in Part Two, Formulating Technological Innovation Strategy.


Part 1: Industry Dynamics of Technological Innovation
Chapter 2 Sources of Innovation
Chapter 3 Types and Patterns of Innovation
Chapter 4 Standards Battles, Modularity, and Platform Competition
Chapter 5 Timing of Entry
Part 2: Formulating Technological Innovation Strategy
Chapter 6 Defining the Organization's Strategic Direction
Chapter 7 Choosing Innovation Projects
Chapter 8 Collaboration Strategies
Chapter 9 Protecting Innovation
Part 3: Implementing Technological Innovation Strategy
Chapter 10 Organizing for Innovation
Chapter 11 Managing the New Product Development Process
Chapter 12 Managing New Product Development Teams
Chapter 13 Crafting a Deployment Strategy
Feedback

Chapter **Two**

Sources of Innovation

The Rise of Cultured Meat[a]

In late 2017, Microsoft founder Bill Gates and a group of other high-powered investors—who comprise Breakthrough Energy Ventures, such as Amazon's Jeff Bezos, Alibaba's Jack Ma, and Virgin's Richard Branson—announced their intention to fund a San Francisco–based start-up called Memphis Meats with an unusual business plan: It grew "clean" meat using stem cells, eliminating the need to breed or slaughter animals. The company had already produced beef, chicken, and duck, all grown from cells.[b]

There were many potential advantages of growing meat without animals. First, growth in the demand for meat was skyrocketing due to both population growth and development. When developing countries become wealthier, they increase their meat consumption. While humanity's population had doubled since 1960, consumption of animal products had risen fivefold and was still increasing. Many scientists and economists had begun to warn of an impending "meat crisis." Even though plant protein substitutes like soy and pea protein had gained enthusiastic followings, the rate of animal protein consumption had continued to rise. This suggested that meat shortages were inevitable unless radically more efficient methods of production were developed.

Large-scale production of animals also had a massively negative effect on the environment. The worldwide production of cattle, for example, resulted in a larger emissions of greenhouse gases than the collective effect of the world's automobiles. Animal production is also extremely water intensive: To produce each chicken sold in a supermarket, for example, requires more than 1000 gallons of water, and each egg requires 50 gallons. Each gallon of cow's milk required 900 gallons of water. A study by Oxford University indicated that meat grown from cells would produce up to 96 percent lower greenhouse gas emissions, use 45 percent less energy, 99 percent less land, and 96 percent less water.[c]

Scientists also agreed that producing animals for consumption was simply inefficient. Estimates suggested, for example, that it required roughly 23 calories worth of inputs to produce one calorie of beef. Cultured meat promised to bring that ratio down to three calories of inputs to produce a calorie of beef—more than seven times greater efficiency. Cultured meat also would not contain

antibiotics, steroids, or bacteria such as *E. coli*—it was literally "cleaner," and that translated into both greater human health and lower perishability.

The Development of Clean Meat

In 2004, Jason Matheny, a 29-year-old recent graduate from the John Hopkins Public Health program decided to try to tackle the problems with production of animals for food. Though Matheny was a vegetarian himself, he realized that convincing enough people to adopt a plant-based diet to slow down the meat crisis was unlikely. As he noted, "You can spend your time trying to get people to turn their lights out more often, or you can invent a more efficient light bulb that uses far less energy even if you leave it on. What we need is an enormously more efficient way to get meat."[d]

Matheny founded a nonprofit organization called New Harvest that would be dedicated to promoting research into growing real meat without animals. He soon discovered that a Dutch scientist, Willem van Eelen was exploring how to culture meat from animal cells. Van Eelen had been awarded the first patent on a cultured meat production method in 1999. However, the eccentric scientist had not had much luck in attracting funding to his project, nor in scaling up his production. Matheny decided that with a little prodding, the Dutch government might be persuaded to make a serious investment in the development of meat-culturing methods. He managed to get a meeting with the Netherland's minister of agriculture where he made his case. Matheny's efforts paid off: The Dutch government agreed to invest two million euros in exploring methods of creating cultured meat at three different universities.

By 2005, cultured meat was starting to gather attention. The journal *Tissue Engineering* published an article entitled "In Vitro-Cultured Meat Production," and in the same year, the *New York Times* profiled cultured meat in its annual "Ideas of the Year." However, while governments and universities were willing to invest in the basic science of creating methods of producing cultured meat, they did not have the capabilities and assets needed to bring it to commercial scale. Matheny knew that to make cultured meat a mainstream reality, he would need to attract the interest of large agribusiness firms.

Matheny's initial talks with agribusiness firms did not go well. Though meat producers were open to the idea conceptually, they worried that consumers would balk at cultured meat and perceive it as unnatural. Matheny found this criticism frustrating; after all, flying in airplanes, using air conditioning, or eating meat pumped full of steroids to accelerate its growth were also unnatural.

Progress was slow. Matheny took a job at the Intelligence Advanced Research Projects Activity (IARPA) of the U.S. Federal Government while continuing to run New Harvest on the side. Fortunately, others were also starting to realize the urgency of developing alternative meat production methods.

Enter Sergey Brin of Google

In 2009, the foundation of Sergey Brin, cofounder of Google, contacted Matheny to learn more about cultured meat technologies. Matheny referred

Brin's foundation to Dr. Mark Post at Maastricht University, one of the leading scientists funded by the Dutch government's cultured meat investment. Post had succeeded in growing mouse muscles in vitro and was certain his process could be replicated with the muscles of cows, poultry, and more. As he stated, "It was so clear to me that we could do this. The science was there. All we needed was funding to actually prove it, and now here was a chance to get what was needed."[e] It took more than a year to work out the details, but in 2011, Brin offered Post roughly three quarters of a million dollars to prove his process by making two cultured beef burgers, and Post's team set about meeting the challenge.

In early 2013, the moment of truth arrived: Post and his team had enough cultured beef to do a taste test. They fried up a small burger and split it into thirds to taste. It tasted like meat. Their burger was 100 percent skeletal muscle and they knew that for commercial production they would need to add fat and connective tissue to more closely replicate the texture of beef, but those would be easy problems to solve after passing this milestone. The press responded enthusiastically, and the *Washington Post* ran an article headlined, "Could a Test-Tube Burger Save the Planet?"[f]

Going Commercial

In 2015, Uma Valeti, a cardiologist at the Mayo Clinic founded his own cultured-meat research lab at the University of Minnesota. "I'd read about the inefficiency of meat-eating compared to a vegetarian diet, but what bothered me more than the wastefulness was the sheer scale of suffering of the animals."[g] As a heart doctor, Valeti also believed that getting people to eat less meat could improve human health: "I knew that poor diets and the unhealthy fats and refined carbs that my patients were eating were killing them, but so many seemed totally unwilling to eat less or no meat. Some actually told me they'd rather live a shorter life than stop eating the meats they loved." Valeti began fantasizing about a best-of-both-worlds alternative—a healthier and kinder meat. As he noted, "The main difference I thought I'd want for this meat I was envisioning was that it'd have to be leaner and more protein-packed than a cut of supermarket meat, since there's a large amount of saturated fat in that meat. . . . Why not have fats that are proven to be better for health and longevity, like omega-3s? We want to be not just like conventional meat but healthier than conventional meat."[h]

Valeti was nervous about leaving his successful position as a cardiologist—after all, he had a wife and two children to help support. However, when he sat down to discuss it with his wife (a pediatric eye surgeon), she said, "Look, Uma. We've been wanting to do this forever. I don't ever want us to look back on why we didn't have the courage to work on an idea that could make this world kinder and better for our children and their generation."[i] And thus Valeti's company, which would later be named Memphis Meats, was born.

Building on Dr. Post's achievement, Valeti's team began experimenting with ways to get just the right texture and taste. After much trial and error, and a growing number of patents, they hosted their first tasting event in

December 2015. On the menu: a meatball. This time the giant agribusiness firms took notice.

At the end of 2016, Tyson Foods, the world's largest meat producer, announced that it would invest $150 million in a venture capital fund that would develop alternative proteins, including meat grown from self-reproducing cells. In August of 2017, agribusiness giant Cargill announced it was investing in Memphis Meats, and a few months later in early 2018, Tyson Foods also pledged investment.

That first meatball cost $1200; to make cultured meat a commercial reality required bringing costs down substantially. But analysts were quick to point out that the first iPhone had cost $2.6 billion in R&D—much more than the first cultured meats. Scale and learning curve efficiencies would drive that cost down. Valeti had faith that the company would soon make cultured meat not only competitive with traditional meat, but also more affordable. Growing meat rather than whole animals had, after all, inherent efficiency advantages.

In December of 2020, cultured chicken made by Eat Just became the first cultured meat product in the world to be approved for commercial sale when it was approved by the Singapore Food Agency. The chicken was expected to go on sale in Singaporean restaurants in early 2021. Eat Just had found earlier success by selling Just Eggs (made from mung beans). Its founder, Josh Tetrick, was also in talks with U.S. and European food regulators and noted, "I would imagine what will happen is the U.S., Western Europe and others will see what Singapore has been able to do, the rigors of the framework that they put together. And I would imagine that they will try to use it as a template to put their own framework together."[j]

Some skeptics believed the bigger problem was not production economies or regulatory approval, but consumer acceptance: Would people be willing to eat meat grown without animals? Sergey Brin, Bill Gates, Jeff Bezos, Jack Ma, and Richard Branson were willing to bet that they would. As Branson stated in 2017, "I believe that in 30 years or so we will no longer need to kill any animals and that all meat will either be clean or plant-based, taste the same and also be much healthier for everyone."[k] Bruce Friedrich, executive director of the nonprofit The Good Food Institute agrees, noting "As nations race to divorce meat production from industrial animal agriculture, countries that delay their investment in this bright food future risk getting left behind."[l]

Discussion Questions

1. What were the potential advantages of developing cultured meat? What were the challenges of developing it and bringing it to market?
2. What kinds of organizations were involved in developing cultured meat? What were the different resources that each kind of organization brought to the innovation and what were their motives?
3. What are the challenges to gaining wide market acceptance of cultured meat, and how could these organizations facilitate that? Can you think of other products or services that faced similar adoption challenges?

[a] Adapted from a New York University teaching case by Paul Shapiro and Melissa Schilling.
[b] Zack Friedman, "Why Bill Gates and Richard Branson Invested in 'Clean' Meat," *Forbes*, August 2017.
[c] Hanna L. Tuomisto and M. Joost Teixeira de Mattos, "Environmental Impacts of Cultured Meat Production," *Environmental Science and Technology* 14(2011): 6117–2123.
[d] Paul Shapiro, *Clean Meat: How Growing Meat without Animals Will Revolutionize Dinner and the World* (New York: Gallery Books, 2018), 35.
[e] Paul Shapiro, *Clean Meat: How Growing Meat without Animals Will Revolutionize Dinner and the World* (New York: Gallery Books, 2018), 60.
[f] "Could a Test-Tube Burger Save the Planet?" *Washington Post,* August 5, 2013.
[g] Paul Shapiro, *Clean Meat: How Growing Meat without Animals Will Revolutionize Dinner and the World* (New York: Gallery Books, 2018), 113.
[h] Paul Shapiro, *Clean Meat: How Growing Meat without Animals Will Revolutionize Dinner and the World* (New York: Gallery Books, 2018), 115.
[i] Paul Shapiro, *Clean Meat: How Growing Meat without Animals Will Revolutionize Dinner and the World* (New York: Gallery Books, 2018), 118.
[j] Ryan W. Miller, "Lab-Grown 'Chicken Bites'? Cultured Meat Product Gets World's First Regulatory Approval," *USA Today*, December 2, 2020.
[k] Zack Friedman, "Why Bill Gates and Richard Branson Invested in 'Clean' Meat," *Forbes*, August 2017.
[l] Ryan W. Miller, "Lab-Grown 'Chicken Bites'? Cultured Meat Product Gets World's First Regulatory Approval," *USA Today*, December 2, 2020.

OVERVIEW

innovation
The practical implementation of an idea into a new device or process.

Innovation can arise from many different sources. It can originate with individuals, as in the familiar image of the lone inventor or users who design solutions for their own needs. Innovation can also come from the research efforts of universities, government laboratories and incubators, or private nonprofit organizations. One primary engine of innovation is firms. Firms are well suited to innovation activities because they typically have greater resources than individuals and a management system to marshal those resources toward a collective purpose. Firms also face strong incentives to develop differentiating new products and services, which may give them an advantage over nonprofit or government-funded entities.

An even more important source of innovation, however, does not arise from any one of these sources, but rather the linkages between them. Networks of innovators that leverage knowledge and other resources from multiple sources are one of the most powerful agents of technological advance.[1] We can thus think of sources of innovation as composing a complex system wherein any particular innovation may emerge primarily from one or more components of the system or the linkages between them (see Figure 2.1).

In the sections that follow, we will first consider the role of creativity as the underlying process for the generation of novel and useful ideas. We will then consider how creativity is transformed into innovative outcomes by the separate components of the innovation system (individuals, firms, etc.), and through the linkages between different components (firms' relationships with their customers, technology transfer from universities to firms, etc.).

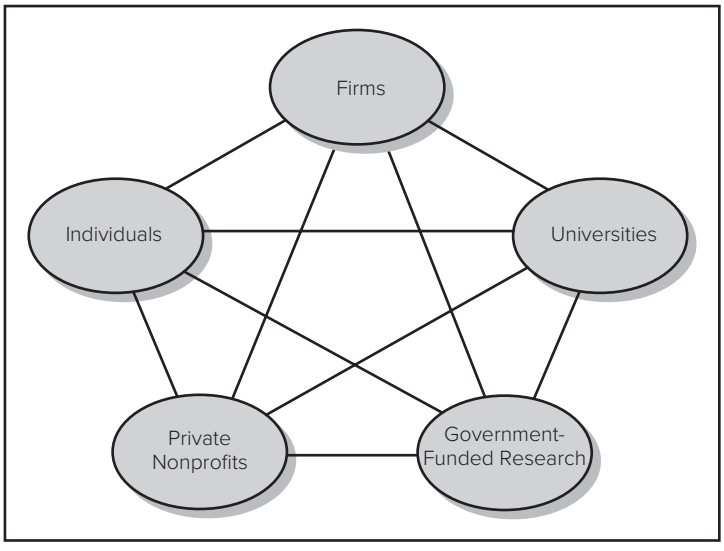


FIGURE 2.1
Sources of Innovation as a System

CREATIVITY

idea
Something imagined or pictured in the mind.

creativity
The ability to produce novel and useful work.

Innovation begins with the generation of new **ideas**. The ability to generate new and useful ideas is termed creativity. **Creativity** is defined as the ability to produce work that is useful and novel. Novel work must be different from work that has been previously produced and surprising in that it is not simply the next logical step in a series of known solutions.[2] The degree to which a product is novel is a function both of how different it is from prior work (e.g., a minor deviation versus a major leap) and of the audience's prior experiences.[3] A product could be novel to the person who made it, but known to most everyone else. In this case, we would call it reinvention. A product could be novel to its immediate audience, yet be well known somewhere else in the world. The most creative works are novel at the individual producer level, the local audience level, and the broader societal level.[4]

Individual Creativity

An individual's creative ability is a function of his or her *intellectual abilities, knowledge, personality, motivation*, and *environment*.

The most important *intellectual abilities* for creative thinking include intelligence, memory, the ability to look at problems in unconventional ways, the ability to analyze which ideas are worth pursuing and which are not, and the ability to articulate those ideas to others and convince others that the ideas are worthwhile. One important intellectual ability for creativity is a person's ability to let their mind engage in a visual mental activity termed *primary process thinking*.[5] Because of its unstructured nature, primary process thinking can result in combining ideas that are not typically related, leading to what has been termed *remote associations* or *divergent thinking*. Sigmund Freud noted that primary process thinking was most likely to occur just

before sleep or while dozing or daydreaming; others have observed that it might also be common when distracted by physical exercise, music, or other activities. Creative people may make their minds more open to remote associations and then mentally sort through these associations, selecting the best for further consideration. Having excellent working memory is useful here too—individuals with excellent working memory may be more likely or more able to search longer paths through the network of associations in their mind, enabling them to arrive at a connection between two ideas or facts that seem unexpected or strange to others.[6] A connection that appears to be random may not be random at all—it is just difficult for other people to see the association because they are not following as long of a chain of associations.

Consistent with this, studies by professors Mathias Benedek and Aljoscha Neubauer found that highly creative people usually follow the same association paths as less creative people—but they do so with such greater speed that they exhaust the common associations sooner, permitting them to get to less common associations earlier than others would.[7] Benedek and Neubauer's research argues that highly creative people's speed of association is due to exceptional working memory and executive control. In other words, the ability to hold many things in one's mind simultaneously and maneuver them with great facileness enables a person to rapidly explore many possible associations.[8]

The impact of *knowledge* on creativity is somewhat double-edged. If an individual has too little knowledge of a field, he or she is unlikely to understand it well enough to contribute meaningfully to it. On the other hand, if an individual knows a field too well, that person can become trapped in the existing logic and paradigms, preventing him or her from coming up with solutions that require an alternative perspective. Thus, an individual with only a moderate degree of knowledge of a field might be able to produce more creative solutions than an individual with extensive knowledge of the field, and breakthrough innovations are often developed by outsiders to a field.[9]

Consider, for example, Elon Musk. Elon Musk developed a city search Web portal called Zip2 in college, then founded an Internet financial payments company that merged with a rival and developed the PayPal financial payment system. Then after selling PayPal, Musk decided to found SpaceX to develop reusable rockets, and also became part of the founding team of Tesla Motors, an electric vehicle company. Tesla subsequently acquired Solar City (a solar panel company that Elon Musk had helped his cousins create) and diversified into energy storage and more. Musk crosses boundaries because he enjoys tackling new, difficult problems. He has been able to be successful in a wide range of industries in part because he challenges the traditional models in those industries.[10] For example, SpaceX was able to dramatically decrease the price of rocket components by building them in-house, and Solar City was able to dramatically increase solar panel adoption by offering a business model based on leasing that gave customers the option of putting no money down and paying for the panels with part of their energy savings.

Another great example is provided by Gavriel Iddan, a guided missile designer for the Israeli military who invented a revolutionary way to allow doctors to see inside a patient's gastrointestinal system. The traditional approach for obtaining images inside the gut is a camera on the end of a long flexible rod. This method is quite uncomfortable, and cannot reach large portions of the small intestine, but it

was the industry standard for many decades. Most gastroenterologists have invested in significant training to use endoscopic tools, and many have also purchased endoscopic equipment for their clinics. Not surprisingly then, most innovation in this domain has focused on incremental improvements in the rod, cameras, and imaging software. Iddan, however, approached the problem of viewing the inside of the gut like a guided missile designer—not a gastroenterologist. He did not have the same assumptions about the need to control the camera with a rod, nor to transmit images with a wire. Instead, he invented a capsule (called the PillCam) with a power source, a light source, and two tiny cameras that the patient can swallow. The patient then goes about her day while the camera pill broadcasts images to a video pack worn by the patient. Roughly eight hours later, the patient returns to the doctor's office to have the images read by a software algorithm that can identify any locations of bleeding (the camera pill exits naturally). The PillCam has proven to be safer and less expensive than traditional endoscopy (the PillCam costs less than $500), and it is dramatically more comfortable. For patients, the camera pill was a no brainer; getting doctors to adopt it has been slower because of their existing investment and familiarity with endoscopy. The PillCam is now sold in more than 60 countries, and several companies now offer competing products. The camera pill is a remarkable solution to a difficult problem, and it is easy to see why it came from an outsider, rather than an endoscope producer.[11]

Outsiders often face resistance and skepticism. People tend to discount generalists and are suspicious of people who engage in activities that seem inconsistent with their identity. Outsiders like Musk, however, bring an advantage that insiders and industry veterans often lack. They aren't trapped by the paradigms and assumptions that have long become calcified in industry veterans, nor do they have the existing investments in tools, expertise, or supplier and customer relationships that make change difficult and unappealing.

The *personality* trait most often associated with creativity is "openness to experience."[12] Openness to experience reflects an individual's use of active imagination, aesthetic sensitivity (e.g., the appreciation for art and literature), attentiveness to emotion, a preference for variety, and intellectual curiosity. It is assessed by asking individuals to rate their degree of agreement or disagreement with statements such as "I have a vivid imagination," "I enjoy hearing new ideas," "I have a rich vocabulary," "I rarely look for deeper meaning in things" (reversed), "I enjoy going to art museums," "I avoid philosophical discussions" (reversed), "I enjoy wild flights of fantasy," and more. Individuals who score high on the openness to experience dimension tend to have great intellectual curiosity, are interested in unusual ideas, and are willing to try new things.

Intrinsic *motivation* has also been shown to be very important for creativity.[13] That is, individuals are more likely to be creative if they work on things they are genuinely interested in and enjoy. In fact, several studies have shown that creativity can be undermined by providing extrinsic motivation such as money or awards.[14] This raises serious questions about the role played by idea collection systems in organizations that offer monetary rewards for ideas. On the one hand, such extrinsic rewards could derail intrinsic motivation. On the other hand, if the monetary rewards are small, such systems may be primarily serving to invite people to offer ideas, which is

a valuable signal about the culture of the firm. More research is needed in this area to know exactly what kind of solicitation for ideas, if any, is most effective.

Finally, to fully unleash an individual's creative potential usually requires a supportive *environment* with time for the individual to explore their ideas independently, tolerance for unorthodox ideas, a structure that is not overly rigid or hierarchical, and decision norms that do not require consensus.[15]

Organizational Creativity

The creativity of the organization is a function of creativity of the individuals within the organization and a variety of social processes and contextual factors that shape the way those individuals interact and behave.[16] An organization's overall creativity level is thus not a simple aggregate of the creativity of the individuals it employs. The organization's structure, routines, and incentives could thwart individual creativity or amplify it.

The most familiar method of a company tapping the creativity of its individual employees is the suggestion box. In 1895, John Patterson, founder of National Cash Register (NCR), created the first sanctioned suggestion box program to tap the ideas of the hourly worker.[17] The program was considered revolutionary in its time. The originators of adopted ideas were awarded $1. In 1904, employees submitted 7000 ideas, of which one-third were adopted. Other firms have created more elaborate systems that not only capture employee ideas, but incorporate mechanisms for selecting and implementing those ideas. Google, for example, utilizes an idea management system whereby employees e-mail their ideas for new products and processes to a company-wide database where every employee can view the idea, comment on it, and rate it (for more on how Google encourages innovation, see the Theory in Action on Inspiring Innovation at Google). Honda of America utilizes an employee-driven idea system (EDIS) whereby employees submit their ideas, and if approved, the employee who submits the idea is responsible for following through on the suggestion, overseeing its progress from concept to implementation. Honda of America reports that more than 75 percent of all ideas are implemented.[18] Bank One, one of the largest holding banks in the United States, has created an employee idea program called "One Great Idea." Employees access the company's idea repository through the company's **intranet**. There they can submit their ideas and actively interact and collaborate on the ideas of others.[19] Through active exchange, the employees can evaluate and refine the ideas, improving their fit with the diverse needs of the organization's stakeholders.

intranet
A private network, accessible only to authorized individuals. It is like the Internet but operates only within ("intra") the organization.

At Bank of New York Mellon they go a step further—the company holds enterprise-wide innovation competitions where employees form their own teams and compete in coming up with innovative ideas. These ideas are first screened by judges at both the regional and business-line level. Then, the best ideas are pitched to senior management in a "Shark Tank" style competition that is webcast around the world. If a senior executive sees an idea they like, they step forward and say they will fund it and run with it. The competition both helps the company come up with great ideas and sends a strong signal to employees about the importance of innovation.[20]

Idea collection systems (such as suggestion boxes) are relatively easy and inexpensive to implement, but are only a first step in unleashing employee creativity. Today companies such as Intel, Motorola, 3M, and Hewlett-Packard go to much greater lengths to tap the creative potential embedded in employees, including investing in

Theory in Action Inspiring Innovation at Google

Google is always working on a surprising array of projects, ranging from the completely unexpected (such as autonomous self-driving cars and solar energy) to the more mundane (such as e-mail and cloud services).[a] In pursuit of continuous innovation at every level of the company, Google uses a range of formal and informal mechanisms to encourage its employees to innovate:[b]

20 Percent Time: All Google engineers are encouraged to spend 20 percent of their time working on their own projects. This was the source of some of Google's most famous products (e.g., Google Mail, Google News).

Recognition Awards: Managers were given discretion to award employees with "recognition awards" to celebrate their innovative ideas.

Google Founders' Awards: Teams doing outstanding work could be awarded substantial stock grants. Some employees had become millionaires from these awards alone.

Adsense Ideas Contest: Each quarter, the Adsense online sales and operations teams reviewed 100 to 200 submissions from employees around the world, and selected finalists to present their ideas at the quarterly contest.

Innovation Reviews: Formal meetings where managers present ideas originated in their divisions directly to founders Larry Page and Sergey Brin, as well as to CEO Eric Schmidt.[c]

[a] D. Bradbury, "Google's rise and rise," *Backbone*, (October 2011): 24–7.
[b] Boris Groysberg, David A. Thomas, and Alison Berkley Wagonfeld, Keeping Google "Googley." *Harvard Business School Case* 9 (2011): 409–39.
[c] J. Kirby, "How Google really does it," *Canadian Business* 82, no. 18 (2009): 54–8.

creativity training programs. Such programs encourage managers to develop verbal and nonverbal cues that signal employees that their thinking and autonomy are respected. These cues shape the culture of the firm and are often more effective than monetary rewards—in fact, as noted previously, sometimes monetary rewards undermine creativity by encouraging employees to focus on extrinsic rather than intrinsic motivation.[21] The programs also often incorporate exercises that encourage employees to use creative mechanisms such as developing alternative scenarios, using analogies to compare the problem with another problem that shares similar features or structure, and restating the problem in a new way. One product design firm, IDEO, even encourages employees to develop mock prototypes of potential new products out of inexpensive materials such as cardboard or styrofoam and pretend to use the product, exploring potential design features in a tangible and playful manner.

TRANSLATING CREATIVITY INTO INNOVATION

Innovation is more than the generation of creative ideas; it is the implementation of those ideas into some new device or process. Innovation requires combining a creative idea with resources and expertise that make it possible to embody the creative idea in a useful form. We will first consider the role of individuals as innovators, including innovation by inventors who specialize in creating new products and processes, and innovation by end users. We then will look at innovation activity that is organized by firms, universities, and government institutions.

The book *Quirky: The Remarkable Story of the Traits, Foibles, and Genius of Breakthrough Innovators Who Changed the World*[a] reports the results of a six-year research study on serial breakthrough innovators, that is, innovators identified in on multiple "most famous innovators" lists and noted for multiple breakthrough innovations. It turns out there are some strong commonalities among serial breakthrough innovators, and some of them are quite surprising. Here are some of the key commonalities noted in the book:

1. **They felt a sense of "separateness" and tended to challenge rules.** Nearly all of the innovators studied exhibited a social detachment, or a sense of not belonging to the social world. This had the surprising benefit of enabling them to challenge assumptions and pursue unconventional paths. As Albert Einstein put it in an autobiographical essay, The World as I See It, "*I have never lost an obstinate sense of detachment. . . Such a person no doubt loses something in the way of geniality and light-heartedness; on the other hand, he is largely independent of the opinions, habits, and judgments of his fellow and avoids the temptation to take his stand on such insecure foundations.*" His ability to reject norms and disregard accepted wisdom is precisely the reason he was able to revolutionize physics. Grace Hopper, a woman who invented one of the first computer programming languages, noted that during her lifetime women were not really supposed to be in business or science. However, because she attained the rank of Admiral in the Navy, companies like IBM and Hewlett Packard were willing to break with tradition to seek her expertise—being an Admiral essentially made her genderless.
2. **They had intense faith in their ability to achieve their objectives.** All of the innovators had extreme levels of something psychologists call "self-efficacy"—a type of confidence specific to overcoming obstacles and achieving goals. Elon Musk's quest to bring humanity to Mars, Nikola Tesla's work on free energy, and Dean Kamen's efforts to create affordable water purification illustrate the power of self-efficacy in its extreme. When people told Kamen that his water purification idea was impossible, he said, "*Don't tell me it's impossible. Tell me you can't do it.*" Things that other people think are impossible may not seem impossible to someone who believes they can overcome any obstacle, and this faith increases their likelihood of tackling a big challenge and sticking with it even in the face of failure or criticism.
3. **They were keenly idealistic.** Another characteristic that fueled the innovators' drive to pursue huge problems with fierce tenacity was their belief that they were pursuing something intrinsically noble and important. This drove them to work extremely long hours, often neglecting family, friends, and health. They poured their money, time, and energy into projects even if they had a high probability of failure. For example, in a 2014 interview on *60 Minutes,* Elon Musk noted that when he started Tesla Motors he had thought it would probably fail. When the interviewer asked him, "But you say you didn't expect the company to be successful? Then why try?" Musk responded, "If something is important enough you should try—even if you think the probable outcome is failure."
4. **They began with modest means and worked very hard for their success.** None of the innovators studied began their careers with significant financial resources. Nikola Tesla, for example, landed in the United States with four cents and a poem, and Benjamin Franklin arrived in New York with enough money for two rolls of bread. Dean Kamen, Thomas Edison, and Steve Jobs started their innovation careers as teenagers in working class families, and Elon Musk started his career by moving to Canada at seventeen, against his father's wishes, and supported himself with odd jobs such as shoveling grain and cleaning the boiler room of a lumber mill. All of the innovators were also happiest when working hard. Marie Curie worked so hard she often fainted at her laboratory bench, having forgotten to eat or rest, and Thomas Edison and Nikola Tesla often worked through the night. But as Edison noted, "Work made the Earth a paradise for me. I never intend to retire."

continued

concluded

5. **They were often self-taught.** Many of the innovators had significantly less formal education than would be expected for the fields in which they worked and made substantial contributions to areas outside of the fields in which they were trained. They were avid self-educators who were described as voracious readers, and several attained exceptional mastery of subjects without earning credentials. For example, Dean Kamen invented the world's first portable drug infusion pump, the world's first portable kidney dialysis machine, and many other extremely important medical inventions (such as the Segway Personal Transporter) without ever earning a college degree. Benjamin Franklin provides an even more extreme example. Having only attended a few years of grammar school, Benjamin Franklin went on to discover many fundamental principles of electricity, invent an improved urinary catheter, and was the first person to chart the Gulf Stream current.

ARE THESE CHARACTERISTICS IMITABLE?

Many of the breakthrough innovators had some personal traits that are difficult to imitate, like exceptional intelligence and memory, or mild manic tendencies. However, understanding *how* these factors helped them to become breakthrough innovators also reveals how anyone can nurture the breakthrough innovation potential in themselves and others. For example, anyone can cultivate a grand ambition that helps them to think bigger and stay motivated. There are a range of ways people can increase self-efficacy in themselves and others. It is also possible to modify cultural norms at schools, work, and home to help people become more comfortable challenging assumptions and mastering new subjects independently.

[a] Melissa A. *Quirky: The Remarkable Story of the Traits, Foibles, and Genius of Breakthrough Innovators Who Changed the World* (New York: Public Affairs, 2018).

The Inventor

The familiar image of the inventor as an eccentric and doggedly persistent scientist may have some basis in cognitive psychology. Analysis of personality traits of inventors suggests these individuals are likely to be interested in theoretical and abstract thinking, and have an unusual enthusiasm for problem solving. One 10-year study of inventors concludes that the most successful inventors possess the following characteristics:

1. They have mastered the basic tools and operations of the field in which they invent, but they have not specialized solely in that field; instead they have pursued two or three fields simultaneously, permitting them to bring different perspectives to each.
2. They are curious and more interested in problems than solutions.
3. They question the assumptions made in previous work in the field.
4. They often have the sense that all knowledge is unified. They seek global solutions rather than local solutions, and are generalists by nature.[22]

These traits are amply demonstrated by people like Albert Einstein (who overthrew several paradigms in physics), Marie Curie (who used her techniques of measuring radioactivity to show that Lord Kelvin was wrong in his calculations of the age of the Earth), and Elon Musk (who refused to listen to people who said that building reusable rockets was impossible). They are also illustrated in the following quotes by Nobel laureates. Sir MacFarlane Burnet, Nobel Prize–winning immunologist, noted, "I think there are dangers for a research man being too well trained in the field he is going to study,"[23] and Peter Debye, Nobel Prize–winning chemist, noted, "At the beginning of the Second World War, R. R. Williams of Bell Labs came to Cornell to try to interest me in

the polymer field. I said to him, 'I don't know anything about polymers. I never thought about them.' And his answer was, 'That is why we want you.'"[24] The global search for global solutions is aptly illustrated by Thomas Edison, who did not set out to invent just a lightbulb: "The problem then that I undertook to solve was . . . the production of the multifarious apparatus, methods, and devices, each adapted for use with every other, and all forming a comprehensive system."[25]

Such individuals may spend a lifetime developing numerous creative new devices or processes, though they may patent or commercialize few. The qualities that make people inventive do not necessarily make them entrepreneurial; many inventors do not actively seek to patent or commercialize their work. Many of the most well-known inventors (e.g., Alexander Graham Bell, Thomas Alva Edison, Albert Einstein, and Benjamin Franklin), however, had both inventive and entrepreneurial traits.[26]

Innovation by Users

Innovation often originates with those who create solutions for their own needs. Users often have both a deep understanding of their unmet needs and the incentive to find ways to fulfill them.[27] While manufacturers typically create new product innovations in order to profit from the sale of the innovation to customers, user innovators often have no initial intention to profit from the sale of their innovation—they create the innovation for their own use.[28] Users may alter the features of existing products, approach existing manufacturers with product design suggestions, or develop new products themselves. For example, the extremely popular small sailboat, the Laser, was designed without any formal market research or concept testing. Instead it was the creative inspiration of three former Olympic sailors, Ian Bruce, Bruce Kirby, and Hans Vogt. They based the boat design on their own preferences: simplicity, maximum performance, transportability, durability, and low cost. The resulting sailboat became hugely successful; during the 1970s and '80s, 24 Laser sailboats were produced daily.[29]

Another dramatic example is the development of Indermil, a tissue adhesive based on Super Glue. Super Glue is a powerful instant adhesive, and while its strength and speed of action were a great asset in most product applications, these features also caused a key product concern—its tendency to bond skin. Managers at Loctite, the company that developed Super Glue, wondered if this tendency could be exploited to develop an alternative to sutures for surgical applications. In the 1970s, the company experimented with developing a version of the adhesive that could be packaged and sterilized, but the project failed and funding for it was canceled. In 1980, the project was resurrected when Loctite was approached by a pharmaceutical company that wanted to collaborate on developing a wound closure product. The two companies spent three years attempting to develop special Super Glues that would degrade quickly in the body, but ultimately shelved the project again. By this point most managers in the company no longer wanted to be involved in developing an alternative to sutures—it was considered far too risky. However, in 1988, Bernie Bolger of Loctite was contacted by Professor Alan Roberts, a worldwide figure in reconstructive surgery. Roberts proceeded to give the managers at Loctite a stunning presentation about doctors who had responded to the Bradford football stadium fire of 1983. Roberts and many other doctors had been called in to carry out surgery and skin grafting in makeshift tents around

the stadium. Because stitching was too slow and skin damage was such that sutures would be ineffective, the doctors had used standard tubes of Super Glue to repair the skin and stick skin grafts in place! Roberts showed pictures of doctors in green garb standing around with Super Glue tubes stuck to their aprons, and pictures of people with large areas of skin missing and then those same people years later, with almost perfect skin repairs. Roberts begged the Loctite managers to continue their work on developing a version of Super Glue for tissue adhesion. Roberts's presentation was so compelling that the company again took up the project, this time with support from the CEO and serious funding. Approval from the U.S. Food and Drug Administration was won in 2002, and by 2003 the product was selling well in over 40 countries.[30]

basic research Research targeted at increasing scientific knowledge for its own sake. It may or may not have any long-term commercial application.

Research and Development by Firms

Across all nations, one of the most obvious sources of firm innovation is the firm's own research and development efforts. In most developed countries, firms account for the majority of R&D performed (see Figure 2.2).

Though the terms *research* and *development* are often lumped together, they actually represent different kinds of investment in innovation-related activities. *Research* can refer to both basic research and applied research. **Basic research** is effort directed at increasing understanding of a topic or field without a specific immediate commercial application in mind. This research advances scientific knowledge, which may (or may not) turn out to have long-run commercial implications. **Applied research** is directed at increasing understanding of a topic to meet a specific need. In industry, this research typically has specific commercial objectives. **Development** refers to activities that apply knowledge to produce useful devices, materials, or processes. Thus, the term *research and development* refers to a range of activities that extend from early exploration of a domain to specific commercial implementations. A firm's R&D intensity (its R&D expenditures as a percentage of its revenues) has a strong positive correlation with its sales growth rate, sales from new products, and profitability.[31]

applied research Research targeted at increasing knowledge for a specific application or need.

development Activities that apply knowledge to produce useful devices, materials, or processes.

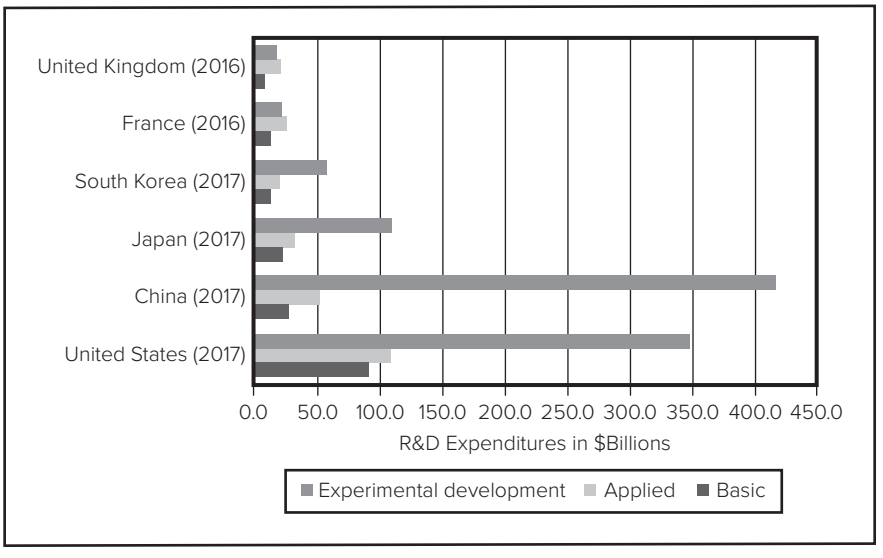


FIGURE 2.2
R&D for Selected Countries by Type, 2017 or most recent year

Sources: National Center for Science and Engineering Statistics, National Science Foundation, National Patterns of R&D Resources (annual series); Organisation for Economic Co-operation and Development, *Main Science and Technology Indicators* (2019/1); United Nations Educational, Scientific and Cultural Organization Institute for Statistics Data Centre.

Figure 2.2 shows the percent of R&D that was basic, applied, or experimental for a selected number of countries in 2017. As shown, most countries spend proportionately more on applied research and experimental development than basic research.

During the 1950s and 1960s, scholars of innovation emphasized a *science-push* approach to research and development.[32] This approach assumed that innovation proceeded linearly from scientific discovery, to invention, to engineering, then manufacturing activities, and finally marketing. According to this approach, the primary sources of innovation were discoveries in basic science that were translated into commercial applications by the parent firm. This linear process was soon shown to have little applicability to real-world products. In the mid-1960s, another model of innovation gained prominence: the *demand-pull* model of research and development. This approach argued that innovation was driven by the perceived demand of potential users. Research staff would develop new products in efforts to respond to customer problems or suggestions. This view, however, was also criticized as being too simplistic. Rothwell, for example, points out that different phases of innovation are likely to be characterized by varying levels of science push and demand pull.[33]

Most current research suggests that firms that are successful innovators utilize multiple sources of information and ideas, including:

- In-house research and development, including basic research.
- Linkages to customers or other potential users of innovations.
- Linkages to an external network of firms that may include competitors, complementors, and suppliers.
- Linkages to other external sources of scientific and technical information, such as universities and government laboratories.[34]

Firm Linkages with Customers, Suppliers, Competitors, and Complementors

Firms often form alliances with customers, suppliers, complementors, and even competitors to jointly work on an innovation project or to exchange information and other resources in pursuit of innovation. Collaboration might occur in the form of alliances, participation in research consortia, licensing arrangements, contract research and development, joint ventures, and other arrangements. The advantages and disadvantages of different forms of collaboration are discussed in Chapter Eight. Collaborators can pool resources such as knowledge and capital, and they can share the risk of a new product development project.

complementors Producers of complementary goods or services (e.g., for video game console producers such as Sony or Nintendo, game developers) are complementors.

The most frequent collaborations are between firms and their customers, suppliers, and local universities (see Figure 2.3).[35] Several studies indicate that firms consider users their most valuable source of new product ideas. The use of such collaborations is consistent across North America, Europe, and Japan, though Japanese firms may be somewhat more likely to collaborate extensively with their customers (see Figure 2.3).

Firms may also collaborate with competitors and complementors. **Complementors** are organizations (or individuals) that produce complementary goods, such as lightbulbs for lamps, chargers for electric vehicles, or applications for smartphones. In some industries, firms produce a range of goods and the line between competitor and complementor can blur.

FIGURE 2.3
Percentage of Companies That Report Extensive Collaboration with Customers, Suppliers, and Universities

Source: E. Roberts, "Benchmarking Global Strategic Management of Technology," *Research Technology Management*, March–April 2001: 25–36.

	North America (%)	Europe (%)	Japan (%)
Collaborates with:			
Customers	44	38	52
Suppliers	45	45	41
Universities	34	32	34

In some circumstances, firms might be bitter rivals in a particular product category and yet engage in collaborative development in that product category or complementary product categories. For instance, Microsoft competes against Rockstar Games in many video game categories, yet also licenses many Rockstar Games to play on its Xbox models. Rockstar is thus both a competitor and complementor to Microsoft. This can make the relationships between firms very complex—firms may have to manage a delicate balance between their roles of competitor versus complementor, or complementors might refuse to cooperate. For example, when Google bought Motorola Mobility in 2011, makers of mobile phone handsets that used Google's Android operating system such as Samsung and HTC were watching closely to see if Google would give Motorola handsets preferential access to Google software. Many analysts speculated that Samsung and HTC would begin developing more phones based on Microsoft's mobile operating system. To avoid the ire and defection of its complementors, Google announced that Motorola would be run as a separate entity and be given no advantages over makers of other Android-powered handsets. Android was to remain an equal-opportunity platform where any handset maker had a shot at making the next great Android phone.[36]

External versus Internal Sourcing of Innovation

Critics have often charged that firms are using external sources of technological innovation rather than investing in original research. But empirical evidence suggests that external sources of information are more likely to be complements to rather than substitutes for in-house research and development. Research by the Federation of British Industries indicated firms that had their own research and development were also the heaviest users of external collaboration networks. Presumably doing in-house R&D helps to build the firm's **absorptive capacity**, enabling it to better assimilate and utilize information obtained externally.[37] Absorptive capacity refers to the firm's ability to understand and use new information (absorptive capacity is discussed in more detail in Chapter Four).

absorptive capacity The ability of an organization to recognize, assimilate, and utilize new knowledge.

Universities and Government-Funded Research

Another important source of innovation comes from public research institutions such as universities, government laboratories, and incubators. A significant share of companies report that research from public and nonprofit institutions enabled them to develop innovations that they would not have otherwise developed.[38]

Universities

Universities in the United States performed $71.4 billion worth of R&D in 2017, making them the second largest performer of R&D in the United States after industry, and making the United States the place where universities spend the most money on R&D, on an absolute basis, in the world (see Figure 2.4). Of that, over $40 billion was for *basic research* (versus *applied research*), making universities the number one performer of basic research in the United States. The nation where universities perform the highest share of R&D, on the other hand, is the France where universities spend $16.8 billion, accounting for 20.7% of total R&D performance in the country. Many universities encourage their faculty to engage in research that may lead to useful innovations. Typically the intellectual property policies of a university embrace both patentable and unpatentable innovations, and the university retains sole discretion over the rights to commercialize the innovation. If an invention is successfully commercialized, the university typically shares the income with the individual inventor(s).[39] To increase the degree to which university research leads to commercial innovation, many universities have established **technology transfer offices**.

technology transfer offices
Offices designed to facilitate the transfer of technology developed in a research environment to an environment where it can be commercially applied.

In the United States, the creation of university technology transfer offices accelerated rapidly after the Bayh–Dole Act was passed in 1980. This act allowed universities to collect royalties on inventions funded with taxpayer dollars. Before this, the federal government was entitled to all rights from federally funded inventions.[40] Several European and Asian countries subsequently followed the U.S. lead and established legislation similar to Bayh–Dole, including Denmark, Austria, Finland, Norway,

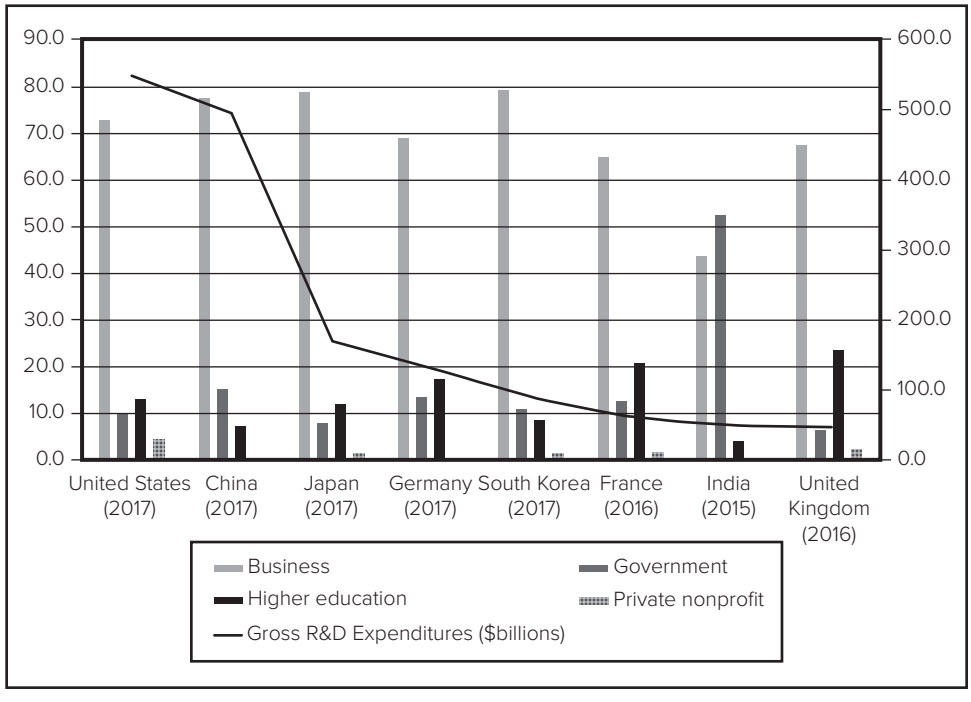


FIGURE 2.4
R&D Expenditures for Top 8 R&D-Performing Countries by Performance Sector, 2017 or most recent year

Sources: National Center for Science and Engineering Statistics, National Science Foundation, National Patterns of R&D Resources (annual series); Organisation for Economic Co-operation and Development, *Main Science and Technology Indicators* (2019/1); United Nations Educational, Scientific and Cultural Organization Institute for Statistics Data Centre.

Germany, France, United Kingdom, Japan, China, and India. Sweden and Italy, on the other hand, still have a policy of "professor's privilege" where university faculty retain sole ownership rights over their inventions. While the revenues from the university technology transfer activities are still quite small in comparison to university research budgets, their importance is growing. Initially, many anticipated that businesses would flock to license the intellectual property created by universities, leading to a substantial flow in licensing revenues. This "if you build it they will come" mindset turned out to be wrong, and licensing revenues were far less than expected. Now universities are taking a much more active role in helping to create start-ups based on their intellectual property, and in proactively forging relationships with the commercial sector.[41] Universities also contribute significantly to innovation through the publication of research results that are incorporated into the development efforts of other organizations and individuals.

Government-Funded Research

science parks Regional districts, typically set up by government, to foster R&D collaboration between government, universities, and private firms.

incubators Institutions designed to nurture the development of new businesses that might otherwise lack access to adequate funding or advice.

Governments of many countries actively invest in research through their own laboratories, the formation of **science parks** and **incubators**, and grants for other public or private research entities. For example, the U.S. Small Business Administration manages two programs that enable innovative small businesses to receive funding from federal agencies such as the Department of Defense, the Department of Energy, the Department of Health and Human Services, and others. The first is the Small Business Innovation Research (SBIR) program. Under the SBIR program, agencies award grants of up to $1,730,000 to small businesses to help them develop and commercialize a new innovation. The second is the Small Business Technology Transfer (STTR) program, which awards grants of up to $1,730,000 to facilitate a partnership between a small business and a nonprofit research institution—its objective is to more fully leverage the innovation that takes place in research laboratories by connecting research scientists with entrepreneurs.

Notable examples of science parks with incubators include:

- Stanford Research Park, established near Stanford University in 1951.
- Research Triangle Park, established in North Carolina in 1959.
- Sophia Antipolis Park, established in Southern France in 1969.
- Cambridge Science Park, established in Cambridge, England, in 1972.

These parks create fertile hotbeds for new start-ups and a focal point for the collaboration activities of established firms. Their proximity to university laboratories and other research centers ensures ready access to scientific expertise. Such centers also help university researchers implement their scientific discoveries in commercial applications.[42] Such parks often give rise to technology clusters that have long-lasting and self-reinforcing advantages (discussed later in the chapter).

Private Nonprofit Organizations

Private nonprofit organizations, such as private research institutes, nonprofit hospitals, private foundations, professional or technical societies, academic and industrial consortia, and trade associations, also contribute to innovation activity in a variety of

complex ways. Many nonprofit organizations perform their own research and development activities, some fund the research and development activities of other organizations but do not do it themselves, and some nonprofit organizations do both in-house research and development and fund the development efforts of others.

INNOVATION IN COLLABORATIVE NETWORKS

As the previous sections indicate, there is a growing recognition of the importance of collaborative research and development networks for successful innovation.[43] Such collaborations include (but are not limited to) joint ventures, licensing and second-sourcing agreements, research associations, government-sponsored joint research programs, value-added networks for technical and scientific interchange, and informal networks.[44] Collaborative research is especially important in high-technology sectors, where it is unlikely that a single individual or organization will possess all of the resources and capabilities necessary to develop and implement a significant innovation.[45]

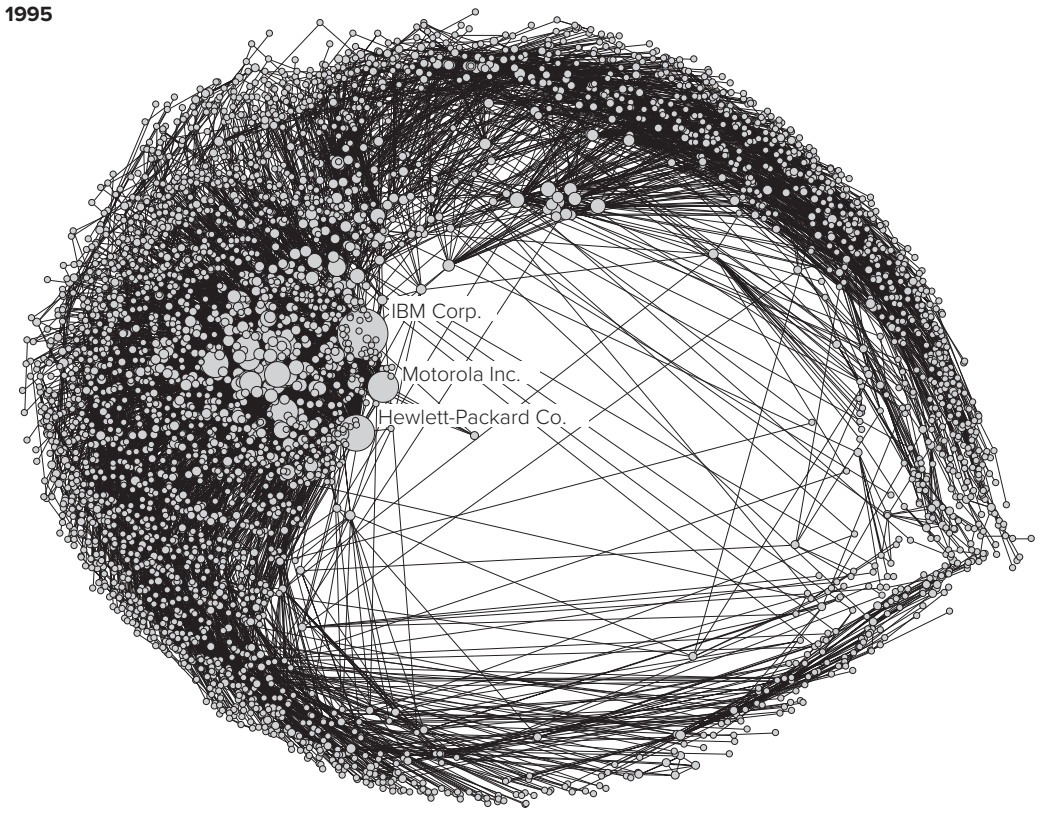
As firms forge collaborative relationships, they weave a network of paths between them that can act as conduits for information and other resources. By providing member firms access to a wider range of information (and other resources) than individual firms possess, interfirm networks can enable firms to achieve much more than they could achieve individually.[46] Thus, interfirm networks are an important engine of innovation. Furthermore, the structure of the network is likely to influence the flow of information and other resources through the network. For example, in a dense network where there are many potential paths for information to travel between any pair of firms, information diffusion should be fairly rapid and widespread.[47]

Figure 2.5 provides pictures of the worldwide technology alliance network in 1995 and in 2000.[48] The mid-1990s saw record peaks in alliance activity as firms scrambled to respond to rapid change in information technologies. This resulted in a very large and dense web of connected firms. The network shown here connects 3856 organizations, predominantly from North America, Japan, and Europe. However, there was a subsequent decline in alliance activity toward the end of the decade that caused the web to diminish in size and splinter apart into two large components and many small components. The large component on the left is primarily made up of organizations in the chemical and medical industries. The large component on the right is primarily made up of organizations in electronics-based industries. If the size and density of the collaboration network influences the amount of information available to organizations that are connected via the network, then the difference between the network shown for 1995 and the network shown for 2000 could have resulted in a substantial change in the amount of information that was transmitted between firms. (The strategic implications for a firm's position within the network are discussed in Chapter Eight.)

Technology Clusters

Sometimes geographical proximity appears to play a role in the formation and innovative activity of collaborative networks. Well-known regional clusters such as Silicon Valley's semiconductor firms, lower Manhattan's multimedia cluster, and the

FIGURE 2.5
The Global Technology Collaboration Network, 1995 and 2000[49]

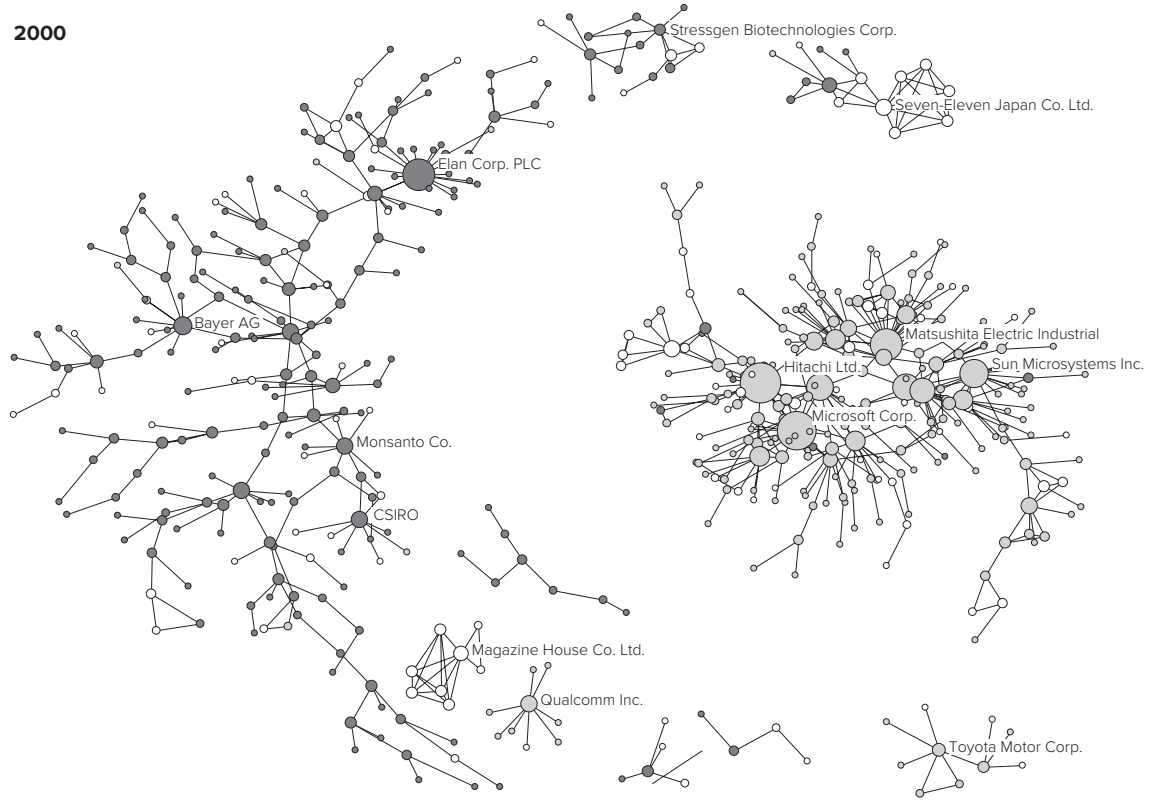


technology clusters
Regional clusters of firms that have a connection to a common technology, and may engage in buyer, supplier, and complementor relationships, as well as research collaboration.

Modena, Italy, knitwear district aptly illustrate this point. This has spurred considerable interest in the factors that lead to the emergence of a cluster. City and state governments, for example, might like to know how to foster the creation of a technology cluster in their region in order to increase employment, tax revenues, and other economic benefits. For firms, understanding the drivers and benefits of clustering is useful for developing a strategy that ensures the firm is well-positioned to benefit from clustering.

Technology clusters may span a region as narrow as a city or as wide as a group of neighboring countries.[50] Clusters often encompass an array of industries that are linked through relationships between suppliers, buyers, and producers of complements. One primary reason for the emergence of regional clusters is the benefit of proximity in knowledge exchange. Though advances in information technology have made it easier, faster, and cheaper to transmit information great distances, several studies indicate that knowledge does not always transfer readily via such mechanisms.

FIGURE 2.5 ***Continued***



complex knowledge Knowledge that has many underlying components, or many interdependencies between those components, or both.

tacit knowledge Knowledge that cannot be readily codified (documented in written form).

Proximity and interaction can directly influence firms' ability and willingness to exchange knowledge. First, knowledge that is **complex** or **tacit** may require frequent and close interaction to be meaningfully exchanged.[51] Firms may need to interact frequently to develop common ways of understanding and articulating the knowledge before they are able to transfer it.[52] Second, closeness and frequency of interaction can influence a firm's *willingness* to exchange knowledge. When firms interact frequently, they can develop trust and reciprocity norms. Firms that interact over time develop greater knowledge of each other, and their repeated interactions give them information as to the likelihood of their partner's behaving opportunistically. A shared understanding of the rules of engagement emerges, wherein each partner understands its obligations with respect to how much knowledge is exchanged, how that knowledge can be used, and how the firms are expected to reciprocate.[53]

Firms that are proximate thus have an advantage in sharing information that can lead to greater innovation productivity. This can, in turn, lead to other self-reinforcing geographical advantages. A cluster of firms with high innovation productivity can lead to more new firms starting up in the immediate vicinity and can attract other firms to the area.[54] As firms grow, divisions may be spun off into new firms, entrepreneurial employees may start their own enterprises, and supplier and distributor

markets emerge to service the cluster. Successful firms also attract new labor to the area and help to make the existing labor pool more valuable by enabling individuals to gain experience working with the innovative firms. The increase in employment and tax revenues in the region can lead to improvements in infrastructure (such as roads and utilities), schools, and other markets that service the population (shopping malls, grocery stores, health-care providers, etc.). The benefits firms reap by locating in close geographical proximity to each other are known collectively as **agglomeration economies**.[55]

agglomeration economies
The benefits firms reap by locating in close geographical proximity to each other.

There are also some downsides to geographical clustering. First, the proximity of many competitors serving a local market can lead to competition that reduces their pricing power in their relationships with both buyers and suppliers. Second, close proximity of firms may increase the likelihood of a firm's competitors gaining access to the firm's proprietary knowledge (this is one of the mechanisms of technology spillovers, discussed in the next section). Third, clustering can potentially lead to traffic congestion, inordinately high housing costs, and higher concentrations of pollution.[56]

A big part of the reason that technologies are often regionally localized is that technological knowledge is, to a large extent, held by people, and people are often only reluctantly mobile. In a well-known study, Annalee Saxenian found that engineers in Silicon Valley were more loyal to their craft than to any particular company, but they were also very likely to stay in the region even if they changed jobs.[57] This was due in part to the labor market for their skills in the region, and in part to the disruption in an individual's personal life if he or she were to move out of the region. Thus, if for some reason an innovative activity commences in a geographic locale, the knowledge and expertise that accumulates might not spread readily into other geographic locales, leading to a localized cluster of technological expertise.[58]

Studies have indicated that while many innovative activities appear to have some geographic component, the degree to which innovative activities are geographically clustered depends on things such as:

- The nature of the technology, such as its underlying knowledge base or the degree to which it can be protected by patents or copyright, and the degree to which its communication requires close and frequent interaction.
- Industry characteristics, such as the degree of market concentration or stage of the industry life cycle, transportation costs, and the availability of supplier and distributor markets.
- The cultural context of the technology, such as the population density of labor or customers, infrastructure development, or national differences in the way technology development is funded or protected.

For example, one study that examined the spatial distribution of technology sectors in different countries found that pharmaceutical development was highly clustered in the United Kingdom and France, but much more spatially diffused in Italy and Germany.[59] The same study found, however, that the manufacture of clothing demonstrated high clustering in Italy, but not in France, Germany, or the United Kingdom. While the clustering of pharmaceutical development may have been influenced by the national

Research Brief Do Grand Innovation Prizes Work?

The patent system, and the resulting opportunity for inventors to profit from a temporary monopoly on their inventions, is generally considered to be the most important incentive for innovation. However, sometimes governments, foundations, or firms offer large cash prizes for a specific innovation outcome—often termed a "Grand Innovation Prize." How do we know when a Grand Innovation Prize might be a better—or complementary—incentive for innovation?

Research has shown that prizes tend to be useful in areas where traditional monetization methods might be weak (e.g., where patents might not be particularly effective), slow (e.g., when development of the innovation might take so long as to make it economically unattractive), or where benefits of the innovation might be so widespread as to lead to underinvestment in a public good.[a] For example, in early 2021, the Xprize Foundation organization was administering a range of grand prize competitions, including two different prize competitions related to carbon capture. One, the NRG COSIA Carbon Xprize would be awarded based on contestants' ability to both remove CO_2 from the atmosphere and put it to use in valuable products such as synthetic fuel or carbon fiber. A total of 48 teams had entered the competition at its launch and nine remained in the final round, vying for two prizes worth $10 million each. A second competition, sponsored by Elon Musk, offered prizes totaling $100 million to contestants who could demonstrate the ability to permanently sequester CO_2 in a process that could be scaled up to multiple gigatons. Both contests sought to provide a public good by helping to slow or stop climate change.[b]

A prize can induce a wide range of participants to enter and yield a diverse array of solutions proposed. Prizes focus energy, effort, and investment toward problems, and many of those efforts may yield advances even if their teams do not win the prize. Furthermore, a prize can have great leverage—the collective investment spent by participants to compete for the prize may dramatically exceed the prize itself. For example, one of the first well-known innovation prizes was the Orteig Prize, a $25,000 prize offered in 1919 by French hotelier Raymond Orteig for the first nonstop flight between New York City and Paris. In 1927, Charles Lindbergh (considered an underdog in the competition) won the prize in a modified single-engine Ryan aircraft called the Spirit of St. Louis. In total, nine teams spent $400,000 in pursuit of the Orteig Prize. Similarly, in 1996, entrepreneur Peter Diamandis offered a $10-million prize to the first team that could build and fly a three-passenger vehicle 100 kilometers into space twice within two weeks. The contest, later named the Ansari XPRIZE for Suborbital Spaceflight, induced 26 teams from seven nations to invest more than $100 million in pursuit of the $10 million reward. On October 4, 2004, Mojave Aerospace Ventures won the Ansari XPRIZE by completing the flight in their spacecraft *SpaceShipOne*.

In addition to the prize money, winning a prize offers accolades, fame, and legitimacy. All of these things can be useful for the team's future commercialization efforts. Furthermore, for many prize competitions, the contestants keep their intellectual property rights in addition to the prize reward (this is the case, for example, with all Xprize foundation competitions), so in essence, the prize is sometimes the subsidization of an economic venture that just needs a boost.

[a] Alberto Galasso, Matthew Mitchell, and Gábor Virag, "A Theory of Grand Innovation Prizes," *Research Policy* 47 (2018): 343–63; Fiona Murray et al., "Grand Innovation Prizes: A Theoretical, Normative and Empirical Evaluation," *Research Policy* 41 (2012): 1779–92; Alexander Tabarrok, "Grand Innovation Prizes to Address Pandemics: A Primer," *Special Edition Policy Brief,* April 8, 2020, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3571437; and Michael J. Burstein and Fiona E. Murray, "Innovation Prizes in Practice and Theory," *Harvard Journal of Law & Technology* 29 (2016): 402–51.

[b] XPRIZE, accessed March 3, 2021, https://www.xprize.org.

systems of research funding and the need to share complex technological expertise, the formation of textile clusters may have been due more to cultural factors that influenced the historical rise of industrial districts.

Technological Spillovers

technological spillovers
A positive externality from R&D resulting from the spread of knowledge across organizational or regional boundaries.

While the work on technology clusters has tended to emphasize the "stickiness" of knowledge, a related body of research has focused on explaining the spread of knowledge across organizational or regional boundaries. This topic is known as **technological spillovers**. Technological spillovers occur when the benefits from the research activities of one firm (or nation or other entity) spill over to other firms (or nations or other entities). Spillovers are thus a positive externality of research and development efforts. Evidence suggests that technology spillovers are a significant influence on innovative activity. For example, in a series of studies conducted in the 1980s and 1990s, Adam Jaffe and his coauthors found that both a firm's patenting activities and profits were influenced by the R&D spending of other firms and universities in its geographical region.[60]

Whether R&D benefits will spill over is partially a function of the strength of protection mechanisms such as patents, copyrights, and trade secrets (methods of protecting innovation are discussed in greater detail in Chapter Nine). Since the strength of protection mechanisms varies significantly across industries and countries, the likelihood of spillovers varies also.[61] The likelihood of spillovers is also a function of the nature of the underlying knowledge base (e.g., as explained in the previous section, tacit knowledge may not flow readily across firm boundaries) and the mobility of the labor pool.[62]

Summary of Chapter

1. Creativity is the underlying process for innovation. Creativity enables individuals and organizations to generate new and useful ideas. Creativity is considered a function of intellectual abilities, knowledge, thinking styles, personality traits, intrinsic motivation, and environment.
2. Innovation sometimes originates with individual inventors. The most prolific inventors tend to be trained in multiple fields, be highly curious, question previously made assumptions, and view all knowledge as unified. The most well-known inventors tend to have both inventive and entrepreneurial traits.
3. Innovation can also originate with users who create solutions to their own needs.
4. Firms' research and development is considered a primary driver of innovation. In most countries, firms spend significantly more on R&D than government institutions spend on R&D, and firms consider their in-house R&D their most important source of innovation.
5. Firms often collaborate with a number of external organizations (or individuals) in their innovation activities. Firms are most likely to collaborate with customers, suppliers, and universities, though they also may collaborate with competitors, producers of complements, government laboratories, nonprofit organizations, and other research institutions.
6. Many universities have a research mission, and in recent years universities have become more active in setting up technology transfer activities to directly commercialize the inventions of faculty. Universities also contribute to innovation through the publication of research findings.

7. Government also plays an active role in conducting research and development (in its own laboratories), funding the R&D of other organizations, and creating institutions to foster collaboration networks and to nurture start-ups (e.g., science parks and incubators). In some countries, government-funded research and development exceeds that of industry-funded research.
8. Private nonprofit organizations (such as research institutes and nonprofit hospitals) are another source of innovation. These organizations both perform their own R&D and fund R&D conducted by others.
9. Probably the most significant source of innovation does not come from individual organizations or people, but from the collaborative networks that leverage resources and capabilities across multiple organizations or individuals. Collaborative networks are particularly important in high-technology sectors.
10. Collaboration is often facilitated by geographical proximity, which can lead to regional technology clusters.
11. Technology spillovers are positive externality benefits of R&D, such as when the knowledge acquired through R&D spreads to other organizations.

Discussion Questions

1. What are some of the advantages and disadvantages of (*a*) individuals as innovators, (*b*) firms as innovators, (*c*) universities as innovators, (*d*) government institutions as innovators, (*e*) nonprofit organizations as innovators?
2. What traits appear to make individuals most creative? Are these the same traits that lead to successful inventions?
3. Could firms identify people with greater capacity for creativity or inventiveness in their hiring procedures?
4. To what degree do you think the creativity of the firm is a function of the creativity of individuals, versus the structure, routines, incentives, and culture of the firm? Provide an example of a firm that does a particularly good job at nurturing and leveraging the creativity of its individuals.
5. Several studies indicate that the use of collaborative research agreements is increasing around the world. What are some reasons collaborative research is becoming more prevalent?

Suggested Further Reading

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Chapter **Three**

Types and Patterns of Innovation

Innovating in India: The chotuKool Project

Godrej & Boyce, founded in India in 1897, sold a range of products to the Indian market including household appliances, office furniture, and industrial process equipment. In recent years, international competitors such as Haier and Samsung were cutting deep into Godrej's market share for household appliances such as refrigerators, washing machines, and air conditioners, and management knew that to preserve the company would require innovative solutions.

One such solution was the chotuKool, a small, portable refrigerator. Though around the world refrigeration was considered a mature technology, in rural India as many as 90 percent of families could not afford household appliances, did not have reliable access to electricity, and had no means of refrigeration. This significantly limited the kinds of foods they could eat and how they could be prepared. Finding a way to provide refrigeration to this segment of the population offered the promise of both a huge market and making a meaningful difference in people's quality of life. As noted by Navroze Godrej, Directed of Special Projects at Godrej, "We imagined we would be making a shrunken down version of a refrigerator. Make it smaller, make it cheaper. And we had preconceived notions of how to build a brand that resonated with these users through big promotions and fancy ad campaigns."

These assumptions would turn out to be wrong. First, as Godrej's team looked at the options of how to reduce the cost of a conventional compressor-based refrigerator, they quickly realized that they could not reduce its cost by enough to make a meaningful difference.[a] Second, they discovered that having the refrigerator be lightweight was more important than they had previously thought because many rural Indians lived migratory lives, moving to follow the availability of work. Third, because of the lack of refrigeration, most people were in the habit of cooking just enough for the day, and thus had relatively low refrigeration capacity needs. Fourth, of those few rural Indians that did

have refrigerators, many did not plug them in for most of the day for fear of them being damaged by power surges. As Godrej notes, "We were surprised by many things, we were shocked by many things . . . we realized our original hypothesis was quite wrong."[b]

Based on these insights, the company designed a small and portable refrigerator based on thermoelectric cooling (rather than compressor technology). Thermoelectric cooling was the cooling method used in laptops; it involved running a current between two semiconductors. It was far more expensive on a per-unit-of-cooling basis, but it had much lower power requirements and could be used on a much smaller scale than compressor cooling. This enabled Godrej to make a very small, lightweight refrigerator with a relatively low price (35–40 percent cheaper than traditional refrigerators). It also lowered the power costs of operating a refrigerator, and made the refrigerator able to operate for several hours on a 12-volt battery, making it much more adaptable to situations where power was unreliable.

In Godrej's initial plan for the chotuKool, the refrigerators would be cherry red and look like coolers. Soon, however, managers at chotuKool realized that if the refrigerators were just perceived as inexpensive alternatives to refrigerators, they had the potential to be stigmatizing for consumers who, in turn, would not talk about them to their friends. This was a serious problem because the company had counted on word of mouth to spread information about the refrigerators deep into rural communities. To get people to talk about the coolers, they needed to be aspirational—they needed to be *cool*.

Godrej decided to revamp the design of the coolers, giving them a more sophisticated shape and making them customizable (buyers could choose from over 100 decorative skin colors for the chotuKool).[c] They also decided to market the refrigerators to the urban affluent market in addition to the rural market, as adoption by the urban affluent market would remove any stigma associated with buying them. To attract this market, they positioned the refrigerators as perfect for picnics, parties, offices, dorm rooms, use in cars, and so on.

To get the chotuKool to rural customers would require a dramatically different distribution system than Godrej had traditionally used. However, building out a distribution system into rural communities would prohibitively raise the cost of chotuKool, potentially rendering the product nonviable. The development team was initially stumped. Then one day G. Sunderraman, vice president of Godrej and leader of the chotuKool project, happened to inquire with a university official about obtaining college application forms for his youngest son and the official pointed out that Sunderraman could get the forms at any post office. At that moment, Sunderraman realized that the post office, which had offices in every rural area of India, could be an ideal distribution channel for the chotuKool.[d] It was a very novel proposition, but India Post agreed to the collaboration and soon chotuKools were available in all post offices in the central region of India.[e] As Sunderraman noted, "The India Post network is very well spread in India and is about three or four times larger than the best logistic suppliers."[f]

The chotuKool won several design awards in its first years, and after selling 100,000 units in its second year *Fast Company* gave Godrej its "Most Innovative Company" award. Godrej and Sunderraman were disappointed to discover that it was not as rapidly adopted by rural poor households as they had hoped; the roughly $50 price was still too expensive for most poor rural families in India. However, the chotuKool turned out to be much more popular than anticipated among hotels, food stalls, flower shops, and other small stores because it enabled these small stores to offer higher valued products (such as cold drinks) or to keep products fresh longer, thereby increasing their profits. The chotuKool also became a popular lifestyle product among the urban affluent population who began to widely use them in their cars.

Godrej's experience developing and launching the chotuKool had provided many lessons. They had learned that to radically reduce the cost of a product might require completely rethinking the technology—sometimes even in ways that initially seemed more expensive. They learned that customers who had adapted their way of life to the lack of a technology (like refrigeration) might not adopt that technology even if it was made markedly less expensive. Finally, they learned not to underestimate the value of making a product work for multiple market segments, including those that might not be initially obvious as customers. Though some people considered chotuKool a failure because it had not achieved its original objective of wide adoption by the rural poor, Godrej (and many others) considered it a success: The product expanded Godrej's market share, penetrated new market segments in which Godrej had not formerly competed, and demonstrated Godrej's innovative capabilities to the world.

Discussion Questions

1. What were the pros and cons of attempting to develop a refrigerator for India's rural poor?
2. What product and process innovations did the chotuKool entail? Would you consider these incremental or radical? Architectural or component? Competence enhancing or competence destroying?
3. Did the chotuKool pose a threat of disrupting the traditional refrigerator market? Why or why not?
4. Is there anything you think Godrej should have done differently to penetrate the market of rural poor families in India?
5. What other products might the lessons Godrej learned with chotuKool apply to?

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OVERVIEW

The previous chapters pointed out that technological innovation can come from many sources and take many forms. Different types of technological innovations offer different opportunities for organizations and society, and they pose different demands upon producers, users, and regulators. While there is no single agreed-upon taxonomy to describe different kinds of technological innovations, in this chapter we will review several dimensions that are often used to categorize technologies. These dimensions are useful for understanding some key ways that one innovation may differ from another.

technology trajectory The path a technology takes through its lifetime. This path may refer to its rate of performance improvement, its rate of diffusion, or other change of interest.

The path a technology follows through time is termed its **technology trajectory**. Technology trajectories are most often used to represent the technology's rate of performance improvement or its rate of adoption in the marketplace. Though many factors can influence these technology trajectories (as discussed in both this chapter and the following chapters), some patterns have been consistently identified in technology trajectories across many industry contexts and over many periods. Understanding these patterns of technological innovation provides a useful foundation that we will build upon in the later chapters on formulating technology strategy.

The chapter begins by reviewing the dimensions used to distinguish types of innovations. It then describes the s-curve patterns so often observed in both the rate of technology improvement and the rate of technology diffusion to the market. In the last section, the chapter describes research suggesting that technological innovation follows a cyclical pattern composed of distinct and reliably occurring phases.

TYPES OF INNOVATION

Technological innovations are often described using dimensions such as *radical* versus *incremental.* Different types of innovation require different kinds of underlying knowledge and have different impacts on the industry's competitors and customers. Four of the dimensions most commonly used to categorize innovations are described here: product versus process innovation, radical versus incremental, competence enhancing versus competence destroying, and architectural versus component.

Product Innovation versus Process Innovation

Product innovations are embodied in the outputs of an organization—its goods or services, even if those products are services. For example, Snapchat's filters and special effects that enable users to augment their photos are product innovations. Process innovations are innovations in the way an organization conducts its business, such as in the techniques of producing or marketing goods or services. For example, Elon Musk's use of automation for most of the production process for the Model 3 with giant robots is a process innovation. Process innovations are often oriented toward improving the effectiveness or efficiency of production by, for example, reducing defect rates or increasing the quantity that may be produced in a given time. For example, a process innovation at a biotechnology firm might entail developing a genetic algorithm that can quickly search a set of disease-related genes to identify a target for therapeutic intervention.

In this instance, the process innovation (the genetic algorithm) can speed up the firm's ability to develop a product innovation (a new therapeutic drug).

New product innovations and process innovations often occur in tandem. First, new processes may enable the production of new products. For example, as discussed later in the chapter, the development of new metallurgical techniques enabled the development of the bicycle chain, which in turn enabled the development of multiple-gear bicycles. Second, new products may enable the development of new processes. For example, the development of advanced workstations has enabled firms to implement computer-aided manufacturing processes that increase the speed and efficiency of production. Finally, a product innovation for one firm may simultaneously be a process innovation for another. For example, when United Parcel Service (UPS) helps a customer develop a more efficient distribution system, the new distribution system is simultaneously a product innovation for UPS and a process innovation for its customer.

Though product innovations are often more visible than process innovations, both are extremely important to an organization's ability to compete. Throughout the remainder of the book, the term *innovation* will be used to refer to both product and process innovations.

Radical Innovation versus Incremental Innovation

radical innovation
An innovation that is very new and different from prior solutions.

incremental innovation
An innovation that makes a relatively minor change from (or adjustment to) existing practices.

One of the primary dimensions used to distinguish types of innovation is the continuum between radical versus incremental innovation. A number of definitions have been posed for **radical innovation** and **incremental innovation**, but most hinge on the degree to which an innovation represents a departure from existing practices.[1] Thus, radicalness might be conceived as the combination of *newness* and the degree of *differentness.* A technology could be new to the world, new to an industry, new to a firm, or new merely to an adopting business unit. A technology could be significantly different from existing products and processes or only marginally different. The most radical innovations would be new to the world and exceptionally different from existing products and processes. The introduction of wireless telecommunication products aptly illustrates this—it embodied significantly new technologies that required new manufacturing and service processes. Incremental innovation is at the other end of the spectrum. An incremental innovation might not be particularly new or exceptional; it might have been previously known to the firm or industry, and involve only a minor change from (or adjustment to) existing practices. For example, changing the screen of a cell phone to make it more crack resistant or offering a new service plan with better international texting rates would represent incremental innovation.

The radicalness of innovation is also sometimes defined in terms of risk. Since radical innovations often embody new knowledge, producers and customers will vary in their experience and familiarity with the innovation, and in their judgment of its usefulness or reliability.[2] The development of third-generation (3G) telephony is illustrative. 3G wireless communication technology utilizes broadband channels. This increased bandwidth gave mobile phones far greater data transmission capabilities that enabled activities such as videoconferencing and accessing the most advanced Internet sites. For companies to develop and offer 3G wireless telecommunications service required a significant investment in new networking equipment and an infrastructure capable of carrying a much larger bandwidth of signals. It also required developing phones

with greater display and memory capabilities, and either increasing the phone's battery power or increasing the efficiency of the phone's power utilization. Any of these technologies could potentially pose serious obstacles. It was also unknown to what degree customers would ultimately value broadband capability in a wireless device. Thus, the move to 3G required managers to assess several different risks simultaneously, including technical feasibility, reliability, costs, and demand.

Finally, the radicalness of an innovation is relative, and may change over time or with respect to different observers. An innovation that was once considered radical may eventually be considered incremental as the knowledge base underlying the innovation becomes more common. For example, while the first steam engine was a monumental innovation, today its construction seems relatively simple. Furthermore, an innovation that is radical to one firm may seem incremental to another. Although both Kodak and Sony introduced digital cameras for the consumer market within a year of each other (Kodak's DC40 was introduced in 1995, and Sony's Cyber-Shot Digital Still Camera was introduced in 1996), the two companies' paths to the introduction were quite different. Kodak's historical competencies and reputation were based on its expertise in chemical photography, and thus the transition to digital photography and video required a significant redirection for the firm. Sony, on the other hand, had been an electronics company since its inception, and had a substantial level of expertise in digital recording and graphics before producing a digital camera. Thus, for Sony, a digital camera was a straightforward extension of its existing competencies.

competence-enhancing and competence destroying innovation
A competence-enhancing innovation builds on existing knowledge and skills whereas a competence-destroying innovation renders existing knowledge and skills obsolete. Whether an innovation is competence enhancing or competence destroying depends on whose perspective is being taken. An innovation can be competence enhancing to one firm, while competence destroying for another.

Competence-Enhancing Innovation versus Competence-Destroying Innovation

Innovations can also be classified as **competence enhancing** versus **competence destroying**. An innovation is considered to be competence enhancing from the perspective of a particular firm if it builds on the firm's existing knowledge base. For example, each generation of Apple's iPhone (e.g., iPhone 12, iPhone SE, and iPhone X) builds on the technology underlying the previous generation. Thus, while each generation embodies innovation, these innovations leverage Apple's existing competencies, making them more valuable.

An innovation is considered to be competence destroying from the perspective of a particular firm if the technology does not build on the firm's existing competencies or renders them obsolete. For example, from the 1600s to the early 1970s, no self-respecting mathematician or engineer would have been caught without a slide rule. Slide rules are lightweight devices, often constructed of wood, that use logarithm scales to solve complex mathematical functions. They were used to calculate everything from the structural properties of a bridge to the range and fuel use of an aircraft. Specially designed slide rules for businesses had, for example, scales for doing loan calculations or determining optimal purchase quantities. During the 1950s and 1960s, Keuffel & Esser was the preeminent slide-rule maker in the United States, producing 5000 slide rules a month. However, in the early 1970s, a new innovation relegated the slide rule to collectors and museum displays within just a few years: the inexpensive handheld calculator. Keuffel & Esser had no background in the electronic components that made electronic calculators possible and was unable to transition to the new technology. By 1976, Keuffel & Esser withdrew from the market.[3] Whereas the inexpensive handheld calculator built

on the existing competencies of companies such as Hewlett-Packard and Texas Instruments (and thus for them would be competence enhancing), for Keuffel & Esser, the calculator was a competence-destroying innovation.

Developments in digitization and artificial intelligence provide another illustrative example. In the last 20 years, digital platforms like Zillow (an online platform for real estate shopping), E*Trade (an online stock brokerage), and Expedia (an online travel booking platform) have begun providing functions that were once provided by people (real estate agents, stockbrokers, and travel agents). These platforms enable customers to access more information and conduct more rapid and detailed searches. Many platforms also enable customers to be algorithmically matched to products that fit their preferences. In many cases, the digital platforms provide better functionality than the human-mediated services could have provided. In the past, human providers relied on their experience and relationships to excel in these fields. For example, a real estate agent might spend significant time attending open houses in a neighborhood, looking at houses, and networking with other agents, to know about house availability in a given neighborhood. A travel agent might build up a personal knowledge base of hotels and cruise lines that proven popular with their clients in the past, enabling them to recommend these choices to new clients with similar preferences. However, the ability of a digital interface to store more data, more rapidly filter searches, and provide matches with more detailed information has proven to be a competence-destroying innovation for many human providers in these industries. Humans cannot process or store as much data as computers, and as computers become better able to make the match between product and customer, the value of agent's personal knowledge base is diminished. The human providers who will succeed in these industries will be those that focus on providing value-added services that the digital platforms cannot (yet) provide.

Architectural Innovation versus Component Innovation

Most products and processes are hierarchically nested systems, meaning that at any unit of analysis, the entity is a system of components, and each of those components is, in turn, a system of finer components, until we reach some point at which the components are elementary particles.[4] For example, a bicycle is a system of components such as a frame, wheels, tires, seat, brakes, and so on. Each of those components is also a system of components: The seat might be a system of components that includes a metal and plastic frame, padding, a nylon cover, and so on.

component (or modular) innovation
An innovation to one or more components that does not significantly affect the overall configuration of the system.

An innovation may entail a change to individual components, to the overall architecture within which those components operate, or both. An innovation is considered a **component innovation** (or **modular innovation**) if it entails changes to one or more components, but does not significantly affect the overall configuration of the system.[5] In the example above, an innovation in bicycle seat technology (such as the incorporation of gel-filled material for additional cushioning) does not require any changes in the rest of the bicycle architecture.

architectural innovation
An innovation that changes the overall design of a system or the way its components interact with each other.

In contrast, an **architectural innovation** entails changing the overall design of the system or the way that components interact with each other. An innovation that is strictly architectural may reconfigure the way that components link together in the system, without changing the components themselves.[6] Most architectural innovations, however, create changes in the system that reverberate throughout its design,

requiring changes in the underlying components in addition to changes in the ways those components interact. Architectural innovations often have far-reaching and complex influences on industry competitors and technology users.

For example, when SpaceX set out to make space travel significantly less expensive, they developed a number of component and architectural innovations. Their development of PICA-X, a heat-shielding material that can enable a rocket to withstand the re-entry from orbit, is an example of a component innovation. It is substituted for existing material on the rocket's exterior. Being able to re-use a rocket is a tremendous source of cost savings. SpaceX also reduced the cost of their rockets by designing them to use only SpaceX-designed Merlin engines, an architectural innovation. Previous NASA designed rockets such as Atlas V used up to three kinds of engines, each tailored for a specific phase of flight, for example, RD-180 engines that burned kerosene in the first stage, solid-fuel strap-on boosters at liftoff, and a liquid hydrogen engine in the final phase of flight. Using three kinds of rockets helped to optimize the performance of the Atlas V, but significantly increased its cost. By contrast, having all of the engines on the SpaceX rocket be the same reduced the amount of tooling and the number of processes required, resulting in what Musk called "huge cost savings."[7] This change was an architectural change that both substituted in a new component (the Merlin engines) and changed the overall design by making it more modular (we discuss modularity at greater length in Chapter Four).

For a firm to initiate or adopt a component innovation may require that the firm have knowledge only about that component. However, for a firm to initiate or adopt an architectural innovation typically requires that the firm have architectural knowledge about the way components link and integrate to form the whole system. Firms must be able to understand how the attributes of components interact, and how changes in some system features might trigger the need for changes in many other design features of the overall system or the individual components. Modularity, and its role in the creation of platform ecosystems, is discussed in greater detail in Chapter Four.

Using the Dimensions

Though the dimensions described above are useful for exploring key ways that one innovation may differ from another, these dimensions are not independent, nor do they offer a straightforward system for categorizing innovations in a precise and consistent manner. Each of the above dimensions shares relationships with others—for example, architectural innovations are often considered more radical and more competence destroying than component innovations. Furthermore, how an innovation is described on a dimension often depends on who is doing the describing and with what it is being compared. An all-electric vehicle, for example, might seem like a radical and competence destroying innovation to a manufacturer of internal combustion engines, but to a customer who only has to change how they fuel/charge the vehicle, it might seem like an incremental and competence-enhancing innovation. Thus, while the dimensions above are valuable for understanding innovation, they should be considered relative dimensions whose meaning is dependent on the context in which they are used.

We now will turn to exploring patterns in technological innovation. Numerous studies of innovation have revealed recurring patterns in how new technologies emerge, evolve, are adopted, and are displaced by other technologies. We begin by examining technology s-curves.

TECHNOLOGY S-CURVES

Both the rate of a technology's performance improvement and the rate at which the technology is adopted in the marketplace repeatedly have been shown to conform to an s-shape curve. Though s-curves in technology performance and s-curves in technology diffusion are related (improvements in performance may foster faster adoption, and greater adoption may motivate further investment in improving performance), they are fundamentally different processes. S-curves in technology improvement are described first, followed by s-curves in technology diffusion. This section also explains that despite the allure of using s-curves to predict when new phases of a technology's life cycle will begin, doing so can be misleading.

S-Curves in Technological Improvement

Many technologies exhibit an s-curve in their performance improvement over their lifetimes.[8] When a technology's performance is plotted against the amount of effort and money invested in the technology, it typically shows slow initial improvement, then accelerated improvement, then diminishing improvement (see Figure 3.1). Performance improvement in the early stages of a technology is slow because the fundamentals of the technology are poorly understood. Great effort may be spent exploring different paths of improvement or different drivers of the technology's improvement. If the technology is very different from previous technologies, there may be no evaluation routines that enable researchers to assess its progress or its potential. Furthermore, until the technology has established a degree of legitimacy, it may be difficult to attract other researchers to participate in its development.[9] However, as scientists or firms gain a deeper understanding of the technology, improvement begins to accelerate. The technology begins to gain legitimacy as a worthwhile endeavor, attracting other developers. Furthermore, measures for assessing the technology are developed, permitting researchers to target their attention toward those activities that reap the greatest improvement per unit of

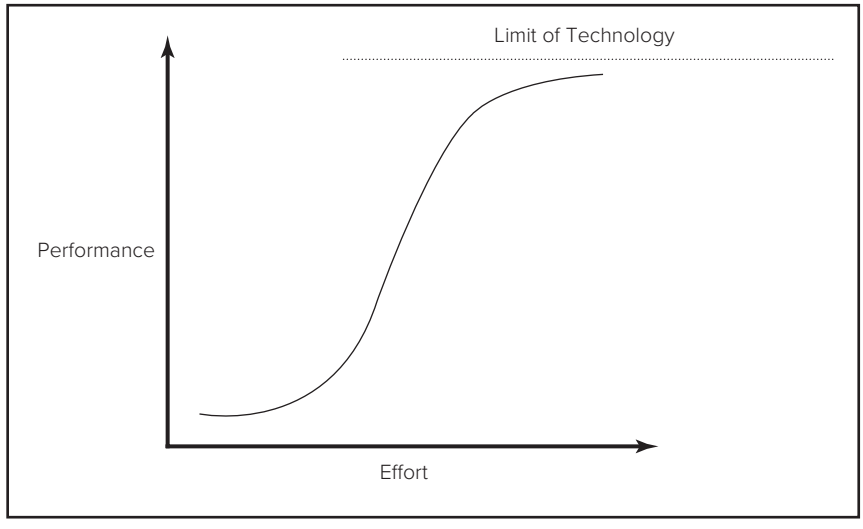


FIGURE 3.1
S-Curve of Technology Performance

effort, enabling performance to increase rapidly. However, at some point, diminishing returns to effort begin to set in. As the technology begins to reach its inherent limits, the cost of each marginal improvement increases, and the s-curve flattens.

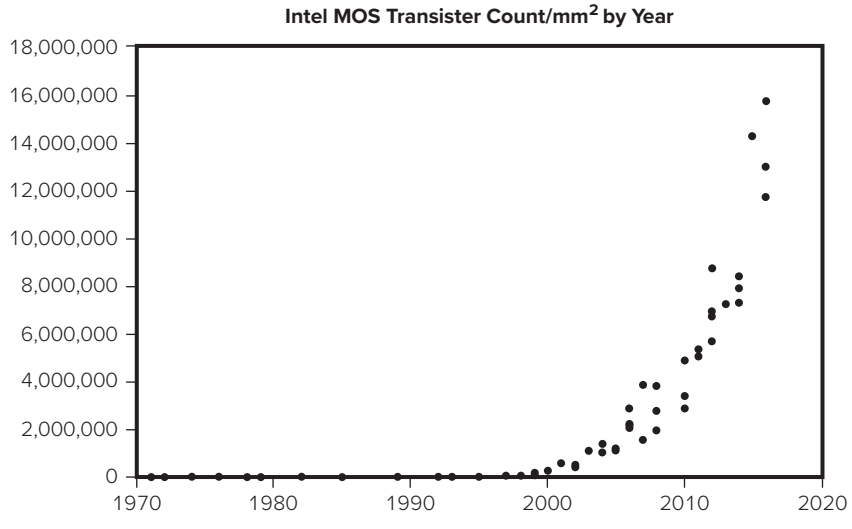
Often a technology's s-curve is plotted with performance (e.g., speed, capacity, or power) against time, but this must be approached with care. If the effort invested is not constant over time, the resulting s-curve can obscure the true relationship. If effort is relatively constant over time, plotting performance against time will result in the same characteristic curve as plotting performance against effort. However, if the amount of effort invested in a technology decreases or increases over time, the resulting curve could appear to flatten much more quickly, or not flatten at all. For instance, one of the more well-known technology trajectories is described by an axiom that became known as Moore's law. In 1965, Gordon Moore, cofounder of Intel, noted that the density of transistors on integrated circuits had doubled every year since the integrated circuit was invented. Figure 3.2 shows Intel's microprocessor transistor density from 1970 to 2020 and reveals a sharply increasing performance curve.

However, Intel's rate of investment (research and development dollars per year) also increased rapidly over that time frame. Not all of Intel's R&D expense goes directly to improving microprocessor power, but it is reasonable to assume that Intel's investment specifically in microprocessors would exhibit a similar pattern of increase. Figure 3.3 shows that the big gains in transistor density have come at a big cost in terms of effort invested. Though the curve does not yet resemble the traditional s-curve, its rate of increase is not nearly as sharp as when the curve is plotted against years.

discontinuous technology
A technology that fulfills a similar market need by building on an entirely new knowledge base.

Technologies do not always get the opportunity to reach their limits; they may be rendered obsolete by new, **discontinuous technologies**. A new innovation is discontinuous when it fulfills a similar market need, but does so by building on an entirely new knowledge base.[10] For example, the switches from propeller-based planes to jets,

FIGURE 3.2
Intel Transistor Density by Year, 1970–2020



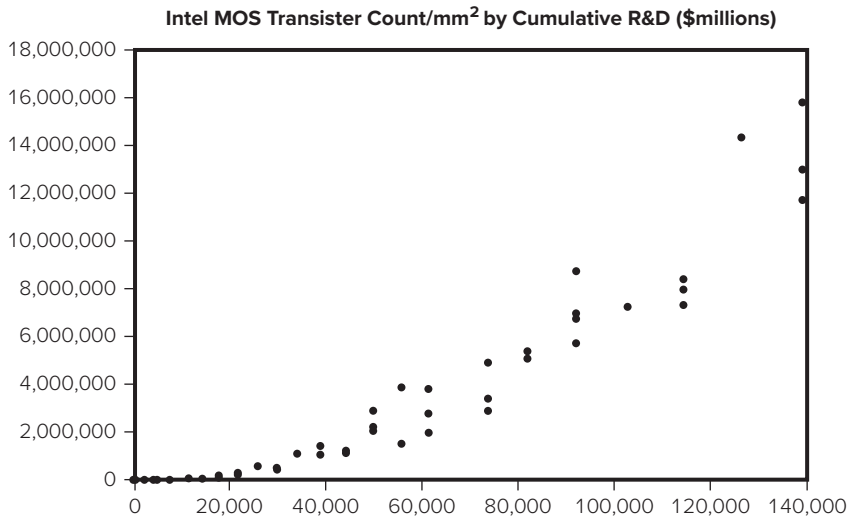


FIGURE 3.3
Intel Transistor Density by Cumulative R&D Spend ($millions), 1970–2020

from silver halide (chemical) photography to digital photography, from carbon copying to photocopying, and from audio on compact discs to MP3 were all technological discontinuities.

Initially, the technological discontinuity may have lower performance than the incumbent technology. For instance, one of the earliest automobiles, introduced in 1771 by Nicolas Joseph Cugnot, was never put into commercial production because it was much slower and harder to operate than a horse-drawn carriage. It was three-wheeled, steam-powered, and could travel at 2.3 miles per hour. A number of steam- and gas-powered vehicles were introduced in the 1800s, but it was not until the early 1900s that automobiles began to be produced in quantity.

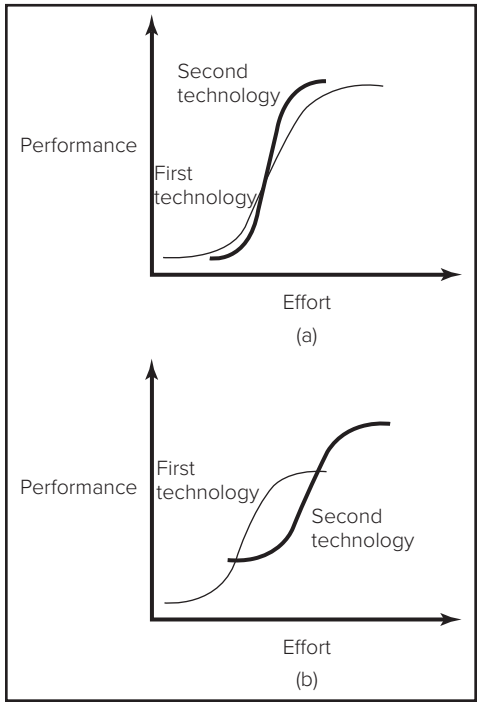


FIGURE 3.4
Technology S-Curves—Introduction of Discontinuous Technology

In early stages, effort invested in a new technology may reap lower returns than effort invested in the current technology, and firms are often reluctant to switch. However, if the disruptive technology has a steeper s-curve (see Figure 3.4a) or an s-curve that increases to a higher performance limit (see Figure 3.4b), there may come a time when the returns to effort invested in the new technology are much higher than effort invested in the incumbent technology. New firms entering the industry are likely to choose the disruptive technology, and incumbent firms face the difficult choice of trying to

extend the life of their current technology or investing in switching to the new technology. If the disruptive technology has much greater performance potential for a given amount of effort, in the long run it is likely to displace the incumbent technology, but the rate at which it does so can vary significantly.

S-Curves in Technology Diffusion

technology diffusion
The spread of a technology through a population.

S-curves are also often used to describe the diffusion of a technology. Unlike s-curves in technology performance, s-curves in **technology diffusion** are obtained by plotting the cumulative number of adopters of the technology against time. This yields an s-shape curve because adoption is initially slow when an unfamiliar technology is introduced to the market; it accelerates as the technology becomes better understood and utilized by the mass market, and eventually the market is saturated so the rate of new adoptions declines. For instance, when electronic calculators were introduced to the market, they were first adopted by the relatively small pool of scientists and engineers. This group had previously used slide rules. Then the calculator began to penetrate the larger markets of accountants and commercial users, followed by the still larger market that included students and the general public. After these markets had become saturated, fewer opportunities remained for new adoptions.[11]

One rather curious feature of technology diffusion is that it typically takes far more time than information diffusion.[12] For example, Mansfield found that it took 12 years for half the population of potential users to adopt industrial robots, even though these potential users were aware of the significant efficiency advantages the robots offered.[13] If a new technology is a significant improvement over existing solutions, why do some firms shift to it more slowly than others? The answer may lie in the complexity of the knowledge underlying new technologies, and in the development of complementary resources that make those technologies useful. Although some of the knowledge necessary to utilize a new technology might be transmitted through manuals or other documentation, other aspects of knowledge necessary to fully realize the potential of a technology might be built up only through experience. Some of the knowledge about the technology might be *tacit* and require transmission from person to person through extensive contact. Many potential adopters of a new technology will not adopt it until such knowledge is available to them, despite their awareness of the technology and its potential advantages.[14]

Furthermore, many technologies become valuable to a wide range of potential users only after a set of complementary resources are developed for them. For example, while the first electric light was invented in 1809 by Humphry Davy, an English chemist, it did not become practical until the development of bulbs within which the arc of light would be encased (first demonstrated by James Bowman Lindsay in 1835) and vacuum pumps to create a vacuum inside the bulb (the mercury vacuum pump was invented by Herman Sprengel in 1875). These early lightbulbs burned for only a few hours. Thomas Alva Edison built on the work of these earlier inventors when, in 1880, he invented filaments that would enable the light to burn for 1200 hours. The role of complementary resources and other factors influencing the diffusion of technological innovations are discussed further in Chapters Four, Five, and Thirteen.

Finally, it should be clear that the s-curves of diffusion are in part a function of the s-curves in technology improvement: As technologies are better developed, they become more certain and useful to users, facilitating their adoption. Furthermore, as

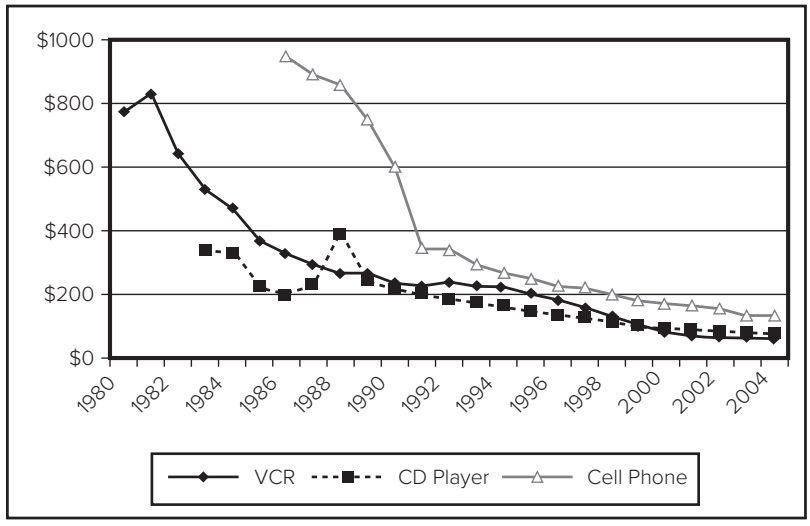


FIGURE 3.5
Average Sales Prices of Consumer Electronics

Source: Consumer Electronics Association.

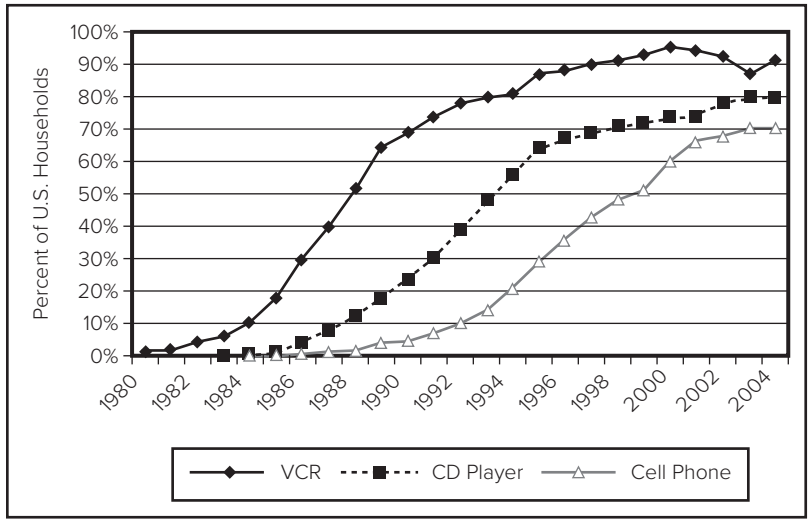


FIGURE 3.6
Penetration of Consumer Electronics

Source: Consumer Electronics Association.

learning-curve and scale advantages accrue to the technology, the price of finished goods often drops, further accelerating adoption by users. For example, as shown in Figures 3.5 and 3.6, drops in average sales prices for video recorders, compact disc players, and cell phones roughly correspond to their increases in household penetration.

S-Curves as a Prescriptive Tool

Several authors have argued that managers can use the s-curve model as a tool for predicting when a technology will reach its limits and as a prescriptive guide for whether and when the firm should move to a new, more radical technology.[15] Firms can use data on the investment and performance of their own technologies, or data on the

overall industry investment in a technology and the average performance achieved by multiple producers. Managers could then use these curves to assess whether a technology appears to be approaching its limits or to identify new technologies that might be emerging on s-curves that will intersect the firm's technology s-curve. Managers could then switch s-curves by acquiring or developing the new technology. However, as a prescriptive tool, the s-curve model has several serious limitations.

Limitations of S-Curve Model as a Prescriptive Tool

First, it is rare that the true limits of a technology are known in advance, and there is often considerable disagreement among firms about what a technology's limits will be. Second, the shape of a technology's s-curve is not set in stone. Unexpected changes in the market, component technologies, or complementary technologies can shorten or extend the life cycle of a technology. Furthermore, firms can influence the shape of the s-curve through their development activities. For example, firms can sometimes stretch the s-curve through implementing new development approaches or revamping the architecture design of the technology.[16]

Christensen provides an example of this from the disk-drive industry. A disk drive's capacity is determined by its size multiplied by its areal recording density; thus, density has become the most pervasive measure of disk-drive performance. In 1979, IBM had reached what it perceived as a density limit of ferrite-oxide–based disk drives. It abandoned its ferrite-oxide–based disk drives and moved to developing thin-film technology, which had greater potential for increasing density. Hitachi and Fujitsu continued to ride the ferrite-oxide s-curve, ultimately achieving densities that were eight times greater than the density that IBM had perceived to be a limit.

Finally, whether switching to a new technology will benefit a firm depends on a number of factors, including (*a*) the advantages offered by the new technology, (*b*) the new technology's fit with the firm's current abilities (and thus the amount of effort that would be required to switch, and the time it would take to develop new competencies), (*c*) the new technology's fit with the firm's position in complementary resources (e.g., a firm may lack key complementary resources, or may earn a significant portion of its revenues from selling products compatible with the incumbent technology), and (*d*) the expected rate of diffusion of the new technology. Thus, a firm that follows an s-curve model too closely could end up switching technologies earlier or later than it should.

TECHNOLOGY CYCLES

The s-curve model above suggests that technological change is cyclical: Each new s-curve ushers in an initial period of turbulence, followed by rapid improvement, then diminishing returns, and ultimately is displaced by a new technological discontinuity.[17] The emergence of a new technological discontinuity can overturn the existing competitive structure of an industry, creating new leaders and new losers. Schumpeter called this process *creative destruction*, and argued that it was the key driver of progress in a capitalist society.[18]

Several studies have tried to identify and characterize the stages of the technology cycle in order to better understand why some technologies succeed and others fail, and whether established firms or new firms are more likely to be successful in introducing

Research Brief The Diffusion of Innovation and Adopter Categories

S-curves in technology diffusion are often explained as a process of different categories of people adopting the technology at different times. One typology of adopter categories that gained prominence was proposed by Everett M. Rogers.[a] Figure 3.7 shows each of Rogers's adopter categories on a technology diffusion s-curve. The figure also shows that if the non cumulative share of each of these adopter groups is plotted on the vertical axis with time on the horizontal axis, the resulting curve is typically bell shaped (though in practice it may be skewed right or left).

INNOVATORS

Innovators are the first individuals to adopt an innovation. Extremely adventurous in their purchasing behavior, they are comfortable with a high degree of complexity and uncertainty. Innovators typically have access to substantial financial resources (and thus can afford the losses incurred in unsuccessful adoption decisions). Though they are not always well integrated into a particular social system, innovators play an extremely important role in the diffusion of an innovation because they are the individuals who bring new ideas into the social system. Rogers estimated that the first 2.5 percent of individuals to adopt a new technology are in this category.

EARLY ADOPTERS

The second category of adopters is the early adopters. Early adopters are well integrated into their social system and have the greatest potential for opinion leadership. Early adopters are respected by their peers and know that to retain that respect they must make sound innovation adoption decisions and they are typically willing to invest substiantial effort in researching their adoption decisions. Other potential adopters look to early adopters for information and advice; thus early adopters make excellent missionaries for new products or processes. Rogers estimated that the next 13.5 percent of individuals to adopt an innovation (after innovators) are in this category.

EARLY MAJORITY

Rogers identifies the next 34 percent of individuals in a social system to adopt a new innovation as the early majority. The early majority adopts innovations slightly before the average member of a social system. These adopters are more risk averse than the earlier two categories, and must be convinced that the product is popular and safe before adopting. They are typically not opinion leaders, but they interact frequently with their peers.

LATE MAJORITY

The next 34 percent of the individuals in a social system to adopt an innovation are the late majority, according to Rogers. Like the early majority, the late majority constitutes one-third of the individuals in a social system. Those in the late majority approach innovation with a skeptical air and may not adopt the innovation until they feel pressure from their peers. The late majority may have scarce resources, thus making them reluctant to invest in adoption until most of the uncertainty about the innovation has been resolved.

LAGGARDS

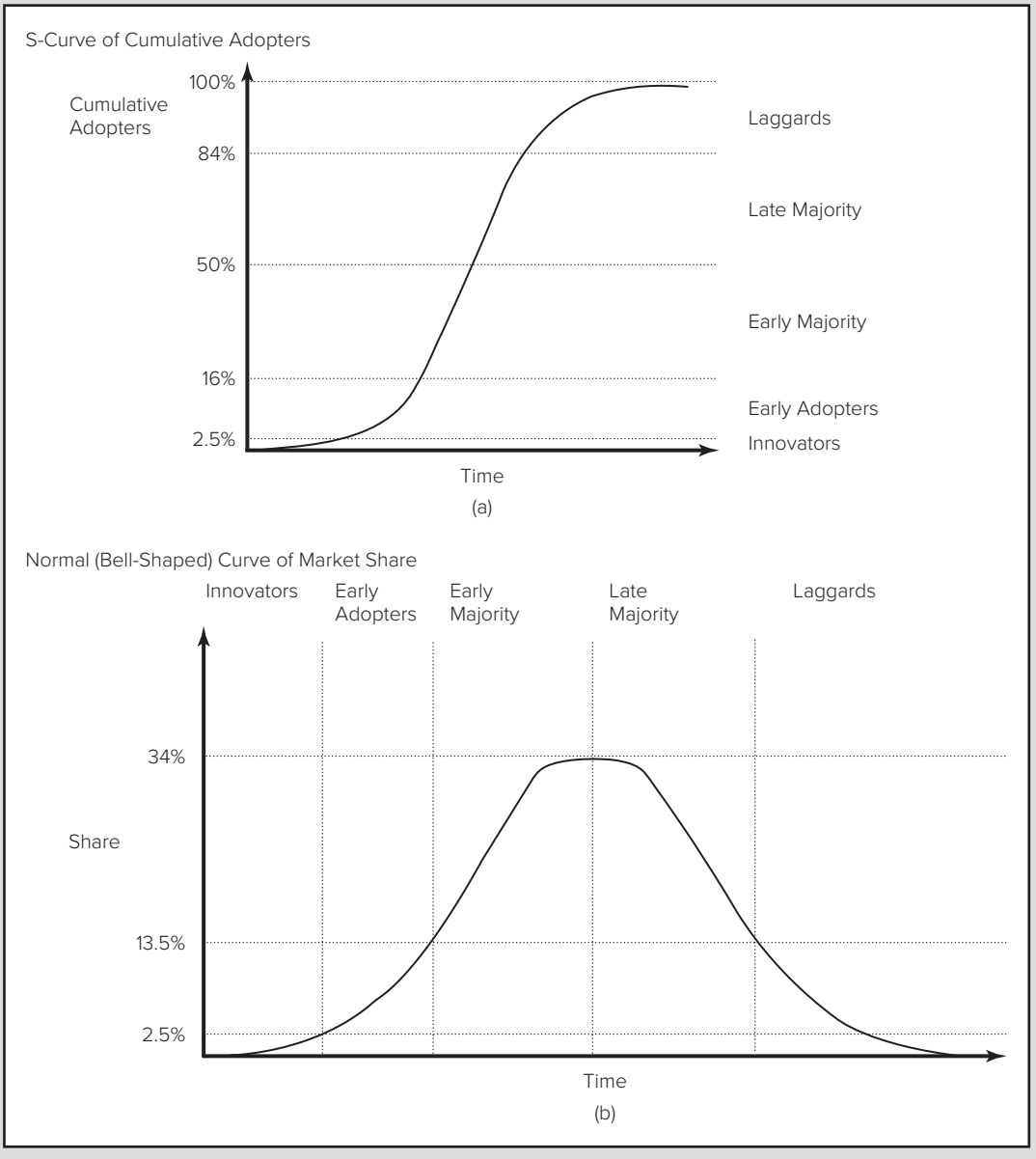
The last 16 percent of the individuals in a social system to adopt an innovation are termed *laggards*. They may base their decisions primarily upon past experience rather than influence from the social network, and they possess almost no opinion leadership. They are highly skeptical of innovations and innovators, and they must feel certain that a new innovation will not fail before adopting it.

[a] Everett M. Rogers, *Diffusion of Innovations*, 5th ed. (New York: Free Press, 2003).

continued

concluded

FIGURE 3.7
Technology Diffusion S-Curve with Adopter Categories



Theory in Action "Segment Zero"—A Serious Threat to Microsoft?

From 1980 to 2012, Microsoft was entrenched as the dominant personal computer operating system, giving it enormous influence over many aspects of the computer hardware and software industries. Though competing operating systems had been introduced during that time (e.g., Unix, Geoworks, NeXTSTEP, Linux, and the Mac OS), Microsoft's share of the personal computer operating system market held stable at roughly 85 percent throughout most of that period. In 2013, however, Microsoft's dominance in computer operating systems was under greater threat than it had ever been. A high-stakes race for dominance over the next generation of computing was well underway, and Microsoft was not even in the front pack.

"SEGMENT ZERO"

As Andy Grove, former CEO of Intel, noted in 1998, in many industries—including microprocessors, software, motorcycles, and electric vehicles—technologies improve faster than customer demands of those technologies increase. Firms often add features (speed, power, etc.) to products faster than customers' capacity to absorb them. Why would firms provide higher performance than that required by the bulk of their customers? The answer appears to lie in the market segmentation and pricing objectives of a technology's providers. As competition in an industry drives prices and margins lower, firms often try to shift sales into progressively higher tiers of the market. In these tiers, high-performance and feature-rich products can command higher margins. Though customers may also expect to have better-performing products over time, their ability to fully utilize such performance improvements is slowed by the need to learn how to use new features and adapt their work and lifestyles. Thus, while both the trajectory of technology improvement and the trajectory of customer demands are upward sloping, the trajectory for technology improvement is steeper (for simplicity, the technology trajectories are drawn in Figure 3.8 as straight lines and plotted against time in order to compare them against customer requirements).

In Figure 3.8, the technology trajectory begins at a point where it provides performance close to that demanded by the mass market, but over time it increases faster than the expectations of the mass market as the firm targets the high-end market. As the price of the technology rises, the mass market may feel it is overpaying for technological features it does not value. In Figure 3.9, the low-end market is not being served; it either pays far more for technology that it does not need or goes without. It is this market that Andy Grove, former CEO of Intel, refers to as segment zero.

For Intel, segment zero was the market for low-end personal computers (those less than $1000). While segment zero may seem unattractive in terms of margins, if it is neglected, it can become the breeding ground for companies that provide lower-end versions of the technology. As Grove notes, "The overlooked, underserved, and seemingly unprofitable end of the market can provide fertile ground for massive competitive change."[a]

FIGURE 3.8
Trajectories of Technology Improvement and Customer Requirements

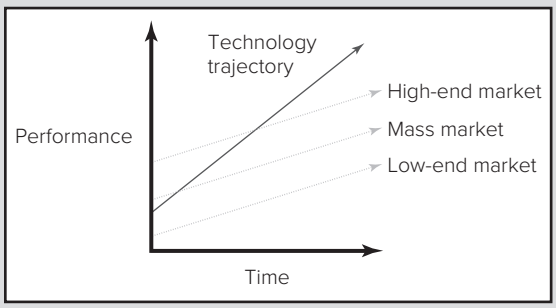
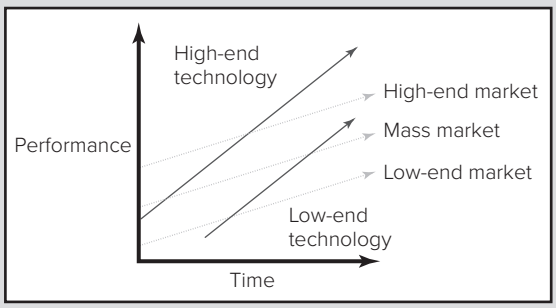


FIGURE 3.9
Low-End Technology's Trajectory Intersects Mass Market Trajectory



continued

concluded

As the firms serving low-end markets with simpler technologies ride up their own trajectories (which are also steeper than the slope of the trajectories of customer expectations), they can eventually reach a performance level that meets the demands of the mass market, while offering a much lower price than the premium technology (see Figure 3.9). At this point, the firms offering the premium technology may suddenly find they are losing the bulk of their sales revenue to industry contenders that do not look so low end anymore. For example, by 1998, the combination of rising microprocessor power and decreasing prices enabled personal computers priced under $1000 to capture 20 percent of the market.

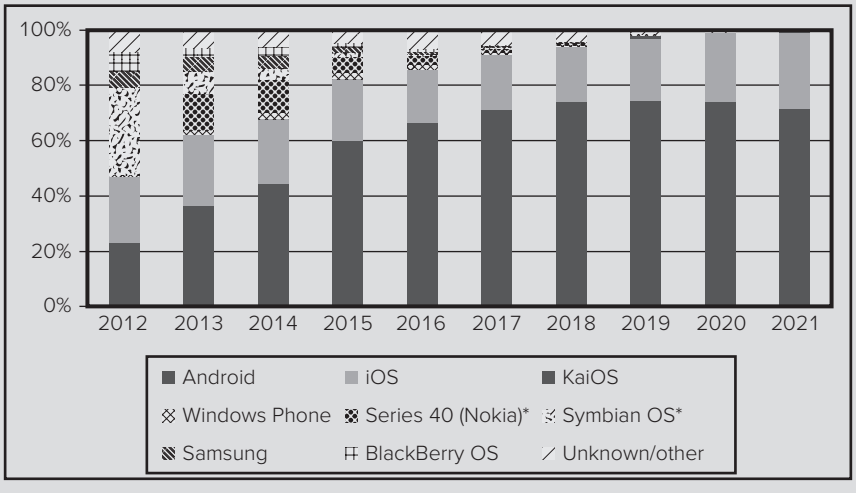
THE THREAT TO MICROSOFT

So where was the "segment zero" that could threaten Microsoft? Look in your pocket. In 2021, Apple's iPhone operating system (iOS) and Google's Android collectively controlled over 99 percent of the worldwide market for mobile operating systems, and mobile had increasingly become the dominant computing format (see Figure 3.10).[b] The iOS and Android interfaces offered a double whammy of beautiful aesthetics and remarkable ease of use. The applications business model used for the phones was also extremely attractive to both developers and customers, and quickly resulted in enormous libraries of applications that ranged from the ridiculous to the indispensible.

From a traditional economics perspective, the phone operating system market should not be that attractive to Microsoft—people do not spend as much on the applications, and the carriers have too much bargaining power, among other reasons. However, those smartphone operating systems soon became tablet operating systems, and tablets were rapidly becoming fully functional computers. Suddenly, all of that mindshare that Apple and Google had achieved in smartphone operating systems was transforming into mindshare in personal computer operating systems. Despite years of masterminding the computing industry, Microsoft's dominant position was at risk of evaporating.

[a] Andrew S. Grove, "Managing Segment Zero," *Leader to Leader,* 1999, 11.
[b] Statista, 2021.

FIGURE 3.10
Mobile Operating System Market Share 2012–2021



or adopting a new technology.[19] One technology evolution model that rose to prominence was proposed by Utterback and Abernathy. They observed that a technology passed through distinct phases. In the first phase (what they termed the *fluid phase*), there was considerable uncertainty about both the technology and its market. Products or services based on the technology might be crude, unreliable, or expensive, but might

suit the needs of some market niches. In this phase, firms experiment with different form factors or product features to assess the market response. Eventually, however, producers and customers begin to arrive at some consensus about the desired product attributes, and a **dominant design** emerges.[20] The dominant design establishes a stable architecture for the technology and enables firms to focus their efforts on process innovations that make production of the design more effective and efficient or on incremental innovations to improve components within the architecture. Utterback and Abernathy termed this phase the *specific phase* because innovations in products, materials, and manufacturing processes are all specific to the dominant design. For example, in the United States, the vast majority of energy production is based on the use of fossil fuels (e.g., oil and coal), and the methods of producing energy based on these fuels are well established. On the other hand, technologies that produce energy based on renewable resources (e.g., solar, wind, and hydrogen) are still in the fluid phase. Organizations such as Royal Dutch/Shell, General Electric, and Ballard Power are experimenting with various forms of solar photocell technologies, wind-turbine technologies, and hydrogen fuel cells in efforts to find methods of using renewable resources that meet the capacity and cost requirements of serving large populations.

dominant design
A product design that is adopted by the majority of producers, typically creating a stable architecture on which the industry can focus its efforts.

Building on the Utterback and Abernathy model, Anderson and Tushman studied the history of the U.S. minicomputer, cement, and glass industries through several cycles of technological change. Like Utterback and Abernathy, Anderson and Tushman found that each technological discontinuity inaugurated a period of turbulence and uncertainty (which they termed the *era of ferment*) (see Figure 3.11). The new technology might offer breakthrough capabilities, but there is little agreement about what the major subsystems of the technology should be or how they should be configured together. Furthermore, as later researchers noted, during the era of ferment different stakeholders might have different concepts of what purpose the technology should serve, or how a business model might be built around it.[21] Thus, while the new technology displaces the old (Anderson and Tushman refer to this as *substitution*), there is considerable design competition as firms experiment with different forms of the technology. Just as in the Utterback and Abernathy model, Anderson and Tushman

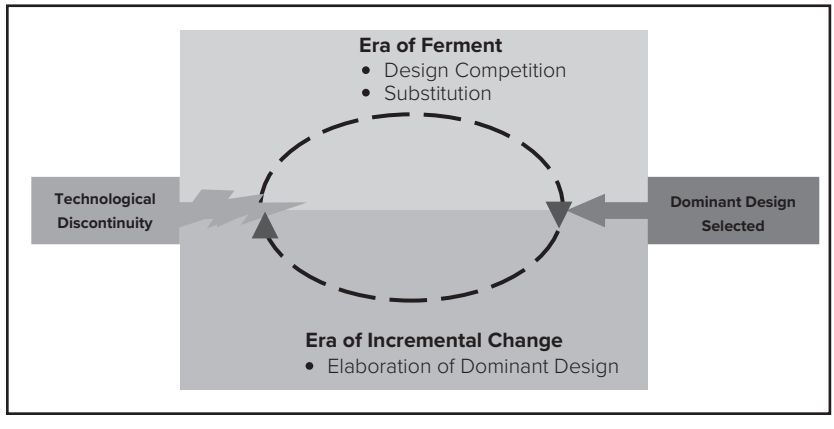


FIGURE 3.11
The Technology Cycle

found that a dominant design always arose to command the majority of the market share unless the next discontinuity arrived too soon and disrupted the cycle, or several producers patented their own proprietary technologies and refused to license to each other. Anderson and Tushman also found that the dominant design was never in the same form as the original discontinuity, but it was also never on the leading edge of the technology. Instead of maximizing performance on any individual dimension of the technology, the dominant design tended to bundle together a combination of features that best fulfilled the demands of the majority of the market.

In the words of Anderson and Tushman, the rise of a dominant design signals the transition from the era of ferment to the *era of incremental change.*[22] In this era, firms focus on efficiency and market penetration. Firms may attempt to achieve greater market segmentation by offering different models and price points. They may also attempt to lower production costs by simplifying the design or improving the production process. This period of accumulating small improvements may account for the bulk of the technological progress in an industry, and it continues until the next technological discontinuity.

Understanding the knowledge that firms develop during different eras lends insight into why successful firms often resist the transition to a new technology, even if it provides significant advantages. During the era of incremental change, many firms cease to invest in learning about alternative design architectures and instead invest in refining their competencies related to the dominant architecture. Most competition revolves around improving components rather than altering the architecture; thus, companies focus their efforts on developing component knowledge and knowledge related to the dominant architecture. As firms' routines and capabilities become more and more wedded to the dominant architecture, the firms become less able to identify and respond to a major architectural innovation. For example, the firm might establish divisions based on the primary components of the architecture and structure the communication channels between divisions on the basis of how those components interact. In the firm's effort to absorb and process the vast amount of information available to it, it is likely to establish filters that enable it to identify the information most crucial to its understanding of the existing technology design.[23] As the firm's expertise, structure, communication channels, and filters all become oriented around maximizing its ability to compete in the existing dominant design, they become barriers to the firm's recognizing and reacting to a new technology architecture.

While many industries appear to conform to this model in which a dominant design emerges, there are exceptions. In some industries, heterogeneity of products and production processes are a primary determinant of value, and thus a dominant design is undesirable.[24] For example, art and cuisine may be examples of industries in which there is more pressure to do things differently than to settle upon a standard.

Summary of Chapter

1. Different dimensions have been used to distinguish types of innovation. Some of the most widely used dimensions include product versus process innovation, radical versus incremental innovation, competence-enhancing versus competence-destroying innovation, and architectural versus component innovation.
2. A graph of technology performance over cumulative effort invested often exhibits an s-shape curve. This suggests that performance improvement in a new technology is

initially difficult and costly, but, as the fundamental principles of the technology are worked out, it then begins to accelerate as the technology becomes better understood, and finally diminishing returns set in as the technology approaches its inherent limits.

3. A graph of a technology's market adoption over time also typically exhibits an s-shape curve. Initially the technology may seem uncertain and there may be great costs or risks for potential adopters. Gradually, the technology becomes more certain (and its costs may be driven down), enabling the technology to be adopted by larger market segments. Eventually the technology's diffusion slows as it reaches market saturation or is displaced by a newer technology.
4. The rate at which a technology improves over time is often faster than the rate at which customer requirements increase over time. This means technologies that initially met the demands of the mass market may eventually exceed the needs of the market. Furthermore, technologies that initially served only low-end customers (segment zero) may eventually meet the needs of the mass market and capture the market share that originally went to the higher-performing technology.
5. Technological change often follows a cyclical pattern. First, a technological discontinuity causes a period of turbulence and uncertainty, and producers and consumers explore the different possibilities enabled by the new technology. As producers and customers begin to converge on a consensus of the desired technological configuration, a dominant design emerges. The dominant design provides a stable benchmark for the industry, enabling producers to turn their attention to increasing production efficiency and incremental product improvements. This cycle begins again with the next technological discontinuity.
6. The first design based on the initial technological discontinuity rarely becomes the dominant design. There is usually a period in which firms produce a variety of competing designs of the technology before one design emerges as dominant.
7. The dominant design rarely embodies the most advanced technological features available at the time of its emergence. It is instead the bundle of features that best meets the requirements of the majority of producers and customers.

Discussion Questions

1. What are some reasons that established firms might resist adopting a new technology?
2. Are well-established firms or new entrants more likely to (*a*) develop and/or (*b*) adopt new technologies? Why?
3. Think of an example of an innovation you have studied at work or school. How would you characterize it on the dimensions described at the beginning of the chapter?
4. What are some reasons that both technology improvement and technology diffusion exhibit s-shape curves?
5. Why do technologies often improve faster than customer requirements? What are the advantages and disadvantages to a firm of developing a technology beyond the current state of market needs?
6. In what industries would you expect to see particularly short technology cycles? In what industries would you expect to see particularly long technology cycles? What factors might influence the length of technology cycles in an industry?

Suggested Further Reading

Classics

Clayton M. Christensen, *The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail* (Boston: Harvard Business School Publishing, 1997).

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Chapter **Four**

Standards Battles, Modularity, and Platform Competition

Netflix and the Battle of the Streaming Services

By early 2021, the widespread penetration of broadband internet had led many consumers worldwide to subscribe to video streaming services. In fact, in the United States, over half of consumers had abandoned their traditional cable television subscriptions (and often also their landline phone lines) in favor of adopting purely internet-delivered video entertainment. Most of these services offered a combination of third-party movies and television content that could be streamed on demand, and some of the services also had significant libraries of original content.

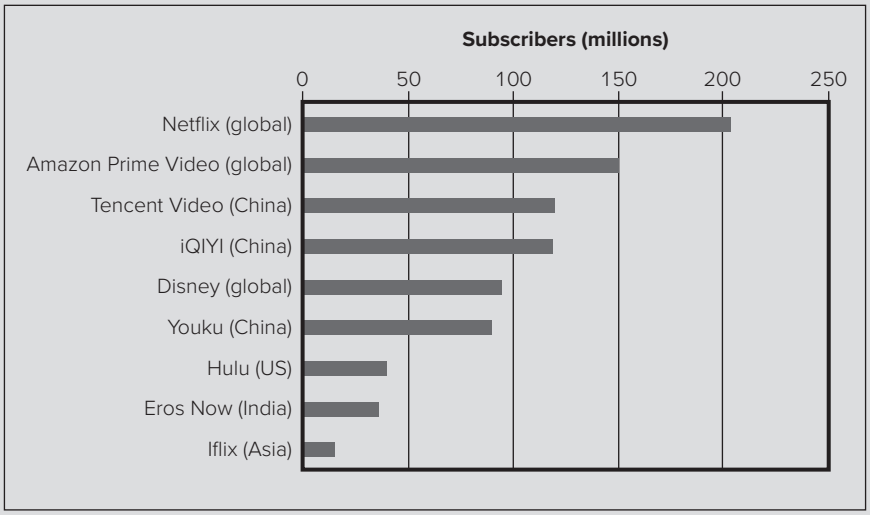
As of 2020, if people had been on the fence about adopting a video streaming service, COVID-19 removed their uncertainty: A theaters were forced to close, major streaming platforms became the only place to see the newest movies. Global box office revenues plummeted from $42.3 billion in 2019 to a paltry $12 billion in 2020, while streaming subscriptions rose 26 percent to 1.1 billion subscribers. In the United States, a survey from Sykes found that 71.7 percent of Americans had started new streaming subscriptions, and a survey by Deloitte found that U.S. households paid for an average of five streaming subscriptions, up from three in 2019.[a]

Globally, the overwhelming leader was Netflix (see Figure 1). In 2021, Netflix had over 200 million subscribers in over 190 countries. It had earned $25 billion in revenues in 2020, and rapid growth in both domestic and international subscribers had fueled intense investor enthusiasm, causing its market capitalization to reach $239 billion and making it one of the fastest growing stocks on the market.[b]

When Netflix was founded in 1997, its business model was to rent and sell movies on DVDs by mail. Customers could browse and select movies online,

FIGURE 1
The World's Most Popular Video Streaming Services

Source: Data compiled from multiple sources, including *Statista,* www.nationalthailand.com, www.cordcutternews.com, and company reports.



and those movies would be mailed out to the customer who would then mail the movies back after watching. Though it initially started with a per-movie rental fee like its largest bricks-and-mortar rival, Blockbuster, it soon moved to a subscription fee. Customers could choose among plans with different prices based on how many movies they wanted to rent simultaneously, and they could keep movies out as long as they wanted without late fees. The subscription plan was a hit, and by 2005, the company was shipping out over a million DVDs a day.[c]

One of the most compelling features of the Netflix site was its recommender system. As people rented movies, Netflix prompted them to review the movies they had already seen. It thus steadily accrued a massive database about correlations among movie preferences that it could use to make movie suggestions to users. For example, if a user gave a five-star rating to "Journey to the Center of the Earth," the system would suggest they might also like "The Mummy," "Indiana Jones and Kingdom of the Crystal Skull," and "Inkheart."

The service turned out to be enormously popular and soon sounded a death knell for bricks-and-mortar video stores. By having centralized inventory and shipping movies to people, Netflix could offer a much wider selection than physical stores could offer, and its scale meant it could both negotiate better prices on content and invest in value-added services for customers like the review and recommender systems mentioned previously, online movie trailers, and more. Importantly, Netflix was also a key channel for films by small independent filmmakers to reach audiences, enabling the company to forge relationships that would prove to be increasingly valuable as time passed.

In 2007, Netflix began offering movie streaming, which rapidly grew to be the preferred mode of movie rental consumption. Then in 2011, the company began acquiring original content for exclusive distribution on Netflix, starting with the series *House of Cards* and *Lilyhammer*. By 2013, it had moved into co-producing original content with production houses such as Marvel Television, and Dreamworks. Then in 2017, it opened Netflix Studios and began recruiting some of television's most successful writers and producers to start making original content in house.[d]

For a movie rental service to vertically integrate into developing its own content seemed a peculiar move. Making films and television shows required fundamentally different technology, equipment, personnel, and expertise than distributing films and television shows. What could a specialist in media distribution know about media production? A lot, it turns out.

Netflix's rapidly growing datasets meant that it knew which customers liked which films, which genres were growing, which new stars were gaining followings, which new production houses were gaining traction, and more. The relationships it had cultivated with small independent filmmakers and budding actors helped ensure the firm's access to a pipeline of new creative talent and helped build goodwill toward the company. Sean Fennessey, a writer for pop culture website *The Ringer,* explained how important Netflix was to frustrated filmmakers who could not raise enough support to get a major studio movie off the ground, "To the creators stifled by the rise of Hollywood's all-or-nothing focus on franchise films, Netflix felt like salve on an open wound."[e]

Netflix also used its massive distribution reach to promote its original content, building audiences for its series and crafting its reputation as a first-tier production house. As put by Ted Sarandos, Netflix Chief Content Officer, "the way we can secure those shows is having a great reputation with talent, having a brand people want to be associated with, and a good track record of delivering."

Furthermore, while most film studios needed to profit directly from their films, Netflix profited in multiple ways from its content: Having popular exclusive shows helped attract and retain subscribers and having both a large audience and a powerful library of original content gave it more bargaining power when negotiating license fees for content produced by others. Collectively, it was a powerful advantage.

Netflix planned to spend about $19 billion on original exclusive content in 2021, making it one of the largest film production companies in the world. Other streaming services were following suit, with Amazon Prime Video, Hulu, and HBO spending $6 billion, $3 billion, and $1.5 billion respectively on creating original content for their streaming services. Companies that have traditionally competed in content, like Disney, Universal, Paramount, and others were countering by offering their own streaming services. Some of the companies were forming interesting alliances—for example, Disney + and Hulu announced a discounted bundle that would also include ESPN, while Warner Brothers announced it would make all its movies available on HBO Max (a move that drew ire both from the

AMC Theater chain[f] and some filmmakers that felt their films would be sold short because of the decision).[g] The streaming market was rapidly becoming crowded and complicated.

When companies made content exclusive to their channel, they were gambling that the boost in subscriptions would be worth more than having wider distribution of that content—a tradeoff that could often be hard to calculate and predict. How many subscriptions would consumers be willing to pay for? How many exclusive hit shows did a service have to offer to convince a particular market segment to subscribe? Could content makers earn more by making their content available to all the platforms? These questions and more would determine the future of the video streaming services.

Discussion Questions

1. Based on your use of streaming services, do you feel the streaming platforms differ in terms stand-alone value (e.g., fidelity, ease of use, recommender systems, etc.)?
2. Again based on your use, how do you feel the streaming services differ based on the types and quality of content (the complementary good) they provide?
3. What factors do you think influence a consumer's choice of whether to subscribe to a streaming service? How important are the standalone features? How important is the content?
4. What are the pros and cons of having content be exclusive to a particular service?
5. Do you think the market will eventually choose a few services as "winners" and the other services will exit (or combine with the winners), or will the market continue to support many different services?

[a] Phil Hall, "The State of Streaming in 2021: A Fight for Content and Eyeballs," Benzinga, March 19, 2021.

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[d] Lucas Shaw, "Netflix Aims to Make the Next Great Hollywood Studio," *Chicago Tribune,* September 15, 2017.

[e] Sean Fennessey, "Netflix and Shill," *The Ringer,* April 20, 2018.

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OVERVIEW

The previous chapter described recurrent patterns in technological innovation, and one of those patterns was the emergence of a dominant design. As Anderson and Tushman pointed out, the technology cycle almost invariably exhibits a stage in

which the industry selects a **dominant design**. Once this design is selected, producers and customers focus their efforts on improving their efficiency in manufacturing, delivering, marketing, or deploying this dominant design, rather than continue to develop and consider alternative designs. In this chapter, we first will examine why industries experience strong pressure to select a single technology design as dominant and the multiple dimensions of value that will shape which technology designs rise to dominance. We will then look at why and how modularity and platform competition emerges in some industries.

WHY DOMINANT DESIGNS ARE SELECTED

dominant design
A single product or process architecture that dominates a product category—usually 50 percent or more of the market. A dominant design is a *de facto standard,* meaning that while it may not be officially enforced or acknowledged, it has become a standard for the industry.

increasing returns
When the *rate of return* (not just gross returns) from a product or process increases with the size of its installed base.

Why do many markets coalesce around a single dominant design rather than support a variety of technological options? One primary reason is that many industries exhibit **increasing returns to adoption**, meaning that the more a technology is adopted, the more valuable it becomes.[1] Complex technologies often exhibit increasing returns to adoption in that the more they are used, the more they are improved. A technology that is adopted usually generates revenue that can be used to further develop and refine the technology. Furthermore, as the technology is used, greater knowledge and understanding of the technology accrue, which may then enable improvements both in the technology itself and in its applications. Finally, as a technology becomes more widely adopted, complementary assets are often developed that are specialized to operate with the technology. These effects can result in a self-reinforcing mechanism that increases the dominance of a technology regardless of its superiority or inferiority to competing technologies. Two of the primary sources of increasing returns are (1) learning effects and (2) network externalities.

Learning Effects

Ample empirical evidence shows that the more a technology is used, the more it is developed and the more effective and efficient it becomes.[2] As a technology is adopted, it generates sales revenues that can be reinvested in further developing and refining the technology. Furthermore, as firms accumulate experience with the technology, they find ways to use the technology more productively, including developing an organizational context that improves the implementation of the technology. Thus, the more a technology is adopted, the better it should become.

One example of learning effects is manifest in the impact of cumulative production on cost and productivity—otherwise known as the *learning curve.* As individuals and producers repeat a process, they learn to make it more efficient, often producing new technological solutions that may enable them to reduce input costs or waste rates. Organizational learning scholars typically model the learning curve as a function of cumulative output: performance increases, or cost decreases, with the number of units of production, usually at a decreasing rate (see Figure 4.1). For example, in studies of industries as diverse as aircraft production and pizza franchises, researchers have consistently found that the cost of producing a unit (e.g., a pizza or an airplane) falls as the number of units produced increases.

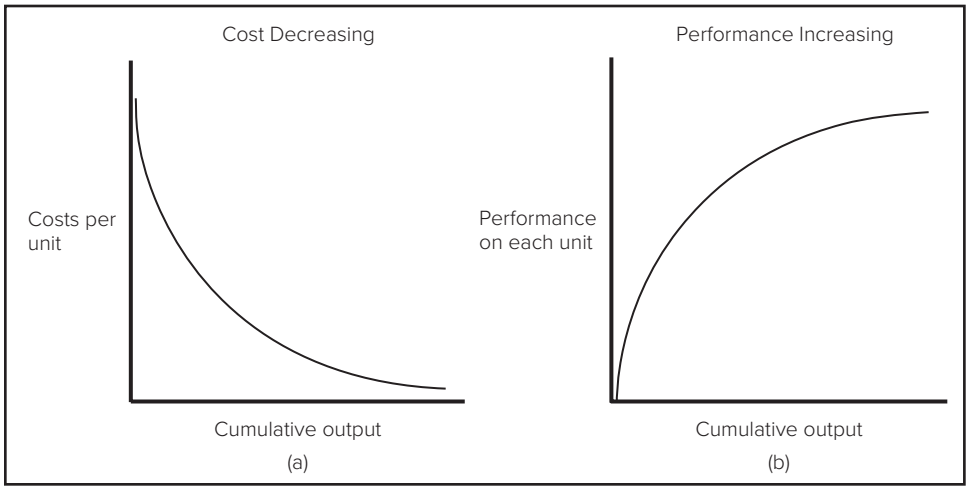


FIGURE 4.1
Standard Learning-Curve Forms

The standard form of the learning curve is formulated as $y = ax^{-b}$, where y is the number of direct labor hours required to produce the xth unit, a is the number of direct labor hours required to produce the first unit, x is the cumulative number of units produced, and b is the learning rate. This pattern has been found to be consistent with production data on a wide range of products and services, including the production of automobiles, ships, semiconductors, pharmaceuticals, and even heart surgery techniques.[3] Learning curves have also been identified by using a variety of performance measures, including productivity, total costs per unit, accidents per unit, and waste per unit.[4]

Though learning curves are found in a wide range of organizational processes, there are substantial differences in the rates at which organizations learn.[5] Both managers and scholars are very interested in understanding why one firm reaps great improvement in a process while another exhibits almost no learning. Many studies have examined reasons for this variability, including looking at how the firm's learning rate is affected by process-improvement projects, intentional innovation, or contact with customers and suppliers.[6] The results suggest the learning rate can be influenced by factors such as the nature of the task, firm strategy, and the firm's prior experience.

Prior Learning and Absorptive Capacity

absorptive capacity
The ability of an organization to recognize, assimilate, and utilize new knowledge.

A firm's investment in prior learning can accelerate its rate of future learning by building the firm's absorptive capacity.[7] **Absorptive capacity** refers to the phenomenon whereby as firms accumulate knowledge, they also increase their future ability to assimilate information. A firm's prior related experience shapes its ability to recognize the value of new information, and to utilize that information effectively. For example, in developing a new technology, a firm will often try a number of unsuccessful configurations or techniques before finding a solution that works well. This experimentation builds a base of knowledge in the firm about how key components behave, what alternatives are more likely to be successful than others, what types of projects the firm is most successful at, and so on. This knowledge base enables the firm to more rapidly assess the value of related new materials, technologies, and methods. The effects of absorptive

capacity suggest that firms that develop new technologies ahead of others may have an advantage in staying ahead. Firms that forgo investment in technology development may find it very difficult or expensive to develop technology in a subsequent period. This explains, in part, why firms that fall behind the technology frontier find it so difficult to catch up.

At the aggregate level, the more firms that are using a given technology and refining it, the more absorptive capacity that is being generated related to that technology, making development of that technology (and related technologies) more effective and efficient. Furthermore, as firms develop complementary technologies to improve the productivity or ease of utilization of the core technology, the technology becomes more attractive to other firms. In sum, learning effects suggest that early technology offerings often have an advantage because they have more time to develop and become enhanced than subsequent offerings. (However, as we shall discuss in Chapter Five, it is also possible to be *too early* to a market!)

network externalities
Also termed *positive consumption externalities*, this is when the value of a good to a user increases with the number of other users of the same or similar good.

Network Externalities

Many markets are characterized by **network externalities**, or positive consumption externalities.[8] In a market characterized by network externalities, the benefit from using a good increases with the number of other users of the same good. The classic examples of markets demonstrating network externality effects are those involving physical networks, such as railroads or telecommunications. Railroads are more valuable as the size of the railroad network (and therefore the number of available destinations) increases. Similarly, a telephone is not much use if only a few people can be called with it—the amount of utility the phone provides is directly related to the size of the network.

installed base
The number of users of a particular good. For instance, the installed base of a particular video game console refers to the number of those consoles that are installed in homes worldwide.

Network externalities can also arise in markets that do not have physical networks. For example, a user's benefit from using a good may increase with the number of users of the same good when compatibility is important. The number of users of a particular technology is often referred to as its **installed base**. A user may choose a computer platform based on the number of other users of that platform, rather than on the technological benefits of a particular platform, because it increases the ease of exchanging files. For example, many people choose a computer that uses the Windows operating system and an Intel microprocessor because the "Wintel" (*Windows* and *Intel*) platform has the largest installed base, thus maximizing the number of people with which the user's files will be compatible. Furthermore, the user's training in a particular platform becomes more valuable as the size of the installed base of the platform increases. If the user must invest considerable effort in learning to use a computer platform, the user will probably choose to invest this effort in learning the format he or she believes will be most widely used.

complementary goods
Additional goods and services that enable or enhance the value of another good. For example, the value of a video game console is directly related to the availability of complementary goods such as video games, peripheral devices, and services such as online gaming.

Network externalities also arise when **complementary goods** are important. Many products are functional or desirable only when there is a set of complementary goods available for them (videotapes for VCRs, film for cameras, etc.). Some firms make both a good and its complements (e.g., Kodak produced both cameras and film), whereas others rely on other companies to provide complementary goods or services for their products (e.g., computer manufacturers often rely on other vendors to supply service and software to customers). Products that have a large installed base are likely to attract more developers of complementary goods. This is demonstrated in the

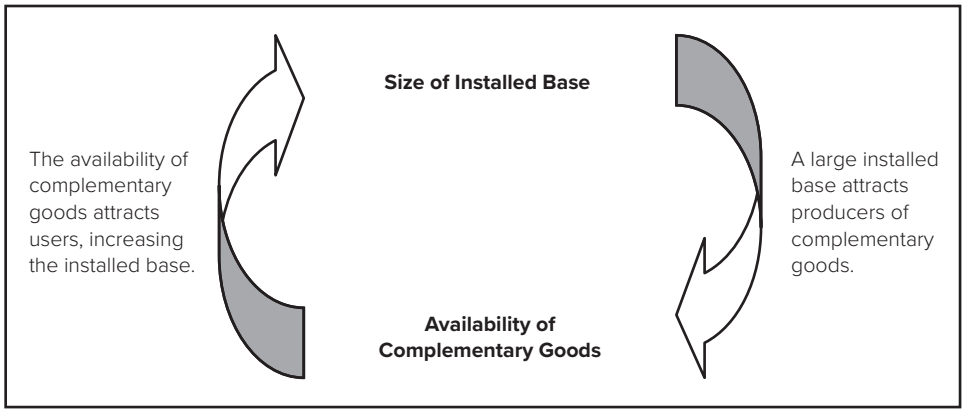


FIGURE 4.2
The Self-Reinforcing Cycle of Installed Base and Availability of Complementary Goods

Theory in Action about Microsoft: Once the Windows operating system had the largest installed base, most producers of complementary software applications chose to design their products to be optimized to work with Windows. Since the availability of complementary goods will influence users' choice among competing platforms, the availability of complementary goods influences the size of the installed base. A self-reinforcing cycle ensues (see Figure 4.2).

The effect of this cycle is vividly demonstrated by Microsoft's dominance of the operating system market, and later the graphical user interface market, as discussed in the Theory in Action on the rise of Microsoft. Microsoft's early advantage in installed base led to an advantage in the availability of complementary goods. These network externality advantages enabled Windows to lock several would-be contenders such as Geoworks and NeXT (and, some would argue, Apple) out of the market.

Firms can also attempt to influence the selection of a dominant design by building coalitions around a preferred technology.[9] This is aptly illustrated in the opening case. While the preceding has emphasized the emergence of dominant designs through market forces, occasionally a dominant design is put in place through government regulation.

Government Regulation

In some industries, the consumer welfare benefits of having compatibility among technologies have prompted government regulation, and thus a legally induced adherence to a dominant design. This has often been the case for the utilities, telecommunications, and television industries, to name a few.[10] For example, in 1953, the U.S. Federal Communications Commission (FCC) approved the National Television Systems Committee (NTSC) color standard in television broadcasting to ensure that individuals with monochrome television sets would be able to receive the color television programs broadcast by networks (though they would see them in black and white). That standard was still in place in 2003. Similarly, in 1998, while a battle was being fought in the United States over wireless technology formats, the European Union (EU) adopted a single wireless telephone standard (the general standard for mobile communications, or GSM). By choosing a uniform standard, the EU could avoid the proliferation of incompatible

Theory in Action The Rise of Microsoft

From the early 1980s to the decade beginning in 2010, Microsoft's Windows controlled an overwhelming share of the personal computer operating system market. An operating system is the main program on a computer, which enables it to run other programs. Operating systems are responsible for recognizing the input from a keyboard, sending output to the display, tracking files and directories on the disk drives, and controlling peripheral devices. Because the operating system determines how other software applications must be designed, Microsoft's dominance in the operating system market made it extraordinarily powerful in the software industry. However, Microsoft's emergence as a software superpower was due largely to the unfolding of a unique set of circumstances. Had these events played out differently, Microsoft's dominance might have never been.

In 1980, the dominant operating system for personal computers was CP/M. CP/M was invented by Gary Kildall and marketed by Kildall's company, Digital Research. Kildall had been retained by Intel in 1972 to write software for Intel's 4004, the first true microprocessor in that it could be programmed to do custom calculations. Later that year, Intel began to sell the 8008 to designers who would use it as a computer, and Kildall was hired to write a programming language for the chip, called PL/M (Programming Language/Microcomputers).[a]

Then Memorex and Shugart began offering floppy disks (which IBM had invented) as a replacement for punch cards, and Kildall acquired one of these drives. However, no existing program would make the disk drive communicate with Intel's microprocessor, so he wrote a disk operating system that he called Control Program/Microprocessor (CP/M).[b] CP/M could be adapted to any computer based on Intel microprocessors.

Before 1980, IBM, the world's largest computer producer, had not been interested in developing a personal computer. IBM managers could not imagine the personal computer market ever amounting to more than a small niche of hobbyists. However, when businesses began adopting Apple computers to do basic accounting or word processing, IBM began to get nervous. IBM suddenly realized that the personal computer market might become a significant industry, and if it wanted to be a major player in that market it needed to act fast. IBM's managers did not believe they had time to develop their own microprocessor and operating system, so they based their personal computer on Intel microprocessors and planned to use Kildall's CP/M operating system. There are many stories of why Kildall did not sign with IBM. One story is that Kildall was out flying his plane when IBM came around, and though the IBM managers left their names with Kildall's wife, Dorothy McEwen, they did not state the nature of their business, and Kildall did not get back to them for some time. Another version of the story posits that Kildall was reluctant to become tied into any long-term contracts with the massive company, preferring to retain his independence. Yet a third version claims that Kildall was simply more interested in developing advanced technologies than in the strategic management of the resulting products. Whatever the reason, Kildall did not sign with IBM.

Pressed for time, IBM turned to Bill Gates, who was already supplying other software for the system, and asked if he could provide an operating system as well. Though Gates did not have an operating system at that time, he replied that he could supply one. Gates bought a 16-bit operating system (basically a clone of CP/M) from Seattle Computer Company, and reworked the software to match IBM's machines. The product was called Microsoft DOS. With DOS bundled on every IBM PC (which sold more than 250,000 units the first year), the product had an immediate and immense installed base. Furthermore, the companies that emerged to fulfill the unmet demand for IBM PCs with clones also adopted Microsoft DOS to ensure that their products were IBM PC compatible. Because it replicated CP/M, Microsoft DOS was compatible with the range of software that had been developed for the CP/M operating system. Furthermore, after it was bundled with the IBM PC, more software was developed for the operating system, creating an even wider availability of complementary goods. Microsoft DOS was soon entrenched as the industry standard, and Microsoft was the world's fastest-growing software company.

"We were able to get the technology out into the market early to develop a standard. We were effective in soliciting software vendors to write to that platform to solidify the standard," said B. J. Whalen, Microsoft product manager. "Once you get it going, it's a snowball effect. The more applications you have available for a

continued

concluded

platform, the more people will want to use that platform. And of course, the more people that want to use that platform, the more software vendors will want to write to that platform."

Later Microsoft would develop a graphical interface named Windows that closely replicated the user-friendly functionality of Apple computers. By bundling Windows with DOS, Microsoft was able to transition its base of DOS customers over to the Windows system. Microsoft also worked vigorously to ensure that compatible applications were developed for DOS and Windows, making applications itself and also encouraging third-party developers to support the platform. Microsoft was able to leverage its dominance with Windows into a major market share in many other software markets (e.g., word processing, spreadsheet programs, presentation programs) and influence over many aspects of the computer software and hardware industries. However, had Kildall signed with IBM, or had Compaq and other computer companies been unable to clone the IBM personal computer, the software industry might look very different today.

[a] Paul Korzeniowski, "DOS: Still Thriving after All These Years," *Software Magazine* 10, no. 6 (1990): 83–112.
[b] S. Veit, "What Ever Happened to . . . Gary Kildall?" *Computer Shopper* 14, no. 11 (1994): 608–14.

standards and facilitate exchange both within and across national borders. Where government regulation imposes a single standard on an industry, the technology design embodied in that standard necessarily dominates the other technology options available to the industry. The consumer welfare impact of dominant designs is explored further in the Theory in Action section on Are Winner-Take-All Markets Good for Consumers?

The Result: Winner-Take-All Markets

All these forces can encourage the market toward natural monopolies. While some alternative platforms may survive by focusing on niche markets, the majority of the market may be dominated by a single (or few) design(s). A firm that is able to lock in its technology as the dominant design of a market usually earns huge rewards and may dominate the product category through several product generations. When a firm's technology is chosen as a dominant design, not only does the firm have the potential to earn near-monopoly rents in the short run, but the firm also is in a good position to shape the evolution of the industry, greatly influencing what future generations of products will look like. However, if the firm supports a technology that is not chosen as the dominant design, it may be forced to adopt the dominant technology, effectively forfeiting the capital, learning, and brand equity invested in its original technology. Even worse, a firm may find itself locked out of the market if it is unable to adopt the dominant technology. Such standards battles are high-stakes games—resulting in big winners and big losers.[11]

path dependency When end results depend greatly on the events that took place leading up to the outcome. It is often impossible to reproduce the results that occur in such a situation.

Increasing returns to adoption also imply that technology trajectories are characterized by **path dependency**, meaning that relatively small historical events may have a great impact on the final outcome. Though the technology's quality and technical advantage undoubtedly influence its fate, other factors, unrelated to the technical superiority or inferiority, may also play important roles.[12] For instance, timing may be crucial; early technology offerings may become so entrenched that subsequent technologies, even if considered to be technically superior, may be unable to gain a foothold in the market. How and by whom the technology is sponsored may also impact adoption. If, for example, a large and powerful firm aggressively

sponsors a technology (perhaps even pressuring suppliers or distributors to support the technology), that technology may gain a controlling share of the market, locking out alternative technologies.

The influence of a dominant design can also extend beyond its own technology cycle. As the dominant design is adopted and refined, it influences the knowledge that is accumulated by producers and customers, and it shapes the problem-solving techniques used in the industry. Firms will tend to use and build on their existing knowledge base rather than enter unfamiliar areas.[13] This can result in a very "sticky" technological paradigm that directs future technological inquiry in the area.[14] Thus, a dominant design is likely to influence the nature of the technological discontinuity that will eventually replace it.

Such winner-take-all markets demonstrate very different competitive dynamics than markets in which many competitors can coexist relatively peacefully. These markets also require very different firm strategies to achieve success.[15] Technologically superior products do not always win—the firms that win are usually the ones that know how to manage the multiple dimensions of value that shape design selection.

MULTIPLE DIMENSIONS OF VALUE

The value a new technology offers a customer is a composite of many different things. We first consider the value of the stand-alone technology, and then show how the stand-alone value of the technology combines with the value created by the size of the installed base and availability of complementary goods.[16] In industries characterized by increasing returns, this combination will influence which technology design rises to dominance.

A Technology's Stand-Alone Value

A technology's standalone value refers to the things it can do or its sources of appeal that are *not* due to its installed base or available complements. For example, a videogame console's standalone value might be a function of its speed, storage capacity, ease of use, or economical price, etc. A music streaming service's standalone value might be a function of whether it is easy to use, enables playlists, has a good recommender system, offers album details, or enables the users to download songs for offline play, etc.

The value a new technology offers to customers can be driven by many different things, such as the functions it enables the customer to perform, its aesthetic qualities, and its ease of use. To help managers identify the different aspects of utility a new technology offers customers, W. Chan Kim and Renee Mauborgne developed a "Buyer Utility Map."[17] They argue that it is important to consider six different utility levers, as well as six stages of the buyer experience cycle, to understand a new technology's utility to a buyer.

The stages they identify are *purchase, delivery, use, supplements, maintenance,* and *disposal.* The six utility levers they consider are *customer productivity, simplicity, convenience, risk, fun and image,* and *environmental friendliness.* Creating a grid with stages and levers yields a 36-cell utility map (see Figure 4.3). Each cell provides an opportunity to offer a new value proposition to a customer.

FIGURE 4.3
The Buyer Utility Map with Toyota Prius Example

Source: Adapted from *Harvard Business Review.* Exhibit from "Knowing a Winning Business Idea When You See One," by W. C. Kim and R. Mauborgne, September–October 2000.

	Purchase	Delivery	Use	Supplements	Maintenance	Disposal
Customer productivity	Price of Prius slightly higher than comparable nonhybrid models		Offers speed and power comparable to nonhybrid models	Can stop less often for gas, saving money and time		
Simplicity	Buyer may feel less able to assess value of vehicle		Operates like a regular combustion engine vehicle	Refuels like a regular combustion engine vehicle		Hybrids have larger batteries that would have to be recycled and disposed of at end of life
Convenience		Will be sold through traditional dealer channels	Does not have to be plugged into electrical outlet	Can purchase fuel at regular gas stations	Maintenance is similar to regular combustion engine vehicle	
Risk			Buyer might face a higher risk of product failure because it embodies a new technology		Buyer might have difficulty finding replacement parts because of new technology	Prius might be more difficult to resell or have lower resell value
Fun and image		Connotes image of environmental responsibility				
Environmental friendliness	Buyers feel they are helping support the development of more environmentally friendly cars		Emits lower levels of pollutants	Requires less use of fossil fuels		

A new technology might offer a change in value in a single cell or in a combination of cells. For example, when retailers establish an online ordering system, the primary new value proposition they are offering is greater *simplicity* in the *purchase* stage. On the other hand, as shown in Figure 4.3, the introduction of the Toyota Prius hybrid-electric vehicle offered customers greater productivity (in the form of gas savings), image benefits, and environmental friendliness in the customer's use, supplements, and maintenance stages, while providing the same simplicity and convenience of regular gasoline-only–powered vehicles.

Kim and Mauborgne's model is designed with an emphasis on consumer products, but their mapping principle can be easily adapted to emphasize industrial products or different aspects of buyer utility. For example, instead of having a single entry for customer productivity, the map could have rows for several dimensions of productivity such as speed, efficiency, scalability, and reliability. The map provides a guide for managers to consider multiple dimensions of technological value and multiple stages of the customer experience. Finally, the new benefits have to be considered with respect to the cost to the customer of obtaining or using the technology—it is the ratio of benefits to cost that determines value.

Network Externality Value

In industries characterized by network externalities, the value of a technological innovation to users will be a function not only of its stand-alone benefits and cost, but also of the value created by the size of its installed base and the availability of complementary goods (see Figure 4.4a).[18] Thus, the value to consumers of using the Windows operating system is due in part to the technology's stand-alone value (e.g., the ability of the operating system to make it easy for consumers to use the computer), the installed base of the operating system (and thus the number of computers with which the user can easily interact), and the availability of compatible software. Visualizing the value of technological innovations in this way makes it clear why even innovations that offer significant improvements in technological functionality often fail to displace existing technologies that are already widely adopted: Even if a new innovation has a significant advantage in functionality, its overall value may be significantly less than the incumbent standard. This situation is poignantly illustrated in the case of NeXT computers. In 1985, Steve Jobs and five senior managers of Apple Computer founded NeXT Incorporated. They unveiled their first computer in 1988. With a 25-MHz Motorola 68030 and 8 MB of RAM, the machine was significantly more powerful than most other personal computers available. It offered advanced graphics capability and even ran an object-oriented operating system (called NextStep) that was considered extremely advanced. However, the machine was not compatible with the IBM-compatible personal computers

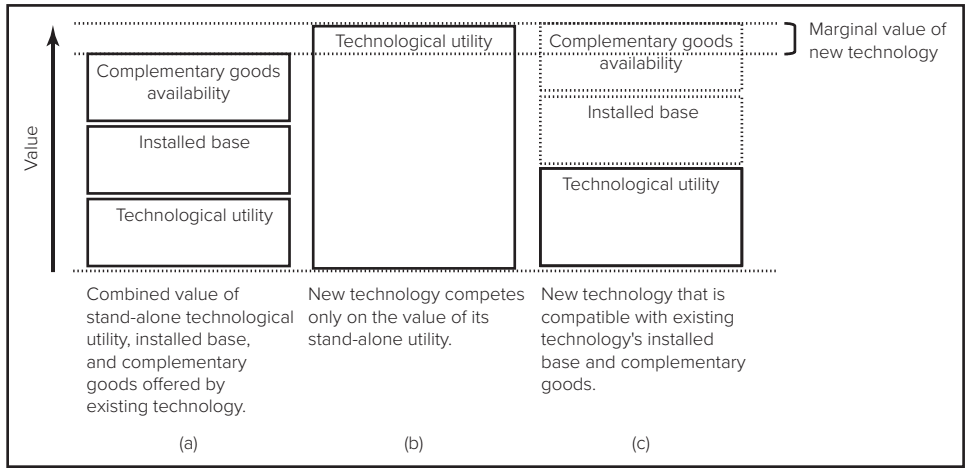


FIGURE 4.4
Components of Value

(based on Intel's microprocessors and Microsoft's operating system) that had become the dominant standard. The machine thus would not run the vast majority of software applications on the market. A small contingent of early adopters bought the NeXT personal computers, but the general market rejected them because of a dire lack of software and uncertainty about the company's viability. The company discontinued its hardware line in 1993 and ceased development of NextStep in 1996.

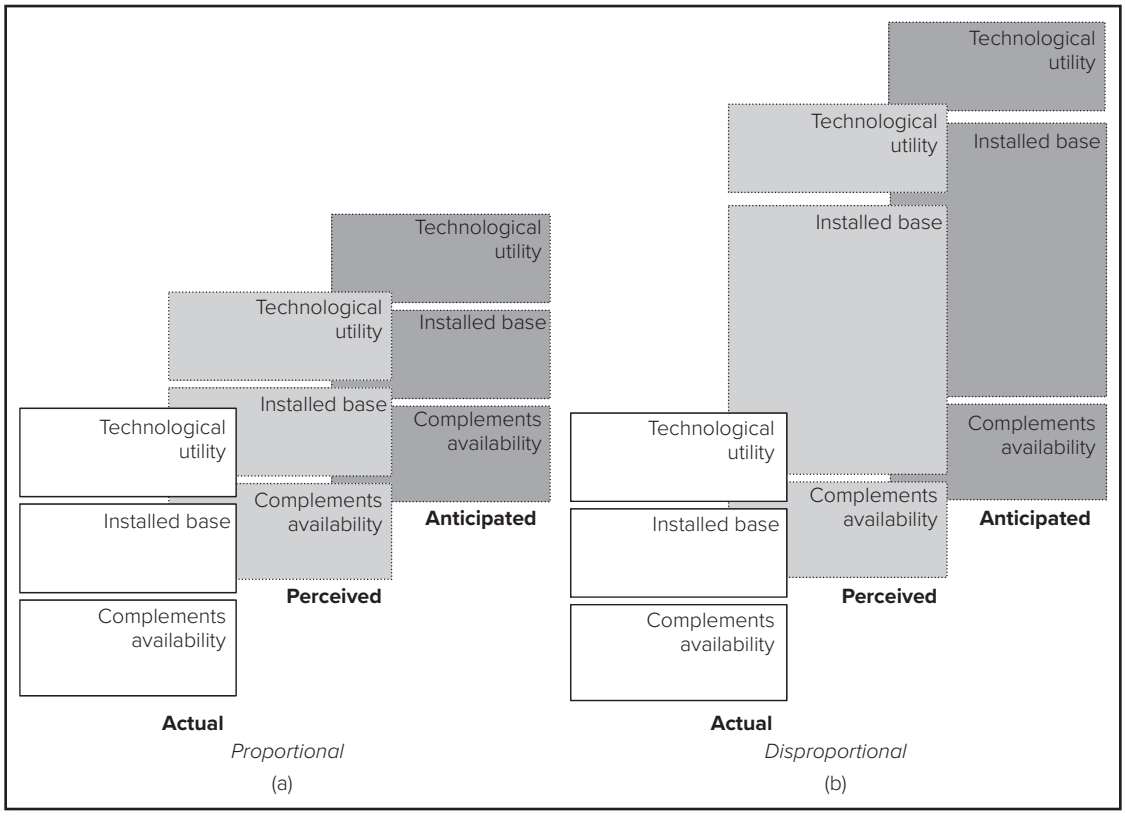
A similar battle was playing out in 2021 between smartphone operating systems, though in this case there were two contenders who were more evenly matched: Apple's iOS and Google's Android. Both companies offered smartphone operating systems with intuitive, powerful, and aesthetically pleasing interfaces (technological utility). Both were aggressively building communities of applications providers that provided large ranges of interesting and/or useful applications (complementary goods). Both were also trying to build installed base through aggressive marketing and distribution. Market share estimates of the two systems varied widely based on the timing of the data announcements, the geographical scope considered, and the product scope considered, but in early 2021 it was clear that Apple and Google were in a head-to-head battle for dominance and other mobile operating systems like Rim's BlackBerry OS and Microsoft's Windows Mobile were locked out (for more on this, see the section on the "Segment Zero" threat to Microsoft in Chapter Three).

As shown in Figure 4.4b, it is not enough for a new technology's stand-alone utility to exceed that of the incumbent standard. The new technology must be able to offer greater overall value. For the new technology to compete on its stand-alone utility alone, that utility must be so great that it eclipses the combined value of an existing technology's stand-alone utility, its installed base, and its complementary goods.

In some cases, the new technology may be made compatible with the existing technology's installed base and complementary goods as in Figure 4.4c. In this case, a new technology with only a moderate functionality advantage may offer greater overall value to users. Sony and Philips employed this strategy with their high-definition audio format, Super Audio CD (SACD), a high-density multichannel audio format based on a revolutionary "scalable" bit-stream technology known as Direct Stream Digital (DSD). Anticipating that users would be reluctant to replace their existing compact disk players and compact disk music collections, Sony and Philips made the new Super Audio CD technology compatible with existing compact disk technology. The Super Audio CD players included a feature that enables them to play standard CDs, and the recorded Super Audio CDs included a CD audio layer in addition to the high-density layer, enabling them to be played on standard CD systems. Customers can thus take advantage of the new technology without giving up the value of their existing CD players and music libraries.

When users are comparing the value of a new technology to an existing technology, they are weighing a combination of objective information (e.g., actual technological benefits, actual information on installed base or complementary goods), subjective information (e.g., perceived technological benefits, perceived installed base or complementary goods), and expectations for the future (e.g., anticipated technological benefits, anticipated installed base and complementary goods). Thus, each of the primary value components described above also has corresponding perceived or anticipated value components (see Figure 4.5). In Figure 4.5a, the perceived and anticipated value components map proportionately to their corresponding actual components.

FIGURE 4.5
Actual, Perceived, and Expected Components of Value



However, as depicted in Figure 4.5b, this need not be the case. For instance, perceived installed base may greatly exceed actual installed base, or customers may expect that a technology will eventually have a much larger installed base than competitors and thus the value accrued from the technology's installed base is expected to grow much larger than it is currently.

Firms can take advantage of the fact that users rely on both objective and subjective information in assessing the combined value offered by a new technology. For example, even a technology with a small installed base can achieve a relatively large mind share through heavy advertising by its backers. Producers can also shape users' expectations of the future installed base and availability of complements through announcements of preorders, licensing agreements, and distribution arrangements. For example, when Sega and Nintendo were battling for dominance in the 16-bit video game console market, they went to great lengths to manage impressions of their installed base and market share, often to the point of deception. At the end of 1991, Nintendo claimed it had sold 2 million units of the Super Nintendo Entertainment System in the U.S. market. Sega disagreed, arguing that Nintendo had sold 1 million units at most. By May 1992, Nintendo was claiming a 60 percent share of the 16-bit market, and Sega was claiming a 63 percent share![19]

Since perceived or expected installed base may drive subsequent adoptions, a large perceived or expected installed base can lead to a large actual installed base.

Such a tactic also underlies the use of "vaporware"—products that are not actually on the market and may not even exist but are advertised—by many software vendors. By building the impression among customers that a product is ubiquitous, firms can prompt rapid adoption of the product when it actually is available. Vaporware may also buy a firm valuable time in bringing its product to market. If other vendors beat the firm to market and the firm fears that customers may select a dominant design before its offering is introduced, it can use vaporware to attempt to persuade customers to delay purchase until the firm's product is available. The video game console industry also provides an excellent example here. When Sega and Sony introduced their 32-bit video game consoles (the Saturn and PlayStation, respectively), Nintendo was still a long way from introducing its next-generation console. In an effort to forestall consumer purchases of 32-bit systems, Nintendo began aggressively promoting its development of a 64-bit system (originally named Project Reality) in 1994, though the product would not actually reach the market until September 1996. The project underwent so many delays that some industry observers dubbed it "Project Unreality."[20] Nintendo was successful in persuading many customers to wait for its Nintendo 64, and the system was ultimately relatively successful.

Nintendo, however, was never able to reclaim dominance over the video game industry. By the time the Nintendo 64 had gained significant momentum, Sony was developing its even more advanced PlayStation 2. Sony's experience in VCRs and compact disks had taught it to manage the multiple dimensions of value very well: Sony's PlayStation 2 offered more than double the processing power of the Nintendo 64, it was backward compatible (helping the PlayStation 2 tap the value of customers' existing PlayStation game libraries), and Sony sold it for a price that many speculated was less than the cost of manufacturing the console ($299). Sony also invested heavily to ensure that many game titles would be available at launch, and it used its distribution leverage and advertising budget to ensure the product would seem ubiquitous at its launch.

Competing for Design Dominance in Markets with Network Externalities

Graphs illustrate how differing technological utilities and network externality returns to installed base or market share impact the competition for design dominance. The following figures examine whether network externalities create pressure for a single dominant design versus a few dominant designs by considering the rate at which value increases with the size of the installed base, and how large of an installed base is necessary before most of the network externality benefits are achieved. As explained earlier, when an industry has network externalities, the value of a good to a user increases with the number of other users of the same or similar good. However, it is rare that the value goes up linearly—instead, the value is likely to increase in an s-shape as shown in Figure 4.6a. Initially, the benefits may increase slowly. For example, whether a cell phone can reach 1 percent of the population or 5 percent is fairly insignificant—the reach of the phone service has to become much wider before the phone has much value. However, beyond some threshold level, the network externality returns begin to increase rapidly, until at some point, most of the benefits have been obtained and the rate of return decreases.

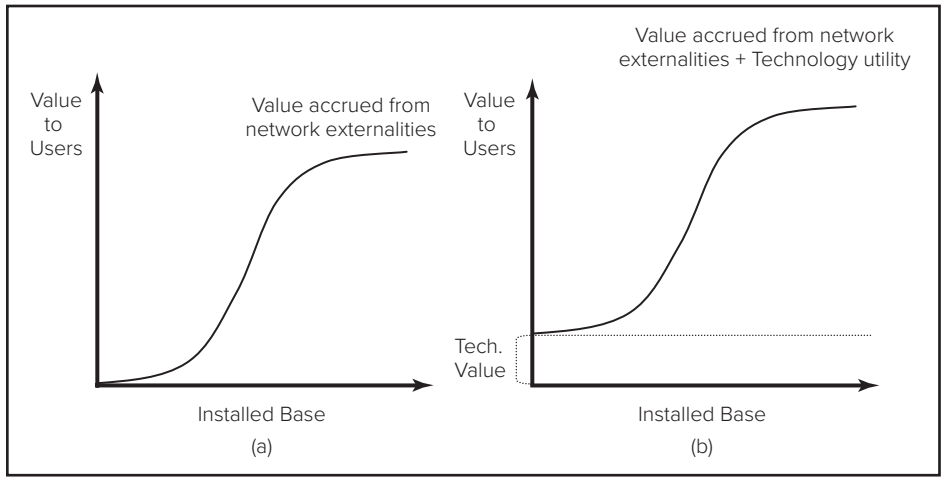


FIGURE 4.6
Network Externality Returns to Market Share

Consider the example of operating systems at the beginning of the chapter: If an operating system has too small of an installed base, few software developers will write applications for it and thus it will be of little value to consumers. An increase from a 1 percent market share to a 2 percent market share makes little difference—developers are still unlikely to be attracted to the platform. Once the operating system exceeds some threshold level of adoption, however, it becomes worthwhile to develop software applications for it, and the value of the operating system begins to increase rapidly. Once the operating system achieves a large share of the market, the user has probably obtained most of the network externality value. There is likely to be a large range of quality software available for the operating system, and incremental increases in available software have less marginal impact on the value reaped by the customer.

Next we consider the stand-alone functionality of the technology. In Figure 4.6b, a base level of technological utility has been added to the graph, which shifts the entire graph up. For example, an operating system that has an exceptionally easy-to-use interface makes the technology more valuable at any level of installed base. This becomes relevant later when two technologies that have different base levels of technological utility are considered.

When two technologies compete for dominance, customers will compare the overall value yielded (or expected) from each technology, as discussed in the previous section. In Figure 4.7, two technologies, A and B, each offer similar technological utility, and have similarly shaped network externality returns curves. To illustrate the competitive effects of two technologies competing for market share, the graphs in Figure 4.7 are drawn with market share on the horizontal axis instead of installed base. Furthermore, the curve for B is drawn with the market share dimension reversed so that we can compare the value offered by the two different technologies at different market share splits, that is, when A has a 20 percent market share, B has an 80 percent market share, and so on. This graph shows that at every point where A has less than 50 percent market share (and thus B has greater than 50 percent market share), B will yield greater overall value, making B more attractive to customers. On the other hand, when A has

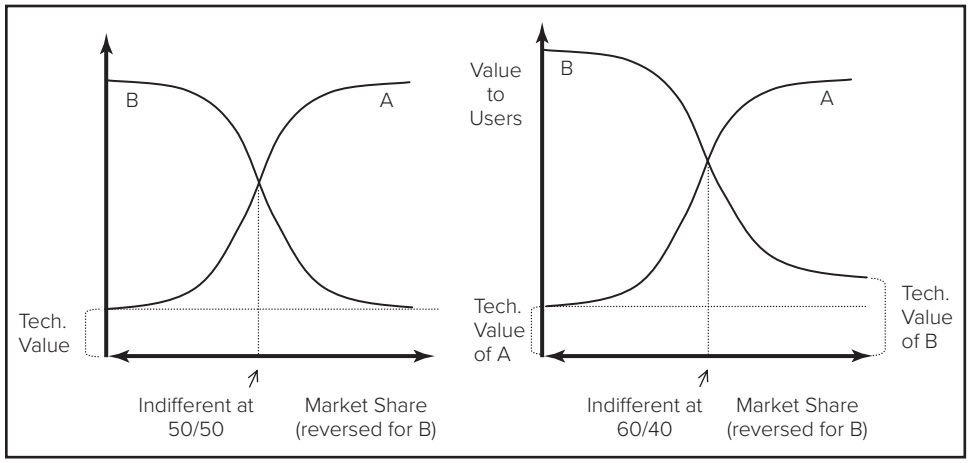


FIGURE 4.7
Network Externality Returns and Technological Utility: Competing Designs

greater than 50 percent market share (and B thus has less than 50 percent market share), A yields more overall value. When each technology has exactly 50 percent market share, they yield the same overall value and customers will be indifferent between them. However, if both technologies earn similar network externality returns to market share, but one technology offers greater stand-alone utility, the indifference point will be shifted in its favor. In the right-hand graph in Figure 4.7, technology B offers a greater level of stand-alone technological utility, shifting its overall value curve up. In this graph, technology A must have greater than 60 percent market share (and B must have less than 40 percent market share) for A to offer more overall value than B.

Another interesting scenario arises when customers attain their desired level of network externality benefits at lower levels of market share, depicted graphically in Figure 4.8. In this graph, the curves flatten out sooner, implying that the maximum amount of network externality value is obtained by customers at lower levels of market share.

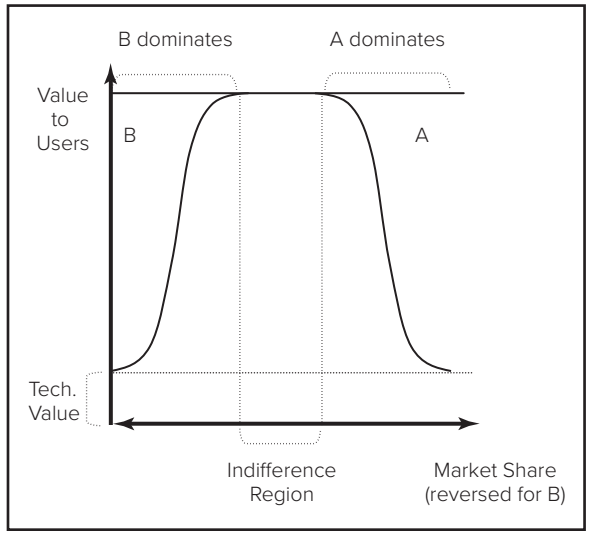


FIGURE 4.8
Network Externality Value Is Fully Tapped at Minority Market Share Levels

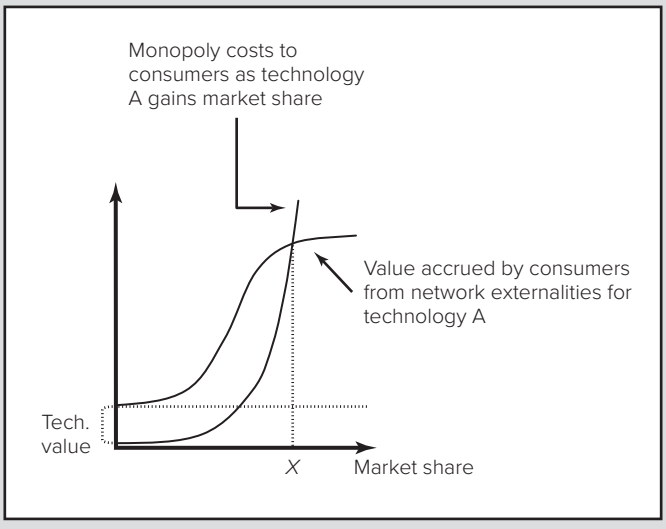
Theory in Action Are Winner-Take-All Markets Good for Consumers?

Traditionally, economics has emphasized the consumer welfare benefits of competitive markets; however, increasing returns make this a complicated issue. This is exemplified by the antitrust suits brought against Microsoft. While some analysts argued that Microsoft had clearly engaged in anticompetitive behavior and had damaged consumers in its quest to dominate the personal computer operating system market, others argued that Microsoft had behaved appropriately, and that its overwhelming share of the personal computer operating system market was good for consumers since it created greater compatibility among computers and more software applications. So how does a regulatory body decide when a firm has become too dominant? One way to think about this is to compare the value customers reap from network externalities at different levels of market share with the corresponding monopoly costs. Network externality returns refers to the value customers reap as a larger portion of the market adopts the same good (e.g., there is likely to be greater availability of complementary goods, more compatibility among users, and more revenues can be channeled into further developing the technology). Monopoly costs refer to the costs users bear as a larger portion of the market adopts the same good (e.g., a monopolist may charge higher prices, there may be less product variety, and innovation in alternative technologies may be stifled). Network externality returns to market share often exhibit the s-shape described in the previous section. Monopoly costs to market share, however, are often considered to be exponentially increasing. Plotting them on the same graph (as in Figure 4.9) reveals how network externality benefits and monopoly costs trade-off against each other.

In Figure 4.9, so long as technology A's market share remains less than *X*, the combination of technological utility and network externality benefits exceeds the monopoly costs, even if *X* represents a very large share of the market. However, as technology A's market share climbs beyond *X*, the monopoly costs now exceed the value of the technology utility and network externality benefits. A number of factors can shift where these two curves cross. If the technology utility for A were higher, the curves would cross at a point greater than *X*. If the network externality returns curve began to flatten at a lower market share (as was demonstrated earlier with the video game console industry), then the curves would cross at a market share less than *X*.

The steepness of the monopoly cost curve is largely a function of the firm's discretionary behavior. A firm can choose not to exploit its monopoly power, thus flattening

FIGURE 4.9
Network Externality Benefits and Monopoly Costs



continued

concluded

the monopoly costs curve. For instance, one of the most obvious assertions of monopoly power is typically exhibited in the price charged for a good. However, a firm can choose not to charge the maximum price that customers would be willing to pay for a good. For example, many people would argue that Microsoft does not charge the maximum price for its Windows operating system that the market would bear. However, a firm can also assert its monopoly power in more subtle ways, by controlling the evolution of the industry through selectively aiding some suppliers or complementors more than others, and many people would argue that in this respect, Microsoft has taken full advantage of its near-monopoly power.

In this case, customers may face a relatively large indifference region within which neither technology clearly dominates. This may be the case with the video game console industry: While customers may experience some network externality benefits to a console having significant share (more game titles, more people to play against), those benefits might be achieved by a console without attaining a majority of the market. For example, even with Sony, Microsoft, and Nintendo splitting the game console market, there is still an abundance of game titles for all three consoles and a significant pool of people to play games against. Such markets may not experience great pressure to select a single dominant design; two or more platforms may successfully coexist.

MODULARITY AND PLATFORM COMPETITION

As noted earlier in the chapter, in many industries the value of a product system is strongly related to the number and quality of complements available for it. A smartphone operating system, for example, is only as good as the applications available, and a music streaming service is only as valuable as the number and quality and music titles it offers. In markets like these, rather than having a single firm produce both the platform (e.g., the smartphone operating system) and all of the complements (e.g., applications for the smartphone), industry players will instead often use *modularity* to create a *platform ecosystem* where many different firms contribute to the product system.[21]

Modularity

modularity
The degree to which a system's components can be separated and recombined.

To understand **modularity**, consider a product like a computer: A computer is a bundle of components that includes a microprocessor, a monitor, a keyboard, and more. Some of these components can be bought separately and assembled by the user, and some are typically bought preassembled. Each of those components can also be thought of as a bundle. The monitor, for example, includes circuitry, plastic housing, power input, and so on. Most of these are not sold separately to the end user, but they might be mixed and matched by a manufacturer to achieve different end products.

Products may be made increasingly modular both by expanding the range of compatible components (increasing the range of possible product configurations), and by uncoupling integrated functions within components (making the product modular at a finer level) (see Figure 4.10). For example, a smartphone manufacturer might originally offer only proprietary phones where they have produced both the hardware and software, and integrate them tightly into a single product configuration (Figure 4.10, panel A). However, greater market demand for flexibility might induce manufacturers to begin offering

FIGURE 4.10
Modularity and Platform Ecosystems

A. Traditional integrated product bundle:
- Provider tries to meet buyers needs itself
- No customization, no external compatibility

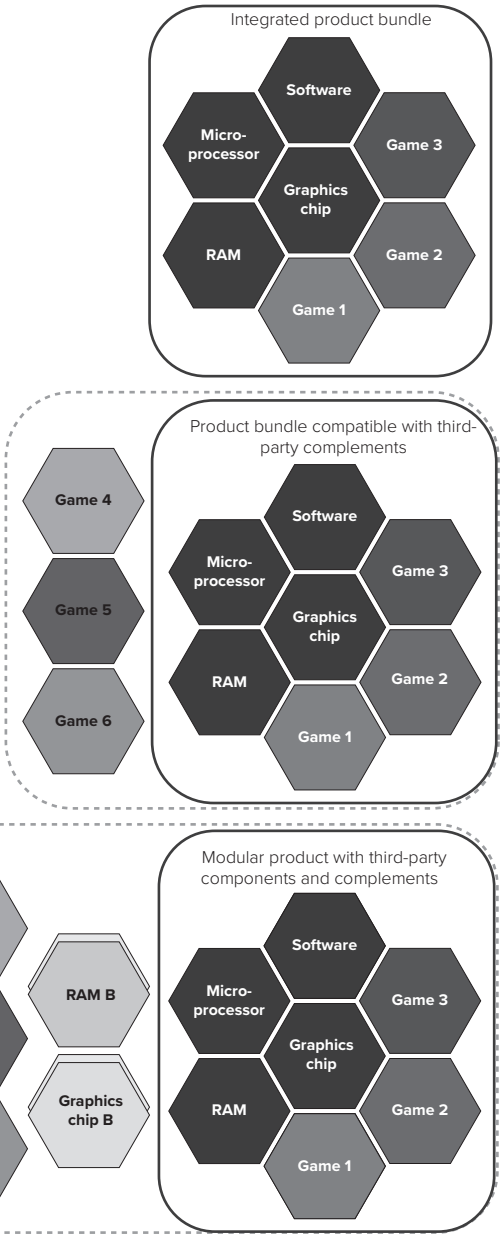
Example: Nokia E90 Communicator

B. Product bundle with third-party complements:
- Compatibility with third-party choices expands options for customers

Example: Apple iPhone

C. Product bundle with third-party components and complements:
- Customer has even greater range of configuration choices

Example: Android device



phones with a few different product configurations, each composed of the firm's own components. Should customers prefer to be able to combine phones with peripheral devices or applications developed by other producers, smartphone makers may eventually "open" their systems up, creating a standard by which other developers can create products that are compatible with the phones (Figure 4.10, panel B). Smartphone makers

may even decide to uncouple their operating systems from the hardware so that consumers can use the operating system on devices made by other manufacturers (Figure 4.10, panel C). In each of these stages, the product has become increasingly modular.

The majority of products are modular at some level. For example, many stereos, shelving systems, and bicycles allow the customer to mix and match components from different vendors. There are a great many more products that are modular in that the customer can choose components but not assemble the end product themselves. For example, when you buy a car, you can often choose an engine size, upholstery options, automatic steering or transmission, stereo system, tires, roof racks, security systems, and so on, but the automaker assembles the configuration for you. Even aircraft are offered this way: though Boeing and Airbus manufacture airframes, they do not manufacture engines. Engines are produced by companies such as GE, Pratt Whitney, or Rolls Royce. The engines are generally designed to be used in a few different aircraft models. Similarly, the aircraft is not committed to a single engine, though they may designate a "launch engine" as a preferred choice. The aircraft customer typically makes the final decision on which engine will be used in the aircraft.

Tightly integrated (i.e., nonmodular) product systems and modular systems have different kinds of advantages. A tightly integrated product system might have components that are customized to work together, which may enable a level of performance that more standardized components cannot achieve. The producer of a tightly integrated system also has more control over the end product, which can enable them to better monitor quality and reliability. For years this was the reason Steve Jobs gave for not making Apple computers as modular as Windows-based computers—he believed that by controlling all of the components and most of the software, Apple computers could achieve greater functionality and be more reliable. An integrated product may also be more attractive to a customer that does not want to choose or assemble components themselves.

platform ecosystems
Ecosystem is a contraction of *ecological* and *system*, and refers to a system where elements share some form of mutual dependence. A platform in this context is a stable core that mediates the relationship between a range of components, complements, and end users. Thus *platform ecosystem* refers to a system of mutually dependent entities mediated by a stable core.

Modular products, on the other hand, often offer more choices over function, design, scale, and other features, enabling the customer to choose a product system that more closely suits their needs and preferences. Second, because components are reused in different combinations, this can achieve product variety while still allowing scale economies in manufacturing the individual components. This is known as *economies of substitution.*[22]

Modularity becomes increasingly valuable in a product system when there are (a) diverse technological options available to be recombined, and (b) heterogeneous customer preferences.[23] For example, there is a very wide range of applications available for smartphones, and customers are very heterogeneous in the applications they want on their smartphones. This increases the value of being able to pick and choose your own customized mix of applications that go on your smartphone. This example also reveals how pressure for modularity can lead to platform ecosystems, as discussed below.

Platform Ecosystems

The word *ecosystem* comes from biology and is a contraction of *ecological system.* This means that when we use the term *ecosystem*, we are usually referring to entities that have some degree of mutual dependence. In a **platform ecosystem**, some core part of a product (such as a video game console or music streaming service) mediates

the relationship between a wide range of other components or complements (such as video games or music titles) and prospective end users.[24]

The rise of the Internet and digitalization has facilitated the reorganization of many industries around platforms and has transformed many firms into either platform sponsors or producers of complementary goods for a platform ecosystem. Mobile operating systems like iOS and Android, ride hailing services like Uber and Lyft, social networking services like Facebook and WeChat, and music and video streaming services like Spotify, Netflix, and Tencent Video are but a few examples of industries wherein platforms have become the dominant business model, and firms that successfully operate in these platform ecosystems often serve millions of end users.[25]

A platform's boundaries can be well-defined with a stable set of members dedicated wholly to that platform, or they can be amorphous and changing, with members entering and exiting freely, and participating in multiple platforms simultaneously. For example, consider the difference between the television/movie streaming services HBO on Demand and Amazon Prime. HBO on Demand exists to serve only HBO content up to consumers. The shows available are tightly controlled, and there is limited entry and exit of show producers. The Amazon Prime ecosystem is much more open. In fact, just about any content producer—including individual independent filmmakers—can make their content available on Amazon Prime.

Because it is the overall appeal of the ecosystem that attracts end users to the platform, the success of individual members depends, at least in part, upon the success of other members of the ecosystem—even those with which they may be simultaneously competing. Furthermore, in many platforms there are switching costs that make it difficult or costly to change ecosystems. Platforms and their complements often make investments in co-specialization or sign exclusivity agreements that bind them into stickier, longer-term relationships than the market contracts used in typical reseller arrangements. A video game that has been made for the Microsoft Xbox, for example, cannot be played on a PlayStation console unless a new version of it is made (and the game producer may have signed a contract with Microsoft that prohibits this).

A platform ecosystem is thus characterized by relationships that are neither as independent as arms-length market contracts nor as dependent as those within a hierarchical organization. It is, in essence, a hybrid organizational form.[26] It strikes a compromise between the loose coupling of a purely modular system and the tight coupling of a traditional integrated product. It enables customers to mix and match some components and complements, while still enabling some co-specialization and curation of the complements and components available for the system (see Figure 4.11).

Once we understand that platforms are like a compromise between pure modularity and pure integration, it becomes easier to understand when platforms will be desirable in a market. First, platforms will be more valuable than a tightly integrated product when (a) customers are diverse and want more choices than a single firm can provide, (b) when third-party options are diverse and high quality, (c) when compatibility with third-party products can be made seamless without integration, and/or (d) when the platform sponsor is powerful enough that it can retain control over quality and the overall product architecture without producing the complements itself. Looking at it from the other direction, platforms will be more valuable than a purely modular system when (a) complements are nonroutine purchases with uncertainty (and thus the customer prefers to have some

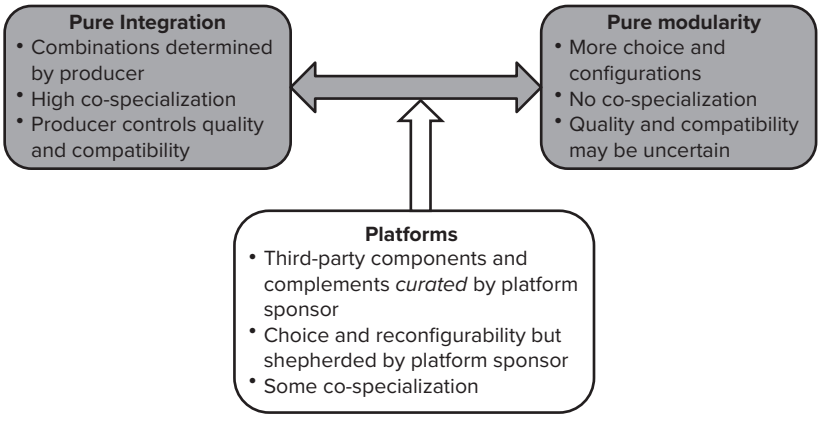


FIGURE 4.11
Platforms as a Compromise between Pure Modularity and Pure Integration

shepherding by the platform sponsor), (b) when some integration between the platform and its complements provides performance advantages, and/or (c) when important components of the ecosystem require subsidization (e.g., the market is unlikely to provide all the complements the end customer needs at adequate quality or value).

Video game systems are an iconic example of platform ecosystems. Consoles need to launch with high-quality games. Since it is difficult to induce game developers to make games for a console that has not yet been widely adopted, most game console producers must produce games themselves (or subsidize their production) to ensure that high-quality games are available when the console launches. On the other hand, end users want more games than just those produced by the console producer, so console producers like Microsoft, Sony, and Nintendo also license third-party developers to produce games for their consoles. They carefully screen the licensed games for quality and compatibility, and they may require the game developers to sign exclusivity agreements or to customize the games for the console. The console maker may also manage the end users' awareness and perception of the games in the ecosystem by giving "Best" of awards to particular games, by bundling particular games with the console at point of purchase, or by featuring particular games in its marketing. These strategies enable the console producer to actively manage the overall value created by its ecosystem.[27]

Summary of Chapter

1. Many technologies demonstrate increasing returns to adoption, meaning that the more they are adopted, the more valuable they become.
2. One primary source of increasing returns is learning-curve effects. The more a technology is produced and used, the better understood and developed it becomes, leading to improved performance and reduced costs.
3. Another key factor creating increasing returns is network externality effects. Network externality effects arise when the value of a good to a user increases with the size of the installed base. This can be due to a number of reasons, such as need for compatibility or the availability of complementary goods.

4. In some industries, the consumer welfare benefits of having a single standard have prompted government regulation, such as the European Union's mandate to use the GSM cellular phone standard.
5. Increasing returns can lead to winner-take-all markets where one or a few companies capture nearly all the market share.
6. The value of a technology to buyers is multidimensional. The stand-alone value of a technology can include many factors (productivity, simplicity, etc.) and the technology's cost. In increasing returns industries, the value will also be significantly affected by the technology's installed base and availability of complementary goods.
7. Customers weigh a combination of objective and subjective information. Thus, a customer's perceptions and expectations of a technology can be as important as (or more important than) the actual value offered by the technology.
8. Firms can try to manage customers' perceptions and expectations through advertising and public announcements of preorders, distribution agreements, and so on.
9. The combination of network externality returns to market share and technological utility will influence at what level of market share one technology will dominate another. For some industries, the full network externality benefits are attained at a minority market share level; in these industries, multiple designs are likely to coexist.
10. In markets where customers have heterogeneous preferences and there are many potential technological options available, firms might use modularity to enable customers to mix and match components, producing a wider array of end products.
11. Platform ecosystems are an example of modularity in action. A stable core platform (e.g., a smartphone operating system) may mediate the relationship between many complements producers (e.g., applications developers) and end users.

Discussion Questions

1. What are some of the sources of increasing returns to adoption?
2. What are some examples of industries not mentioned in the chapter that demonstrate increasing returns to adoption?
3. What are some of the ways a firm can try to increase the overall value of its technology and its likelihood of becoming the dominant design?
4. What determines whether an industry is likely to have one or a few dominant designs?
5. Are dominant designs good for consumers, competitors, complementors, and suppliers?

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Chapter **Five**

Timing of Entry

Failure to Launch at Uber Elevate[1]

In April of 2017, Jeff Holden, the chief product officer at Uber Technologies Inc. announced a radically new product: uberAIR, an on-demand air transportation service that would be part of its broader "flying car" project, Uber Elevate.

> "On-demand aviation, has the potential to radically improve urban mobility, giving people back time lost in their daily commutes. . .Just as skyscrapers allowed cities to use limited land more efficiently, urban air transportation will use three-dimensional airspace to alleviate transportation congestion on the ground. A network of small, electric aircraft that take off and land vertically (called Vertical Take-off and Landing, or VTOL, and pronounced vee-tol), will enable rapid, reliable transportation between suburbs and cities and, ultimately, within cities."[a]

Uber's on-demand ride-sharing service had seriously disrupted traditional taxi and livery, and induced many people to eschew car ownership altogether. However, that service was based on an innovative business model and a software application—it did not require technological advances in either automobiles or driving infrastructure. Uber Elevate was almost the opposite: It would leverage Uber's existing business model and software programs but would require major technological development in air transportation technology, infrastructure for air traffic control, and a network of landing pads. It was an ambitious project, to put it mildly.

Uber's Rise

Uber was founded in 2009 as a taxi-like ride-sharing service. Customers could request a ride using a smartphone application, and the software would notify Uber drivers in the area of the request. When one accepted, it would show the customer the car's approach on a map in real-time. The driver would take the user to their destination, and payment would occur automatically using the customer's credit card information that was stored online.

Perhaps the most unique part of the business model was that Uber drivers did not technically work for Uber. Drivers were independent contractors; they only needed a smartphone, a driver's license, a car, insurance, and a clean driving record to qualify to become an Uber driver.[b] A dynamic pricing model raised

prices when demand was high and drivers were few, and the higher prices, in turn, lured more Uber drivers to start accepting ride requests.

Over time, the company added different classes of services (such as UberPool, Uber X, Uber XL, and Uber Select), and different services such as food delivery (UberEats), freight service (Uber Freight), and pet transport (Uber Pets). It also had development underway for its own fleet of autonomous cars.

Though the firm had endured numerous early conflicts with taxi unions and some highly public scandals involving one of the company's founders, Travis Kalanick, the service was a huge success. By 2015, it had completed its one billionth ride, making it the second largest ride-sharing service worldwide, after Didi Chuxing in China. By 2021, it was operating in more than 900 cities in 69 countries worldwide. Its 2020 revenue was $11.14 billion, and in May 2021, its market capitalization was over $86 billion.

Opportunities and Challenges for uberAIR

At the time of Uber's announcement of their new air service, there were more than 70 companies developing electric VTOLs (eVTOLS—electric vertical take-off and landing aircraft), including Karem Aircraft, Embraer, Aurora Flight Sciences, and Bell Helicopter. However, there were many obstacles that had to be overcome to make Uber's air taxi ambitions a reality. First, an eVTOL used an enormous amount of energy and would be heavily reliant on advances on battery development and charging infrastructure. Second, there would be numerous legal and safety issues to be worked out pertaining to air traffic control, pilot training and licensing, compatibility with city infrastructures, noise, adverse weather, and more.

Last but not least was cost. The technology to vertically take off and land already existed in the form of helicopters, but most people have never ridden in one because it is an extremely expensive mode of transportation, estimated to be at least $8.93 per passenger per mile. How would uberAIR be different? First, electric propulsion was expected to be much more fuel efficient and require less maintenance. Second, the much smaller eVTOLs could land at flexible "skyports" rather than the large helipads or airports that helicopters used. Uber estimated that its initial operating costs would be $5.73 per passenger per mile, and with efficient pooling, it believed it could get the operating costs down to $1.84 per passenger per mile. As people began to use eVTOLs in large numbers, scale economies would also drive down the cost of producing the eVTOLs themselves. The final major cost-cutting measure would be to eliminate pilots with fully autonomous eVTOLs, saving training costs, salaries, and making room for an additional passenger. Uber estimated that with fully autonomous operation at scale, the long-run operating costs of uberAIR could be as low as 44 cents per passenger mile—less than the operating cost of many cars.[c]

Uber's CEO, Dara Khosrowshahi was initially doubtful about the project. However, after several rounds of discussion on the economics of it, he began to be persuaded. "For me the 'aha' moment came when I started understanding that Uber isn't just about cars," Khosrowshahi said. "Ultimately where we want to go is about urban mobility and urban transport, and being a solution for the cities in which we operate."[d]

In 2018, the company publicly announced that it planned to have commercial deployment of the service by 2023. However, Uber's businesses had yet to become profitable and investors were beginning to get impatient with Uber's grand ambitions. After racking up massive losses in 2019 and 2020 (of 8.5 and 6.8 billion, respectively), the company began paring back its projects. In early December 2020, it announced it would sell off its autonomous vehicles division to Aurora Innovation and its air taxi division to Joby Aviation, in efforts to get to profitability by 2021.[e] Uber had previously invested $50 million in Joby Aviation, and although the terms of the deal by which it transferred its Elevate assets to Joby were not disclosed, it was revealed that Uber agreed to invest an additional $75 million in Joby. The deal also called for Joby and Uber to integrate their respective services into each other's applications to create "seamless integration between ground and air travel for future customers."[f]

Battle for the Skies

As of mid-2021, no company had yet launched an air taxi service using eVTOLs[g], but a few appeared to be getting close. Notably, German company Lilium announced that it would begin commercial inter-city transportation with its five-seat, all-electric Lilium Jet beginning in 2025. Another German company, Volocopter, was betting that it could beat Lilium to the commercial market with its VoloCity, a two-person craft that looked like an egg suspended under a halo of 18 rotors designed for intra-city travel.[h] Guangzhou-based EHang also looked like it was making great strides, but then in early 2021, it was the subject of a scandal when an investment group accused the company of inflating its value "with a collection of lies about its products, manufacturing, revenues, partnerships and potential regulatory approval."[i] Finally, in the United States, both Palo Alto-based Archer and Santa Cruz-based Joby Aviation were promising to launch commercial operations by 2024.[j] All of these companies had already repeatedly demonstrated their aircraft; the hurdles were now largely based on regulation and infrastructure.

Discussion Questions

1. Will there be increasing returns to adoption for an early mover in air taxi service? If so, what will they be?
2. What are the disadvantages of entering the air taxi market early?
3. What are the important complementary goods and enabling technologies for the air taxi market? Are they available in sufficient quality and economy?
4. Was Uber well positioned to be a dominant player in this market? What resources would it have needed to be successful?
5. Overall, would you say Uber's entry into the air taxi market was too early, too late, or about right?
6. Based on the stage of development of eVTOLs and the complementary and enabling technologies and infrastructure, what companies do you think stand the best chance of success in this market?

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OVERVIEW

The previous chapter pointed out that some industries are characterized by increasing returns to adoption, meaning that the more a technology is adopted, the more valuable it becomes. In such industries, timing can be crucial—a technology that is adopted earlier than others may reap self-reinforcing advantages such as greater funds to invest in improving the technology, greater availability of complementary goods, and less customer uncertainty. On the other hand, the same factors that cause increasing returns to adoption may make very early technologies unattractive: If there are few users of the technology or availability of complementary goods is poor, the technology may fail to attract customers. A number of other first-mover advantages, and disadvantages, can shape how timing of entry is related to likelihood of success.

first movers
The first entrants to sell in a new product or service category.

early followers
Entrants that are early to market, but not first.

late entrants
Entrants that do not enter the market until the time the product begins to penetrate the mass market or later.

Entrants are often divided into three categories: **first movers** (or pioneers), which are the first to sell in a new product or service category; **early followers** (also called early leaders), which are early to the market *but not first*; and **late entrants**, which enter the market when or after the product begins to penetrate the mass market. The research on whether it is better to be a first mover, early follower, or late entrant yields conflicting conclusions. Some studies that contrast early entrants (lumping first movers and early followers together) with late entrants find that early entrants have higher returns and survival rates, consistent with the notion of first-mover (or at least early-mover) advantage.[2] However, other research has suggested the first firm to market is often the first to fail, causing early followers to outperform first movers.[3] Still other research contends the higher returns of being a first mover typically offset the survival risk.[4] A number of factors influence how timing of entry affects firm survival and profits. In this chapter, we will first examine first-mover advantages and disadvantages. We will then look more closely at what factors determine the optimal timing of entry, and its implications for a firm's entry strategy.

FIRST-MOVER ADVANTAGES

Being a first mover may confer the advantages of brand loyalty and technological leadership, preemption of scarce assets, and exploitation of buyer switching costs.[5] Furthermore, in industries characterized by increasing returns, early entrants may accrue learning and network externality advantages that are self-reinforcing over time.[6]

Brand Loyalty and Technological Leadership

The company that introduces a new technology may earn a long-lasting reputation as a leader in that technology domain. Such a reputation can help sustain the company's image, brand loyalty, and market share even after competitors have introduced comparable products. The organization's position as technology leader also enables it to shape customer expectations about the technology's form, features, pricing, and other characteristics. By the time later entrants come to market, customer requirements may be well established. If aspects that customers have come to expect in a technology are difficult for competitors to imitate (e.g., if they are protected by patent or copyright, or arise from the first mover's unique capabilities), being the technology leader can yield sustained **monopoly rents**. Even if the technology characteristics are imitable, the first mover has an opportunity to build brand loyalty before the entry of other competitors.

monopoly rents
The additional returns (either higher revenues or lower costs) a firm can make from being a monopolist, such as the ability to set high prices, or the ability to lower costs through greater bargaining power over suppliers.

Preemption of Scarce Assets

Firms that enter the market early can preemptively capture scarce resources such as key locations, government permits, patents, access to distribution channels, and relationships with suppliers.

For example, companies that wish to provide any wireless communication service must license the rights to broadcast over particular radio frequencies from the government. In the United States, the Federal Communications Commission (FCC) is primarily responsible for allotting rights to use bands of radio frequencies (known as the spectrum) for any wireless broadcasting. The FCC first allocates different portions of the spectrum for different purposes (digital television broadcasting, third-generation wireless telecommunication, etc.) and different geographic areas. It then auctions off rights to use these segments to the highest bidders. This means that early movers in wireless services can preemptively capture the rights to use portions of the wireless spectrum for their own purposes, while effectively blocking other providers. By 2003, the proliferation of wireless services had caused the spectrum to become a scarce commodity, and the FCC was under pressure to allow the holders of wireless spectrum rights to sublet unused portions of their spectrum to other organizations.

Exploiting Buyer Switching Costs

Once buyers have adopted a good, they often face costs to switch to another good. For example, the initial cost of the good is itself a switching cost, as is the cost of complements purchased for the good. Additionally, if a product is complex, buyers must spend time becoming familiar with its operation; this time investment becomes a switching

cost that deters the buyer from switching to a different product. If buyers face switching costs, the firm that captures customers early may be able to keep those customers even if technologies with a superior value proposition are introduced later. This is often the reason given for the dominance of the QWERTY typewriter keyboard. In 1867, Christopher Sholes began experimenting with building a typewriter. At that time, letters were struck on paper by mechanical keys. If two keys were struck in rapid succession, they often would jam. Key jamming was a particularly significant problem in the 1800s because typewriters then were designed so that keys struck the back side of the paper, making it impossible for users to see what they were typing. The typist thus might not realize he or she had been typing with jammed keys until after removing the page. Scholes designed his keyboard so that commonly used letter combinations were scattered as widely as possible over the keyboard. The QWERTY keyboard also puts a disproportionate burden on the left hand (3000 English words can be typed with the left hand alone, while only 300 can be typed with the right hand alone). This positioning of keys would slow the typing of letter combinations, and thus reduce the likelihood of jamming the keys.[7]

Over time, many competing typewriter keyboards were introduced that boasted faster typing speeds or less-tiring typing. For example, the Hammond and Blickensderfer "Ideal" keyboard put the most commonly used letters in the bottom row for easy access, and used only three rows total. Another example, the Dvorak keyboard, placed all five vowels and the three most commonly used consonants in the home row, and common letter combinations required alternating hands frequently, reducing fatigue. However, QWERTY's early dominance meant typists were trained only on QWERTY keyboards. By the time Dvorak keyboards were introduced in 1932, tens of millions of typists were committed to QWERTY keyboards—the switching costs of learning how to type all over again were more than people were willing to bear.[8] Even after daisy-wheel keys (and later, electronic typewriters) removed all possibility of jamming keys, the QWERTY keyboard remained firmly entrenched. August Dvorak is said to have died a bitter man, claiming, "I'm tired of trying to do something worthwhile for the human race. They simply don't want to change!"[9]

Reaping Increasing Returns Advantages

In an industry with pressures encouraging adoption of a dominant design, the timing of a firm's investment in new technology development may be particularly critical to its likelihood of success. For example, in an industry characterized by increasing returns to adoption, there can be powerful advantages to being an early provider; a technology that is adopted early may rise in market power through self-reinforcing positive feedback mechanisms, culminating in its entrenchment as a dominant design. Intel is an apt example of this.

Intel's Ted Hoff invented the first microprocessor in 1971, and in 1975, Bill Gates and Paul Allen showed that it could run a version of BASIC that Gates had written. Gates's BASIC became widely circulated among computer enthusiasts, and as BASIC was adopted and applications developed for it, the applications were simultaneously optimized for Intel's architecture. IBM's adoption of Intel's 8088 microprocessor in its PC introduction secured Intel's dominant position, and each of Intel's subsequent generations of products has set the market standard.[10]

FIRST-MOVER DISADVANTAGES

Despite the great attention that first-mover advantages receive, there are also arguments for not entering a market too early. In a historical study of 50 product categories, Gerard Tellis and Peter Golder found that market pioneers have a high failure rate—roughly 47 percent—and that the mean market share of market pioneers is 10 percent.[11] By contrast, early leaders (firms that enter aftermarket pioneers but assume market leadership during the early growth phase of the product life cycle) averaged almost three times the market share of market pioneers.[12] Tellis and Golder point out that the market may often perceive first movers to have advantages because it has misperceived who the first mover really was. For example, while today few people would dispute Procter & Gamble's claim that it "created the disposable diaper market,"[13] in actuality, Procter & Gamble entered the disposable market almost 30 years after Chux, a brand owned by a subsidiary of Johnson & Johnson. In the mid-1960s, *Consumer Reports* ranked both products as best buys. However, over time Pampers became very successful and Chux disappeared, and eventually people began to reinterpret history.

Other studies have found that first movers earn greater revenues than other entrants, but that they also face higher costs, causing them to earn significantly lower profits in the long run.[14] First movers typically bear the bulk of the research and development expenses for their product or service technologies, and they must also often pay to develop suppliers and distribution channels, plus consumer awareness. A later entrant often can capitalize on the research and development investment of the first mover, fine-tune the product to customer needs as the market becomes more certain, avoid any mistakes made by the earlier entrant, and exploit **incumbent inertia**.[15] Later entrants can also adopt newer and more efficient production processes while early movers are either stuck with earlier technologies or must pay to rebuild their production systems.[16]

incumbent inertia
The tendency for incumbents to be slow to respond to changes in the industry environment due to their large size, established routines, or prior strategic commitments to existing suppliers and customers.

Research and Development Expenses

Developing a new technology often entails significant research and development expenses, and the first to develop and introduce a technology typically bears the brunt of this expense. By the time a firm has successfully developed a new technology, it may have borne not only the expense of that technology but also the expense of exploring technological paths that did not yield a commercially viable product. This firm also typically bears the cost of developing necessary production processes and complementary goods that are not available on the market. Since the new product development failure rate can be as high as 95 percent, being the first to develop and introduce an unproven new technology is expensive and risky.

By contrast, later entrants often do not have to invest in exploratory research. Once a product has been introduced to the market, competitors can often ascertain how the product was created. The later entrant can also observe the market's response to particular features of the technology and decide how to focus its development efforts. Thus, the later entrant can both save development expense and produce a product that achieves a closer fit with market preferences.

Undeveloped Supply and Distribution Channels

When a firm introduces a new-to-the-world technology, often no appropriate suppliers or distributors exist. The firm may face the daunting task of developing and producing its own supplies and distribution service, or assisting in the development of supplier and developer markets. For example, when DEKA Research began developing its self-balancing IBOT wheelchair, it needed a type of ball bearing for which there were no suppliers. DEKA was forced to develop a machine to mold the bearings. According to Dean Kamen, the company's founder, "Nobody here planned to invent new ball bearings, but in order to make this engine practical we have to develop a bearing technology that doesn't exist."[17]

Immature Enabling Technologies and Complements

enabling technologies Component technologies that are necessary for the performance or desirability of a given innovation.

When firms develop technologies, they often rely on other producers of **enabling technologies**. For instance, for Uber to launch its UberAIR transportation service required the development of economical aircraft capable of taking off and landing vertically, advances in battery technology that would increase the range of the aircraft while keeping costs low, a charging infrastructure to charge the aircraft, a network of "skyports" for landing and takeoff, and pilots for the aircraft (until they could be made fully autonomous). Uber was thus heavily reliant on numerous third parties to make its project feasible.

As discussed in Chapter Four, many products also require complementary goods to be useful or valuable. Computers need software, cameras need film, automobiles need service, gasoline, and roads. When new technologies are introduced to a market, important complements may not yet be fully developed, thus hindering adoption of the innovation. The development of vehicles powered by hydrogen fuel cells provides an excellent example of how a lack of complementary technologies and infrastructure can pose serious obstacles for early movers.

Uncertainty of Customer Requirements

A first mover to the market may face considerable uncertainty about what product features customers will ultimately desire and how much they will be willing to pay for them. For a very new product technology, market research may be of little help. Customers may have little idea of the value of the technology or the role it would play in their lives. As a consequence, first movers may find that their early product offerings must be revised as the market begins to reveal customer preferences.

For instance, when Kodak introduced the 8-mm video camera in the late 1980s, it expected that customers would flock to the design's smaller size and superior recording ability. Instead, consumers rejected the product. The 8-mm video cameras were more expensive, and consumers had not yet recognized a need for this product and were unsure of what value it could provide. Kodak decided to withdraw from the market. However, by the early 1990s, consumers had become more comfortable with the concept of 8-mm video camera technology, and several competitors (most notably Sony) successfully entered this market.

First movers have an opportunity to shape customer preferences by establishing the precedent for product design in the newly emerging market and by investing in customer education. Customer education efforts are expensive, however. If the product is slow to begin to reap revenues for the sponsoring firm, it may collapse under the weight of its R&D and marketing expenses. Figure 5.1 provides a number of product categories with their first movers, prominent followers, and which of these were ultimately more successful.

FIGURE 5.1
First Movers and Followers—Who Wins?

Source: Robert M. Grant, *Contemporary Strategy Analysis* (Malden, MA: Blackwell Publishers, 1998); David Teece, *The Competitive Challenge: Strategies for Industrial Innovation and Renewal* (Cambridge, MA: Ballinger, 1987); and Melissa A. Schilling, "Technology Success and Failure in Winner-Take-All Markets: Testing a Model of Technological Lock Out," *Academy of Management Journal* 45 (2002): 387–98.

Product	First Mover	Notable Follower(s)	The Winner
8-mm video camera	Kodak	Sony	Follower
Disposable diaper	Chux	Pampers Kimberly Clark	Followers
Float glass	Pilkington	Corning	First mover
Groupware	Lotus	AT&T	First mover
Instant camera	Polaroid	Kodak	First mover
Microprocessors	Intel	AMD Cyrix	First mover
Microwave	Raytheon	Samsung	Follower
Personal computer	MITS (Altair)	Apple IBM	Followers
Personal computer operating system	Digital Research	Microsoft (MS-DOS)	Follower
Smartphones	IBM (Simon)	Apple Nokia	Followers
Social networking sites	SixDegrees.com	MySpace Facebook	Followers
Spreadsheet software	VisiCalc	Microsoft (Excel) Lotus	Followers
Video game console	Magnavox	Atari Nintendo	Followers
Web browser	NCSA Mosaic	Netscape Microsoft (Internet Explorer)	Followers
Word processing software	MicroPro (WordStar)	Microsoft (MS Word) WordPerfect	Followers
Workstation	Xerox Alto	Sun Microsystems Hewlett-Packard	Followers

FACTORS INFLUENCING OPTIMAL TIMING OF ENTRY

In very early market stages, a technology may be underdeveloped and its fit with customer needs unknown. In late market stages, a technology may be well understood, but competitors may have already captured controlling shares of the market. How does a firm decide whether to attempt to pioneer a technology category or to wait while others do so? The answer will depend on several factors, including customer certainty, the margin of improvement offered by the new technology, the state of enabling technologies and complementary goods, the threat of competitive entry, the degree to which the industry exhibits increasing returns, and the firm's resources.

1. How certain are customer preferences?

When new-to-the-world technologies are first developed, customers may have difficulty understanding the technology and its role in their life. Both producers and customers may face considerable ambiguity about the importance of various features of the technology. As producers and customers gain experience with the technology, features that initially seemed compelling may turn out to be unnecessary, and features that had seemed unimportant may turn out to be crucial. For example, many of the companies that raced to establish an online presence in the e-commerce frenzy of the late 1990s

believed that their Web sites needed exciting graphics and sounds to be competitive. Graphics and sound, however, turned out to be the downfall of many early Web sites. Many customers did not have high-speed Internet access or computers with enough processing power to quickly download the Web sites, making multimedia Web sites an annoyance rather than an attraction.

The reverse scenario is demonstrated in Sony's introduction of the PlayStation 2. When Sony introduced its multifeatured PlayStation 2, many industry analysts believed that Sony had overestimated consumer interest in having a game console that would play music CDs or DVD movies. It turned out, however, that Sony may have *underestimated* the desirability of these features. Video game consoles are typically sold at cost (or at a loss) in order to rapidly build an installed base. Profits are then made on game royalties. However, when consumers realized that the PlayStation 2 was a very affordable combination of game console and high-quality DVD player, many consumers bought the system for its DVD capabilities first and game capabilities second. Many of these consumers bought very few games, causing Sony's strategy of subsidizing the console with the intention of making money on the games to backfire. Observing this, Microsoft disabled DVD playback on its Xbox unless consumers purchased an add-on DVD playback kit.

Not all pioneers face customer uncertainty—some innovations are developed in response to well-understood customer needs. Customer requirements may have been long known even if the method of meeting them was not. For example, the developers of Tagamet (a medication for patients with chronic heartburn or ulcers) faced very little customer uncertainty. Customers wanted an affordable, easy-to-use solution to their stomach discomfort. Once a method of achieving this objective had been developed, tested, and approved, its developers raced the product to market in hopes of patenting it and securing market share ahead of competing products. Other things being equal, less customer uncertainty favors earlier timing of entry.[18]

2. How much improvement does the innovation provide over previous solutions?

The degree to which the technology represents an improvement over previous technologies increases a firm's likelihood of successful early entry. That is, when a technology makes a dramatic improvement over previous generations or different technologies that serve similar functions, it will more rapidly gain customer acceptance. There will be less ambiguity about the value of the technology and more early adoptions (as well as more support by complementary goods providers); as a consequence, customer expectations should become known sooner, and adoptions should be more rapid.[19]

3. Does the innovation require enabling technologies, and are these technologies sufficiently mature?

As mentioned earlier, many innovations rely on crucial enabling technologies to ensure their performance. A high-definition television set is of little value if networks are incapable of broadcasting in high definition; cellular phones or portable stereos would have little value if small and long-lasting batteries were unavailable. A developer must identify which enabling technologies will affect the performance of the new innovation and assess the degree to which those technologies are mature enough (or *will be* mature enough) to deliver the desired performance. More mature enabling technologies allow earlier entry; less mature enabling technologies may favor waiting for enabling technologies to be further developed.

4. Do complementary goods influence the value of the innovation, and are they sufficiently available?

If the value of an innovation hinges critically on the availability and quality of complementary goods, then the state of complementary goods determines the likelihood of successful entry. Not all innovations require complementary goods, and many more innovations can utilize existing complementary goods. For example, though numerous innovations in 35-mm cameras have been introduced in the last few decades, almost all have remained compatible with standard rolls of 35-mm film; thus availability of that complementary good was ensured. If, on the other hand, the innovation requires the development of new complementary goods, then a pioneer must find a way to ensure their availability. Some firms have the resources and capabilities to develop both a good and its complements, while others do not. If the firm's innovation requires complementary goods that are not available on the market, and the firm is unable to develop those complements, successful early entry is unlikely.

5. How high is the threat of competitive entry?

If there are significant entry barriers or few potential competitors with the resources and capabilities to enter the market, the firm may be able to wait while customer requirements and the technology evolve. Over time, one would expect customer expectations to become more certain, enabling technologies to improve, and support goods and services to be developed, thus increasing the likelihood that sponsored technologies will possess a set of attributes that meet consumer demands. However, if the technology proves to be valuable, other firms are also likely to be attracted to the market. Thus, if entry barriers are low, the market could quickly become quite competitive, and entering a market that has already become highly competitive can be much more challenging than entering an emerging market.[20] Margins may already have been driven down to levels that require competitors to be highly efficient, and access to distribution channels may be limited. If the threat of competitive entry is high, the firm may need to enter earlier to establish brand image, capture market share, and secure relationships with suppliers and distributors. This is discussed further in the Research Brief "Whether and When to Enter?"

6. Is the industry likely to experience increasing returns to adoption?

In industries that have increasing returns to adoption due to strong learning curve effects or network externalities, allowing competitors to get a head start in building an installed base can be very risky. If a competitor's offering builds a significant installed base, the cycle of self-reinforcing advantages could make it difficult for the firm to ever catch up. Furthermore, if there are forces encouraging adoption of a single dominant design, a competitor's technology may be selected. If protection mechanisms such as patents prevent the firm from offering a compatible technology, the firm may be locked out.[21]

7. Can the firm withstand early losses?

As was discussed earlier, a first mover often bears the bulk of the expense and risk of developing and introducing a new innovation. First movers thus often need significant amounts of capital that either is available internally (in the case of large firms) or can be accessed externally (e.g., through the debt or equity markets). Furthermore, the

Research Brief Whether and When to Enter?

In a study of 30 years of data on entry into the medical diagnostic imaging industry, Will Mitchell examined the factors that drive whether and when a firm that is an incumbent in one subfield of an industry chooses to enter a newly emerging subfield of the industry.[a] For instance, what determines whether and when a manufacturer of conventional X-ray machines decides to go into magnetic resonance imaging (MRI) equipment? While new goods offer opportunities for growth, they can cannibalize existing products, and they also require an investment in new skills. Incumbents often are slow to enter new technical subfields.[b] They may be intentionally waiting for industry turbulence to subside, or they may be unintentionally slowed by factors that create inertia, such as the difficulty in altering well-established routines and strategic commitments to the firm's existing supplier and customer base.

Mitchell pointed out that entry barriers and imitability of a new product (e.g., whether it can be effectively protected by patents) interact to create different incentives for timing. First, if only one firm can produce an inimitable good, it can enter if and when it wants. However, if several firms could produce a good that will subsequently be inimitable, they may race to do so to capture the market. In such a circumstance, being early confers a significant advantage. Finally, if the good is expected to be highly imitable (e.g., if it would be difficult to effectively protect with patents because competitors could easily invent around the patent), then firms will prefer to wait while others bear the expense of developing and introducing the good. There are disincentives to being early to market.[c]

Mitchell found that firms that had more specialized assets that would be useful in the new subfield (e.g., a well-established distribution system that could be used for the new imaging equipment) were more likely to enter the new subfield. A firm was also more likely to enter if the products it currently produced were threatened by the new products (i.e., if the new technology was likely to displace the firm's current technology as the dominant choice in the market). Furthermore, the incumbent was likely to enter *earlier* if its core products were threatened and there were several potential rivals.

[a] Will Mitchell, "Whether and When? Probability of Incumbent's Entry into Emerging Technical Subfields," *Administrative Science Quarterly* 38 (1989): 208–30.

[b] Frederic M. Scherer, *Industrial Market Structure and Economic Performance*, 2nd ed. (Chicago: Rand McNally, 1980).

[c] Michael Katz and Carl Shapiro, "Technology Adoption in the Presence of Network Externalities," *Journal of Political Economy* 94 (1986): 822–41.

first mover must be able to withstand a significant period with little sales revenue from the product. Even in the case of successful new technologies, often a considerable period elapses between the point at which a first mover introduces a new innovation and the point at which the innovation begins to be adopted by the mass market. The s-curve shape of technology diffusion (discussed in Chapters Three and Thirteen) illustrates this aptly. New innovations tend to be adopted very slowly at first, while innovators and early adopters try the technology and communicate their experience to others. This slow initial takeoff of new innovations has caused the demise of many start-up firms. For example, in the personal digital assistant (PDA) industry—the precursor to smartphones—start-ups such as GO Corporation and Momenta had received accolades for their technology designs, but were unable to withstand the long period of market confusion about PDAs and ultimately ran out of capital. Companies such as IBM and Compaq survived because they were large and diversified, and thus not reliant on PDA

revenues. Palm was a relatively late mover in the PDA industry so it did not have to withstand as long of a takeoff period, but even Palm was forced to seek external capital and was acquired by U.S. Robotics, which was later bought by 3COM.

On the other hand, firms with significant resources also may be able to more easily catch up to earlier entrants.[22] By spending aggressively on development and advertising, and leveraging relationships with distributors, a late entrant may be able to rapidly build brand image and take market share away from earlier movers. For example, though Nestlé was very late to enter the freeze-dried coffee market with Taster's Choice, the company was able to use its substantial resources to both develop a superior product and rapidly build market awareness. It was thus able to quickly overtake the lead from General Foods' Maxim.[23]

8. Does the firm have resources to accelerate market acceptance?

A firm with significant capital resources not only has the capability to withstand a slow market takeoff, but also can invest such resources in accelerating market takeoff. The firm can invest aggressively in market education, supplier and distributor development, and development of complementary goods and services. Each of these strategies can accelerate the early adoption of the innovation, giving the firm much greater discretion over entering early.[24] These strategies are discussed in more detail in Chapter Thirteen. Thus, a firm's capital resources can give it some influence on the shape of the adoption curve.

9. Is the firm's reputation likely to reduce the uncertainty of customers, suppliers, and distributors?

In addition to capital resources, a firm's reputation and credibility can also influence its optimal timing of entry.[25] A firm's reputation can send a strong signal about its likelihood of success with a new technology. Customers, suppliers, and distributors will use the firm's track record to assess its technological expertise and market prowess. Customers may use the firm's reputation as a signal of the innovation's quality, and thus face less ambiguity about adopting the innovation. A firm with a well-respected reputation for successful technological leadership is also more likely to attract suppliers and distributors.[26] This was aptly demonstrated in Microsoft's entry into the video game console industry: Despite having little experience in producing hardware, suppliers and distributors eagerly agreed to work with Microsoft because of its track record in personal computing. Other things being equal, an entrant with a strong reputation can attract adoptions earlier than entrants without strong reputations.

STRATEGIES TO IMPROVE TIMING OPTIONS

As should now be clear, managing the timing of entry into the market is a complex matter. If the technology has a clear advantage to consumers, entering the market early may give the entrant a path-dependent advantage that is nearly impossible for competitors to overcome. If, on the other hand, a firm enters a market very early and the advantages of the technology are not very clear to consumers, there is a strong possibility that the technology will receive a tepid welcome. Confounding this risk is the

fact that watchful competitors may be able to use the firm's failure to their advantage, refining the technology the firm has introduced to the market and making any corrections necessary to improve the technology's market acceptance. The later entrant may be able to enter at a lower cost because it can capitalize on the research and development of the early firm, and use knowledge of the market gained from observing the early entrant's experience.

In the above, it is assumed that timing of entry is a matter of *choice* for the firm. However, implicit in this assumption is a corollary assumption that the firm is capable of producing the technology at any point in the time horizon under consideration. For this to be true, the firm must possess the core capabilities required to produce the technology to consumer expectations, or be able to develop them quickly. Furthermore, if the firm intends to refine an earlier entrant's technology and beat the earlier entrant to market with a new version of this technology, it must have *fast-cycle development processes.* If a firm has very fast development processes, the firm not only has a better chance at being an early entrant, but it can also use experience gained through customers' reactions to its technology to quickly introduce a refined version of its technology that achieves a closer fit with customer requirements. In essence, a firm with very fast development deployment processes should be able to take advantage of both first- and second-mover advantages. The research on new product development cycle time indicates that development time can be greatly shortened by using strategic alliances, cross-functional new product development teams, and **parallel development processes**. Chapter Eleven will deal specifically with how firms can ensure that their innovations are deployed rapidly to the market.

parallel development process
When multiple stages of the new product development process occur simultaneously.

Summary of Chapter

1. A first mover may be able to build brand loyalty and a reputation for technological leadership, preemptively capture scarce resources, and exploit buyer switching costs.
2. First movers may also benefit from increasing returns to adoption due to learning curve effects and network externalities.
3. Some studies, however, argue that first movers may have higher failure rates. First movers have to bear the brunt of R&D expenses and may face considerable consumer ambiguity. Second movers can capitalize on the R&D and marketing efforts of the first mover, producing a technology that costs less to develop and that corrects for any of the first mover's mistakes.
4. First movers may also face poorly developed supplier markets, distribution channels, and availability of complementary goods, all of which can increase the challenge of successfully launching their new product or service. Enabling technologies may also be immature, hindering the new technology's performance.
5. The biggest disadvantage many first movers face is uncertainty over customer requirements. Customers themselves may be uncertain about what features or form they desire in a new innovation. A firm may have to withstand significant losses before customer preferences become more certain.
6. The optimal timing of entry is thus a function of several factors, including the margin of advantage offered by the new innovation, the state of enabling technologies

and complements, the state of customer expectations, the threat of competitive entry, whether the industry faces increasing returns, and a firm's resources.

7. Firms that have fast-cycle development processes have more options when it comes to timing. Not only does a fast-cycle developer have an advantage in introducing innovations earlier, but it also can be its own fast follower by quickly introducing refined versions of its own technology.

Discussion Questions

1. What are some advantages of entering a market early? Are there any advantages to entering a market late?
2. Name a successful (*a*) first mover, (*b*) early follower, and (*c*) late entrant. Identify unsuccessful examples of each.
3. What factors might make some industries harder to pioneer than others? Are there industries in which there is no penalty for late entry?

Suggested Further Reading

Classics

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Endnotes

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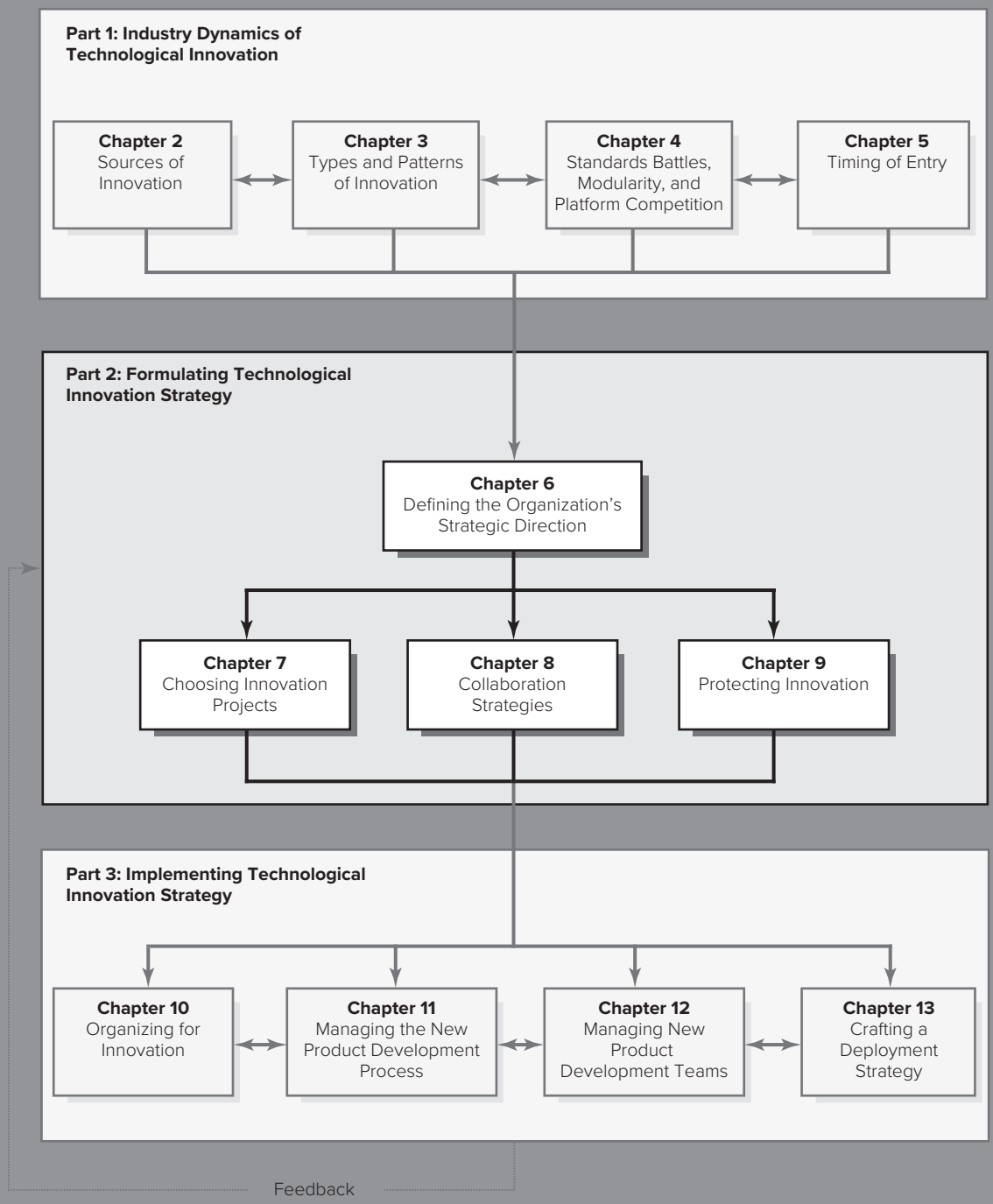
Part **Two**

Formulating Technological Innovation Strategy

In this section, we will cover the key aspects of formulating a technological innovation strategy, including:

- Assessing the firm's position and defining its strategic direction.
- Choosing innovation projects in which to invest, including both quantitative and qualitative valuation techniques.
- Deciding whether and how the firm will collaborate on development activities, choosing a collaboration mode, and choosing and monitoring partners.
- Crafting a strategy for protecting—or diffusing—a technological innovation through such methods as patents, trademarks, copyrights, and trade secrets.

Formulating Technological Innovation Strategy



Chapter **Six**

Defining the Organization's Strategic Direction

Tesla in 2021

In 2021, Tesla was one of the most valuable and talked about companies in the world. What started as an unlikely and risky venture to produce an all-electric luxury sports car had grown into a company with $31.5 billion in sales and a market capitalization of almost $640 billion. The company delivered just under a half million cars in 2020 (despite a temporary factory shutdown due to COVID-19) and turned its first full year of profit the same year—$721 million.[a]

Tesla operated its business in two main segments: (1) Automotive, which included auto manufacturing (i.e., Model S, Model X, Model 3, and Model Y), operation of its own dealerships, after-sales service, and vehicle insurance; and (2) Energy generation and storage, which included the assembly and sale of solar panels and solar roof tiles and energy storage systems (i.e., Powerwall, Powerpack, and Megapack). Tesla additionally announced several planned future products, including a consumer truck (Cybertruck), a new version of its sports car (Tesla Roadster), new commercial trucks, and a full self-driving system.

Tesla's founder, Elon Musk, had made bold moves and encountered plenty of skepticism about his prospects for success in the auto industry. Musk's confidence had, however, paid off. The company's ramp up in production sales had been nothing short of astonishing, and by 2021 Tesla was an iconic and globally revered brand, and Musk himself had become one of the wealthiest people in the world. Perhaps more importantly, Tesla's success had changed the public perception of electric vehicles from being the underpowered and nerdy small vehicles that were offered in the 1990s, to stylish and powerful cars that offered luxury, convenience, and environmental friendliness. In demonstrating the viability of electric vehicles to regulators, it had also changed the playing field—policy makers began to more vigorously impose limits on carbon emissions while creating other incentives for electric vehicle adoption, and this in turn helped to prompt the entire auto industry to begin to pivot away from gas-powered cars.

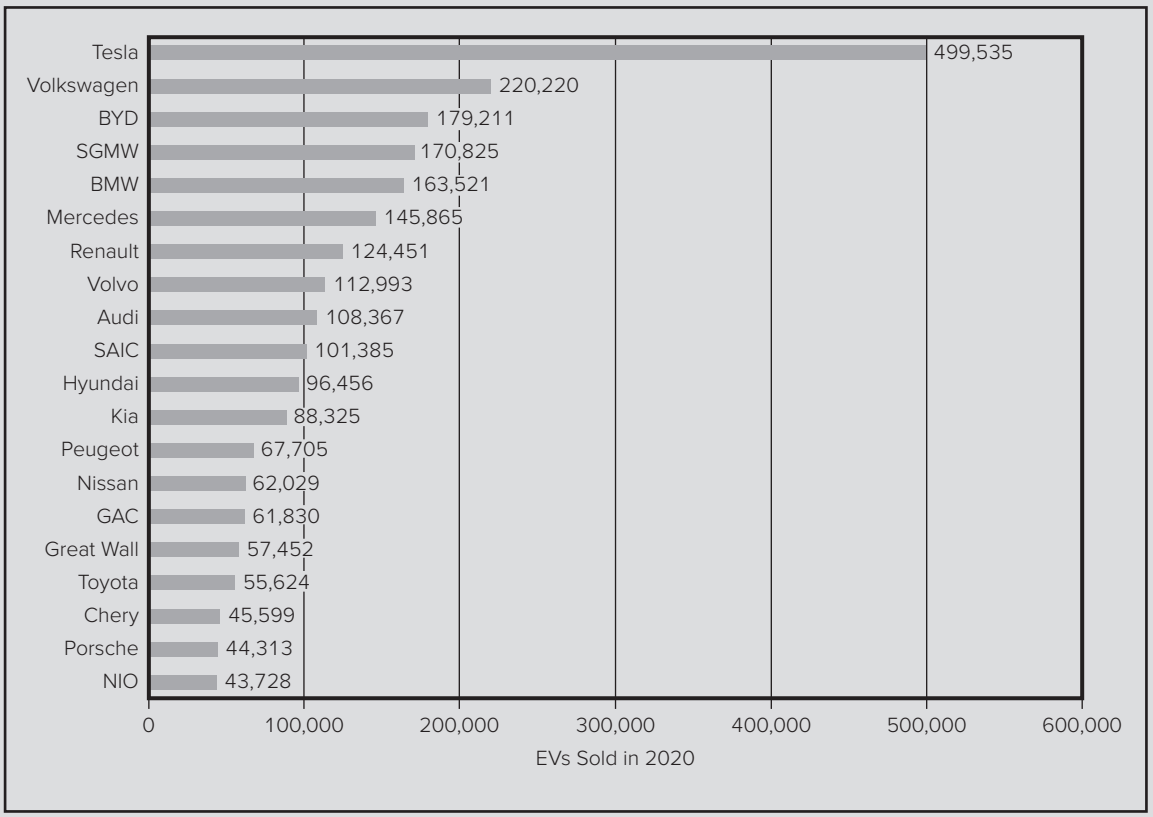
Notably, part of what made Tesla profitable in 2020 was $1.6 billion in revenues made from selling regulatory credits to rival auto makers that needed them to comply with emissions regulations. That, however, was likely to change. Competition was heating up in the electric vehicle market, with nearly all incumbent automakers and many startups launching electric cars and trucks (see Figure 1). As other automakers made significant inroads into the electric vehicle market, Tesla's positioning would have to shift from being a unique technological disruptor, to a company that competed on the traditional dimensions of quality and value. Would Tesla be able to maintain its lead in electric vehicles once the entire auto industry began to vigorously compete in this market?

History of Tesla

In the year 2003, an engineer named Martin Eberhard was looking for his next big project. A tall, slim man with a mop of gray hair, Eberhard was a serial entrepreneur who had launched a number of start-ups, including a company called

FIGURE 1
2020 World Plug-in Electric Vehicle Sales (units sold)

Source: Cleantechnica, February 4, 2021.



NuvoMedia, which he sold to Gemstar in a $187 million deal. Eberhard was also looking for a sports car that would be environmentally friendly—he had concerns about global warming and U.S. dependence on the Middle East for oil. When he didn't find the car of his dreams on the market he began contemplating building one himself, even though he had zero experience in the auto industry. Eberhard noticed that many of the driveways that had a Toyota Prius hybrid electric vehicle (or "dork mobile" as he called it) also had expensive sports cars in them—making Eberhard speculate that there could be a market for a high-performance environmentally friendly car. As explained by Eberhard, "It was clear that people weren't buying a Prius to save money on gas—gas was selling close to inflation–adjusted all-time lows. They were buying them to make a statement about the environment."[b]

Eberhard began to consider a range of alternative fuel options for his car: hydrogen fuel cells, natural gas, diesel. However, he soon concluded the highest efficiency and performance would come from a pure electric vehicle. Luckily for Eberhard, Al Cocconi (founder of AC Propulsion and one of the original engineers for GM's ill-fated EV-1) had concluded the same thing and had produced a car called the tzero. The tzero could go from 0 to 60 miles per hour in 4.1 seconds, but it was powered with extremely heavy lead-acid batteries, limiting its range to about 60 miles between charges. Eberhard approached Cocconi with the idea of using the lighter lithium-ion batteries, which offered six times more energy per pound. Cocconi was eager to try out the idea (he had, in fact, been experimenting with lithium-ion batteries himself), and the resulting lithium-ion-based tzero accelerated to 60 miles per hour in 3.6 seconds, and could travel more than 300 miles. Eberhard licensed the electric-drive-train technology from AC Propulsion, and founded his company, Tesla Motors (named after Nikola Tesla, a late 19th century and early 20th century inventor who developed, among other things, the AC electrical systems used in the United States today).[c]

Meanwhile, there was another entrepreneur—one with much deeper pockets—also interested in developing electric vehicles based on the tzero: Elon Musk. In 2002, Elon Musk was a 31-year-old South African living in California, who had founded a company that ultimately became PayPal. After selling PayPal to eBay in 2002 for $1.5 billion, he started a company called SpaceX with the ambitious goal of developing cheap, consumer space travel. (SpaceX's Dragon spacecraft ultimately made history in May of 2012 by becoming the first commercial vehicle to launch and dock at the International Space Station.[d]) Musk's assertive style, and his astonishing record of high-tech entrepreneurship, made him one of the inspirations for the portrayal of Tony Stark character in Jon Favreau's *Iron Man* movies.

Like Eberhard, Musk thought electric cars were the key to the United States achieving energy independence, and he approached Cocconi about buying the tzero. Tom Gage, who was then AC Propulsion's CEO, suggested that Musk collaborated with Eberhard. After a two-hour meeting in February of 2004, Musk agreed to fund Eberhard's plan with $6.3 million. He would be the company's chairman and Eberhard would serve as CEO.

The Roadster

The first Tesla prototype, named the Roadster, was based on the $45,000 Lotus Elise, a fast and light sports car that seemed perfect for the creation of Eberhard and Musk's grand idea (see Figure 2A). The car would have 400 volts of electric potential, liquid-cooled lithium-ion batteries, and a series of silicon transistors that would give the car acceleration so powerful the driver would be pressed back against their seat.[e] It would be about as fast as a Porsche 911 Turbo, would not create a single emission, and would get about 220 miles on a single charge from the kind of outlet you would use to power a washing machine.[f]

After a series of clashes between Musk and Eberhard that led to delays in launching the Roadster, Eberhard was pushed out of the company. The Roadster missed its deadline for beginning production at the Lotus facility, triggering a penalty built into the manufacturing contract Eberhard had signed with Lotus: a $4 million fee. However, when the car finally launched in 2008, the enthusiastic response it received was astonishing—it boasted an all-star list of celebrities with reservations to buy, and everywhere the Roadster drove, people stopped to stare.[g]

The Model S

Musk's ambitions did not stop at a niche high-end car. He wanted to build a major U.S. auto company—a feat that had not been successfully accomplished since the 1920s. To do so, he knew he needed to introduce a less expensive car that could attract a higher volume of sales, if not quite the mass market. In June of 2008, Tesla announced the Model S, a high-performance all-electric sedan that would sell for a price ranging from $57,400 to $77,400, and compete against cars like the BMW 5 Series (see Figure 2B). The car would have an all-aluminium body and a range of up to 300 miles per charge.[h] Estimates suggested that the Model S cost $500 million to develop, however offsetting that cost was a $465 million loan Tesla received from the U.S. government to build the car, as part of the U.S. government's initiative to promote the development of technologies that would help the United States to achieve energy independence.[i]

To build the car, Tesla bought a recently closed automobile factory in Fremont California that had been used for the New United Motor Manufacturing Inc. (NUMMI) venture between Toyota and General Motors. The factory, which was capable of producing 1000 cars a week, was far bigger than Tesla's immediate needs and would give the company room to grow. Furthermore, though the plant and the land it was on had been appraised at around $1 billion before NUMMI was shut down, Tesla was able to snap up the idled factory for $42 million.[j] Tesla also used the factory to produce battery packs for Toyota's RAV4, and a charger for a subcompact Daimler AG electric vehicle. These projects supplemented Tesla's income while also helping it to build scale and learning-curve efficiencies in its technologies.

In the first quarter of 2013, Tesla announced its first quarterly profit. The company had taken in $562 million in revenues, and reported an $11.2 million profit.

FIGURE 2A
Tesla Roadster

Ian Giblin



FIGURE 2B
Tesla Model S

Melissa Schilling



FIGURE 2C
Tesla Model X

Melissa Schilling



FIGURE 2D
Tesla Model 3

Melissa Schilling



Then more good news came: the Model S had earned *Consumer Reports'* highest rating, and had outsold similarly priced BMW and Mercedes models in the first quarter.[k] In May of 2013, the company raised $1 billion by issuing new shares, and then surprised investors by announcing that it had paid back its government loan. After repaying the loan, Tesla had about $679 million in cash. Musk had announced confidently that he felt it was his obligation to pay back taxpayer money as soon as possible, and that the company had sufficient funds now to develop its next generation of automobiles without the loan, and without issuing further shares.[l]

Model X

The Model X, unveiled in 2015, was designed as a high-end sport utility vehicle (SUV) that seats seven. It had several distinctive features that set it apart from the crowded luxury SUV market (Figure 2C). In addition to being all-electric and able to go from 0 to 60 miles per hour in just 3.2 seconds, it featured a panoramic windshield and distinctive gull-wing doors (that open upward rather than swinging out) that open automatically in response to the driver's approach. "It will triangulate my position," Musk said, "It will open the front door without touching. When you sit down, it will close the door."[m] The Model X had a range of about 250 miles (like the Model S) but could tow 5000 pounds. Its selling price would start at $70,000 but could exceed $100,000 depending on the options selected.

In the United States, the mid-size luxury SUV market was about five times the size of the high-end luxury sedan market, and the Model X rapidly attracted a long waiting list of people who placed deposits for the car. Musk projected a fast production ramp up, with goals of producing 85,000 to 90,000 Model X and S vehicles in 2017. Analysts at the time doubted that production could be ramped up so quickly, but despite several supplier parts shortages, Tesla's estimates ended up being very close to the mark: The company produced a total of 83,922 cars in 2017.[n]

Reviews of the car were mixed. *Consumer Reports* found the car disappointing, citing rear doors that were prone to pausing, the car's limited cargo capacity, and a ride that was "too firm and choppy for a $110,000 car."[o] *Car and Driver*'s review also expressed some doubts about the wing doors, but gave the car overall a rating of five out of five stars, stating, "There are no other electric SUVs at the moment. And even against fossil-fuel-fed SUVs, the Tesla's effortless performance and efficiency can't be matched."[p]

By the end of 2016, the Model X had accumulated total sales of 25,524, ranking it seventh among the best-selling plug-in cars in the world (notably, cumulative sales of Tesla Model S reached 158,159 by the end of 2016, making it the second best-selling plug-in car in the world, behind only the Nissan Leaf).[q] By the end of 2017, cumulative sales of the Model X reached approximately 72,059 units.[r] Both the Model S and X, however, would soon be dwarfed in sales by Musk's newest project: Model 3.

Model 3

To achieve Musk's goal of making a real dent in fossil fuel use, Tesla needed a truly mass-market car. Thus, in the fall of 2016, he announced the Model 3, a mid-size all-electric four-door sedan with a range of 220 to 310 miles (depending on the battery option), and a base price of $35,000 (see Figure 2D). Within a week, Tesla had received 325,000 reservations for the car, ranking it among the most sought-after cars in the world. A review in *Road and Track* said that the "Model 3 proves that Tesla is thinking far beyond the edges of the Model S and X. Stepping out of the 3, you realize that, as far as the S and X pushed the envelope, they were always meant as intermediaries, stepping stones designed to draw people away from comfortable convention and into the future of the

automobile."[s] *Popular Mechanics* gave the car its 2018 Car of the Year award, and *Automobile Magazine* gave it the 2018 Design of the Year Award.

The company announced an extremely ambitious production ramp up plan, with a goal of being able to produce 500,000 total units (across all three models) by the end of 2018. This required a massive expansion in production capacity that many experts viewed as unattainable in such a short time frame. The Model 3 would also incorporate new hardware and software to enable automated driving that created significant new design and production challenges. By early 2018, it was clear that Model 3 production was well behind Musk's initial ambitious projections, and criticism from analysts and the press were coming at a furious pace. As one analyst at Cowen and Co noted, "Tesla needs to slow down and more narrowly focus its vision and come up for a breath of fresh air. . . . Elon Musk needs to stop over promising and under delivering."[t]

The breakneck pace at which Musk sought to ramp up production of the Model 3 put major strain on the company. Giant tents were set up in the parking lots to accommodate a new production line, sparking incredulous comments from auto analysts. For example, Karl Brauer, executive publisher of *Autotrader* and *Kelley Blue Book,* told *The Verge,* "I can't say for sure, but I would guess it's currently the only vehicle assembly for a high-volume car going on under a tent right now," and analyst Rebecca Lindland quipped, "Apparently Tesla is furthering its prep for Mars occupation."[u] The period also put the company under acute financial stress, with Musk noting that the company had come perilously close to bankruptcy as noted in the Tweet in Figure 3.

Though it took longer to achieve Musk's goal of more than 5000 cars per week than initially hoped, Tesla did hit the production target by April 1, 2019 (see Figure 4), an undeniably impressive exponential ramp up.

Other Projects

In September of 2012, Tesla had also begun building out its own charging network to serve Tesla customers, and by the end of February 2021, it operated 2564 charging stations (a total of 23,277 Superchargers) worldwide. The buildout of the charging network was designed to directly facilitate adoption of electric



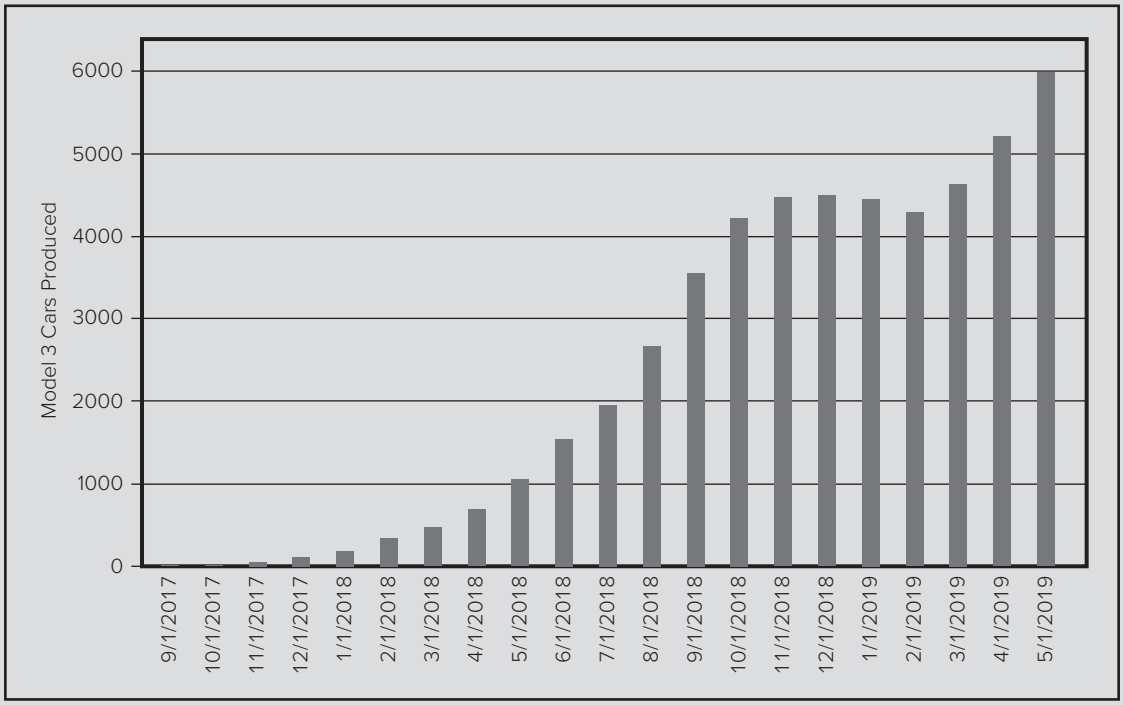
FIGURE 3
Elon Musk Tweet from November 3, 2020

Source: Twitter, https://twitter.com/elonmusk/status/1323640901248393217?ref_src=twsrc%5Etfw%7Ctwcamp%5Etweetembed%7Ctwterm%5E1323640901248393217%7Ctwgr%5E%7Ctwcon%5Es1_&ref_url=https%3A%2F%2F.

Twitter

FIGURE 4
Model 3 Weekly Production Rate based on Three-Week Trailing Average

Source: "Tesla Model 3 Tracker," *Bloomberg,* https://www.bloomberg.com/graphics/tesla-model-3-vin-tracker.



vehicles by reducing consumers' "range anxiety," and thus even though it was an unusual move, it was not that surprising. However, Tesla also began to make even more unusual moves into energy production and storage. In November of 2016, Tesla acquired SolarCity, a company that leases and installs solar panels, for $2.6 billion. Solar City was founded in 2006 by Peter and Lyndon Rive, Elon Musk's cousins. Musk had sketched out the concept for the company around the time of Tesla's founding and had helped his cousins start the company. He also served as its Chairman of the Board. The company had an innovative business model that enabled consumers to have solar panels installed on their roofs with no upfront costs and to pay instead for the power generated by the panels at a price that was comparable to or less than the price they would normally pay for electricity.

In the same month that the Solar City acquisition plan was finalized, Elon Musk announced that the company would begin producing house roofs made entirely from solar panels. "I think this is a really fundamental part of achieving differentiated product strategy, where you have a beautiful roof," Musk said. "It's not a thing on the roof. It is the roof."[v]

In 2016, Tesla opened Gigafactory 1—a giant lithium-ion battery factory built near Reno, Nevada, with its partner Panasonic. Musk justified the vertical

integration move by arguing that the Gigafactory 1 would ultimately drive battery production costs down by as much as 30 percent. In addition to producing batteries for Tesla automobiles, the factory would build Powerwall and Powerpack energy storage devices. The Powerwall was a device for consumers to store solar energy at home. The Powerpack enabled industrial users to manage variable energy needs and provided a source for backup power.

By early 2018, Tesla had also built a new Gigafactory 2 in Buffalo, New York, and in 2019 opened "Giga Shanghai, the first automotive factory in China fully owned by a foreign company.[w] In 2020, Tesla also began construction on a new Gigafactory in Berlin and began deliveries of its fifth vehicle model, the Model Y, a compact crossover utility vehicle with a starting price of $39,990. Musk also noted that he expected Tesla to have a $25,000 electric vehicle by the end of 2023[x] and that the company was also working on an autonomous taxi service.[y]

Tesla's Future

Tesla's cars had rapidly attracted a large and loyal fan base, and sales were growing at an impressive rate. However, designing and launching multiple major car platforms while building a large-scale battery company, a network of charging stations, and operating Solar City was a lot for a company to take on in its first fifteen years. This left some analysts scratching their heads. Was Tesla trying to do too much too quickly?

Discussion Questions

1. What were Musk's and Eberhard's goals in founding Tesla?
2. How would you characterize competition in the auto industry?
3. What do you think are Tesla's core competencies? Does it have any sources of sustainable competitive advantage?
4. What is your assessment of Tesla's moves into (a) mass-market cars, (b) batteries (car batteries and Powerwall), (c) solar panels? Please consider both the motivation for the moves, and the opportunities and challenges for Tesla to compete in these businesses.
5. Do you think Tesla will be profitable in all of these businesses? Why or why not?
6. What do you think Tesla's (or Elon Musk's) strategic intent is?

[a] Rebecca Elliott, "Tesla Posts First Full Year of Profitability," *The Wall Street Journal,* January 27, 2021.

[b] Michael V. Copeland, "Tesla's Wild Ride," *Fortune* 158, no. 2 (2008): 82–94.

[c] Ibid.

[d] John Boudreau, "In a Silicon Valley Milestone, Tesla Motors Begins Delivering Model S Electric Cars," *Oakland Tribune*, June 24, 2012, Breaking News Section.

[e] Michael V. Copeland, "Tesla's Wild Ride," *Fortune* 158, no. 2 (2008): 82–94.

[f] Alex Williams, "Taking a Tesla for a Status Check in New York," *New York Times*, July 19, 2009, ST.7.

[g] Ibid.

[h] Mike Ramsey, "Tesla Sets 300-Mile Range for Second Electric Car," *Wall Street Journal (Online)*, March 7, 2011.

[i] Steven Overly, "This Government Loan Program Helped Tesla at a Critical Time. Trump Wants to Cut It," Washington Post, March 16, 2017.

[j] Anonymous. "Idle Fremont Plant Gears Up for Tesla," *Wall Street Journal (Online)*, October 20, 2010.

[k] Michael Levi, "How Tesla Pulled Ahead of the Electric-Car Pack," *Wall Street Journal*, June 21, 2013, A.11.

[l] Joseph B. White, "Corporate News: Electric Car Startup Tesla Repays U.S. Loan," *Wall Street Journal* (May 23, 2013): B.3.

[m] Jerry Hirsch and Russ Mitchell, "Model X: Under the Hood of Tesla's SUV Strategy," *Los Angeles Times* September 29, 2015.

[n] Tesla Q4 Production and Delivery Report (January 3, 2016).

[o] Tesla Model X Review: Fast and Flawed. Consumer Reports (2016).

[p] Tesla Model S. *Car and Driver* (May, 2016).

[q] Jeff Cobb, "Tesla Model S Is World's Best-Selling Plug-in Car for Second Year in a Row," *HybridCars* (January 26, 2017).

[r] Estimate based on Tesla quarterly production and delivery reports for quarter 1-4 in 2017. In some years Tesla only provides rounded numbers for breakdown between Model X and Model S, thus only an approximate number can be given here.

[s] Bob Sorokanich, "Tesla Model 3: The Road & Track Review," *Road and Track*, January 12, 2018.

[t] Arjun Panchadar, "Tesla Must Stop Overpromising, Could Need More Finance: Analysts," *Reuters Business News*, November 2, 2017.

[u] Andrew J. Hawkins and Sean O'Kane, "Tesla Built a Giant Tent to Expand Production of the Model 3," *The Verge,* June 22, 2018.

[v] Oliver Milman, "Elon Musk Leads Tesla Effort to Build House Roofs Entirely Out of Solar Panels," *The Guardian,* August 19, 2016.

[w] D. Hull and C. Zhang, "Elon Musk Set Up His Shanghai Gigafactory in Record Time," *Bloomberg,* October 23, 2019.

[x] N. Chokshi, "Elon Musk Promises to Make a $25,000 Car (in 3 Years)," *The New York Times, ISSN 0362-4331.* September 22, 2020.

[y] A. Durkee, "Elon Musk Wants His Own 'Robot Taxis' to Replace Uber and Lyft," *Vanity Fair,* April 22, 2019.

OVERVIEW

The first step in formulating a company's technological innovation strategy is to assess its current position and define its strategic direction for the future. This chapter reviews some of the basic tools used in strategic analysis to assess the firm's current position and help chart its direction for the future. These tools help the manager answer such questions as:

- What threats and opportunities are most pressing in the firm's environment?
- What are the firm's key strengths and weaknesses?
- Does the firm have any sources of sustainable competitive advantage?
- What are the firm's core competencies, and what kind of value propositions do those core competencies offer to customers? How do managers want those value propositions to evolve?
- What key resources and capabilities does the firm need to develop or acquire to meet its long-term objectives?

The outputs of the analytical tools in this chapter are crucial inputs for the tools used in Chapter Seven, Choosing Innovation Projects. A coherent technological innovation

strategy both leverages and enhances the firm's existing competitive position, and it provides direction for the future development of the firm. Formulating a technological innovation strategy first requires an accurate appraisal of where the firm currently is. It then requires articulating an ambitious strategic intent—one that creates a gap between a company's existing resources and capabilities and those required to achieve its intent.[1] The ability of the firm to cohesively leverage all its resources around a unified vision can enable it to create a competitive advantage that is very difficult for competitors to imitate.[2]

ASSESSING THE FIRM'S CURRENT POSITION

To assess the firm's current position in the marketplace, it is useful to begin with some standard tools of strategic analysis for analyzing the external and internal environment of the firm.

External Analysis

The two most commonly used tools for analyzing the external environment of the firm include Porter's five-force model and stakeholder analysis.

Porter's Five-Force Model

In this model, the attractiveness of an industry and a firm's opportunities and threats are identified by analyzing five forces plus the role of complements (see Figure 6.1).[3]

While the five-force model was originally developed to assess industry attractiveness (i.e., "Is this a desirable industry in which to compete?"), in practice the model is often used to assess a specific firm's external environment (i.e., "What factors in the firm's external environment create threats and opportunities for the firm?"). The difference

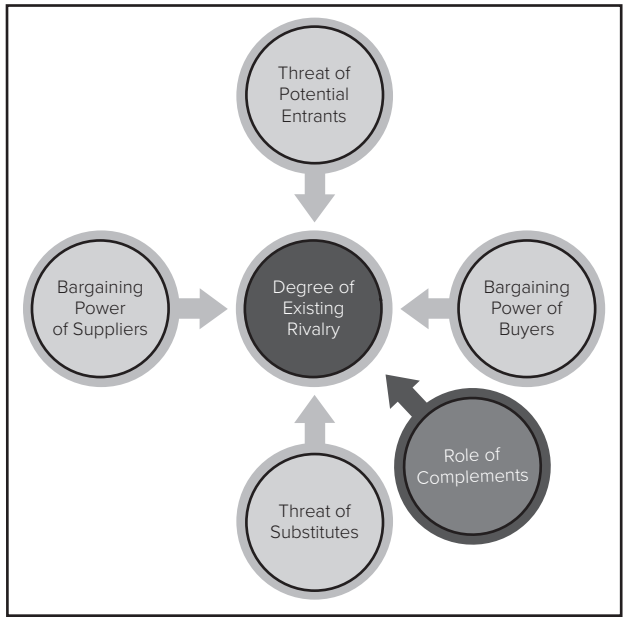


FIGURE 6.1
Porter's Five Forces Plus Complements

Source: https://www.youtube.com/watch?v=W0hySWGHKj4&t=12s

between these two approaches is subtle but important. In the former approach, the analysis focuses on the industry level, treating all competitors as roughly the same, and its objective is to ascertain whether the industry as a whole will tend to be profitable. In the latter approach, the analysis may take the perspective of a particular firm, often identifying ways in which the external forces differentially affect the firm vis-à-vis its competitors, and its objective is to identify threats and opportunities for the firm.[4] For example, an external analysis of the discount retailing industry that is focused only on industry attractiveness might conclude that the industry is relatively unattractive given high price competition and limited opportunities to differentiate. An external analysis of the discount retailing industry focused on Walmart, on the other hand, might conclude that while the industry is a difficult one in which to be profitable, Walmart is likely to be more profitable than its competitors because its scale, its use of advanced technology for inbound and outbound logistics, and its location strategies give it considerable bargaining power over both suppliers and buyers. The latter approach will be emphasized here because it better suits our purpose of helping a particular firm to chart its strategic direction.

The five forces are:

1. **The degree of existing rivalry.** An industry's degree of rivalry is influenced by a number of factors. First, the number and relative size of competitors will shape the nature of rivalry. In general, the more firms competing that are of comparable size, the more competitive the industry will be. There are, however, exceptions to this generality. For example, **oligopolistic industries** (those that have a few large competitors) can be fiercely competitive if firms choose to engage in price wars (as happened in the personal digital assistant industry). On the other hand, oligopolistic industries can have a low degree of rivalry if the competitors choose to avoid competing head-to-head in the same market segments, or if they engage in tacit price collusion. Rivalry is also influenced by the degree to which competitors are differentiated from each other. For example, if competitors are highly differentiated, they will experience less direct rivalry because their products are likely to appeal to different market segments. For example, even though Genzyme operated in the extremely competitive biotech industry, its unique focus on orphan drugs meant that it typically was not competing head-on with other rivals for its customers. This enabled it to charge much higher margins on its products. Demand conditions also influence degree of rivalry. When demand is increasing, there are more revenues to go around and firms will experience less competitive pressure. On the other hand, when demand is declining, firms have to compete for a shrinking pool of revenues, and competition can become very aggressive. In declining industries, high **exit barriers** (fixed capital investments, emotional attachment to the industry, etc.) can also intensify rivalry by making firms reluctant to abandon the industry.
2. **Threat of potential entrants.** The threat of potential entrants is influenced by both the degree to which the industry is likely to attract new entrants (e.g., is it profitable, growing, or otherwise alluring?) and the height of **entry barriers**. Entry barriers can include such factors as large start-up costs, brand loyalty, difficulty in gaining access to suppliers or distributors, government regulation, threat of

oligopolistic industries
Highly consolidated industries with a few large competitors.

exit barriers
Costs or other commitments that make it difficult for firms to abandon an industry (large fixed-asset investments, emotional commitment to the industry, etc.).

entry barriers
Conditions that make it difficult or expensive for new firms to enter an industry (government regulation, large start-up costs, etc.).

retaliation by existing competitors, and many others. While profitability and growth may attract new entrants, entry barriers will deter them. For example, while high projected growth in the smartphone market attracts potential entrants to this market, the challenge of competing against large, well-established, and efficient competitors such as Nokia and Ericsson deters many entrants. To effectively compete against these companies requires that an entrant be able to manufacture, advertise, and distribute on a large scale, suggesting significant start-up costs for an entrant to achieve a competitive position. However, some of these capabilities could be obtained through partnerships with other firms, such as having contract manufacturers handle production and having mobile phone service providers handle distribution, thereby lowering start-up costs.

3. **Bargaining power of suppliers.** The degree to which the firm relies on one or a few suppliers will influence its ability to negotiate good terms. If there are few suppliers or suppliers are highly differentiated, the firm may have little choice in its buying decision, and thus have little leverage over the supplier to negotiate prices, delivery schedules, or other terms. On the other hand, if suppliers are very abundant and/or are not highly differentiated, the firm may be able to force the suppliers to bid against one another for the sale. The amount the firm purchases from the supplier is also relevant. If the firm's purchases constitute the bulk of a supplier's sales, the supplier will be heavily reliant upon the firm and the supplier will have little bargaining power. Likewise, if the supplier's sales constitute a large portion of the firm's purchases, the firm will be heavily reliant upon the supplier and the supplier will have more bargaining power. For example, manufacturers that sell to Walmart often have little bargaining power because Walmart's enormous volume often makes its purchases a significant portion of a manufacturer's yearly sales. Walmart's suppliers typically have little bargaining power. When Intel sells to personal computer manufacturers, on the other hand, it typically wields considerable supplier bargaining power. When computer manufacturers consider potential suppliers for microprocessors, they often have little choice but to go with Intel—many consumers demand that their systems include Intel microprocessors, and the majority of personal computer hardware and software has been optimized for the Intel architecture. If the firm faces **switching costs** that make it difficult or expensive to change suppliers, this will also increase the supplier's bargaining power. Finally, if the firm can backward **vertically integrate** (i.e., produce its own supplies), this will lessen supplier bargaining power, and if the supplier can threaten to forward vertically integrate into the firm's business, this will increase the supplier's bargaining power.

switching costs
Factors that make it difficult or expensive to change suppliers or buyers, such as investments in specialized assets to work with a particular supplier or buyer.

vertical integration
Getting into the business of one's suppliers (backward vertical integration) or one's buyers (forward vertical integration). For example, a firm that begins producing its own supplies has practiced backward vertical integration, and a firm that buys its distributor has practiced forward vertical integration.

4. **Bargaining power of buyers.** Many of the same factors that influence the bargaining power of suppliers have an analogous role with the bargaining power of buyers. The degree to which the firm is reliant on a few customers will increase the customer's bargaining power, and vice versa. If the firm's product is highly differentiated, buyers will typically experience less bargaining power, and if the firm's product is undifferentiated, buyers will typically experience greater bargaining power. If buyers face switching costs, this is likely to lower their bargaining power, and if the firm faces switching costs to work with other buyers, this will increase the buyer's bargaining power. Finally, if the buyers can threaten to backward vertically

integrate, this will increase their bargaining power, and if the firm can threaten to forward vertically integrate, it will lower customer bargaining power.

5. **Threat of substitutes.** Substitutes are products or services that are not considered competitors, but fulfill a strategically equivalent role for the customer. For example, Starbucks may consider other coffeehouses as competitors, but other social destinations (such as bars or restaurants) or beverages (such as soft drinks or beer) as substitutes. The more potential substitutes there are, and the closer they are in function to the firm's product or service, the greater the threat of substitution. Furthermore, the threat of substitutes will also be shaped by the relative price. For example, while traveling by bus versus air is not particularly comparable in terms of speed, traveling by bus is often considerably less expensive; thus, it poses a threat of substitution, particularly for shorter distances to be traveled. Note that distinguishing between a competitor and a substitute depends on how the industry is defined. For example, if one considers the *airline* industry as the unit of analysis, then bus service is a substitute for airlines. However, if one were considering the *transportation* industry as the unit of analysis, then bus services would be competitors of airlines.

complements Products or services that enhance the usefulness or desirability of another product.

Recently, Porter has acknowledged the role of **complements**.[5] As has been discussed in several of the earlier chapters, complements are products that enhance the usefulness or desirability of a good. For example, software is an important complement for computers, and gasoline is an important complement for automobiles. The availability, quality, and price of complements will influence the threats and opportunities posed by the industry. It is important to consider (1) how important complements are in the industry, (2) whether complements are differentially available for the products of various rivals (impacting the attractiveness of their goods), and (3) who captures the value offered by the complements. For example, desktop printer manufacturers such as Hewlett Packard and Lexmark make a considerable portion of their desktop printing profits from the ink cartridges that consumers have to replace when empty. The printer manufacturers thus design the printer cartridges to be specific to each printer model, avoiding standardized designs that would facilitate consumers purchasing printer cartridges from other vendors for their Hewlett Packard and Lexmark printers. The market for ink cartridges is so lucrative, however, that a number of third-party vendors have emerged that either clone Hewlett Packard and Lexmark cartridges or offer to refill the consumer's empty cartridges with ink.

Stakeholder Analysis

stakeholder Any entity that has an interest ("stake") in the organization.

Stakeholder models are often used for both strategic and normative purposes. A strategic *stakeholder analysis* emphasizes the stakeholder management issues that are likely to impact the firm's financial performance, while a *normative stakeholder analysis* emphasizes the stakeholder management issues the firm ought to attend to due to their ethical or moral implications.[6] Typically, the first step of a stakeholder analysis is to identify all the parties that will be affected by the behavior of the firm (and thus have a "stake" in the firm). For each party, the firm identifies what that stakeholder's interests are, what resources they contribute to the organization, what claims they are likely to make on the organization, and which will be most important from the firm's perspective. Stakeholders include (but are not limited to) stockholders, employees, customers, suppliers, lenders, the local community, government, and rivals (see Figure 6.2).

FIGURE 6.2
Stakeholder Analysis

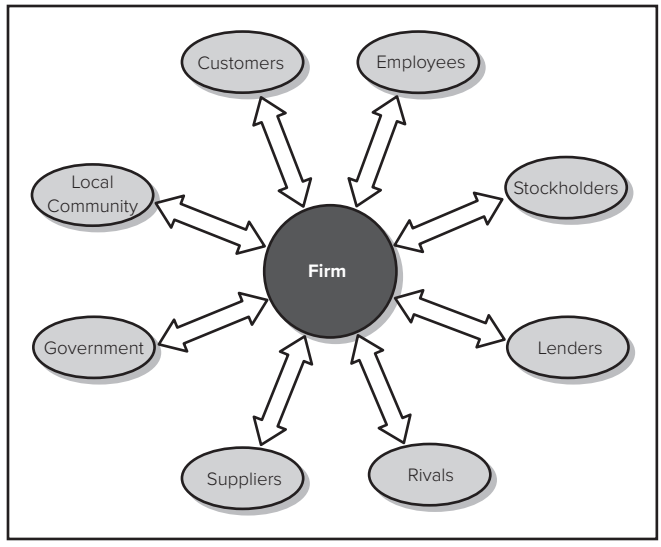
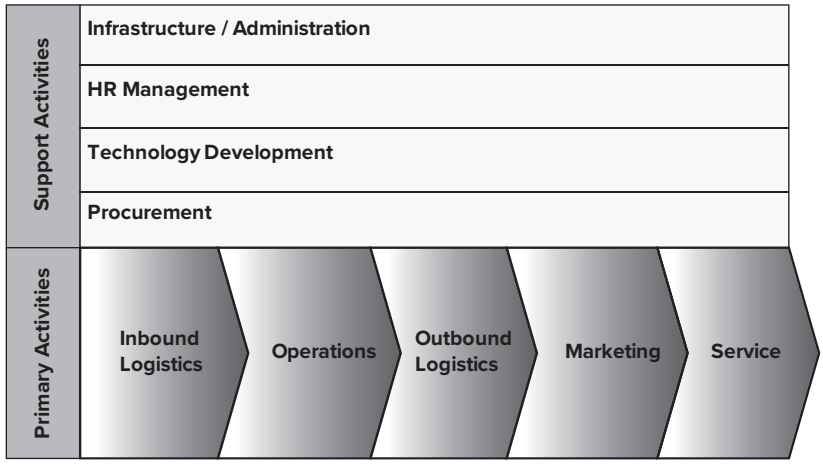


FIGURE 6.3
Porter's Value Chain

Source: Michael Porter, *Competitive Advantage: Creating and Sustaining Superior Performance*.



Internal Analysis

The analysis of the internal environment of the firm most often begins with identifying the firm's strengths and weaknesses. Sometimes this task is organized by examining each of the activities of the value chain (see Figure 6.3).[7] In Michael Porter's model of a value chain, activities are divided into primary activities and support activities. Primary activities include *inbound logistics* (all activities required to receive, store, and disseminate inputs), *operations* (activities involved in the transformation of inputs into outputs), *outbound logistics* (activities required to collect, store, and distribute outputs), *marketing and sales* (activities to inform buyers about products and services and to induce their purchase), and *service* (after-sales activities required to keep the product or

service working effectively). Support activities include *procurement* (the acquisition of inputs, but not their physical transfer, as that would be covered in inbound logistics), *human resource management* (activities such as recruiting, hiring, training, and compensating personnel), *technology development* (activities involved in developing and managing equipment, hardware, software, procedures, and knowledge necessary to transform inputs into outputs), and *infrastructure* (functions such as accounting, legal counsel, finance, planning, public affairs, government relations, quality assurance, and general management necessary to ensure smooth functioning of the firm). This generic model can be adapted to better fit a particular firm's needs. For example, for a biotechnology firm or software developer, research and development is likely to be a primary activity and inbound logistics may be insignificant.

Each activity can then be considered from the point of view of how it contributes to the overall value produced by the firm, and what the firm's strengths and weaknesses are in that activity. For example, Figure 6.4 illustrates what a value-chain analysis might look like for Take-Two Interactive Software, which produces the Grand Theft Auto video game. In the figure, research and development is considered a primary activity, but the support activity of the technology development is not considered. Because all the game manufacturing is performed by the console producers rather than by Take-Two its primary technology activities center on design of the games, which is covered in the research and development section.

FIGURE 6.4
Example of Value-Chain Analysis for Take-Two Interactive Software

Value-Chain Activity	Strengths	Weaknesses
Inbound Logistics Insignificant; few inputs necessary.		
Research & Development Ability to incorporate state-of-the-art graphics capabilities, sound, and innovative themes significantly differentiates the product in the eyes of the consumer.	Take-Two game maximized the polygon processing potential of the video game console, making play more lifelike. Parallel development processes kept the development cycles short.	Lack of experience in developing online games could become major liability if the market for playing games over the Internet takes off.
Operations Company focuses on producing a few very high-quality games, introducing a new version every year. Once designed, games are manufactured by the video game console producers (e.g., Sony).	Concentrating on a few games enables firm to focus significant resources on ensuring that the game is a hit.	Concentrating on a few games could be risky—if a game fails, the company may have no revenues to support operations. Take-Two is completely reliant on console manufacturers for the right to develop compatible games and for manufacture of games.

continued

concluded

Value-Chain Activity	Strengths	Weaknesses
Outbound Logistics Products are sold through game retailers (e.g., Gamestop), general merchandisers (e.g., Best Buy), and occasionally through bundling arrangements with video console producers, avoiding expense of maintaining the company's own retail stores. On initial launch, Take-Two had signed an exclusive deal with Sony to offer Grand Theft Auto exclusively for PlayStation 2.	Existing retailers already have excellent market penetration, providing rapid and wide deployment of the games. Since Sony PlayStation 2 was the No. 1 video game console, signing an exclusive with Sony enabled Take-Two to tap a large market.	Using retailers gives company little discretion over store placement, promotion, and pricing.
Marketing Helps build customer awareness of products, builds brand image, accelerates sales. Uses Web sites, advertisements in gaming magazines, billboards. Grand Theft Auto is targeted toward the adult market.	GTA had successfully established an image as being leading edge, and was the No. 1 game in 2002. In 2008, Grand Theft Auto: San Andreas made the "Top 20 console games of all times" list by selling 12 million copies.	Some consumers, retailers, and regulatory agencies criticized Grand Theft Auto's violence and sexual images, potentially tarnishing the company's image.
Service Phone line for technical support helps customers resolve problems in using the product.	Take-Two has had relatively few returns or warranty problems.	
Firm Infrastructure Legal department negotiates license rights for games.		Copyright infringement suits by other game producers are becoming more frequent.
Human Resource Management Hiring and retaining skilled and creative developers is crucial for the production of high-quality games. Company had 2002 full-time employees in 2007.	Employees are not unionized. Employee stock option plan improves loyalty and morale.	
Procurement Necessary to acquire rights to use copyright-protected characters and music.	Thus far, Take-Two has been very successful in obtaining rights to use copyrighted materials.	

Sources: S. Balasubramanian, A. Kim, L. Lei, and S. Singh, "Beyond the Mayhem: Take-Two Interactive Software," New York University teaching case, 2003; www.Take2games.com.

Once the key strengths and weaknesses are identified, the firm can assess which strengths have the potential to be a source of sustainable competitive advantage. This helps the firm gain valuable perspective on which of its activities and resources should be further leveraged in its articulation of its strategic intent for the future.

To be a potential source of sustainable competitive advantage, resources must be *rare, valuable, durable,* and *inimitable.*[8] Resources that are rare and valuable may yield

Research Brief Using Big Data to Guide Innovation

In the last decade, massive advancements in **digitalization** and social media, combined with the proliferation of mobile devices, has led to the creation of huge datasets—colloquially known as "big data."[a] Advanced software can then be used to find correlations and identify potential causal relationships that conventional market research would have never revealed. In general, people have assumed that these large datasets should yield a treasure trove of insights about how to create better products and improve profitability. The near-real-time access to customer search and feedback can uncover hidden patterns,[b] and provide information both about how to improve existing products and tap new potential market opportunities. Big data was thus anticipated to uniformly improve product development, leading to ever closer matches between the goods suppliers provide and customers seek,[c] and managers around the globe rushed to create big data analytics programs at their firms. In fact, an analysis report from Dresner Advisory Services found that by 2020, 53 percent of all companies were adopting big data analytics.

digitalization
The conversion of text, pictures, or sound into a digital form that can be processed by a computer.

The results of applying big data to new product development, however, have been more mixed.[d] While studies have shown that companies like Amazon, eBay, Target, and Ford have been successful in using big data to improve forecasts of customer needs and design better products,[e] other studies have highlighted the costs and risks of big data, including high direct costs of maintaining and securing the data, reputation risks if data is misused, and bias that can occur in its collection.[f] Firms that lack the ability to adequately process the data and verify its accuracy and representativeness could find that big data is at best a distraction, and at worst, seriously misleading.[g]

Professor Foster Provost, a noted researcher in big data analytics, notes that a key mistake managers often make is to think, "How should I mine the data I have?" or to plan on collecting a large data repository and hope that it reveals the problems the firm should solve.[h] Instead, he argues, managers should start by thinking about what the most important problems or opportunities are that the firm is facing and invest in collecting the kind of data that will help address those problems and opportunities. In his book, *Data Science for Business,*[i] he offers some tips for making data science valuable in the organization:

1. **Treat data as assets.** This allows you to not just think about mining the data that you have, but to consider (a) how to invest in data, and (b) how to get return on your data assets. In most cases, the raw data are not ready to help the organization. They must be structured and cleaned; new data must be engineered from the raw data.
2. **Managers must take the building and composition of the data science team very seriously.** There is a very large difference in what a firm can and will achieve based on the expertise and experience of the data science team. A key challenge here is having enough expertise to evaluate whether someone is a great data scientist. If the firm does not already have top data science expertise, it may have trouble both attracting and evaluating top talent.
3. **Differentiate between the business opportunities and challenges facing the firm, and the data analytics questions that you pose (i.e., they are not the same).** Can you explain why the particular data analytics questions being used are in service of addressing some important business challenge?
4. **Assess how representative, accurate, and reliable the data are.** Managers must make sure to understand the processes of generating the data and take them into account. In general, you should assume the data are biased; it is almost certain that the data are not a large random sample from exactly the population that the firm intends to examine. How and why were the data generated? What are the potential sources of error and bias, and what are the implications for the analyses that should be conducted?

continued

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[a] Michael Cox and David Ellsworth "*Application-controlled demand paging for out-of-core visualization. IEEE Access 97CB36155(1997):* 235–44; Daniel Trabucchi, Tommaso Buganza, and Elena Pellizzoni, "Give Away Your Digital Services," *Research-Technology Management* 60, no. 2 (2017): 43–52; Daniel Trabucchi, Tommaso Buganza, Claudio Dell'Era, and Elena Pellizzoni, "Exploring the Inbound and Outbound Strategies Enabled by User Generated Big Data: Evidence from Leading Smartphone Applications," *Creativity and Innovation Management* 27, no. 1 (2018): 42–55.

[b] Jovan Milenkovic, 2019. "30 Eye-Opening Big Data Statistics for 2020: The Patterns Are Everywhere." Kommando Tech, 2019, https://kommandotech.com/statistics/big-data-statistics.

[c] Gianvito Lanzolla, et al., "Digital Transformation: What Is New if Anything? Emerging Patterns and Management Research," *Academy of Management Discoveries:* 341–50.

[d] Francesco Cappa et al., "Big Data for Creating and Capturing Value in the Digitalized Environment: Unpacking the Effects of Volume, Variety, and Veracity on Firm Performance," *Journal of Product Innovation Management* 38, no. 1 (2020): 49–67.

[e] Hsinchun Chen et al., "Business Intelligence and Analytics: From Big Data to Big Impact," *MIS Quarterly* 36, no. 4 (2012): 1165–88; Amir Gandomi and Murtaza Haider, "Beyond the Hype: Big Data Concepts, Methods, and Analytics," *International Journal of Information Management* 35, no. 2 (2015): 137–44; Samuel A. Wamba et al., "Big Data Analytics and Firm Performance: Effects of Dynamic Capabilities," *Journal of Business Research* 70(2017): 356–65; Sunil Erevelles, Nobuyuki Fukawa, and Linda Swayne, "Big Data Consumer Analytics and the Transformation of Marketing," *Journal of Business Research* 69, no. 2 (2016): 897–904.

[f] Francesco Cappa et al., "Big Data for Creating and Capturing Value in the Digitalized Environment: Unpacking the Effects of Volume, Variety, and Veracity on Firm Performance," *Journal of Product Innovation Management* 38, no. 1 (2020): 49–67; Daniel Trabucchi, Tommaso Buganza, and Elena Pellizzoni, "Give Away Your Digital Services," *Research-Technology Management* 60, no. 2: 43–52; Tim Harford, "Big Data: Are We Making a Big Mistake?" *Financial Times,* 2014, https://www.ft.com/content/21a6e7d8-b479-11e3-a09a-00144feabdc0.

[g] Daan Van Knippenberg et al., "From the Editors Information, Attention, and Decision Making," *Academy of Management Journal* 58, no. 3 (2015): 649–57; Kimberly A. Whitler, "Why Too Much Data Is a Problem and How to Prevent It," *Forbes,* 2018, https://www.forbes.com/sites/kimberlywhitler/2018/03/17/why-too-much-data-is-a-problem-and-how-to-prevent-it/#348d5d77755f.

[h] Foster Provost and Jim Euchner, "What Managers Need to Know about Big Data," *Research-Technology Management,* (May–June 2017): 11–7.

[i] Foster Provost and Tom Fawcett, *Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking* (O'Reilly Media, 2013).

tacit resources
Resources of an intangible nature (such as knowledge) that cannot be readily codified.

socially complex resources
Resources or activities that emerge through the interaction of multiple individuals.

causal ambiguity
The relationship between a resource and the outcome it produces is poorly understood (the *causal* mechanism is *ambiguous*).

a competitive advantage, but that advantage will not be sustainable if the firm is incapable of keeping the resources, or if other firms are capable of imitating them. For example, a positive brand image can be a rare and valuable resource, but it requires ongoing investment to sustain. If a firm lacks the capital to reinvest in its brand image, it will erode. Furthermore, many valuable resources are quickly imitated by other firms. Technological advances are reverse-engineered, skillful marketing campaigns are copied, innovative human resource practices are adopted by imitators, and so on. Some resources, however, are not readily imitable. For example, if valuable resources are **tacit** (i.e., they cannot be readily codified in written form), path dependent (i.e., they are dependent on a particular historical sequence of events), **socially complex** (i.e., they arise through the complex interaction of multiple people), or **causally ambiguous** (i.e., it is unclear how the resource gives rise to value), they will be extremely difficult to imitate.[9] For example, *talent* is typically considered to be a tacit and causally ambiguous resource. It is thought to be an inherent trait that cannot be trained, and the mechanisms by which individuals acquire it or tap it are poorly understood. A first-mover advantage is a path-dependent advantage that cannot be copied—once a firm has become the first mover in a category, other firms no longer have the opportunity to be first. Once the firm has established a baseline internal analysis, it can move on to identifying its core competencies and formulate its strategic intent.

IDENTIFYING CORE COMPETENCIES AND DYNAMIC CAPABILITIES

Core Competencies

A company's core competencies are typically considered to be those that differentiate it strategically. A core competency is more than just a core technology. A core competency arises from a firm's ability to combine and harmonize multiple primary abilities in which the firm excels into a few key building blocks of specialized expertise. Competencies often combine different kinds of abilities, such as abilities in managing the market interface (e.g., advertising, distribution), building and managing an effective infrastructure (e.g., information systems, logistics management), and technological abilities (e.g., applied science, process design).[10] This combination and harmonization of multiple abilities make core competencies difficult to imitate. Consider, for example, Sony's core competency in miniaturization.[11] This competency arises from the harmonization of multiple technologies (liquid crystal displays, semiconductors, etc.) and is leveraged into multiple markets (televisions, radios, personal digital assistants, etc.). A firm's core competencies also depend on building high-quality relationships across different functions and business units.

Prahalad and Hamel compare core competencies to roots, from which grow core products such as major components or subassemblies. Core products, in turn, give rise to business units, whose fruits are the various end products of the company (see Figure 6.5).

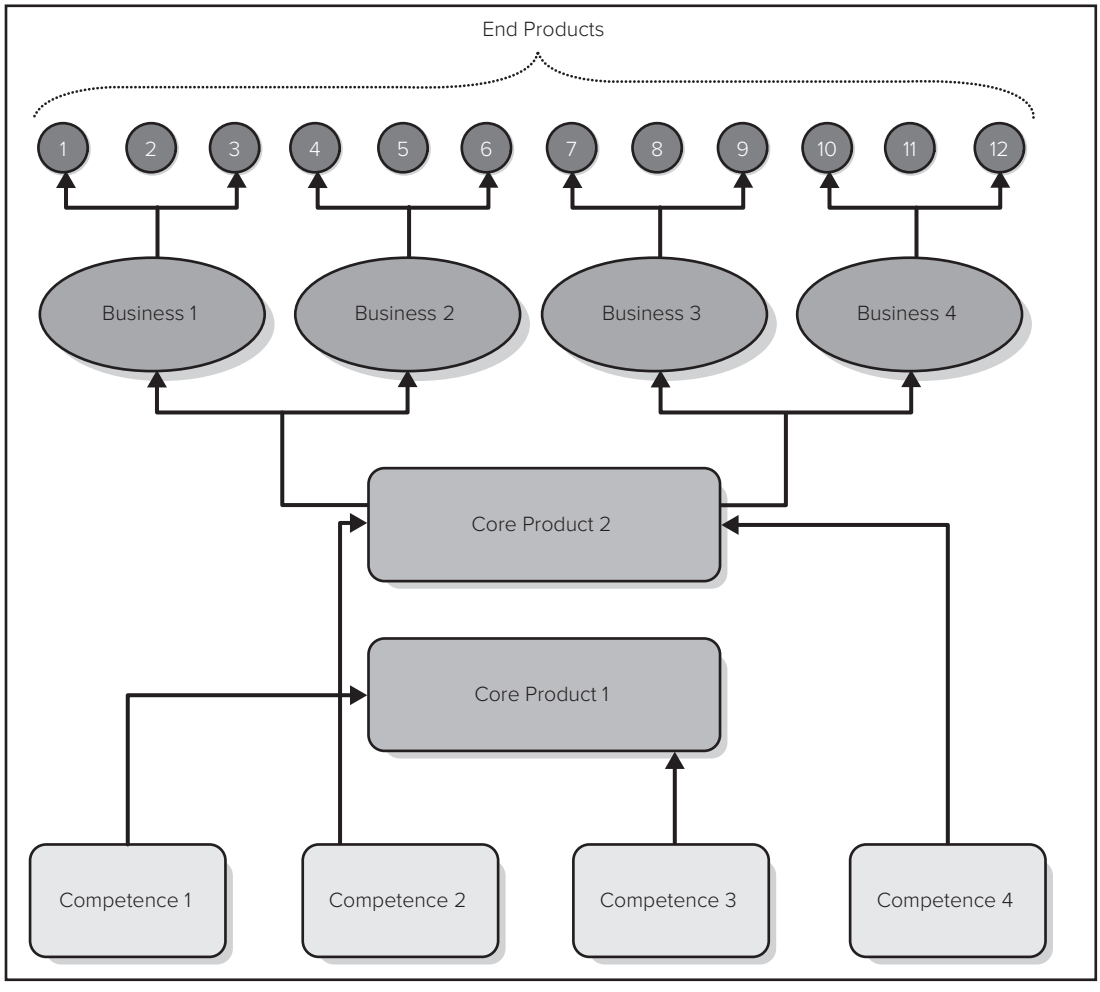
Several core competencies may underlie an individual business unit, and several business units may draw upon the same core competency. This indicates the organization's structure and incentives must encourage cooperation and exchange of resources across strategic business unit boundaries. If managers or resources are wed too closely to their business units, there will be underinvestment in the development and leverage of core competencies.[12] Prahalad and Hamel go so far as to argue that strategic business units should be expected to bid for people in the firm who have particular skills to contribute to a project. Instead of viewing individuals as being employed by a particular strategic business unit, individuals should be considered corporate assets that can be redeployed across the organization.

Prahalad and Hamel offer the following tests to identify the firm's core competencies:

1. Is it a significant source of competitive differentiation? Does it provide a unique signature to the organization? Does it make a significant contribution to the value a customer perceives in the end product? For example, Sony's skills in miniaturization have an immediate impact on the utility customers reap from its portable products.
2. Does it transcend a single business? Does it cover a range of businesses, both current and new? For example, Honda's core competence in engines enables the company to be successful in businesses as diverse as automobiles, motorcycles, lawn mowers, and generators.
3. Is it hard for competitors to imitate? In general, competencies that arise from the complex harmonization of multiple technologies will be difficult to imitate. The competence may have taken years (or decades) to build. This combination of resources and embedded skills will be difficult for other firms to acquire or duplicate.

FIGURE 6.5
Visualizing the Firm's Core Competencies, Core Products, Business Units, and End Products

Source: C. K. Prahalad and G. Hamel, "The Core Competence of the Corporation," *Harvard Business Review*, May–June 1990.



According to Prahalad and Hamel, few firms are likely to be leaders in more than five or six core competencies. If a company has compiled a list of 20 to 30 capabilities, it probably has not yet identified its true core competencies. By viewing the business as a portfolio of core competencies, managers are better able to focus on value creation and meaningful new business development, rather than cost cutting or opportunistic expansion.[13]

The Risk of Core Rigidities

Sometimes the very things that a firm excels at can enslave it, making the firm rigid and overly committed to inappropriate skills and resources.[14] Incentive systems may

evolve that favor activities that reinforce the firm's core competencies. The organizational culture may reward employees who are most closely connected to core competencies with higher status and better access to other organizational resources. While these systems and norms can prove beneficial in reinforcing and leveraging the firm's existing core competencies, they can also inhibit the development of new core competencies. For example, a firm's emphasis on a scientific discipline that is central to its core competency can make the firm less attractive to individuals from other disciplines. Rewards for engaging in core competency activities can discourage employees from pursuing more exploratory activities. Finally, as noted in Chapter Four, knowledge accumulation tends to be very path dependent. Firms that have well-developed knowledge sets along a particular trajectory may find it very hard to assimilate or utilize knowledge that appears unrelated to that trajectory, potentially limiting the firm's flexibility.[15]

Dynamic Capabilities

In fast-changing markets, it can be extremely useful for a firm to develop a core competency in responding to change. Whereas in Prahalad and Hamel's model, core competencies relate to sets of specific core products, it is also possible for a firm to develop core competencies that are not specific to any set of technologies or products, but rather to a set of abilities that enable it to quickly reconfigure its organizational structure and routines in response to new opportunities.[16] Such competencies are termed **dynamic capabilities**. Dynamic capabilities enable firms to quickly adapt to emerging markets or major technological discontinuities. For example, Corning has made its own evolvability one of its most important core competencies. It invests heavily in research in areas likely to provide scientific breakthroughs (such as opal glasses and their solvents). It develops pilot plants to experiment with new products and production processes.[17] It even manages its relationships with alliance partners not as individual relationships focused on particular projects, but rather as an integrative and flexible system of capabilities that extend the firm's boundaries.[18]

dynamic capabilities
A set of abilities that make a firm more agile and responsive to change.

STRATEGIC INTENT

A firm's purpose is to create value. This entails more than just improving operations or cutting costs; it means leveraging corporate resources to create more performance for customers, more well-being for employees, and more returns for shareholders. This is accomplished through developing new businesses and markets, and leveraging corporate resources, all guided by the firm's *strategic intent.*[19]

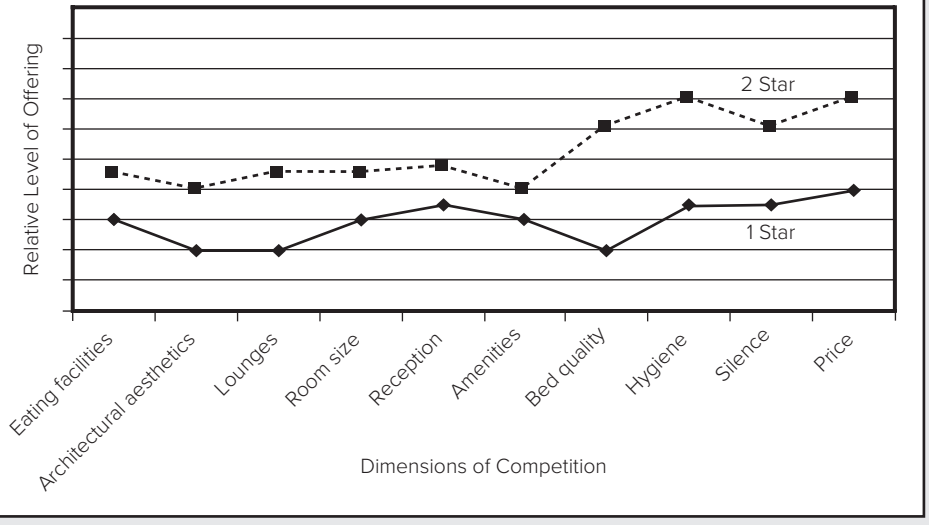
A company's strategic intent is a long-term goal that is ambitious, builds upon and stretches the firm's existing core competencies, and draws from all levels of the organization. Hamel and Prahalad's examples include Canon's obsession with overtaking Xerox in copiers, Apple's mission of ensuring that every individual has a personal computer, and Yahoo's goal of becoming the world's largest Internet shopping mall. Typically, a strategic intent looks 10 to 20 years ahead and establishes clear milestones for employees to target.[20] This forward-looking orientation is crucial; without it companies can easily become focused on markets they have served in

Research Brief Blue Ocean Strategy

In a series of articles and a book[a], Renée Mauborgne and W. Chan Kim describe firms who crafted what they call "blue ocean" strategies by innovating in a way that allowed them to enter untapped market space. In most industries, the rules of the game are well-understood and accepted, and firms compete by trying to outdo each other on the accepted dimensions of competition. Each firm hopes to capture a greater share of existing demand, and as the industry becomes crowded, the likelihood of firm profits or growth diminishes. Cut-throat competition turns the ocean bloody (also known as "red ocean"). Blue oceans refer to untapped market space that firms create by redefining the dimensions of competition. They are uncharted, and there are no (or few) competitors. Blue ocean strategies are thus fundamentally about differentiation through innovation.

Mauborgne and Chan suggest that firms can identify "blue ocean" strategies by first using a visualization tool, the "strategy canvas," to help them understand how different players are competing in an industry, and how they might choose to compete differently. The horizontal axis lists the factors that the industry competes on/invests in, and the vertical axis indicates "high" or "low." Managers can then plot "**value curves**" for different product offerings. For example, you might draw a graph as indicated in the figure below.

Red Ocean Strategy	**Blue Ocean Strategy**
Compete in existing market space	Create uncontested market space
Beat the competition	Make the competition irrelevant
Exploit existing demand	Create and capture new demand
Make the value-cost trade-off	Break the value-cost trade-off
Align the whole system of a firm's activities with its strategic choice of differentiation *or* low cost.	Align the whole system of a firm's activities in pursuit of differentiation *and* low cost.



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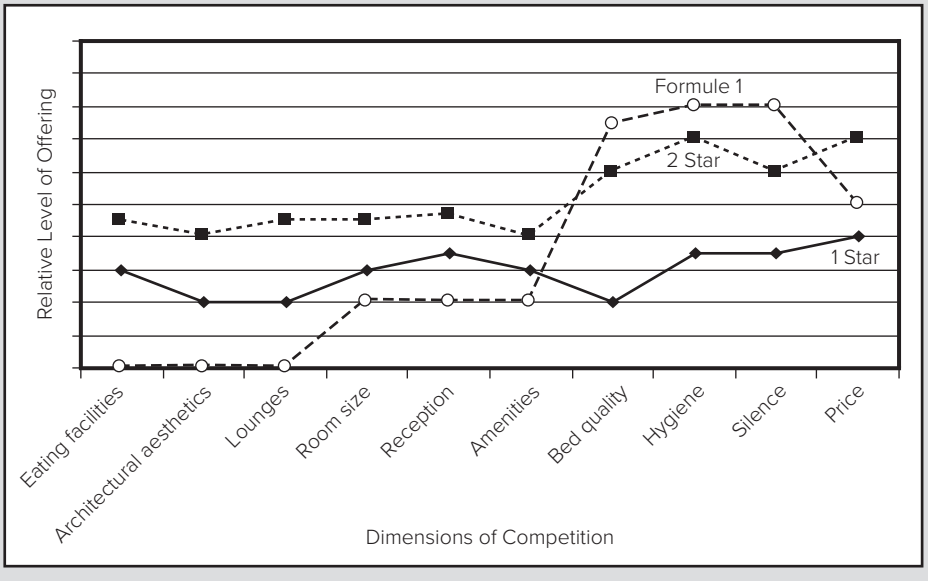
Managers can then challenge the industry's strategic logic by asking the following four questions:

1. Which of the factors that the industry takes for granted should be *eliminated?*
2. Which factors should be *reduced well below* the industry's standard?
3. Which factors should be *raised well above* the industry's standard?
4. Which factors should be *created* that the industry has never offered?

For example, returning to the hotel example Formule 1 found a successful market space by rejecting the idea that all hotel customers need eating facilities, lounges, and large rooms. Instead, some customers would prefer a hotel that skimped on these things and instead provided very quiet and clean rooms with high-quality beds at a moderate price:

Mauborgne and Chan argue that the Nintendo Wii, Cirque du Soleil, and Southwest Airlines' business model are all examples of successful "blue ocean" strategies.

[a] Adapted from W. Chan Kim and Renee Mauborgne, *Blue Ocean Strategy* (Boston: Harvard Business School Press, 2005).



the past. Focusing on the firm's existing markets results in the development of products and services that meet current market requirements, rather than future market requirements. Successful and innovative firms question existing price–performance assumptions. They lead customers by developing and introducing products that extend well beyond current market requirements and help mold the market's expectations for the future.[21]

Once the strategic intent has been articulated, the company should be able to identify the resources and capabilities required to close the gap between the strategic intent and

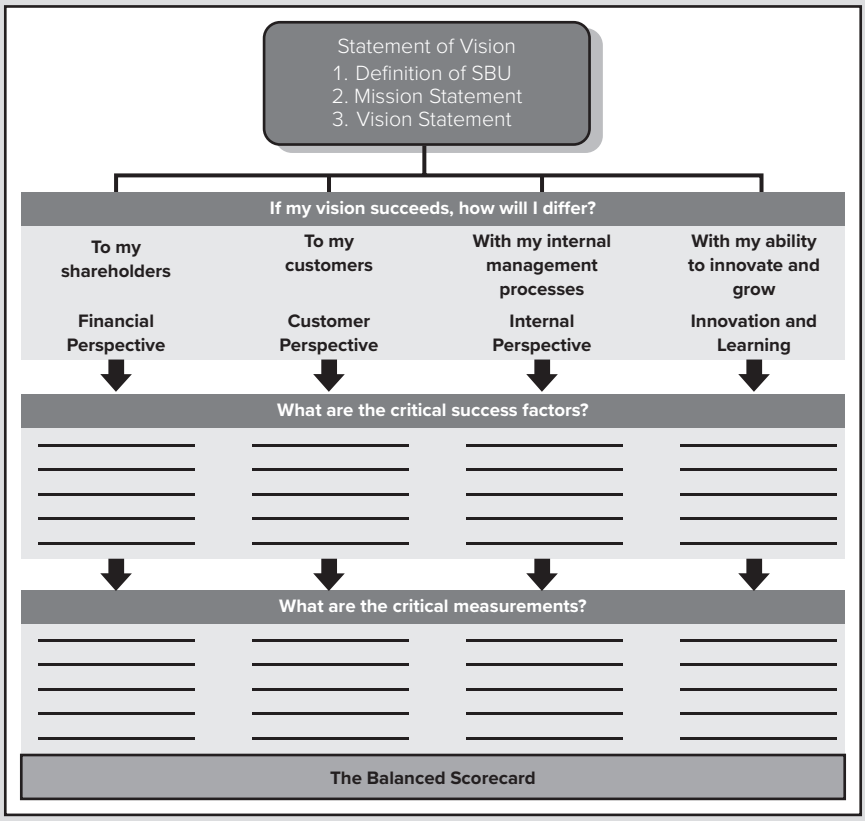
Theory in Action The Balanced Scorecard

Robert Kaplan and David Norton point out that a firm's methods of measuring performance will strongly influence whether and how the firm pursues its strategic objectives. They argue that effective performance measurement must be more than simple reliance on financial indicators; it must be a coherent and integral part of the management process. They proposed a method, the "balanced scorecard," that they argue can motivate breakthrough improvements in product, process, customer, and market development.[a] The balanced scorecard (see Figure 6.6) emphasizes four perspectives the firm should take in formulating goals that target critical success factors and in defining measures:

1. **Financial perspective.** Goals might include such things as "meet shareholder's expectations" or "double our corporate value in seven years." Measures might include return on capital, net cash flow, and earnings growth.
2. **Customer perspective.** Goals might be to "improve customer loyalty," "offer best-in-class customer service," or "increase customer satisfaction." Measures might include market share, percentage of repeat purchases, customer satisfaction surveys, and so on.
3. **Internal perspective.** Goals might include such things as "reduce internal safety incidents," "build best-in-class franchise teams," or "improve

FIGURE 6.6
The Balanced Scorecard

Source: Robert Kaplan and David Norto, "Putting the Balanced Scorecard to Work," *Harvard Business Review*, September–October 1993.



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inventory management." Measures might include the number of safety incidents per month, franchise quality ratings, stockout rates, and inventory costs.

4. **Innovation and learning perspective.** Goals might include such things as "accelerate and improve new product development" or "improve employee skills." Measures might include the percentage of sales from products developed within the past five years, average length of the new product development cycle, or employee training targets.

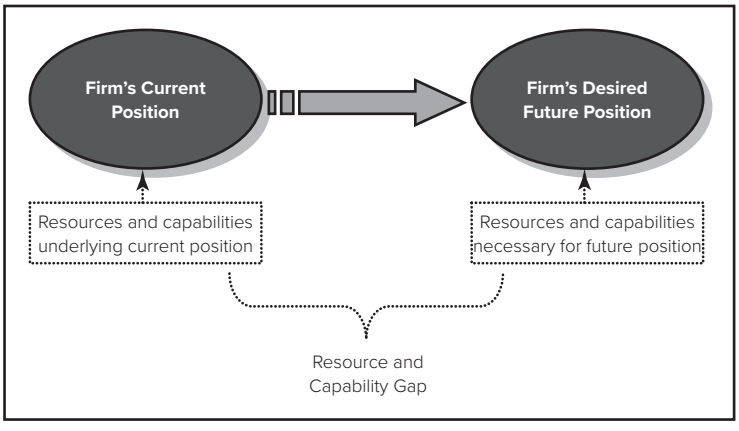
Kaplan and Norton acknowledge that the balanced scorecard model often has to be adapted to fit different markets and businesses, but many firms (including IBM, Philips Electronics, Apple, and Advanced Micro Electronics) in many different industries (including electronics, petrochemicals, and health care) are finding the balanced scorecard useful.[b] In fact, a 2002 survey by Bain & Company found that approximately 50 percent of Fortune 1000 companies in the United States and 40 percent in Europe use some version of the balanced scorecard.[c]

[a] Robert Kaplan and David Norton, "Putting the Balanced Scorecard to Work," *Harvard Business Review* (September–October 1993): 134–47; and Robert Kaplan and David Norton, "The Balanced Scorecard—Measures That Drive Performance," *Harvard Business Review* (January–February 1992): 71–80.

[b] Kaplan and Norton, "Putting the Balanced Scorecard to Work."

[c] Andra Gumbus and Bridget Lyons, "The Balanced Scorecard at Philips Electronics," *Strategic Finance* 84, no. 5 (2002): 45–9.

FIGURE 6.7
Identifying the Resource and Capability Gap



the current position (see Figure 6.7). This includes identifying any technological gap. Articulating the company's strategic intent enables the company to focus its development efforts and choose the investments necessary to develop strategic technologies and incorporate them into the company's new products.[22] Many companies are now pairing the articulation of their strategic intent with a multidimensional performance measurement system, such as the balanced scorecard, as discussed in the Theory in Action section.

Summary of Chapter

1. The first step in establishing a coherent strategy for the firm is assessing the external environment. Two commonly used models of external analysis are Porter's five-force model and stakeholder analysis.
2. Porter's five-force model entails assessing the degree of existing rivalry, threat of potential entrants, bargaining power of suppliers, bargaining power of customers,

and threat posed by substitutes. Recently Porter added a sixth force, the role of *complements.*

3. Stakeholder analysis involves identifying any entity with an interest in the firm, what it wants from the company, and what claims it can make on the company.
4. To analyze the internal environment, firms often begin by identifying strengths and weaknesses in each activity of the value chain. The firm can then identify which strengths have the potential to be a source of sustainable competitive advantage.
5. Next the firm identifies its core competencies. Core competencies are integrated combinations of abilities that distinguish the firm in the marketplace. Several core competencies may underlie each business unit, and several business units may draw upon the same core competency.
6. Sometimes core competencies can become core rigidities that limit the firm's ability to respond to a changing environment.
7. Dynamic capabilities are competencies that enable a firm to quickly reconfigure the firm's organizational structure or routines in response to change in the firm's environment or opportunities.
8. A firm's strategic intent is the articulation of an ambitious long-term (10 to 20 years out) goal or set of goals. The firm's strategic intent should build upon and stretch its existing core competencies.
9. Once the firm articulates its strategic intent, managers should identify the resources and capabilities that the firm must develop or acquire to achieve its strategic intent.
10. The balanced scorecard is a measurement system that encourages the firm to consider its goals from multiple perspectives (financial, customer, business process, and innovation and learning) and establish measures that correspond to each of those perspectives.

Discussion Questions

1. What is the difference between a strength, a competitive advantage, and a sustainable competitive advantage?
2. What makes an ability (or set of abilities) a core competency?
3. Why is it necessary to perform an external and internal analysis before the firm can identify its true core competencies?
4. Pick a company you are familiar with. Can you identify some of its core competencies?
5. How is the idea of *strategic intent* different from models of strategy that emphasize achieving a fit between the firm's strategies and its current strengths, weaknesses, opportunities, and threats (SWOT)?
6. Can a strategic intent be too ambitious?

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Chapter **Seven**

Choosing Innovation Projects

Where Should We Focus Our Innovation Efforts? An Exercise.[a]

Technological change can make the world feel very uncertain—how can we know what customers will want, or competitors will do, in the future? The future of a technology, however, is not as unknowable as it might seem. If you can get the big picture of the dimensions along which a technology is improving, and where the big payoffs are still yet to be reaped, you can gain insight into where the next big breakthroughs are likely to, or should, be. This can help managers understand where to focus their R&D efforts, and to anticipate the moves of competitors.

The story of Sony's SACD provides a great example. In the mid-1990s, the compact disc market was approaching saturation and both the consumer electronics and recording industries were eager to introduce a next-generation audio format that would usher in a new era of growth. In 1996, Hitachi, JVC, Mitsubishi, Toshiba, Universal Music Group, Time Warner, and others formed a consortium to back a new standard, DVD audio, that offered superior fidelity and surround sound. The plan was twofold: to drive consumers to upgrade their players and libraries and to do an end-run around Sony and Philips which owned the compact disc standard and extracted a licensing fee for every CD and player sold.

Sony and Philips, however, were not going to go down without a fight. They counterattacked with their own new format, Super Audio CD, an extension of the old one which would have allowed them to control the royalties for the new discs and players. The industry gave a collective groan; manufacturers, distributors, and consumers all stood to lose big if they placed bets on the wrong format. Nonetheless, Sony launched the first super-audio players in late 1999; DVD audio players began to hit the market in mid-2000 and prices for both rapidly fell and converged. A costly format war seemed inevitable.

You may be scratching your head at this point, wondering why you'd never heard about this format war. What happened? MP3 happened. While the consumer

electronics giants were pursuing new heights in audio fidelity, an algorithm that slightly *depressed* fidelity in exchange for reduced audio file size was taking off. Soon after the file-sharing platform Napster launched in 1999, consumers were downloading free music files by the millions and Napster-like services were sprouting up like weeds.

It's easy to think that Sony, Philips, and the DVD Audio consortium just got unlucky. After all, who could have predicted the disruptive arrival of MP3? How could the consumer electronics giants have known that a format trajectory of ever-increasing fidelity would be overtaken by a technology with *less* fidelity? Actually, they all should have seen that the next big breakthrough success was far more likely to be about portability and selection than fidelity.

Here is an exercise that can help you get the "big picture" of technology evolution in an area, and to prioritize innovation investments.

Step One: Identify the Dimensions

Technologies typically progress along several dimensions at once. For example, computers became faster and smaller in tandem; speed was one dimension, size another. Developments in any dimension come with specific costs and benefits and have measurable and changing utility to customers. Identifying the key dimensions that a technology has progressed along is the first step in predicting its future.

One of the best ways to identify these dimensions is to trace the technology's path to date, and for each major change in the technology's form, identify the dimensions that were affected. The further back in time you can go, to the most primitive way of fulfilling the need the technology meets, the easier it will be to see the high-level dimensions along which the technology has changed.

For example, prior to the phonograph, people could hear music or a speech only when and where it was performed. The first phonographs introduced in the 1800s *desynchronize* the time and place of a performance so that it could be heard at any time, anywhere. That was a huge accomplishment. Those phonographs, however, were cumbersome. The sound quality was "tinny" and music selection was extremely limited. When Emile Berliner introduced flat disc-shaped records, it dramatically increased the sound *fidelity,* while dramatically lowering the *cost,* which in turn led to a huge increase in the *selection* of music available. Later, in the 1960s eight-track tape cartridges would bring *portability* to recorded music. Cassette tapes rose to dominance in the 1970s further enhancing *portability,* but also offering *customizability* for the first time—the ability to record your own music, speeches, and more. Then in 1982, Sony and Philips introduced the compact disc standard which offered greater *fidelity* than cassette tapes and rapidly became the dominant format. Tracing this history thus reveals six key dimensions that have driven the evolution of audio format technology: *De synchronization, cost, fidelity, selection, portability, and customizability.*

Now repeat this process in the area you are interested, starting far back in the past and following the path of the technology forward in time. Look for each major inflection point. Try to identify the three to five most important high-level dimensions along which the technology has evolved. Getting to a high level of

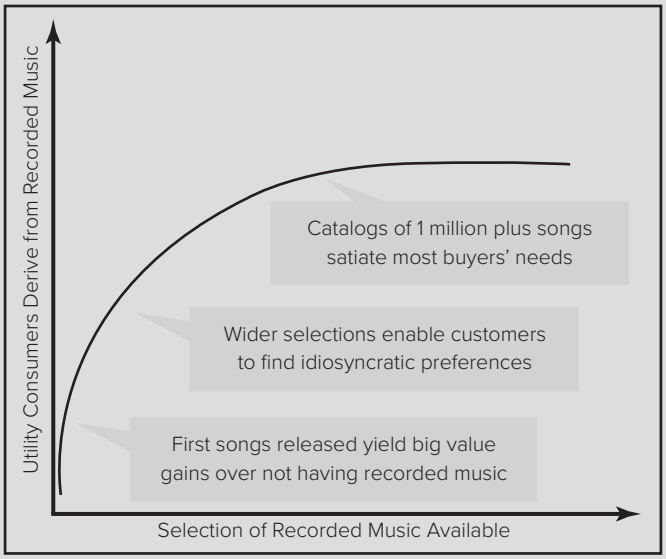


FIGURE 1
A Parabolic Utility Curve for the Selection of Music Available

a dimension is important because it helps you to see the big picture rather than being caught up in the details.

Step Two: Where Are We on the Utility Curve for Each Dimension?

For each dimension, we now want to determine the shape of the utility curve—the plot of the value customers derive from a technology according to its performance on a given dimension—and where we currently are on the curve. This will help reveal where the most opportunity for improvement lies.

For example, the history of audio formats suggests that the *selection* of music available has a concave parabolic utility curve: utility increases with increasing selection, but at a decreasing rate, and not indefinitely (see Figure 1). When there's little music to choose from, even a small increase in *selection* significantly enhances utility. When the first phonographs appeared, for example, there were few recordings to play on them. As more recordings became available, customers eagerly bought them, and the appeal of owning a player grew dramatically. Over the ensuing decades, selection grew exponentially and the utility curve ultimately began to flatten; people still valued new releases, but each new recording added less additional value. Today digital music services like iTunes, Amazon Prime Music, and Spotify offer tens of millions of songs. With this virtually unlimited selection, most customers' appetites are sated—and we are probably approaching the top of the curve.

Now let's consider the *fidelity* dimension, the primary focus of Super Audio CD and DVD audio. Fidelity also likely has a concave parabolic utility curve. The first phonographs had awful fidelity; music sounded thin and tinny, though it was still a remarkable benefit to be able to hear any recorded music at all. The early

improvements in fidelity that records offered made a big difference in people's enjoyment of music. Then along came compact discs which offered even more fidelity, though many people felt that vinyl records were "good enough" and some even preferred their "warmth." The fidelity curve was already leveling out when Sony, Philips, and the DVD Audio consortium introduced their new formats in the early 2000s. Super Audio CD and DVD Audio offered significantly higher fidelity than the compact disc. For example, whereas CDs have a frequency range up to about 20 thousand cycles per second (kHz), the new formats offered ranges that reached 100 kHz. That's an impressive high end—but as human hearing peaks out at about 20 kHz, only the family dog was likely to appreciate it. In 2007, the Audio Engineering Society released results of a year-long trial assessing how well subjects (including professional recording engineers) could detect the difference between Super Audio and regular CDs. Subjects correctly identified the Super Audio CD format only half the time, no better than if they'd been simply guessing!

Had the companies analyzed the utility curve for *fidelity* they could have seen that there was little payoff for investing in this dimension. Meanwhile, even a cursory look at the *portability, selection,* and *cost* curves would have revealed the opportunity there, and MP3 offered big payoffs on all three.

Step Three: Where Should We Invest Our Money and Effort?

Using a matrix like the one below can help us assess which technology-development investments are likely to yield the biggest bang for the buck (see Figure 2). First, list the performance dimensions you've identified as most important to customers. Then, score each dimension on a scale of 1 to 5 in three areas: importance to customers (1 means "not important" and 5 means "very important"); room for improvement (i.e., how far we are from the utility curve flattening out, where 1 means "minor opportunity for improvement" and 5 means "large opportunity for improvement"); and ease of improvement (1 means "very difficult" and 5 means "very easy").

For example, the scores created by a company that makes glucose-monitoring devices are shown in the matrix (Figure 2). The feature most important to customers is *reliability*—reliable measures can mean life or death for a patient. However, existing devices (mostly finger-prick devices that test a drop of blood with a portable reader) are already very reliable, and thus scored low

FIGURE 2
Ranking Areas of Focus

Rank	Dimension	Importance to Buyers (1–5 scale)	Room for Improvement (1–5 scale)	Ease of Improvement (1–5 scale)	Total Score
	Cost	4	2	2	8
	Comfort	4	4	3	11
	Reliability	5	1	1	7
	Ease of use	3	2	3	8

on the "room for improvement" measure. They are also fairly *easy to use,* and reasonably *low cost.* However, they are uncomfortable. *Comfort* is a dimension that is ranked highly by users (though not as high as *reliability*), yet is much further from the optimum. Both *comfort* and *ease of use* are modestly difficult to improve (scoring 3s in this example), yet because *comfort* is both more important to customers than *ease of use,* and further from the optimum, *comfort* ends up with a significantly higher total score. The team that provided the scores for this example ended up proposing that the company develop a patch worn on the skin that would detect glucose levels based on sweat that would be sent via Bluetooth to the user's smartphone.

Notably, we can adjust the scale of the scores to reflect the organization's objectives (e.g., a cash-strapped firm might weigh "ease of improvement" higher, thus we could change the scale to 1–10), and we can repeat this analysis for different market segments that have different ratings for the importance of a dimension or its room for improvement. For example, we could divide the glucose-monitoring device market into a market for children's devices versus adults' devices. *Comfort* and *ease of use* are particularly important for glucose monitors for children. We might thus score *comfort* and *ease of use* as 5 for "importance to customer" in the children's device market, indicating that in this market, these dimensions are often more important than cost.

Shifting the Organization's Focal Point

This exercise can help managers broaden their perspective on their industry, and shift their focus from "this is what we do" to "this is where our market is (or should be) heading." It can also help overcome the bias and inertia that tend to lock an organization's focus on technology dimensions that are less important to consumers than they once were. To be truly innovative, it is not enough for organizations to merely improve what they are doing; they need to be able to look into the future and identify *what could be done* in the industry that is better—much better—than what the industry is doing today.

Discussion Questions

1. Why did the consumer electronic giants invest so much in developing audio technology that exceeded the hearing capacity of humans?
2. Can you think of other companies or industries that are currently investing in technology dimensions where the utility payoff of improving the technology has flattened?
3. Can you come up with an example of a firm that would likely weight "importance to customers" much higher than "ease of improvement"? Can you come up with an example of a firm that would likely weight "ease of improvement" much higher than "importance to customers"?

[a] Adapted from Melissa A. Schilling, "What's Your Best Innovation Bet?" *Harvard Business Review* (July–August 2017): 86–93.

OVERVIEW

Developing innovative new products and services is expensive and time-consuming. It is also extremely risky—most studies have indicated that the vast majority of development projects fail. Firms have to make difficult choices about which projects are worth the investment, and then they have to make sure those projects are pursued with a rigorous and well-thought-out development process. In this chapter, we will explore the various methods used to evaluate and choose innovation projects. The methods range from informal to highly structured, and from entirely qualitative to strictly quantitative. We will start by considering the role of **capital rationing** in the R&D investment decision, and then we will cover various methods used to evaluate projects including strictly quantitative methods, qualitative methods, and approaches that combine quantitative and qualitative techniques.

capital rationing
The allocation of a finite quantity of resources over different possible uses.

THE DEVELOPMENT BUDGET

While many project valuation methods seem to assume that all valuable projects will be funded, most firms face serious constraints in capital and other resources, forcing them to choose between multiple valuable projects (or obtain external financing as discussed in the Theory in Action section on Financing New Technology Ventures). Many firms use a form of *capital rationing* in formulating their new product development plans. Under capital rationing, the firm sets a fixed research and development budget (often some percentage of the previous year's sales), and then uses a rank ordering of possible projects to determine which will be funded. Firms might establish this budget on the basis of industry benchmarks or historical benchmarks of the firm's own performance. To provide a sense of what firms in different industries spend on R&D, Figure 7.1 shows the 10 industries with the highest **R&D intensity** (R&D expenditures as a percentage of sales), based on North American publicly held firms in 2020. Some industries (notably drugs, software, electronics and special industry machinery) spend considerably more of their revenues on R&D than other industries, on average.

R&D intensity
The ratio of R&D expenditures to sales.

There is also considerable variation within each of the industries in the amount that individual firms spend. For example, as shown in Figure 7.2, Johnson & Johnson's R&D intensity was significantly higher than the average for drug producers (46 percent versus 25 percent), whereas Microsoft's is somewhat lower than the industry average (10 percent versus 14 percent). Figure 7.2 also reveals the impact of firm size on R&D budgets: For example, whereas the absolute amount spent on R&D at Samsung surpasses the R&D spent at most other firms by a large amount, Samsung's R&D intensity is relatively low due to its very large sales base.

The rank ordering used in capital rationing may be established by any number of methods, including quantitative methods, such as discounted cash flow analysis or options analysis, or qualitative methods, such as screening questions and portfolio mapping, or a combination of multiple methods. Knowing the requirements, strengths, and weaknesses of each method helps managers make sound decisions about which valuation techniques to employ.

FIGURE 7.1
Top 10 Industries (Three Digit SIC) by R&D Intensity, 2020

Based on Compustat data; only industries with 20 or more firms were included. Sales and R&D were aggregated up to the industry level prior to calculating R&D intensity to minimize the effect of large firm-level outliers.

Rank	Industry Description	Number of U.S. Publicly Held Firms	Industry R&D Spend ($millions)	Industry Revenues ($millions)	Industry R&D Intensity (R&D/ Sales)
1	Pharmaceuticals	429	$166,990	$ 662,138	25%
2	Software and Computer Services	404	$153,091	$1,101,614	14%
3	Electronics	119	$ 56,809	$ 448,089	13%
4	Special Industry Machinery	23	$ 7037	$ 56,523	12%
5	Communications Equipment	47	$ 31,756	$ 363,354	9%
6	Medical Devices	128	$ 16,954	$ 200,426	8%
7	Measuring and Controlling Instruments	62	$ 10,526	$ 146,474	7%
8	Computer Equipment	46	$ 27,807	$ 410,345	7%
9	Automotive Manufacturing	45	$ 53,141	$1,172,034	5%
10	General Industry Machinery	26	$ 1830	$ 71,502	3%

FIGURE 7.2
Top 10 Global R&D Spenders, 2020

Source: Data from Annual Reports

Company	Sales ($billions)	R&D ($billions)	R&D Intensity (R&D/Sales)
Samsung	$197.7	$14.9	8%
Alphabet	$182.5	$14.8	8%
Volkswagen	$254.6	$14.5	6%
Microsoft	$ 143	$14.6	10%
Huawei	$136.2	$12.5	9%
Intel	$ 77.9	$12.1	16%
Apple	$274.5	$10.7	4%
Roche	$ 64.7	$ 9.8	15%
Johnson & Johnson	$ 21.1	$ 9.7	46%
Daimler	$ 190	$ 9.6	5%

While large firms can fund innovation projects internally, new technology start-ups must often obtain external financing. This can sometimes be daunting. Because technology start-ups often have both an unproven technology and an unproven business concept (and sometimes an unproven management team), they typically face a much higher cost of capital than larger competitors, and their options for obtaining capital can be very limited. In the first few stages of start-up and growth, entrepreneurs may have to turn to friends, family, and personal debt. Start-ups might also be able to obtain some initial funding through government agencies. If the idea and the management team seem promising enough, the entrepreneur can tap "angel investors" and venture capitalists as sources of both funds and mentoring.

FAMILY, FRIENDS, AND CREDIT CARDS

When a new venture is starting out, often the technology and/or management is unproven, making the venture appear very risky. In this stage, entrepreneurs must often rely on friends and family members who are willing to provide initial funding either in the form of a loan or an exchange for equity in the company. Alternatively, the entrepreneur may try to obtain debt financing from a local bank. A very large number of start-ups are actually funded with credit cards, resulting in a very high rate of interest!

GOVERNMENT GRANTS AND LOANS

Some new ventures obtain start-up funds from government sources. In the United States, the Small Business Administration (SBA) is designed to foster entrepreneurship and innovation by administering grants, loans, and venture capital programs from many different federal agencies including the Department of Commerce, Department of Agriculture, Department of Energy, NASA, and others. Similarly, in the United Kingdom, the Enterprise Fund administers a series of programs designed to fund small- to medium-size technology firms, and in Germany there are more than 800 federal and state government programs established to finance new firms.[a]

ANGEL INVESTORS

Angel investors are private investors who fund projects without utilizing a venture capital limited partnership structure. They are often wealthy individuals who have been very successful in business, and who enjoy the thrill of vicarious entrepreneurship afforded by investing in—and sometimes mentoring—start-up firms. Angels typically fund projects that are $1 million or less. While angel investors lose money in a significant share of their investments, those investments that pay off can earn very high returns. Angels are usually not listed in public directories, but are identified through professional networks (e.g., through one's former colleagues, professors, or lawyers). A large number of start-ups obtain "seed-stage" (before there is a real product or company organized) financing from angel investors. While it is difficult to get data on angel investing because most of the transactions are not publicly reported, estimates from the Center for Venture Research indicate that angel investors funded 64,380 entrepreneurial ventures in 2016 for a total of $21.3 billion (an average of $330,000 per deal).

VENTURE CAPITAL

For projects that require more than $1 million, entrepreneurs often turn to venture capital, either from independent venture capital firms or from corporate venture capital sources.

Independent venture capital firms manage a pool of funds that they invest in projects they believe to have rapid growth potential. Many venture capital firms specialize in particular industries, making them better able to evaluate the potential of start-ups in that industry. They also prefer to make deals with firms within driving distance so that they can have frequent face-to-face contact.[b]

The venture capital funds are likely to be provided in a complex debt-equity hybrid contract that essentially looks more like equity if the firm performs well, or debt if the firm performs poorly.[c] If and when the business is successful, the venture capitalist can cash out of the investment by negotiating an initial public offering or a buyout of the company by another firm. Venture capitalists are very selective and reject the vast majority of proposals considered. However, for those projects that are funded, the support of the venture capitalist provides a number of valuable benefits including credibility among other investors (and thus better access to capital) and mentoring. While some venture capitalists specialize in providing seed-stage funding, venture capitalists

continued

concluded

are more likely to provide funding during early stages—after the company has been organized and the product has shown early signs of success, but before it is able to sustain its development activities and growth through its own revenues. According to estimates by Crunchbase, in 2017 venture capitalists invested about $213.6 billion on 22,700 deals globally. The most prolific venture capital firm, Sequoia Capital, lead 70 venture rounds in 2017.

STRATEGIC INVESTORS

Strategic investment, such as corporate venture capital, is investment made by a firm in the same or related industry that has a strategic interest in the technology development and may be able to contribute knowledge or other assets as a collaborator, supplier, or customer. For example, the corporate venture capital arm of a firm may wish to take a minority stake in another firm to gain access to cutting-edge technology that they may wish to develop further should it prove technically and commercially promising. Such firms may establish an internal venturing group that is closely tied to the firm's own development operations, or they may create a dedicated external fund that has more independence from the firm's own operations.[d] The benefit of the former structure is that the firm should be better able to use its own expertise and resources to help the new venture succeed. However, under this structure, the entrepreneur may have concerns about the larger firm expropriating the entrepreneur's proprietary technology. Under the latter structure, the independence of the external venture fund provides some reassurance that the entrepreneur's technology will not be stolen, but it also limits the ability of the entrepreneur to leverage any of the larger firm's nonfinancial resources.[e] According to CB Insights, there were 1791 corporate venture capital deals globally in 2017, totaling $31.2 billion (averaging over $17 million per deal). According to CB Insights, the most active corporate venture capital programs in 2017 were GV (formerly Google Ventures), Intel Capital, Salesforce Ventures, Qualcomm Ventures, GE Ventures, Legend Capital, Microsoft Ventures, K Cube Ventures, Fosun RZ Capital, and Samsung Ventures.

[a] Bronwyn H. Hall, "The Financing of Research and Development," *Oxford Review of Economic Policy* 18 (2002): 35–51.

[b] Paul Gompers and Josh Lerner, *The Venture Capital Cycle* (Boston: MIT Press, 2004); Emily Cox Pahnke, Riitta Katila, and Kathleen M. Eisenhardt, "Who Takes You to the Dance? How Partners' Institutional Logics Influence Innovation in Young Firms," *Administrative Science Quarterly* 60 (2015): 596–633.

[c] Hall, "The Financing of Research and Development."

[d] Paul A. Gompers, "Corporations and the Financing of Innovation: The Corporate Venturing Experience," *Economic Review—Federal Reserve Bank of Atlanta* 87, no. 4 (2002): 1–18.

[e] Gary Dushnitsky, "Limitations to External Knowledge Acquisition: The Paradox of Corporate Venture Capital," doctoral dissertation, New York University, 2003.

QUANTITATIVE METHODS FOR CHOOSING PROJECTS

Quantitative methods of analyzing new projects usually entail converting projects into some estimate of future cash returns from a project. Quantitative methods enable managers to use rigorous mathematical and statistical comparisons of projects, though the quality of the comparison is ultimately a function of the quality of the original estimates. The accuracy of such estimates can be questionable—particularly in highly uncertain or rapidly changing environments. The most commonly used quantitative methods include discounted cash flow methods and real options.

Discounted Cash Flow Methods

Many firms use some form of discounted cash flow analysis to evaluate projects. Discounted cash flows are quantitative methods for assessing whether the anticipated future benefits are large enough to justify expenditure, given the risks. Discounted cash flow methods take into account the payback period, risk, and time value of money. The two most commonly used forms of discounted cash flow analysis for

net present value (NPV) The discounted cash inflows of a project minus the discounted cash outflows.

internal rate of return (IRR) The rate of return yielded by a project, normally calculated as the discount rate that makes the net present value of an investment equal zero.

evaluating investment decisions are **net present value (NPV)** and **internal rate of return (IRR)**. Both methods rely on the same basic discounted cash flow mechanics, but they look at the problem from different angles. NPV asks, "Given a particular level of expenditure, particular level(s) and rate of cash inflows, and a discount rate, what is this project worth today?" IRR asks instead, "Given a particular level of expenditure and particular level(s) and rate of cash inflows, what rate of return does this project yield?" For either method, managers must use estimates of the size and timing of expenditures and cash inflows. Both methods enable the decision maker to incorporate some basic measure of risk. For example, riskier projects may be examined by using a higher discount factor in NPV analysis. Managers also often calculate discounted cash flow measures using best-case and worst-case cash flow estimates.

Net Present Value (NPV)

To calculate the NPV of a project, managers first estimate the costs of the project and the cash flows the project will yield (often under a number of different "what if" scenarios). Costs and cash flows that occur in the future must be discounted back to the current period to account for risk and the time value of money. The present value of cash inflows can then be compared to the present value of cash outflows:

$$\text{NPV} = \text{Present value of cash inflow} - \text{Present value of cash outflows}$$

If this value is greater than 0, then the project generates wealth, given the assumptions made in calculating its costs and cash inflows.

To find the present value of cash inflow and outflows, each cash flow must be discounted back to the current period using a discount rate (see Figure 7.3). If there is a single expenditure at the beginning of the project (year 0), the original expenditure can be compared directly to the present value of the future expected cash flows. In the example in Figure 7.3, the present value of the future cash flows (given a discount rate of 6 percent) is $3465.11. Thus, if the initial cost of the project were less than $3465.11, the net present value of the project is positive. If there are cash outflows for multiple

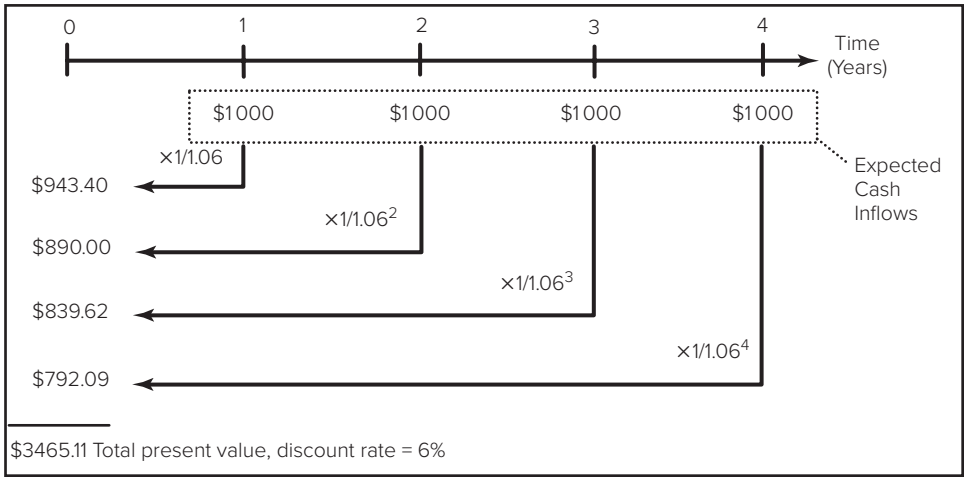


FIGURE 7.3
Example of Present Value of Future Cash Flows

periods (as is common with most development projects), those cash outflows would have to be discounted back to the current period.

If the cash inflows from the development project were expected to be the same each year (as they were in Figure 7.3), we can use the formula for calculating the present value of an annuity instead of discounting each of the cash inflows individually. This is particularly useful when cash inflows are expected for many years. The present value of C dollars per period, for t periods, with discount rate r is given by the following formula:

$$\text{Annuity present value} = C \times \frac{1 - \{1/(1+r)^t\}}{r}$$

This amount can then be compared to the initial investment. If the cash flows are expected in perpetuity (forever), then a simpler formula can be used:

$$\text{Perpetuity present value} = C \times 1/r$$

discounted payback period
The time required to break even on a project using discounted cash flows.

The present value of the costs and future cash flows can also be used to calculate the **discounted payback period** (i.e., the time required to break even on the project using discounted cash flows). Suppose for the example above, the initial investment required was $2000. Using the discounted cash inflows, the cumulative discounted cash flows for each year are:

Year	Cash Flow
1	$ 934.40
2	1833.40
3	2673.02
4	3465.11

Thus, the investment will be paid back sometime between the end of year 2 and the end of year 3. The accumulated discounted cash flows by the end of year 2 are $1833.40, so we need to recover $166.60 in year 3. Since the discounted cash flow expected for year 3 is $839.62, we will have to wait $166.60/$839.61 ≈ 0.20 of a year. Thus, the payback period is just over two years and two months.

Internal Rate of Return (IRR)

The internal rate of return of a project is the discount rate that makes the net present value of the investment zero. Managers can compare this rate of return to their required return to decide if the investment should be made. Calculating the IRR of a project typically must be done by trial and error, substituting progressively higher interest rates into the NPV equation until the NPV is driven down to zero. Calculators and computers can perform this trial and error. This measure should be used cautiously, however; if cash flows arrive in varying amounts per period, there can be multiple rates of return, and typical calculators or computer programs will often simply report the first IRR that is found.

Both net present value and internal rate of return techniques provide concrete financial estimates that facilitate strategic planning and trade-off decisions. They explicitly consider the timing of investment and cash flows, and the time value of money and risk. They can make the returns of the project seem unambiguous, and managers may

find them very reassuring. However, this minimization of ambiguity may be deceptive; discounted cash flow estimates are only as accurate as the original estimates of the profits from the technology, and in many situations it is extremely difficult to anticipate the returns of the technology. Furthermore, such methods discriminate heavily against projects that are long term or risky, and the methods may fail to capture the strategic importance of the investment decision. Technology development projects play a crucial role in building and leveraging firm capabilities, and creating options for the future. Investments in new core technologies are investments in the organization's capabilities and learning, and they create opportunities for the firm that might otherwise be unavailable.[1] Thus, standard discounted cash flow analysis has the potential to severely undervalue a development project's contribution to the firm. For example, Intel's investment in DRAM technology might have been considered a total loss by NPV methods (Intel exited the DRAM business after Japanese competitors drove the price of DRAM to levels Intel could not match). However, the investment in DRAM technology laid the foundation for Intel's ability to develop microprocessors—and this business has proved to be enormously profitable for Intel. To better incorporate strategic implications in the new product development investment decision, some managers and scholars have recently begun promoting the idea of treating new product development decisions as real options, as described below.

Real Options

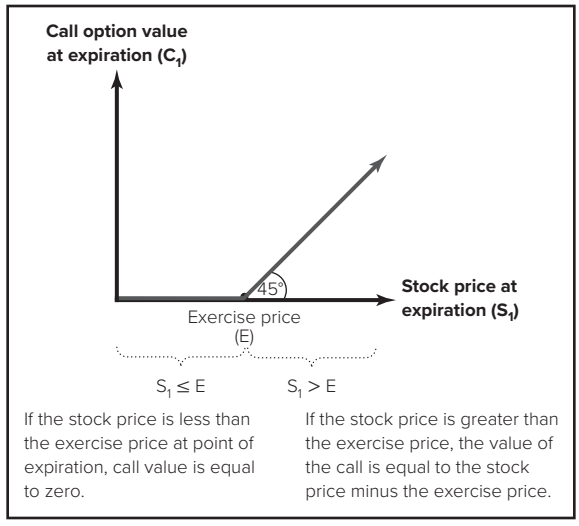
When a firm develops new core technologies, it is simultaneously investing in its own learning and in the development of new capabilities. Thus, development projects can create valuable future opportunities for the firm that would otherwise be unavailable.[2] Even development projects that appear unsuccessful (as Intel's DRAM discussed above) may prove to be very valuable when they are considered from the perspective of the options they create for the future of the firm. Some managers and scholars have begun arguing that new product development decisions should be evaluated as **real options**.

real options The application of stock option valuation methods to investments in nonfinancial assets.

To understand real options, it is first useful to consider the financial model upon which they are based—stock options. A call option on a stock enables an investor to purchase the right to buy the stock at a specified price (the *exercise price*) in the future. If, in the future, the stock is worth more than the exercise price, the holder of the option will typically exercise the option by buying the stock. If the stock is worth more than the exercise price plus the price paid for the original option, the option holder makes money on the deal. If the stock is worth less than the exercise price, the option holder will typically choose not to exercise the option, allowing it to expire. In this case, the option holder loses the amount of money paid for the initial option. If, at the time the option is exercised, the stock is worth more than the exercise price but not more than the exercise price plus the amount paid for the original option, the stockholder will typically exercise the option. Even though the stockholder loses money on the deal (some portion of the price paid for the original option), he or she loses less than if he or she allowed the option to expire (the entire price paid for the original option).

In *real options,* the assets underlying the value of the option are nonfinancial resources.[3] An investor who makes an initial investment in basic R&D or in breakthrough

FIGURE 7.4
The Value of a Call Option at Expiration



technologies is, it is argued, buying a real call option to implement that technology later should it prove to be valuable:

- The cost of the R&D program can be considered the price of a call option.
- The cost of future investment required to capitalize on the R&D program (such as the cost of commercializing a new technology that is developed) can be considered the exercise price.
- The returns to the R&D investment are analogous to the value of a stock purchased with a call option.[4]

As shown in Figure 7.4, the value of a call stock option is zero as long as the price of the stock remains less than the exercise price. If the value of the stock rises above the exercise price, however, the value of the call rises with the value of the stock, dollar for dollar (thus the value of the call rises at a 45-degree angle).[5]

Options are valuable when there is uncertainty, and because technology trajectories are uncertain, an options approach may be useful. Though there has not yet been much empirical work in the area, several authors have developed methodologies and applications of options analysis to valuing technology development investments.[6] Also, some evidence shows that an options approach results in better technology investment decisions than a cash flow analysis approach.[7]

Other authors, however, warn against liberal application of the approach, pointing out that technology investment scenarios often do not conform to the same capital market assumptions upon which the approach is based.[8] For instance, implicit in the value of options is the assumption that one can acquire or retain the option for a small price, and then wait for a signal to determine if the option should be exercised.[9] While this assumption might hold true for an outside firm investing venture capital in another firm's innovation effort, it would be rare for this assumption to hold for a company investing in its own development efforts. In the case of a firm undertaking solo new

product development, it may not be possible to secure this option at a small price; it may require full investment in the technology before a firm can determine if the technology will be successful.[10] Furthermore, while the value of a stock is independent of the call holder's behavior (i.e., the call holder can simply wait and observe whether the value of the stock rises or falls), the value of an R&D investment is not independent of the investor's behavior. A firm's degree of investment, its development capabilities, its complementary assets, and its strategies can all significantly influence the future returns of the development project.[11] Therefore, rather than simply waiting and observing the value of the investment, the investor is an active driver of the value of the investment.

DISADVANTAGES OF QUANTITATIVE METHODS

Quantitative methods for analyzing potential innovation projects can provide concrete financial estimates that facilitate strategic planning and trade-off decisions. They can explicitly consider the timing of investment and cash flows and the time value of money and risk. They can make the returns of the project seem unambiguous, and managers may find them very reassuring. However, this minimization of ambiguity may be deceptive; discounted cash flow estimates are only as accurate as the original estimates of the profits from the technology, and in many situations, it is extremely difficult to anticipate the returns of the technology. As noted by Professor Freek Vermeulen, author of *Business Exposed*, one of the most common mistakes managers make in their innovation strategy is to insist on "seeing the numbers"—for truly innovative products, it is impossible to reliably produce any numbers. It is very difficult to compute the size of a market that does not yet exist.[12] Furthermore, such methods discriminate heavily against projects that are long term or risky, and the methods may fail to capture the strategic importance of the investment decision. Technology development projects play a crucial role in building and leveraging firm capabilities and creating options for the future. Investments in new core technologies are investments in the organization's capabilities and learning, and they create opportunities for the firm that might otherwise be unavailable.

QUALITATIVE METHODS FOR CHOOSING PROJECTS

Most new product development projects require the evaluation of a significant amount of qualitative information. Many factors in the choice of development projects are extremely difficult to quantify, or quantification could lead to misleading results. Almost all firms utilize some form of qualitative assessment of potential projects, ranging from informal discussions to highly structured approaches.

Screening Questions

As a starting point, a management team is likely to discuss the potential costs and benefits of a project, and the team may create a list of screening questions that are used to structure this discussion. These questions might be organized into categories such as

the role of the customer, the role of the firm's capabilities, and the project's timing and cost.[13] Some examples are provided below:

Role of Customer

Market

- Who are the most likely customers of the new product?
- How big is this market? Are there other likely markets for the product?
- What type of marketing will be required to create customer awareness?

Use

- How will customers use the product?
- What new benefits will the product provide the customer?
- What other products are customers likely to consider as substitutes for this product?

Compatibility and Ease of Use

- Will the product be compatible with the customer's existing complements?
- Will the product require significant new learning on the part of the customer?
- How will the customer perceive the product's ease of use?
- Will the product require the customer to bear other costs?

Distribution and Pricing

- Where will the customer buy the product?
- Will the product require installation or assembly?
- How much are customers likely to be willing to pay for the product?

Role of Capabilities

Existing Capabilities

- Does the new project leverage the firm's core competencies or sources of sustainable competitive advantage?
- Will the project render some of the firm's existing competencies obsolete or cannibalize existing products? If so, does the firm have a transition strategy to handle possible cash flow implications?
- Does the firm have the necessary manufacturing capabilities, and if not, will those capabilities be developed in-house or acquired externally (e.g., outsourcing)?
- Will the firm need to hire employees with new skills?

Competitors' Capabilities

- Do one or more competitors have better capabilities for developing this project?
- If the company does not develop this technology, are competitors likely to?
- Will the company be able to protect its intellectual property through patents, copyright, trademarks, or trade secrets?
- Should the firm seek to form a collaboration with a potential competitor?

Future Capabilities

- Will the project help the firm build new capabilities that will allow it to achieve its strategic intent?
- What other products/markets will the new capabilities enable the firm to develop?
- Is this project a platform that will lead to a family of new products?

Project Timing and Cost

Timing

- How long will the project take to complete?
- Is the firm likely to be first to market? Is pioneering the technology a desirable strategy?
- Is the market ready for the product? (For example: Are enabling and complementary technologies well developed? Will customers perceive the value of the technology?)
- If the firm misses its target deadlines, what impact will this have on the potential value of the project?
- Are there already appropriate suppliers and distribution channels?

Cost Factors

- How much will the project cost? What is the potential variability in these costs?
- What will the manufacturing costs be? At what rate are these costs expected to decline with experience?
- Will the firm need to bear other costs related to customer adoption (e.g., production of complements, installation, technical support, etc.)?

After creating a list of questions, managers can use the questions to structure debate about a project, or they can create a scoring mechanism (such as a scaled response to each question such as "Project fits closely with existing competencies" to "Project fits poorly with existing competencies") that can then be weighted according to importance and used in subsequent analysis.

While screening questions such as the ones above do not always provide concrete answers about whether or not to fund a project, they enable a firm to consider a wider range of issues that may be important in the firm's development decisions. Consider Boeing's development of the Sonic Cruiser, a supersonic jet that was designed by Boeing, but never made it off the drawing board. Boeing continued designing the aircraft even after it became clear that the jet would not be profitable because Boeing considered the project necessary for preserving the company's development capabilities.

As noted by Walt Gillette, Boeing's development program manager, "If the company doesn't create a new airplane every 12 to 15 years, the needed skills and experience will be gone. Too many of the people who created the last new airplane will have retired or moved on to other companies, and their skills and experience will not have been passed on to the next generation of Boeing employees."[14] Thus, Boeing's development of the Sonic Cruiser is expected to be valuable to the firm even if the only return from the project is the enhancement of the firm's development capabilities. Such value would be difficult to assess via quantitative methods, but is revealed clearly by qualitative analysis.

The R&D Portfolio

Many companies find it valuable to map their R&D portfolio according to degree of change and timing of cash flows. Managers can use this map to compare their desired balance of projects with their actual balance of projects.[15] It can also help them to identify capacity constraints and better allocate resources.[16] Companies may use an R&D portfolio map (similar to that depicted in Figure 7.5) to aid this process. Four types of development projects commonly appear on this map–advanced R&D, breakthrough, platform, and derivative projects. Over time, a particular technology may migrate through these different types of projects. Advanced R&D projects are the precursor to commercial development projects and are necessary to develop cutting-edge strategic technologies. Breakthrough projects involve development of products that incorporate revolutionary new product and process technologies. For example, while Honda's work on hydrogen fuel cells might be considered an advanced R&D project since it is still a significant distance from a commercial application, the company's development of its original hybrid-electric vehicle, the Insight, would be considered a breakthrough project. The Honda Insight incorporated revolutionary new technology in a commercialized application.

Platform projects typically offer fundamental improvements in the cost, quality, and performance of a technology over preceding generations. Derivative projects involve

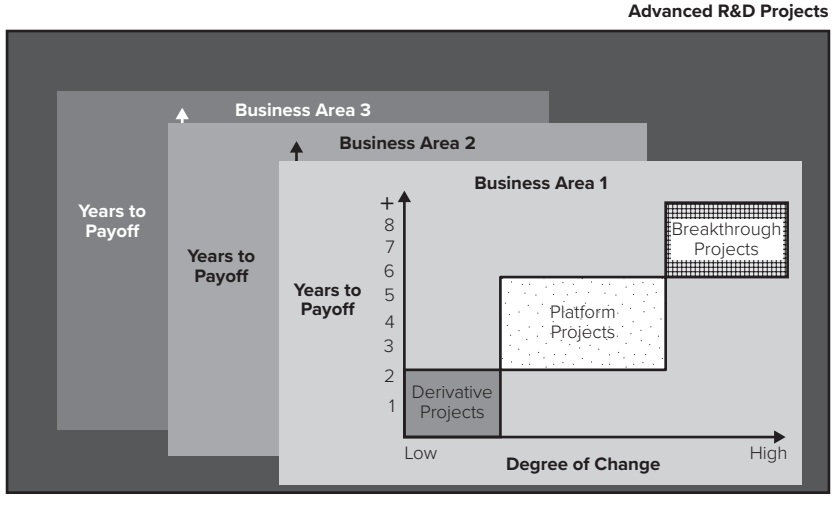


FIGURE 7.5
R&D Portfolio

Source: Adaptation based on "Project Map" in Steven C. Wheelwright and Kim B. Clark from *Revolutionizing Product Development: Quantum Leaps in Speed, Efficiency, and Quality*.

incremental changes in products and/or processes. A platform project is designed to serve a core group of consumers, whereas derivative projects represent modifications of the basic platform design to appeal to different niches within that core group. For example, Hunter's Care Free humidifier is a platform that offers several derivative versions to appeal to different customer segments. The water storage tank comes in different sizes (e.g., 2.0 gallon, 2.5 gallon, and 3.0 gallon); some models include a digital humidistat; and some also include Nite Glo lights. However, all of the models are based on the same Permawick filter design and fan system. Similarly, Toyota's Camry is a platform for a family of car models that includes the Camry LE, Camry SE, and Camry XLE. While all the models are based on the same basic design, each offers a different combination of features to appeal to different market segments. These variations on the Camry theme are derivative products.

Companies that use the R&D portfolio map categorize all their existing projects and projects under consideration by the resources they require (e.g., engineers, time, capital, etc.) and by how they contribute to the company's product line. The company can then map the project types and identify gaps in the development strategy.[17] They may also wish to create separate maps for different business areas of the firm, as shown in Figure 7.5. Managers can also use the map to identify their desired mix of projects, and allocate resources accordingly. The mix of projects represented on such a map should be consistent both with the company's resources, strategic position, and with its strategic intent (as analyzed in Chapter Six). For example, a typical firm experiencing moderate growth might allocate 10 percent of its R&D budget to breakthrough innovation, 30 percent to platform projects, and 60 percent to derivative projects. A firm pursuing more significant growth might allocate higher percentages to breakthrough and platform projects, while a firm that needs to generate more short-term profit might allocate a higher percentage to derivative projects.[18] Respondents to a recent survey administered by the Product Development and Management Association indicated that roughly 8 percent of their projects were breakthrough or advanced R&D projects, 17 percent were platform projects, and 75 percent were derivative projects.

Mapping the company's R&D portfolio encourages the firm to consider both short-term cash flow needs and long-term strategic momentum in its budgeting and planning. For instance, a firm that invests heavily in derivative projects that may be immediately commercialized with little risk may appear to have good returns on its R&D investment in the short run, but then be unable to compete when the market shifts to a newer technology. On the other hand, a firm that invests heavily in advanced R&D or breakthrough projects may be on the leading edge of technology, but run into cash flow problems from a lack of revenues generated from recently commercialized platform or derivative projects. As once noted by Jack Welch, former CEO of General Electric, "You can't grow long term if you can't eat short term. Anyone can manage short. Anyone can manage long. Balancing those two things is what management is."[19]

This is poignantly illustrated in the pharmaceutical industry, where high project failure rates, long product development cycles, and reliance on patent protection can cause a firm to suddenly find it has a devastating gap in its product pipeline. Studies indicate that developing a new drug takes an average of 12 years and can cost as much as $2 billion by the time a product makes it through clinical trials. In 2017, many of the world's largest pharmaceutical companies were facing a "patent cliff" as the patents were expiring on many of their blockbuster drugs, exposing the firms to much heavier competition from

generics. Furthermore, traditional pharma companies were beginning to be exposed to more competition from "biologics," treatments developed by biotech firms. This situation creates intense volatility in the revenues of a firm, which creates enormous pressures on the firm (including managing staffing, manufacturing capacity, managing R&D funds, and more). As a result, many pharmaceutical firms have begun shifting their emphasis to lower volume, specialty drugs that could potentially be more profitable because they would require less marketing investment and face less competition.

Q-Sort

Q-sort is a simple method for ranking objects or ideas on a number of different dimensions. The Q-sort method has been used for purposes as diverse as identifying personality disorders to establishing scales of customer preferences. Individuals in a group are each given a stack of cards with an object or idea on each card. In the case of new product development, each card could identify a potential project. Then a series of project selection criteria are presented (e.g., technical feasibility, market impact, fit with strategic intent), and for each criterion, the individuals sort their cards in rank order (e.g., best fit with strategic intent) or in categories (e.g., technically feasible versus infeasible) according to that criterion. Individuals then compare their rank orderings and use these comparisons to structure a debate about the projects. After several rounds of sorting and debating, the group is expected to arrive at a consensus about the best projects.[20]

COMBINING QUANTITATIVE AND QUALITATIVE INFORMATION

As demonstrated above, both quantitative methods and qualitative methods offer a number of benefits to managers in choosing development projects. Thus, many firms use a combination of methods to arrive at an investment decision.[21] For example, a firm might have screening questions that require quantitative analysis in addition to qualitative responses. Firms might also use quantitative methods to estimate the cash flows anticipated from a project when balancing their R&D portfolio on a project map. There are also valuation techniques that attempt to translate qualitative assessments into quantitative measures, such as conjoint analysis and data envelopment analysis, as discussed below.

Conjoint Analysis

conjoint analysis
A family of techniques that enables assessment of the weight individuals put on different attributes of a choice.

Conjoint analysis is a family of techniques (including discrete choice, choice modeling, hierarchical choice, trade-off matrices, and pairwise comparisons) used to estimate the specific value individuals place on some attribute of a choice, such as the relative value of features of a product or the relative importance of different outcomes of a development project. While individuals may find it very difficult to accurately assess the weight they put on individual attributes of a decision, conjoint analysis enables these weights to be derived statistically. Conjoint analysis enables a subjective assessment of a complex decision to be decomposed into quantitative scores of the relative importance of different criteria.

The most common use of conjoint analysis is to assess the relative importance to customers of different product attributes—these values can then be used in development and pricing decisions. For example, potential customers might be given a series of cards describing different models of a camera with different features and prices. The individuals are then asked to rate each in terms of its desirability (e.g., on a scale of 1 to 10) or

Theory in Action Courtyard by Marriott

In the mid-1980s, Marriott was facing a nearly saturated market for full-service, upscale hotels. Marriott's managers knew that the only way to sustain the company's 20 percent annual sales growth rate was to expand its product line. Marriott's management believed there was a market opportunity in the moderately priced ($35 to $60 a night) hotel category. The dominant chains in this category (e.g., Holiday Inn, Howard Johnson) tended to receive poor marks for customer satisfaction, and Marriott's managers believed that a chain that provided newer facilities and more consistent service would be enthusiastically received by customers. They also knew, however, that the company's most valuable resource was its well-established brand name, and they were reluctant to hastily affix the Marriott name to a scaled-down version of its hotels.

The company constructed a carefully structured plan for evaluating potential designs for a midprice hotel line. The company first conducted focus groups to identify different customer segments and the major factors that influenced their purchase decisions. The factors identified included external surroundings, room, food, lounge, services, leisure activities, and security. Within each major factor, there were several specific attributes that could take on multiple levels of product or service quality. For example, within the "services" factor, one of the attributes was "reservations," and the company had two levels: "call the hotel directly" or "call an 800 reservation number." A sample of hotel customers was selected to participate in a research group whereby each participant was given a fictional $35 with which to build his or her own hotel. Each participant was given seven cards (one card for each of the major factors identified above). On the card was listed each of the specific attributes and their possible levels, along with a price for each level. The participants would evaluate one card at a time, selecting the features they desired. If participants went over the $35 budget, they were required to eliminate some features or choose a less expensive level of service. This technique helped management understand customer priorities and the trade-offs made by different customer segments. On the basis of these priorities, the managers then developed a series of hotel profiles offering different combinations of features and levels of service. Participants were then asked to rate each of these profiles. The managers could then use regression to assess how different levels of service within a specific attribute influenced customer ratings of the hotel overall. For example, as illustrated in Figure 7.6, after the hotel profiles are rated by the participants, the values for the levels in the profiles and the participant ratings can be entered into a spreadsheet. The ratings can then be regressed on the attribute levels, yielding a model that estimates the relative importance of each attribute.

On the basis of their conjoint analysis, Marriott's managers came up with the Courtyard concept: relatively small hotels (about 150 rooms) with limited amenities, small restaurants, small meeting rooms, enclosed courtyards, high-security features, well-landscaped exteriors, and significantly lower room rates than its traditional Marriott hotels. Courtyard by Marriott turned out to be very successful. By the end of 2017, there were 1145 Courtyard hotels (508 of those were in the United States), and their average occupancy rate of more than 70 percent was well above the industry average.

continued

asked to order the models in terms of which they would most likely buy. Multiple regression is then used to assess the degree to which each attribute influences the overall rating, resulting in the assignment of specific weights to individual criteria. These weights provide a quantitative assessment of the trade-offs that customers implicitly consider in their evaluation of products. The firm can then use these weights in a series of "what if" scenarios to consider the implications of different product configurations. For example, Marriott used conjoint analysis to determine what features customers would most value in a moderately priced hotel. This analysis enabled Marriott to develop its very successful line of Courtyard by Marriott hotels (see the accompanying Theory in Action).

concluded

FIGURE 7.6
Hotel Profiles and Ratings for Conjoint Analysis

Hotel Profile 1	Hotel Profile 2	Hotel Profile 3	. . .
Reservations			
1–800 number (1)	Call hotel directly (0)	1–800 number (1)	
Room Service			
Full-menu, 24 hours a day (5)	Limited menu, offered 6 A.M. to midnight (3)	No room service (1)	
Newspaper Delivery			
None (0)	Daily (1)	None (0)	
(others)			

Attributes	Reservations	Room Service	Newspaper Delivery	(others)	Overall Rating (1–10)
Participant 1					
Hotel Profile 1	1	5	0		8
Hotel Profile 2	0	3	1		7
Hotel Profile 3	1	1	0		5
Participant 2					
Hotel Profile 1	1	5	0		7
Hotel Profile 2	0	3	1		9
Hotel Profile 3	1	1	0		4
(others)					

Source: Adapted from R. J. Thomas, New Product Success Stories (New York: John Wiley & Sons, 1995).

Data Envelopment Analysis

data envelopment analysis (DEA)
A method of ranking projects based on multiple decision criteria by comparing them to a hypothetical efficiency frontier.

Data envelopment analysis (DEA) is a method of assessing a potential project (or other decision) using multiple criteria that may have different kinds of measurement units.[22] For instance, for a particular set of potential projects, a firm might have cash flow estimates, a ranking of the project's fit with existing competencies, a ranking of the project's potential for building desired future competencies, a score for its technical feasibility, and a score for its customer desirability. Each of these measures captures something that is qualitatively different, and the numbers assigned to them are based on different units of measure. While the first measure is in dollars and is a nearly continuous measure, the second two measures are rank orders and thus are categorical measures with little information about what the difference is between one level of rank and another. The last two measures are scores that might be based on a ranking system or scaling system (e.g., a Likert measure that goes from one to seven).

FIGURE 7.7
DEA Ranking of the Advanced Technologies Group's 10 Most Attractive Projects and Every 50th Project Thereafter

Source: Adapted from J. D. Linton, S. T. Walsch, and J. Morabito, "Analysis, Ranking and Selection of R&D Projects in a Portfolio," *R&D Management* 32, no. 2 (2002): 139–48.

Rank	Intellectual Property	Product Market	Investment	Cash Flow (most likely)	Cash Flow (optimistic)	Cash Flow (pessimistic)
1	2.25	1.5	$4322	$1,296,700	$1,353,924	$1,184,192
2	1.5	1.5	850	525,844	551,538	493,912
3	1.5	1.5	1	4	4	3
4	2.25	2.25	478	545,594	822,164	411,082
5	1.5	1.5	1	15	15	11
6	1.5	2.25	65	89,144	178,289	0
7	1.5	1.5	1068	685,116	1,027,386	342,558
8	1.5	1.5	4	3766	4707	2824
9	1.5	1.5	20	4800	4800	−96
10	1.5	2.25	2	23	27	18
50	1.5	2.25	9	116	139	93
100	1.5	1.5	15	60	72	48
150	2.25	2.25	40	5531	13,829	2766
200	2.25	1.5	38	90	135	45

efficiency frontier
The range of hypothetical configurations that optimize a combination of features.

Data envelopment analysis uses linear programming to combine these different measures from the projects to create a hypothetical **efficiency frontier** that represents the best performance on each measure. It can also consider which measures are inputs (such as costs) versus outputs (expected benefits). It then measures the distance of each project from this frontier to give it an efficiency value. These values can then be used to rank-order the projects or identify projects that clearly dominate others.[23] For example, Figure 7.7 shows a DEA ranking of projects evaluated by the Advanced Technologies Group of Bell Laboratories. The Advanced Technologies Group chose to evaluate projects in terms of three measures of discounted cash flows expected from the project (most likely, optimistic scenario, pessimistic scenario), the investment required, and each project's desirability from the perspective of intellectual property benefits and product market benefits. For the latter two measures, projects were given a score of 1, 1.5, or 2.25 based on the group's model for intellectual property and product market benefits. These scores reflect this particular group's scoring system—it would be just as appropriate to use a scaled measure (e.g., 1 = very strong intellectual property benefits, to 7 = no intellectual property benefits), or other type of measure used by the firm. As shown, the DEA method enabled Bell Laboratories to rank-order different projects, despite the fact that they offered different kinds of benefits and risks.

The biggest advantage of DEA is that it enables comparisons of projects using multiple kinds of measures. However, just as with several of the methods described previously, the results of DEA are only as good as the data utilized. Managers bear the responsibility of determining which measures are most important to include and of ensuring that the measures are accurate.

Summary of Chapter

1. Firms often use a combination of quantitative and qualitative methods to evaluate which projects should be funded. Though some methods assume that all valuable projects will be funded, resources are typically constrained and firms must use capital rationing.
2. The most commonly used quantitative methods of evaluating projects are discounted cash flow methods such as net present value (NPV) or internal rate of return (IRR). While both methods enable the firm to create concrete estimates of returns of a project and account for the time value of money, the results are only as good as the cash flow estimates used in the analysis (which are often unreliable). Both methods also tend to heavily discount long-term or risky projects and can undervalue projects that have strategic implications that are not well reflected by cash flow estimates.
3. Some firms now use a real options approach to assess projects. Real options better account for the long-run strategic implications of a project. Unfortunately, many new product development investment decisions do not conform to the assumptions inherent in an options valuation approach.
4. One commonly used qualitative method of assessing development projects is to subject the project to a series of screening questions that consider the project from multiple angles. These questions may be used merely to structure the discussion of a project or to create rating scales that are then utilized in an approach that combines qualitative and quantitative assessment.
5. A company's portfolio of projects typically includes projects of different types (e.g., advanced R&D, breakthrough, platform, and derivative projects) that have different resource requirements and different rates of return. Companies can use a project map to assess what their balance of projects is (or should be) and allocate resources accordingly.
6. Q-sort is a qualitative method of assessing projects whereby individuals rank each project under consideration according to a series of criteria. Q-sort is most commonly used to provide a format for discussion and debate.
7. Conjoint analysis is a method of converting qualitative assessments of a choice into quantitative weights of the different criteria underlying the choice. It is most often used for assessing how customers value different product attributes.
8. Data envelopment analysis (DEA) is another method that combines qualitative and quantitative measures. DEA enables projects that have multiple criteria in different measurement units to be ranked by comparing them to a hypothetical efficiency frontier.

Discussion Questions

1. What are the advantages and disadvantages of discounted cash flow methods such as NPV and IRR?
2. For what kind of development projects might a real options approach be appropriate? For what kind of projects would it be inappropriate?
3. Why might a firm use both qualitative and quantitative assessments of a project?

4. Identify a development project you are familiar with. What methods do you believe were used to assess the project? What methods do you believe *should have been* used to assess the project?
5. Will different methods of evaluating a project typically yield the same conclusions about whether to fund its development? Why or why not?

Suggested Further Reading

Classics

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Chapter **Eight**

Collaboration Strategies

Zeta Energy and "The Holy Grail" of Batteries

Introduction

In 2014, Charles Maslin, a lawyer and investor with an avid interest in technology, decided to place a major bet on innovations in battery technology. Like many others, Maslin believed that the key to reducing environmental degradation and climate change was better batteries. Batteries that were lower cost, higher capacity, and sustainably manufactured could enable more people to use solar power in their homes and could make electric vehicles (that had the potential of being powered by renewable energy sources) competitive with vehicles that used fossil fuels. Maslin also knew that a better battery technology could have enormous commercial potential. Maslin thus began funding and licensing research being done on an advanced electrode material that had been developed at Rice University. Maslin saw the potential of this electrode to be used as an anode for lithium metal batteries and began supporting a research group led by world renown chemist and nanotechnologist Dr. James Tour. Soon after, Maslin hired two of the group's postdoctoral fellows, Dr. Abdul-Rahman Raji and Dr. Rodrigo Salvatierra, and founded the company Zeta Energy, LLC, with the intention of producing and selling extremely advanced lithium metal batteries.

The Technology

By 2020, the lithium-ion rechargeable battery technology that had come to dominate rechargeable applications like electric vehicles, laptops, smartphones, and more, was clearly mature.[a] There was widespread consensus that a breakthrough innovation in battery technology would be necessary to significantly advance applications in these and other markets. By this time, it was also already well understood that lithium *metal* batteries—those that contained lithium in metal form instead of just lithium ions in an electrolyte—offered significant potential advantages in terms of how much energy a battery could hold for a given volume or weight.[b] However, lithium metal batteries had not been widely commercialized because they suffered from some serious problems that needed to first be solved.[c] One of the most significant of these was dendrites. In a lithium metal battery, dendrites (pointy projections of lithium particles) tended to form on

the anode, similar to how a stalagmite forms on the floor of a cave. These projections could ultimately penetrate the separator between the anode and cathode, causing short circuits in the battery. Dendrites limited both the ability to recharge lithium metal batteries and created safety hazards. A short circuit in a battery could result in a fire or an explosion, for example.[d]

To overcome this problem, the Zeta Energy team had created anodes with a graphene-carbon nanotube structure. The nanotubes looked like a shag carpet of tiny tubes that could hold a tremendous amount of lithium while preventing the formation of dendrites.[e] This would enable a lithium metal battery to simultaneously achieve both an extremely high energy density and have a low rate of degradation.

To realize the full potential of their anode, however, the team realized they also needed a better cathode. Sulfur cathodes had been of considerable interest in the scientific community because sulfur is cheap and abundant, and offered the theoretical potential of increasing the energy capacity of a battery up to 500 percent compared to conventional lithium-ion battery materials.[f] However, several challenges to realizing this potential had limited their use. Namely, sulfur cathodes suffered from a polysulfide shuttle effect that resulted in "leakage" of active material, leading to degradation of the battery.[g] Sulfur also expanded as it took on lithium ions, creating instability in the battery's structure. As a result of both effects, lithium sulfur batteries had a history of poor "cyclability," meaning that they could not be recharged very many times, making them commercially nonviable for most applications.[h] The Zeta Energy team, however, had developed a way of polymerizing sulfur that they believed solved these problems.[i] The company held numerous patents (and had many more pending) on both the carbon graphene nanotube anode, and their polymerized sulfur cathode.

The combination of a high-density anode that was dendrite free, and a sulfur cathode with excellent "cyclability" achieved what *Forbes* termed "The Holy Grail"[j] in battery technology. As an added bonus, Zeta's battery did not use cobalt. Eliminating cobalt from the production of batteries was highly desirable since cobalt was primarily mined in the Democratic Republic of Congo, a country that was notorious for using child labor in cobalt mines.[k]

Ramping Up

Zeta Energy had received early funding from Maslin himself and investments from friends, family, and angel investors. However, to demonstrate the potential of Zeta's battery at commercial scale required significant capital—on the order of $100 million—to build the required factory and equipment. Though early testing had shown Zeta's technology to be very promising, there were other potentially promising battery technologies also emerging in the market, including lithium silicon batteries, solid state batteries, and other lithium sulfur batteries. This made it challenging to secure sizeable investments—the future of battery technology appeared very uncertain to many would-be investors. To make matters more complicated, Maslin was reluctant to approach venture capitalists (VC), because he knew venture capital firms would insist on taking significant control of the firm.

As of 2020, Maslin deliberated over a few options for continuing Zeta's progress. Each entailed difficult choices. First, another startup had approached Zeta about a

potential collaboration. This startup was also developing a lithium sulfur battery but had not overcome the polysulfide shuttle problem and was eager to gain access to Zeta's technology. The company had secured some early large investments from a chemical firm that produced (among other things) electrolytes. Second, Zeta had been approached by several potential customers of its battery, including a European automotive firm, a British consumer electronics firm and the U.S. Army, but in all those cases Zeta would have to find a way to first fund its own lab-scale factory to produce enough demonstration batteries for these companies to test. Third, Maslin could reverse his decision on VC funding and give more control of the company away in exchange for the funds and management support offered by VCs. Finally, Maslin also knew there was a potential to strike a joint venture with an existing large lithium-ion battery firm, a new but well-funded advanced battery firm, or an automotive manufacturer, though Maslin was concerned that working with heavyweight companies in the close and frequent contact of a joint venture could put Zeta's proprietary technology at risk. Specifically, the concern was that a large and skillful partner might improve upon Zeta's manufacturing process in a way that would force a cross-licensing of Zeta's design patents with the large partner's newly developed process patents, ultimately leading to Zeta losing its technological edge.

A Decision to Be Made

In early 2021, Zeta's team (which had now grown to a dozen people) felt a sense of urgency. The company was rapidly depleting its early funds and had made promises to several potential customers about producing demonstration units. It also knew that the longer it waited to bring its products to market, the greater the likelihood that advances by other battery startups would come to market, making it that much harder to get gain traction with investors and customers. Maslin had absolute faith that Zeta had a "holy grail" solution for batteries, but what was the best path to helping the company realize that potential?

Discussion Questions

1. What are the pros and cons of Zeta Energy collaborating with another organization?
2. What type of potential partner offers the most "resource fit" for Zeta Energy? What type of potential partner offers the most "strategic fit" for Zeta Energy? Is there a collaboration partner you would recommend?
3. If Zeta Energy entered into a collaboration, what type of collaboration structure (e.g., R&D alliance, joint venture, licensing, outsourcing, etc.) should it use? What would you recommend to Zeta Energy about governance of the collaboration?

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OVERVIEW

Firms frequently face difficult decisions about the scope of activities to perform in-house, and whether to perform them alone as a solo venture or to perform them collaboratively with one or more partners. As mentioned in Chapter Two, a significant portion of innovation arises not from any single individual or organization, but instead from the collaborative efforts of multiple individuals or organizations. Collaboration can often enable firms to achieve more, at a faster rate, and with less cost or risk than they can achieve alone. However, collaboration also often entails relinquishing some degree of control over development and some share of the expected rewards of innovation, plus it can expose the firm to risk of malfeasance by its partner(s). In this chapter, we will first consider the reasons that a firm might choose to engage in collaborative development or might choose to avoid it. We will then review some of the most common types of collaborative arrangements and their specific advantages and disadvantages.

REASONS FOR GOING SOLO

A firm might choose to engage in solo development of a project for a number of reasons. First, the firm may perceive no need to collaborate with other organizations—it may possess all the necessary capabilities and resources for a particular development project in-house. Alternatively, the firm may prefer to obtain complementary skills or resources from a partner, but there may be no available partner that is appropriate or willing to collaborate.

A firm might also choose to develop a project as a solo venture if it is concerned that collaborating would put its proprietary technologies at risk, or if it seeks to have full control over the project's development and returns. Furthermore, a firm's solo development of a technological innovation might give it more opportunities to build and renew its capabilities.

1. Availability of Capabilities

Whether a firm chooses to partner on a project is largely determined by the degree to which it possesses all of the necessary capabilities in-house and the degree to which one or more potential partners have necessary capabilities. If a firm has all of the necessary capabilities for a project, it may have little need to collaborate with others and may opt to go it alone. Furthermore, if a firm finds that it lacks certain required capabilities but there are also no potential partners with such capabilities, it may be forced to develop the capabilities on its own.

For example, in the late 1970s Monsanto was interested in developing food crop seeds that were genetically modified to survive strong herbicides. Monsanto's Roundup, a powerful herbicide, had been introduced in 1974 and had been remarkably successful. However, Roundup killed almost all plants that it came into contact with and thus had to be applied with great care. If crops could be developed that were genetically modified to resist Roundup, the herbicide could be used more easily and in larger quantities. The biotechnology industry was still quite young, so there were no appropriate partners from which to acquire the necessary technologies. Monsanto decided to pursue the opportunity as a solo internal venture and declared that biotechnology was its new strategic focus.[1] In 1983, Monsanto successfully developed its first transgenic plant, but it would not be until 1995 that it would have its first genetically modified crop seed, Roundup Ready soybeans, approved for commercialization.[2] Though many environmental groups opposed both Roundup and the genetically modified Roundup Ready crops, the combination was enormously successful. By 2002, more than 130 million acres worldwide were planted with Monsanto's Roundup Ready soybean, corn, cotton, and canola seed.[3]

2. Protecting Proprietary Technologies

Firms sometimes avoid collaboration for fear of giving up proprietary technologies. Working closely with a partner might expose the company's existing proprietary technologies to the prying eyes of a would-be competitor. Furthermore, the firm may wish to have exclusive control over any proprietary technologies created during the development project. Consider Zeta Energy's decision about collaborating in its development of a lithium sulfur battery. While collaborating could give Zeta needed capital, testing, manufacturing, and marketing abilities, it also meant that Zeta would have to share the profits and control, and that might put its proprietary technology at risk.

3. Controlling Technology Development and Use

Sometimes firms choose not to collaborate because they desire to have complete control over their development processes and the use of any resulting new technologies. This desire might be for pragmatic reasons (e.g., the new technology is expected to yield high margins and the firm does not wish to share rents with collaborators) or cultural reasons (e.g., a company's culture may emphasize independence and self-reliance). Both of these reasons are demonstrated by Honda in the development of its hybrid-electric

alliance
Alliance is a general term that can refer to any type of relationship between firms. Alliances may be short or long term and may include formally contracted agreements or be entirely informal in nature.

vehicle, the Insight. While other auto manufacturers were enthusiastically forming **alliances** to collaborate on automobile design and the development of more efficient manufacturing processes, Honda was very cautious about forming collaborative relationships. Honda's decision not to join the Alliance of Automobile Manufacturers, the industry trade group that leads the fight against tougher fuel and emissions standards, had both pragmatic and cultural reasons. From a pragmatic standpoint, Honda worried that participating in the trade group would limit its discretion over its development of environmentally friendly automobiles, an area where Honda intended to be the market leader. This decision was reinforced by Honda's culture that emphasized retaining complete control over the firm's technology development and direction. This is illustrated by Honda President Hiroyuki Yoshino's statement, "It's better for a person to decide about his own life rather than having it decided by others."[4]

4. Building and Renewing Capabilities

Firms may also choose to engage in solo development even when partnering could save time or money because they believe that development efforts are key to building and renewing their capabilities. Solo development of a technological innovation challenges the firm to develop new skills, resources, and market knowledge. As noted in Chapter Seven, the potential for creating and enhancing the organization's capabilities may be more valuable than the innovation itself. This is aptly demonstrated in a quote from Walt Gillette of Boeing about the development of the Sonic Cruiser: "Industry experience indicates that if the company doesn't create a new airplane every 12 to 15 years, the needed skills and experience will be gone. Too many of the people who created the last new airplane will have retired or moved on to other companies, and their skills and experience will not have been passed on to the next generation of Boeing employees."[5]

Though there are several reasons a firm might choose to stick with solo development, there are also many reasons for firms to engage in collaborative development, and collaboration appears to be on the rise. In the next sections, we will discuss the advantages of collaboration and the strengths and weaknesses of various types of collaboration.

ADVANTAGES OF COLLABORATING

Collaborating on development projects can offer a firm a number of advantages, including faster speed to market, greater flexibility, learning capabilities from other firms, and building a coalition around a standard.

1. Acquiring Capabilities and Resources Quickly

It is not unusual for a company to lack some of the complementary assets required to transform a body of technological knowledge into a commercial product. Given time, the company can develop such complementary assets internally. However, doing so extends cycle time.[6] Instead, a company may be able to gain rapid access to important complementary assets by entering into strategic alliances or licensing arrangements.[7] For example, when Apple was developing its LaserWriter, a high-resolution laser printer, it did not possess the technological expertise to produce the printer's engine, and developing such capabilities in-house would have taken a long time. Apple persuaded Canon, the market leader in printer engines, to collaborate on the project.[8] With Canon's help, Apple was able to bring the high-quality printer to market quickly.

2. Increasing Flexibility

Obtaining some of the necessary capabilities or resources from a partner rather than building them in-house can help a firm reduce its asset commitment and enhance its flexibility. This can be particularly important in markets characterized by rapid technological change. High-speed technological change causes product markets to rapidly transform. Product life cycles shorten, and innovation becomes the primary driver of competition. When technology is progressing rapidly, firms may seek to avoid committing themselves to fixed assets that may rapidly become obsolete. They may choose to become more narrowly specialized and to use linkages with other specialized firms to access resources they do not possess in-house.

3. Learning from Partners

Collaboration with partners can be an important source of learning for the firm. Close contact with other firms can facilitate both the transfer of knowledge between firms and the creation of new knowledge that individual firms could not have created alone.[9] By pooling their technological resources and capabilities, firms may be able to expand their knowledge bases and do so more quickly than they could without collaboration.

4. Resource and Risk Pooling

One primary reason firms collaborate on a development project is to share the costs and risks of the project. This can be particularly important when a project is very expensive or its outcome highly uncertain.[10]

5. Building a Coalition around a Shared Standard

Firms may also collaborate on a development project when such a collaboration would facilitate the creation of a shared standard. For example, as of early 2018, there were four major electric vehicle charging standards competing in the market: CHAdeMO (used by most of the Japanese electric vehicle manufacturers), SAE Combined Charging System (used by most of the major European electric vehicle manufacturers plus General Motors and Ford in the United States), GB/T (used by Chinese electric vehicle manufacturers such as BYD, SAIC, and Dongfeng), and Tesla Supercharger (used by Tesla). Globally, in 2018 there were about 7000 CCS charging stations, 16,639 CHAdeMO charging stations (mostly in Japan and Europe), 8496 Tesla Superchargers (mostly in the United States), and 127,434 GB/T charging stations (all in China). In late 2017, Daimler, BMW, Ford, and Volkswagen group announced they would form the Charging Interface Initiative (CharIN) to collaborate on building CCS charging stations, hoping to build momentum for the standard. Tesla, meanwhile, had hedged its bets by releasing its patents on the Supercharger standard so that others could adopt it, and joining both the CharIN and CHAdeMO alliances. Furthermore, Tesla had created adapters for its vehicles for both the CHAdeMO and GB/T standards.[11]

TYPES OF COLLABORATIVE ARRANGEMENTS

Collaboration can include partnering with suppliers, customers, competitors, complementors, organizations that offer similar products in different markets, organizations that offer different products in similar markets, nonprofit organizations, government

joint venture
A partnership between two or more firms involving a significant equity stake by the partners and often resulting in the creation of a new business entity.

licensing
A contractual arrangement whereby one organization or individual (the licensee) obtains the rights to use the proprietary technology (or trademark, or copyright, etc.) of another organization or individual (the licensor).

organizations, universities, or others. Collaboration can also be used for many different purposes, including manufacturing, services, marketing, or technology-based objectives. In North America, as many as 23 percent of all alliances are for research and development activities, compared to 14 percent in Western Europe and 12 percent in Asia.[12]

Collaboration arrangements can also take many forms, from very informal alliances to highly structured **joint ventures** or technology exchange agreements (**licensing**). The most common forms of collaborative arrangements used in technological innovation include strategic alliances, joint ventures, licensing, outsourcing, and collective research organizations.

Strategic Alliances

A strategic alliance is a temporary relationship that can take many forms. It can be formalized in a contract or an informal agreement. It can be a short-term agreement or a long-term agreement, and it can include an equity investment made by the partners in each other (termed equity alliances, discussed later in the chapter). Most alliances (with the exception of joint ventures that establish a new legal subsidiary) are considered temporary agreements, and thus offer a good way for firms to flexibly combine their efforts and resources. Firms may use strategic alliances to access a critical capability that is not possessed in-house or to more fully exploit their own capabilities by leveraging them in another firm's development efforts. Firms with different capabilities necessary for developing a new technology or penetrating a new market might form alliances to pool their resources so that collectively they can develop the product or market faster or less expensively. Even firms that have similar capabilities may collaborate in their development activities in order to share the risk of a venture or to speed up market development and penetration. Large firms might form alliances with small firms in order to take a limited stake in the smaller firm's development efforts, while small firms might form alliances with large firms to tap the larger firm's greater capital resources, distribution and marketing capabilities, or credibility.[13] For example, many large pharmaceutical firms have allied with small biotechnology firms for their mutual benefit: The pharmaceutical firms gain access to the drug discoveries of the biotechnology companies, and the biotechnology companies gain access to the capital resources, manufacturing, and distribution capabilities of the pharmaceutical firms.

Through an alliance, firms can establish a limited stake in a venture while maintaining the flexibility to either increase their commitment later or shift these resources to another opportunity.[14] Firms can use alliances to gain an early window on emerging opportunities that they may want to commit to more fully in the future. Alliances also enable a firm to rapidly adjust the type and scale of capabilities the firm can access, which can be very valuable in rapidly changing markets.[15]

Alliances are also used to enable partners to learn from each other and develop new competencies. Alliance partners may hope to transfer knowledge between the firms or to combine their skills and resources to jointly create new knowledge. However, alliance relationships often lack the shared language, routines, and coordination that facilitate the transfer of knowledge—particularly the complex and tacit knowledge that is most likely to lead to sustainable competitive advantages.[16] To use alliances for learning requires a serious commitment of resources, such as a pool of dedicated people willing to travel between the home firm and partner firm, test-bed facilities, and active

procedures for internalizing what has been learned.[17] Alliances can thus also be costly. They require monitoring and coordination. There is also a risk of partners taking the firm's intellectual property for their own advantage.[18]

Yves Doz and Gary Hamel argue that it is useful to categorize a firm's alliance strategy along two dimensions.[19] The first dimension is the degree to which alliances practice **capability complementation** versus **capability transfer**. The second dimension is whether the firm manages each alliance individually or manages a collective network of alliances (see Figure 8.1).

capability complementation
Combining ("pooling") the capabilities and other resources of partner firms, but not necessarily transferring those resources between the partners.

capability transfer
Exchange of capabilities across firms in such a manner that partners can internalize the capabilities and use them independently of the particular development project.

In quadrant A are firms that forge an individual alliance to combine complementary technologies or skills needed for a project. For example, in the mid-1970s, General Electric (GE) and SNECMA (a French jet engine producer) formed a joint venture called CFM International to develop a new jet engine. The venture would combine GE's F101 turbojet with SNECMA's low-pressure fan expertise to create a powerful and fuel-efficient engine. Because the F101 was considered a sensitive military technology by the U.S. Air Force, the venture was set up to carefully avoid the exchange of proprietary technology between the firms. GE would build the F101 portion as a sealed "black box," which could then be shipped to a separate assembly location. The resulting engine, the CFM-56, became the most successful jet engine in the history of aviation.[20]

In quadrant B are firms that use a network of alliances to combine complementary skills and resources. For example, Corning, known primarily as a producer of glass products, has created a web of alliances with partners that have complementary skills in order to extend its glass technology into fields as diverse as medical products, computer products, and fiber optics. Instead of attempting to internalize its partners' technologies, Corning views its relationships with its partners as a form of extended enterprise that forms a flexible and egalitarian network of independent businesses.[21]

In quadrant C are firms that use individual alliances to transfer capabilities between them. Doz and Hamel provide the example of the alliance between JVC and Thomson. While both companies produce VCRs, Thomson wanted to glean product technology and manufacturing expertise from JVC, whereas JVC needed to learn how to penetrate the European market from Thomson. Both sides perceived an equitable opportunity for gain from exchanging capabilities.

FIGURE 8.1
Technology Alliance Strategies

Source: Y. Doz and G. Hamel, 1997, "The Use of Alliances in Implementing Technology Strategies." In M. L. Tushman and P. Anderson, *Managing Strategic Innovation and Change*, 1997.

	Individual Alliance	**Network of Alliances**
Capability Complementation	A GE-SNECMA alliance	B Corning Glass alliances
Capability Transfer	C Thomson-JVC alliance	D Aspla

In quadrant D are firms that use a network of alliances to exchange capabilities and jointly develop new capabilities. The collective research organizations described later in the chapter (including Aspla and the National Center for Manufacturing Sciences) are examples of alliance networks in which a formal body has been created to govern the network. These organizations are designed to enable their member organizations to collectively create, share, and utilize knowledge. In building an alliance portfolio, managers should think carefully about competitive effects, complementing effects, and network structure effects. First, if multiple alliances are serving the same strategic needs, there is a risk of redundant resources investment, or competitive conflict between partners. The costs and benefits of this should be carefully weighed as alliance partners could become adversaries. Second, complementary alliances can be super-additive if carefully managed. For example, a pharmaceutical firm might be using an alliance to develop a drug target with one partner, and another alliance to develop a delivery method for that same drug, enabling it to bring the product to market faster.[22] In this situation, the benefits of each alliance are accentuated by the benefits of the other. Finally, managers should consider how their portfolio of alliances positions them in the web of relationships that connects their firm, their partners, and their partners' partners.[23] Such networks can be very influential in the diffusion of information and other resources, and being positioned well in an alliance network can confer significant advantages (see the Research Brief on "Strategic Positions in Collaborative Networks" later in this chapter).

The opportunities and flexibility that can be gained through using alliances can come at a cost. The potential for opportunism and self-interest exists for all parties of an alliance due to limited levels of mutual commitment.[24] Studies suggests that between 30 percent and 70 percent of alliances fail by neither meeting the goals of the partners nor delivering the operational or strategic benefits for which they were intended.[25] Firms need to be constantly on guard to ensure that the alliance does not inadvertently result in giving too much away to a potential competitor. According to Doz and Hamel, while collegiality between partners can facilitate trust and communication, *too much* collegiality may be a warning sign that information gatekeepers within the firm are not being sufficiently vigilant.[26] Employees at all levels should be regularly informed about what information and resources are off-limits to the partner, and the firm should stringently monitor what information the partner requests and receives.[27]

Joint Ventures

Joint ventures are a particular type of strategic alliance that entails significant structure and commitment. While a strategic alliance can be any type of formal or informal relationship between two or more firms, a joint venture involves a significant equity investment from each partner and often results in establishment of a new separate entity. The capital and other resources to be committed by each partner are usually specified in carefully constructed contractual arrangements, as is the division of any profits earned by the venture.

For example, in April of 2018, Tencent Holdings (a Chinese multinational conglomerate that specializes in Internet-related services such as social media, music streaming, mobile games, and more) announced it was forming a joint venture which

Chinese state-owned Changan Automobile Company to develop Internet-connected autonomous driving technology that would utilize artificial intelligence. Tencent would invest 102 million yuan (about $16.2 million) for a 51 percent share of the venture, and Changan would invest 98 million yuan for the remaining 49 percent share. Changan had already completed a 2000 kilometer road test from Chongqing to Beijing, and said it had received clearance to start testing autonomous vehicles on public roads in the U.S. state of California.[28]

Licensing

Licensing is a contractual arrangement whereby one organization or individual (the *licensee*) obtains the rights to use the proprietary technology (or trademark, copyright, etc.) of another organization or individual (the *licensor*). Licensing enables a firm to rapidly acquire a technology (or other resource or capability) it does not possess. For example, when Microsoft realized it had lost precious time to Netscape and needed to get a Web browser to market fast, it licensed the software it needed to produce Internet Explorer from Spyglass Inc. Microsoft also bought several companies (including Vermeer Technologies, Colusa Software, and eShop Inc.) to provide other Internet utilities.

For the licensor, licensing can enable the firm's technology to penetrate a wider range of markets than it could on its own. For example, Delphi Automotive, a supplier to the automotive industry, had developed a software program that can simulate various aspects of machining, including turning, milling, and drilling. The software enabled manufacturers that do high-volume machining to identify ways of improving their machining processes. Delphi had developed the software for its own use, but then realized it could make more money by licensing the software to others.[29]

Licensing a technology from another firm is typically much less expensive for a licensee than developing a new technology in-house. As discussed in earlier chapters, new product development is both expensive and risky; through licensing, a firm can obtain a technology that is already technically or commercially proven. Though it is often presumed that a technology available for license is an unlikely source of advantage (because it is typically available to many potential licensees), Procter & Gamble's experience shows that this need not be the case. Through its "Connect and Develop" program, it focuses on sourcing ideas and technologies external to the firm that it can then add value to in its labs. Thus, while a licensed technology provides the foundation for a new product, the product that arrives to market typically draws on the deep (and difficult to imitate) expertise and other resources P&G possesses.[30] This approach is emblematic of the "Open Innovation" approach now being used by many firms.[31]

Licensing agreements typically impose many restrictions on the licensee, enabling the licensor to retain control over how the technology is used. However, over time, licensees may gain valuable knowledge from working with the licensed technology that can enable them to later develop their own proprietary technologies. In the long run, the licensor's control over the technology may erode.

Sometimes firms license their technologies to preempt their competitors from developing their own competing technologies. This can be particularly important if competitors are likely to be able to imitate the primary features of the technology or if the industry has strong pressures for the adoption of a single dominant design

(see Chapter Four). By licensing out the technology to potential competitors, the licensor gives up the ability to earn monopoly rents on the technology. However, doing so may prevent potential competitors from developing their own proprietary technologies. Thus, licensing enables a firm to opt for a steady stream of royalties rather than gambling on the big gain—or big loss—of having its technology compete against others for market dominance.

Outsourcing

Firms that develop new technological innovations do not always possess the competencies, facilities, or scale to perform all the value-chain activities for the new innovation effectively or efficiently. Such firms might outsource activities to other firms.

contract manufacturing When a firm hires another firm (often a specialized manufacturer) to manufacture its products.

One common form of outsourcing is the use of contract manufacturers. **Contract manufacturing** allows firms to meet the scale of market demand without committing to long-term capital investments or an increase in the labor force, thus giving the firm greater flexibility.[32] It also enables firms to specialize in those activities central to their competitive advantage while other firms provide necessary support and specialized resources the firm does not possess. Contract manufacturing further enables a firm to tap the greater economies of scale and faster response time of a dedicated manufacturer, thereby reducing costs and increasing organizational responsiveness to the environment.[33] For example, when Apple redesigned a screen for its iPhone just weeks before it was due on the shelves, it was able to call a foreman at a Chinese factory it was working with, who woke up the 8000 workers sleeping in dormitories. The workers were given biscuits and tea, and immediately started a 12-hour shift fitting glass screens into beveled frames. Within 96 hours, the plant was manufacturing more than 10,000 iPhones a day. "The speed is breathtaking," an Apple executive noted. "There's no American plant that can match that." Whereas Apple directly employs 43,000 people in the United States and 20,000 people in other countries, an additional 700,000 people work for Apple's contractors, engineering, building, and assembling Apple products. In response to a query from U.S. President Barack Obama of "What would it take to make iPhones in the United States?" Steve Jobs replied, "Those jobs aren't coming back." Apple executives noted that the vast scale of overseas factories, and the flexibility, diligence, and industrial skills of their workers had outpaced American counterparts. But in response to criticisms about what this had done to employment in the United States, the executives explained, "We sell iPhones in over a hundred countries. . . . Our only obligation is making the best product possible."[34]

Other activities, such as product design, process design, marketing, information technology, or distribution can also be outsourced from external providers. For example, large contract manufacturers such as Flextronics and Solectron now often help firms design products in addition to manufacturing them. Companies such as IBM or Siemens will provide a company with a complete information technology solution, while United Parcel Service will take care of a company's logistics and distribution needs. Outsourcing can have a number of downsides, however. Reliance on outsourcing may cause the firm to forfeit important learning opportunities, potentially putting it at a disadvantage in the long run.[35] By not investing in development of in-house capabilities, a firm might not develop many of the skills and resources related to its products that enable the development of future product platforms. The firm risks becoming hollow.[36]

In fact, Prahalad and Hamel argue that Korean firms such as Goldstar, Samsung, and Daewoo have explicit missions to capture investment initiative away from potential competitors by serving as contract manufacturers for them. This allows the Korean firms to use would-be competitors' funds to accelerate their own competence development, while the competitors' competencies erode.[37]

Outsourcing can also impose significant transaction costs for a firm.[38] Contract manufacturing, for example, requires a well-specified contract: Product design, cost, and quantity requirements must be clearly communicated and generally specified up front. The contracting firm may also have to go to great lengths to protect itself from having any proprietary technology expropriated by the contract manufacturer. In addition, the contract manufacturer may bear significant costs in ramping up production for a particular firm, and must therefore specify the contract to avoid being held up by the contracting firm after the manufacturer has made investments specific to the contract.[39]

Collective Research Organizations

In some industries, multiple organizations have established cooperative research and development organizations such as the Semiconductor Research Corporation or the American Iron and Steel Institute.[40] Collective research organizations may take a number of forms, including trade associations, university-based centers, or private research corporations.

Many of these organizations are formed through government or industry association initiatives. For example, the National Center for Manufacturing Sciences (NCMS) was formed in 1986 by the U.S. Defense Department, the Association for Manufacturing Technology, the Manufacturing Studies Board, General Motors, and 20 other manufacturing companies. Its purpose was to promote collaborations among industry, government, and academic organizations. By 2012, the center had 175 U.S., Canadian, and Mexican corporate members. Typical NCMS projects involve 15 to 20 organizations and run for two to four years.[41]

Other collective research organizations have been formed solely through the initiative of private companies. For example, in 2002, six Japanese electronics manufacturers (Fujitsu, Hitachi, Matsushita Electric Industrial, Mitsubishi Electric, NEC, and Toshiba) set up a collective research company called Aspla to develop designs for more advanced computer chips. Global competition had driven down margins on chips, resulting in major losses for many of the major Japanese electronics makers. Furthermore, research in advanced chip designs had become extremely expensive. The collaborative research organization would enable the companies to share the development expense and help the Japanese semiconductor industry retain its competitive edge. Each of the companies initially invested 150 million yen ($1.3 million) in the organization, and plans were for each to contribute about $85 million annually toward joint research.[42] The Japanese government also agreed to contribute $268 million.

CHOOSING A MODE OF COLLABORATION

Figure 8.2 summarizes some of the trade-offs between solo internal development and various modes of collaboration. Solo internal development is, on average, a relatively slow and expensive way of developing a technology. The firm bears all the costs and

FIGURE 8.2
Summary of Trade-Offs between Different Modes of Development

	Speed	Cost	Control	Potential for Leveraging Existing Competencies	Potential for Developing New Competencies	Potential for Accessing Other Firms' Competencies
Solo Internal Development	Low	High	High	Yes	Yes	No
Strategic Alliances	Varies	Varies	Low	Yes	Yes	Sometimes
Joint Ventures	Low	Shared	Shared	Yes	Yes	Yes
Licensing In	High	Medium	Low	Sometimes	Sometimes	Sometimes
Licensing Out	High	Low	Medium	Yes	No	Sometimes
Outsourcing	Medium/High	Medium	Medium	Sometimes	No	Yes
Collective Research Organizations	Low	Varies	Varies	Yes	Yes	Yes

risks, and may spend considerable time learning about the new technology, refining its designs, and developing production or service processes to implement the new technology. However, a firm that engages in solo internal development retains total control over how the technology is developed and used. Solo internal development also offers great potential for the firm to leverage its existing competencies and to develop new competencies, but offers little to no potential for accessing another firm's competencies. Therefore, solo internal development might make sense for a firm that has strong competencies related to the new technology, has access to capital, and is not under great time pressure.

Because strategic alliances can take many forms, the speed, cost, and degree of control they offer vary considerably. Some strategic alliances may enable a firm to relatively quickly and cheaply gain access to another firm's technology, but give the firm a low level of control over that technology. Other strategic alliances might be aimed at utilizing the firm's own technology in a broader range of markets, which can be fast and cost-effective, and still enable the firm to retain a considerable amount of control. Most alliances offer opportunities for leveraging existing competencies or developing new competencies. Strategic alliances may or may not offer potential for accessing another firm's competencies, depending on the alliance's purpose and structure.

By comparison, a joint venture is much more structured. While a joint venture typically involves developing a new technology and can take almost as long as solo internal development, it may be slightly faster due to the combination of the capabilities of multiple firms. Joint ventures enable partners to share the cost of the development effort, but they must also share control. Because joint ventures typically entail a long-term relationship between two or more firms that results in the development of a new product or business, joint ventures offer great potential for leveraging a firm's existing competencies, developing new competencies, and accessing its partners' competencies.

Joint ventures may be more appropriate than a strategic alliance or solo internal development when the firm places great importance on access to other firms' competencies.

Licensing in technology offers a fast way to access a new technology that is typically lower in cost than developing it internally. The firm typically has limited discretion over what it can do with the technology, however, and thus has a low degree of control. Depending on the firm's capability mix and the nature of what it has licensed, licensing can sometimes offer the potential of leveraging a firm's existing competencies, developing new competencies, and accessing another organization's competencies. For example, many potential drugs or medical therapies are first developed in university research centers or medical schools. Pharmaceutical and biotechnology firms then license the right to explore whether the discovery has potential as a commercially viable medical treatment using their own drug development, testing, and manufacturing capabilities. Licensing the promising compounds or therapies enables the pharmaceutical and biotechnology firms to obtain drug targets quickly, thus helping them keep their pipelines full. It also helps the firms focus their development efforts on projects that have already demonstrated some treatment potential.

Licensing can also be a good way for a firm to obtain enabling technologies that are necessary for its products or services, but that are not central to the firm's competitive advantage. For example, while producers of digital cameras need to be able to incorporate batteries that are long-lasting, light, and affordable in their camera designs, most camera producers do not perceive battery power as being central to their competitive advantage, and thus rely on externally sourced technology to meet this need. Licensing can also be an effective way for a firm that lacks technological expertise to gain initial market entry and experience that it can later build upon in developing its own technological capabilities.

Licensing out a technology offers a fast way for a firm to extend the reach of its technology that is nearly free and offers the potential for royalties. The firm relinquishes some control over the technology, but also retains a moderate amount of control through restrictions in the license agreement. Licensing out a technology explicitly leverages the firm's existing competencies by enabling the technology to be deployed in a wider range of products or markets than the firm participates in itself. It offers little opportunity for developing new competencies, however. Sometimes licensing out a technology is a way of accessing another firm's competencies, as when a firm uses licensing to expand its technology into products or markets in which it has little expertise.

When a firm outsources design, production, or distribution of its technology, it is intentionally giving up a moderate amount of control to rapidly gain access to another firm's expertise and/or lower cost structure. While the firm pays to outsource activities, it typically pays less than it would to develop the capability of performing those activities in-house, and it gains access to those activities more quickly than it could develop them in-house. While outsourcing offers little opportunity for building new competencies, it can leverage the firm's existing competencies by enabling it to focus on those activities in which it earns its greatest returns. For example, Nike's strategy of outsourcing nearly all its athletic shoe production to contract manufacturers in Asia enables Nike to focus on its competitive advantages in design and marketing while tapping the lower labor and capital costs of its manufacturers. Thus, outsourcing might sometimes be appropriate for (*a*) firm activities that are not central to its competitive

advantage, (*b*) activities that would cause the firm to give up crucial flexibility if performed in-house, or (*c*) activities in which the firm is at a cost or quality disadvantage.

Participation in a collective research organization is typically a long-term commitment rather than an effort to rapidly access capabilities or technology. As with strategic alliances, the nature of a firm's participation in a collective research organization can take many forms; thus, cost and control can vary significantly. Collective research organizations can be very valuable ways for the firm to leverage and build upon its existing competencies, as well as to learn from other participating organizations. Though collective research organizations may not yield immediate returns in the form of new products or services, participating in collective research organizations can be extremely useful in industries that have complex technologies and require considerable investment in basic science. By pooling their knowledge and effort, firms in collective research organizations can share the cost and risk of basic research, while accelerating the rate at which it yields useful new solutions.

CHOOSING AND MONITORING PARTNERS

Gaining access to another firm's skills or resources through collaboration is not without risks.[43] It may be difficult to determine if the resources provided by the partner are a good fit, particularly when the resource gained through the collaboration is something as difficult to assess as experience or knowledge. It is also possible that a collaboration partner will exploit the relationship, expropriating the company's knowledge while giving little in return. Furthermore, since managers can monitor and effectively manage only a limited number of collaborations, the firm's effectiveness at managing its collaborations will decline with the number of collaborations to which it is committed. This raises the possibility of not only diminishing returns to the number of collaborations, but also negative returns as the number of collaborations grows too large.[44] These risks can be minimized if the company limits the number of collaborations in which it engages, chooses its partners very carefully, and establishes appropriate monitoring and governance mechanisms to limit opportunism.[45]

Partner Selection

The success of collaborations will depend in large part on the partners chosen. A number of factors can influence how well suited partners are to each other, including their relative size and strength, the complementarity of their resources, the alignment of their objectives, and the similarity of their values and culture.[46] These factors can be boiled down to two dimensions: resource fit and strategic fit.[47]

Resource fit refers to the degree to which potential partners have resources that can be effectively integrated into a strategy that creates value.[48] Such resources may be either complementary or supplementary. Most collaborations are motivated by the need to access resources the firm does not possess; such collaborations are based on the combination of complementary resources. Most of the examples in this chapter have entailed complementary resources, such as the combination of Apple's computer technology with Canon's printer engine technology, or the combination of Sangamo's gene editing technology with the clinical testing and manufacturing expertise of larger

biotechnology and pharmaceutical firms. Other collaborations seek supplementary stocks of resources that are similar to those possessed by the firm. The pooling of supplementary resources can enable partners to achieve market power or economies of scale. For example, British Petroleum and Mobil consolidated many of their operations in Europe to gain economies of scale and lower their cost structure.[49]

Strategic fit refers to the degree to which partners have compatible objectives and styles. The objectives of the partners need not be the same as long as the objectives can be achieved without harming the alliance or the partners. Not knowing a partner's true objectives or forging an alliance with a partner with incompatible objectives can result in conflict, wasted resources, and forfeited opportunities. Das and Teng provide an example of an alliance forged between General Motors and South Korea's Daewoo. While GM desired to use the alliance to drive down costs on its existing automobile models, Daewoo's objective was to develop new technologies and design new models. The alliance ultimately failed because of the incompatibility of GM's cost orientation and Daewoo's R&D orientation.[50]

Firms can also evaluate potential partners using many of the same tools used to evaluate the firm's own position and strategic direction (for a review of these, see Chapter Six). This includes assessing how collaboration with the partner is likely to impact the firm's opportunities and threats in its external environment; its internal strengths, weaknesses, or potential for sustainable competitive advantage; and the firm's ability to achieve its strategic intent.

Impact on Opportunities and Threats in the External Environment

Assessing the collaboration's impact on the firm's opportunities and threats includes asking such questions as:

- How would the collaboration change the bargaining power of customers or suppliers?
- Would the collaboration impact the threat of entry? For example, is the partner likely to become a new competitor? Does the partnership raise barriers to entry for other potential entrants?
- Would the collaboration impact the firm's position vis-à-vis its rivals?
- Would the collaboration influence the availability of complementary goods or the threat of substitutes?

Impact on Internal Strengths and Weaknesses

Assessing the collaboration's impact on the firm's strengths and weaknesses includes asking such questions as:

- How would the collaboration leverage or enhance the firm's strengths? Does the collaboration put any of those strengths at risk?
- How would the collaboration help the firm overcome its weaknesses?
- Is the collaboration likely to yield a position of competitive advantage that is difficult for competitors to imitate? Is such a competitive advantage achievable without collaborating?
- Would the collaboration leverage or enhance the firm's core capabilities?
- Is the collaboration likely to impact the firm's financial strengths or weaknesses?

Research Brief Strategic Positions in Collaborative Networks[a]

A growing body of research suggests that a firm's position within a collaborative network influences its access to information and other resources, and its influence over desired outcomes. For example, a firm with a highly central position in the network is typically expected to have access to a greater amount of information and to be able to access that information more quickly than a firm in a more peripheral role. A firm that occupies a key brokerage role in a network (e.g., a firm that serves as a bridge between two otherwise disconnected groups of firms) is thought to benefit both by having exposure to diverse information (assuming the two groups of firms have quite distinct information resources) and by occupying a key gatekeeping position that controls the flow of information between the two groups. A firm's position within the network may also serve as a valuable signal to other potential partners about the value of its resources. For example, if a firm is young or small but has alliances with important and innovative firms, these alliances can serve as reputation endorsements when the quality of the firm is otherwise difficult to assess.[b] Such endorsements may enhance the firm's likelihood of receiving financing or attracting other important alliances.

Consider Figure 8.3, which shows the "main component" (the largest connected group) of the global technology collaboration network in 1998 (based on R&D alliances, cross-technology transfer agreements, and cross-licensing agreements formed from 1996 to 1998, as reported by SDC's alliance database).[c] The large group on the top of the network is mostly composed of organizations in industries whose underlying technology is electronics-based (computer hardware and software, communication equipment and service, transportation equipment, etc.), and the group on the bottom is dominated by organizations in the chemical and medical-based industries (pharmaceuticals, chemicals, health services, medical equipment, etc.). This grouping also includes a large concentration of educational organizations (primarily universities). As can be seen, some firms (e.g., IBM, Toshiba, Eli Lilly) have significantly more alliances than others. The number of links an organization has in a network is known as its "degree centrality." In general, the degree centrality of an organization tends to be strongly related to its size and prominence. The size and prominence of an organization help to determine how attractive it is to potential partners, and only large organizations typically have the resources necessary to manage a large number of alliances. An organization does not, however, have to be large or prominent to occupy a key brokerage position. Brokerage refers to how crucial an organization is to the transmission of information or other resources through the network. It is often measured with "betweenness centrality," which is the number of times an organization lies on the shortest path between other pairs of organizations. The three organizations with the highest betweenness centrality scores in this network are IBM, Eli Lilly, and PPD (Pharmaceutical Product Development Inc., a contract research organization). PPD had only three alliances during the 1996 to 1998 time period, but Figure 8.3 shows just how important those alliances were to the overall connectivity of the network. IBM's link to PPD and PPD's link to Eli Lilly provide a bridge from the center of the electronics group to the center of the chemical/medical group. This link is one of only three observed bridges between the two groups, and is the most central of those three.

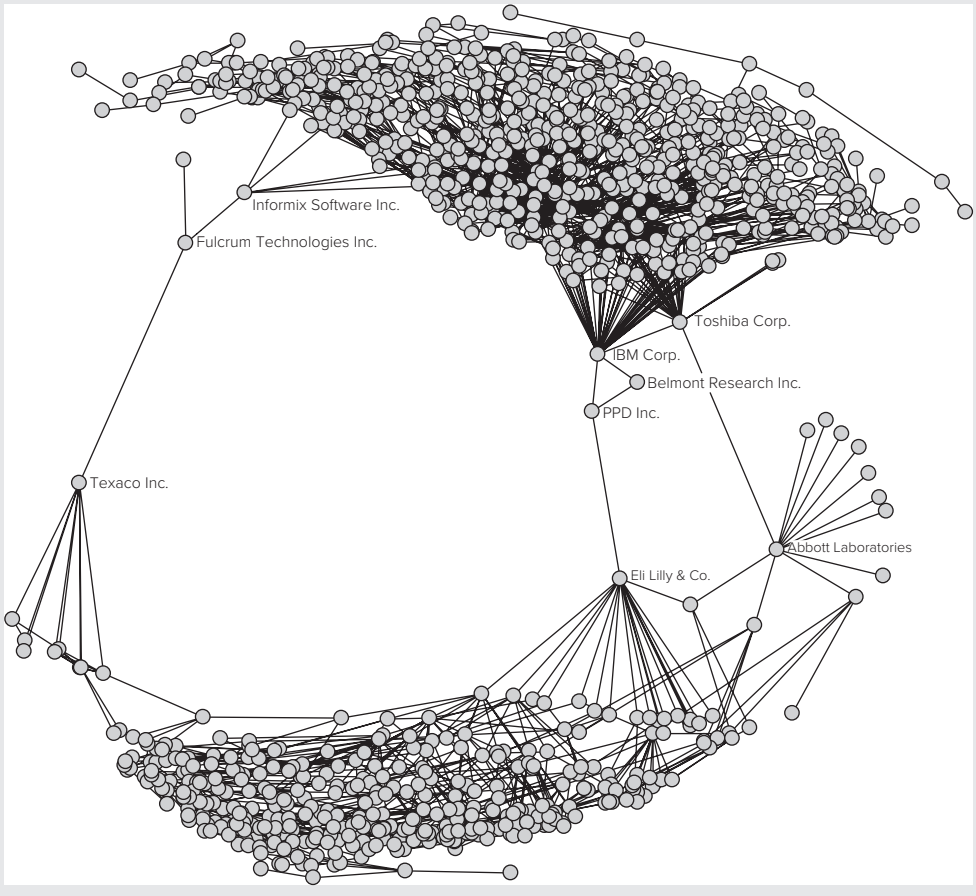
There is still considerable debate about the relative benefits of centrality and brokerage. While many scholars argue that highly central firms have the greatest access to information and influence over information transmission, others argue that highly central firms are constrained by their many relationships to other organizations and suggest that it is better to occupy a brokerage role. There are similar debates about brokerage—while a broker is likely to have access to diverse information and serves as a key gatekeeper for the transmission of information between otherwise

continued

concluded

FIGURE 8.3
The Global Technology Collaboration Network (Main Component) in 1998[d]

Adapted from Schilling, "The Global Technology Collaboration Network: Structure, Trends, and Implications."



disconnected groups, it is unclear to what degree brokers typically benefit from this position. Some have argued that brokers can have difficulty assimilating and utilizing such diverse information, and that it might be better to be fully embedded in one group rather than be the only bridge between multiple groups. In other words, it might be better to have a relationship with a broker than to be the broker. There generally is consensus, however, that it is less desirable to be isolated (i.e., not connected to the network) or in a "pendulum" position (i.e., have only one link, and thus hang from the network like a pendulum).

[a] Adapted from Melissa A. Schilling, "The Global Technology Collaboration Network: Structure, Trends, and Implications," New York University working paper, 2009.
[b] Toby E. Stuart, "Interorganizational Alliances and the Performance of Firms: A Study of Growth and Innovation Rates," *Strategic Management Journal* 21 (2000): 791–811.
[c] Adapted from Schilling, "The Global Technology Collaboration Network: Structure, Trends, and Implications."
[d] Ibid.

Impact on Strategic Direction

Assessing the fit of the collaboration with the firm's strategic direction includes asking such questions as:

- How does this collaboration fit with the firm's statement of strategic intent?
- Is the collaboration likely to help the firm close any resource or technology gap between where it is now and where it would like to be?
- Are the objectives of the collaboration likely to change over time? How are such changes likely to be compatible or incompatible with the firm's strategic direction?

Partner Monitoring and Governance

governance
The act or process of exerting authority and/or control.

Successful collaboration agreements typically have clear, yet flexible, monitoring and **governance** mechanisms.[51] Not surprisingly, the more resources put at risk by the collaboration (e.g., the greater the upfront investment or the more valuable the intellectual property contributed to the collaboration), the more governance structure partner firms are likely to impose on the relationship.[52] There are three main types of governance mechanisms organizations use to manage their collaborative relationships: alliance contracts, equity ownership, and relational governance.[53]

alliance contracts
Legally binding contractual arrangements to ensure that partners (a) are fully aware of their rights and obligations in the collaboration and (b) have legal remedies available if a partner should violate the agreement.

Alliance contracts are legally binding contractual arrangements to ensure that partners (a) are fully aware of their rights and obligations in the collaboration and (b) have legal remedies available if a partner should violate the agreement. Such contracts typically include:

- What each partner is obligated to contribute to the collaboration, including money, services, equipment, intellectual property, and so on.
- How much control each partner has in the arrangement. For example, the contract may stipulate whether partners have the right to admit new partners to the relationship or change the terms of the agreement. It may also stipulate the rights partners will have over any proprietary products or processes developed in the course of the collaboration.
- When and how proceeds of the collaboration will be distributed. For example, the collaboration agreement may stipulate whether cash, intellectual property rights, or other assets will be distributed and the schedule of such distribution.

Such contracts also often include mechanisms for monitoring each partner's adherence to the agreement, such as through regular review and reporting requirements.[54] Some collaboration agreements include provisions for periodic auditing either by the partner organizations or by a third party. Many agreements also include provisions for terminating the relationship if the need for the alliance ends (e.g., if the mission of the alliance is completed or the goals of the partner firms have changed) or partners encounter disputes they cannot resolve.[55] Markets and strategies change over time, and effective collaboration agreements should be flexible enough to be adapted in the event of change and provide a graceful exit strategy for members that no longer wish to participate in the collaboration.

equity ownership
When each partner contributes capital and owns a specified right to a percentage of the proceeds from the alliance.

Many alliances involve shared **equity ownership**, that is, each partner contributes capital and owns a share of the equity in the alliance. Equity ownership helps

relational governance Self-enforcing norms based on goodwill, trust, and reputation of the partners. These typically emerge over time through repeated experiences of working together.

to align the incentives of the partners (because the returns to their equity stake are a function of the success of the alliance) and provides a sense of ownership and commitment to the project that can facilitate supervision and monitoring of the alliance.

Finally, many alliances also rely on **relational governance**. Relational governance is the self-enforcing governance based on the goodwill, trust, and reputation of the partners that is built over time through shared experiences of repeatedly working together. Research suggests that relational governance can help to reduce contracting and monitoring costs of managing an alliance, and facilitate more extensive cooperation, sharing, and learning by alliance partners.[56]

Summary of Chapter

1. A number of factors will influence whether a firm chooses to collaborate on an innovation. Some of the most important include whether the firm (or a potential partner) has the required capabilities or other resources, the degree to which collaboration would make proprietary technologies vulnerable to expropriation by a potential competitor, the importance the firm places on controlling the development process and any innovation produced, and the role of the development project in building the firm's own capabilities or permitting it to access another firm's capabilities.
2. Firms may choose to avoid collaboration when they already possess the necessary capabilities and other resources in-house, they are worried about protecting proprietary technologies and controlling the development process, or they prefer to build capabilities in-house rather than access a partner firm's capabilities.
3. Some of the advantages of collaboration include sharing costs and risks of development, combining complementary skills and resources, enabling the transfer of knowledge between firms and the joint creation of new knowledge, and facilitating the creation of shared standards.
4. The term *strategic alliances* refers to a broad class of collaboration activities that may range from highly structured (e.g., joint ventures) to informal. Strategic alliances can enable simple pooling of complementary resources for a particular project, or they may enable the transfer of capabilities between partners. The transfer of capabilities often requires extensive coordination and cooperation.
5. A joint venture is a partnership between firms that entails a significant equity investment and often results in the creation of a new separate entity. Joint ventures are usually designed to enable partners to share the costs and risks of a project, and they have great potential for pooling or transferring capabilities between firms.
6. Licensing involves the selling of rights to use a particular technology (or other resource) from a licensor to a licensee. Licensing is a fast way of accessing (for the licensee) or leveraging (for the licensor) a technology, but offers little opportunity for the development of new capabilities.
7. Outsourcing enables a firm to rapidly access another firm's expertise, scale, or other advantages. Firms might outsource particular activities so that they can avoid the fixed asset commitment of performing those activities in-house. Outsourcing can give

a firm more flexibility and enable it to focus on its core competencies. Overreliance on outsourcing, however, can make the firm hollow.

8. Groups of organizations may form collective research organizations to jointly work on advanced research projects that are particularly large or risky.
9. Each form of collaboration mode poses a different set of trade-offs in terms of speed, cost, control, potential for leveraging existing competencies, potential for developing new competencies, or potential for accessing another firm's competencies. An organization should evaluate these trade-offs in formulating a collaboration strategy.
10. Successful collaboration requires choosing partners that have both a resource fit and a strategic fit.
11. Successful collaboration also requires developing clear and flexible monitoring and governance mechanisms to ensure that partners understand their rights and obligations, and have methods of evaluating and enforcing each partner's adherence to these rights and obligations.

Discussion Questions

1. What are some advantages and disadvantages of collaborating on a development project?
2. How does the mode of collaborating (e.g., strategic alliance, joint venture, licensing, outsourcing, and collective research organization) influence the success of a collaboration?
3. Identify an example of collaboration between two or more organizations. What were the advantages and disadvantages of collaboration versus solo development? What collaboration mode did the partners choose? What were the advantages and disadvantages of the collaboration mode?
4. If a firm decides it is in its best interest to collaborate on a development project, how would you recommend the firm choose a partner, a collaboration mode, and governance structure for the relationship?

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Chapter **Nine**

Protecting Innovation

The Patent Battle Over CRISPR-Cas9 Gene Editing

On October 7th, 2020, the Royal Swedish Academy of Sciences awarded the Nobel Prize in Chemistry to Professors Emmanuelle Charpentier and Jennifer Doudna for their contribution to the development of a method of genome editing called CRISPR-Cas9. Referring to the method as "genetic scissors," the Nobel Prize committee noted that "This technology has had a revolutionary impact on the life sciences, is contributing to new cancer therapies and may make the dream of curing inherited diseases come true."

Charpentier had been studying *Streptococcus pyogenes* (a bacteria that causes scarlet fever, pneumonia, and numerous other life-threatening infections) when she discovered a previously unknown molecule that is part of bacteria's ancient immune system, CRISPR/Cas. Bacteria can use this immune system to disarm viruses by cleaving their DNA. Charpentier was working at Umeå University in Sweden when she published her discovery in 2011. In that same year, she initiated a collaboration with Jennifer Doudna, a very experienced biochemist at University of California, Berkeley.[a] Together, they were able to recreate the bacteria's genetic scissors and demonstrated that they could be reprogrammed to cut any DNA molecule at a chosen site—they could rewrite the genetic code of a cell!

CRISPR-Cas9 offered a relatively easy way to genetically modify a living cell, offering tremendous hope for addressing disease. The most obvious opportunities were monogenic ("one gene") diseases like cystic fibrosis, Huntington disease, and sickle-cell anemia. However, genetic editing could also be used to address nongenetic diseases. For example, a small percentage of people have a mutation in their *CCR5* gene—a gene that makes a protein on the surface of cells—that makes it difficult for HIV to enter their cells. These people have some natural protection against HIV. Scientists speculated that a gene editing system like CRISPR-Cas9 could be used to give people the mutation to protect against HIV or even to help eliminate HIV in people already living with the disease. In fact, a company called Sangamo was already using a different gene editing system in clinical trials to attempt this feat.[b]

As the example above reveals, CRISPR-Cas9 was not the first method of gene editing to be discovered. However, its mechanism—harnessing an

innate property of bacteria—was vastly simpler than the other methods that were in development. It was so simple, in fact, that researchers and students around the world began rapidly adopting CRISPR-Cas9 in their research programs and the race was on to apply it to curing disease, augmenting crops, and more. By 2020, CRISPR-Cas9 had already been used to genetically modify mosquitos so that they could not carry malaria, and to restore the efficacy of front-line chemotherapies for lung cancer.[c] It was the breakthrough of a lifetime.

But who owned the intellectual property rights to CRISPR-Cas9? That turned out to be a complicated question. Doudna and Charpentier had filed their first CRISPR-Cas9 patent application in May of 2012 at the U.S. Patent and Trademark Office (USPTO), followed by many more applications both at the USPTO, and at patent offices in Europe and Canada, claiming the initial May 2012 priority date. A few months later in December of 2012, another team at the Broad Institute (affiliated with MIT and Harvard) led by Feng Zhang also filed their first (of many) patent application for CRISPR-Cas9 at the USPTO, followed by applications at other patent offices, claiming the December 2012 priority date. Because patent applications are kept secret for the first 18 months after filing, at first neither team realized that they were filing competing patent applications.

What transpired next became one of the most interesting patent battles in recent history. In 2011, the United States had passed the *Leahy-Smith America Invents Act (AIA*) that changed the patent system from a "first to invent" to a "first to file" system of priority. Though "first to invent" is more closely aligned with the underlying intention of the patent system, "first to file" is significantly easier (and cheaper) to verify and had become the dominant system around the world.[d] Doudna and Charpentier had filed their patent applications first, in 2012, and thus understandably expected their patents would receive priority over Zhang's patents. However, the law did not go into effect until March 16, 2013, and Zhang had requested to file under the "first to invent" system, claiming his invention dated back to before the law changed.[e] Zhang had also requested expedited review, which led to his patent applications being granted before the Doudna-Charpentier patent applications. To make matters even more complicated, Doudna and Charpentier's first papers only demonstrated the use of CRISPR-Cas9 in single celled organisms while Zhang's first papers—albeit 7 months later—were the first to show that CRISPR-Cas9 could be used in eukaryotes (multi-celled organisms). Since the most valuable application of gene editing were likely to be applications to multi-celled organisms (like humans!), this was a critical distinction.

After several rounds of appeals filed by Doudna and Charpentier with the U.S. Patent Trial and Appeal Board (PTAB), the PTAB concluded that the patents were not actually in conflict: Doudna and Charpentier would be granted the patent for a critical tool in the use of CRISPR-Cas9 (a single molecule guide RNA which became the standard method of using CRISPR-Cas9) and Zhang would be granted the patent for application to multi-celled organisms. Canada would follow the priority decision made in the United States. In Europe, however, the European

Patent Office drew a different conclusion about priority dates and awarded all of the first generation CRISPR-Cas9 patents to Doudna and Charpentier.[f]

As of 2021, most parties interested in commercializing CRISPR-Cas9 applications needed to obtain licenses from different organizations. For example, to develop CRISPR-Cas9 therapies for humans, rights must be obtained from CRISPR Therapeutics (which had obtained exclusive rights from Emmanuelle Charpentier), Intellia Therapeutics (which obtained technology rights from Jennifer Doudna), and Editas Medicine (which had licensed patents from Feng Zhang).[g] Many people hoped that the organizations would cooperate to form a patent pool under a single organization that could streamline the process of obtaining rights to commercialize CRISPR, and the Broad Institute (with which Zhang was affiliated) was pushing for this outcome, stating, "A complex patent and licensing landscape threatens innovation. The best thing, for the entire field, is for the parties to reach a resolution and for the field to focus on using CRISPR technology to solve today's real-world problems." However, as of April 2021, an agreement had not yet been struck.

Discussion Questions

1. What do you think are some of the challenges with awarding patents based on "first to invent"?
2. What are the pros and cons of Doudna–Charpentier and Zhang pooling their CRISPR-Cas9 patents to make it easier for others to license them?
3. What are some of the potential implications of the fact that the European Patent Office made a different decision about the patent battle than the U.S. Patent and Trademark Office?
4. Do you think patents on a revolutionary technology with medical applications helps to foster the development of lifesaving treatments or slows it down?

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OVERVIEW

A crucial element of formulating a firm's technological innovation strategy is determining whether and how to protect its technological innovation. Traditionally, economics and strategy have emphasized the importance of vigorously protecting an innovation in order to be the primary beneficiary of the innovation's rewards, but the decision about whether and to what degree to protect an innovation is actually complex. Sometimes *not* vigorously protecting a technology is to the firm's advantage—encouraging other producers (and complementary goods providers) to support the technology may increase its rate of diffusion and its likelihood of rising to the position of dominant design. In this chapter, we first will review the factors that shape the degree to which a firm is likely to appropriate the returns from its innovation, and the mechanisms available to the firm to protect its innovation. We then will consider the continuum between a wholly proprietary strategy and a wholly open strategy, examining the trade-offs inherent in decisions about whether (and to what degree) to protect or diffuse a technological innovation. The chapter concludes by listing factors the firm should consider in formulating its protection strategy.

APPROPRIABILITY

appropriability
The degree to which a firm is able to capture the rents from its innovation.

The degree to which a firm can capture the rents from its innovation is termed **appropriability**. In general, the appropriability of an innovation is determined by how easily or quickly competitors can imitate the innovation. The ease with which competitors can imitate the innovation is, in turn, a function of both the nature of the technology itself and the strength of the mechanisms used to protect the innovation.

tacit knowledge
Knowledge that cannot be readily codified or transferred in written form.

socially complex knowledge
Knowledge that arises from the interaction of multiple individuals.

Some technological innovations are inherently difficult for competitors to copy; the knowledge underlying the technology may be rare and difficult to replicate. A firm's unique prior experience or talent pool may give it a foundation of technical know-how that its competitors do not possess. If this knowledge base is **tacit** (i.e., it cannot be readily codified into documents or procedures) or **socially complex** (i.e., it arises through complex interactions between people), competitors will typically find it very difficult to duplicate.[1] For example, a firm that has a team of uniquely talented research scientists may have a rare and difficult-to-imitate knowledge base. While some of the skill of the research scientists may be due to imitable training procedures, *talent* typically implies that an individual (or group) has a natural endowment or ability that is very difficult, if not impossible, to replicate through training. Furthermore, if the unique capabilities of the research team arise in part from the nature of the interactions between the scientists, their performance will be socially complex. Interactions between individuals can significantly shape what each individual perceives, and thus what each individual—and the collective group—discovers or learns. The outcomes of these interactions are path dependent, and thus are idiosyncratic to the combination of individuals, the moment of the interaction, and the nature of the interaction. This means that knowledge can emerge from the interaction of a group that could not be replicated by any individual or any different group.

Many innovations, however, are relatively easy for competitors to imitate. Individuals and firms often employ legal mechanisms to attempt to protect their innovations. Most countries offer legal protection for intellectual property in the form of patent, trademark, copyright, and trade secret laws.

PATENTS, TRADEMARKS, AND COPYRIGHTS

patent
A property right protecting a process, machine, manufactured item (or design for manufactured item), or variety of plant.

trademark
An indicator used to distinguish the source of a good.

copyright
A property right protecting works of authorship.

While patents, copyrights, and trademarks are all ways of protecting intellectual property, they are each designed to protect different things. A **patent** protects an invention, and a **trademark** protects words or symbols intended to distinguish the source of a good. A **copyright** protects an original artistic or literary work. Thus, a typical computer might have components whose designs are protected by patents, logos such as the Starbuck's mermaid that are protected by trademark law, and software that is protected by copyright (though as discussed later in the section on patents, many types of software are now also eligible for patent protection).

The purpose of intellectual property protection is to provide recognition and incentive for creative work. Patents and copyrights, for example, provide a legal means for individuals to protect their creative work and earn rewards from it, in exchange for making the knowledge underlying their work public. Making this knowledge public is important because it helps others to build upon that knowledge, driving technological and social advance. In absence of those protections, people might prefer to keep their discoveries and inventions secret.

Patents

In many countries, inventors can apply for patent protection for their inventions. An invention can be a product, such as a new type of battery, or a process, such as a new way to manufacture bagels. In the United States, patents are categorized into different types such as a *utility* patent for a new and useful process, machine, manufactured item, or combination of materials; a *design* patent for an original and ornamental design for a manufactured item; or a *plant* patent for the discovery and asexual reproduction of a distinct and new variety of plant.

Each country has its own patent system with different requirements, and unless a patent is filed under a regional patent office or an international treaty, the rights it is granted are applicable only in the country in which the patent is filed.

To qualify for a patent, an invention must usually meet the following criteria:

1. It must be *useful* (i.e., it must produce a desirable result, solve a problem, improve on or propose a new use for an existing development or show potential of doing so).
2. It must be *novel* (i.e., it must not already be patented or described in public literature, or be in public use for more than a year).
3. It must *not be obvious* (i.e., a person with experience or skill in the particular art of the patent would not be expected to achieve the same invention with a normal amount of effort).

In most countries, the discovery of scientific principles that pertain to natural laws (e.g., gravity) cannot be patented because they are considered to have always existed. Additionally, the following are not typically patentable:

- Substituting one material for another (e.g., plastic for metal).
- Merely changing the size of an already existing device.
- Making something more portable.
- Substituting an element for an equivalent element.
- Altering an item's shape.

Printed materials are not typically patentable, but it may be possible to protect them by copyright, as discussed in the next section. Historically software algorithms were not considered patentable, but in 1998, a U.S. Supreme Court case upheld a patent on a computerized method of managing mutual funds that relied on software algorithms, unleashing a flood of patent applications for software. From 1997 to 2000, patent filings in the United States for software-enabled methods of doing business increased more than 700 percent.[2] For example, Amazon patented its "1-click" system that streamlines the process by which customers place orders.[3]

Patenting an invention is a serious undertaking. To apply for a patent, the inventor must explain how to make and use the invention, and make claims about what it does that makes it a new invention. Drawings of the new invention are also often required. In the United States, this application is reviewed by a patent examiner who may modify the scope of the claims made by the patent. The patent is then published for a time in which other inventors can challenge the patent grant (if, e.g., they believe that the patent infringes on previously granted patents). If the standards for patentability are met, the patent is then granted. The entire process from application to granting is lengthy. For example, the time from filing to grant is between two and five years in the United States, three and five years in Europe, five and six years in Japan, and five and seven years in India. In industries in which product lifecycles are short, such delays significantly diminish the usefulness of patenting. This has led to a number of proposals for how the patenting system might be reformed to make it more efficient.

The number of patent applications being filed is growing around the world, but is highest (by far) in China, where over 1.4 million patent applications were filed in 2019 alone (see Figure 9.1).

A number of costs are also involved in filing and maintaining a patent. Fees vary by patenting office. In the United States, the entire patenting process for a "small entity" (e.g., an independent inventor, a small business, or a nonprofit organization) costs around $1500 in filing fees (and roughly double that for large entities), and $5000–$15,000 in attorney fees. In most countries, patent protection lasts for 20 years.

In a major study of historical success rates for U.S. patent applications, Michael Carley, Deepak Hegde, and Alan Marco followed the history of the 2.15 million new patent applications filed at the USPTO after 1996 and found that only 55.8 percent of the applications became granted. Patent applications in the "Drugs and Medical Instruments" sector had the lowest success on average (42.8 percent), and applications in the "Electrical and Electronics" had the highest (66.6 percent). They also found that success was generally lower for small firms and that overall, success rates for all types of applications had gone down over time.[4]

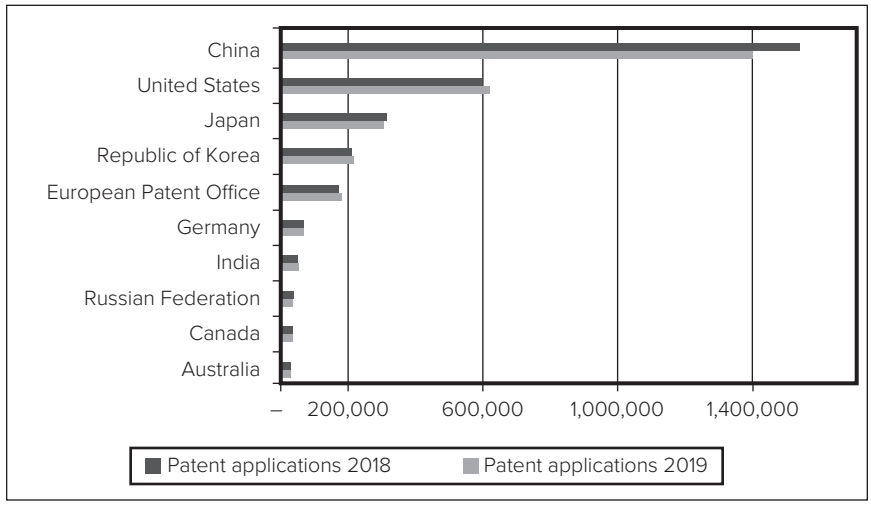


FIGURE 9.1
Patent Applications in 2018 and 2019 in the 10 Largest Patent Offices in the World

Source: World Intellectual Property Organization.

Major International Patent Treaties

There is currently no "world patent," and a patent granted in one country does not automatically provide protection in other countries. In some regions, however, there are regional patent offices (such as the European Patent Office and the Africa Regional Intellectual Property Organization) that grant patents valid in all the member nations of that program.

Significant differences exist in national patent laws. For example, in most countries, publication of information about the invention before applying for a patent will bar the right to a patent, but the United States allows a one-year grace period (i.e., an inventor can publish an invention up to a year before applying for the patent). Thus, if international patent protection will eventually be sought, inventors must uphold the stricter standard of applying for patent before publishing information about the patent, even if they plan to first patent the invention in the United States. Many countries also require that the invention be manufactured in the country in which a patent was granted within a certain time frame (often three years) from the time the patent is granted. This is called the "working requirement," and it effectively prevents inventors from patenting inventions in countries in which they have no intention of setting up production.

Many inventors wish to patent their inventions in many countries simultaneously. To make that easier, several international treaties have been negotiated between countries that seek to harmonize the patent laws around the world. Two of the most significant are the Paris Convention for the Protection of Industrial Property and the Patent Cooperation Treaty.

The *Paris Convention for the Protection of Industrial Property* (also known as the Paris Convention Priority) is an international intellectual property treaty adhered to by 177 countries as of March 2021. Under the Paris Convention, a citizen of any member country may patent an invention in any of the member countries and enjoy the same benefits of patent protection as if the inventor were a citizen of those countries.

That is, the Paris Convention eliminates (for its member countries) any differential patent rights afforded to citizens of the country versus foreign nationals. Furthermore, the treaty also provides the right of "priority" for patents and trademarks. Once an inventor has applied for patent protection in one of the member countries, the inventor may (within a certain time period) apply for protection in all the other member countries. The time period is 12 months for utility patents and six months for design patents and trademarks. Most important, the applications to these later countries will be treated as if they were made on the same date as the first application. This enables the inventor to establish priority over any other patents applied for in those countries after the inventor made the first application. For example, if an inventor applied for a utility patent for an invention in Madagascar in January 2003, and another inventor applied for a patent for a very similar invention in France in June 2003, the Madagascar inventor could have applied for patent protection in France in December 2003 and claim priority over the French invention. The French inventor would have to prove that his or her invention was substantively different from the Madagascar invention, or the French inventor's patent would be denied.

As mentioned previously, in many countries, public disclosure of an invention makes it impossible to subsequently patent that invention. However, with the priority rights established under the Paris Convention, an inventor who patents an invention in one of the member countries can then publicly disclose information about that invention without losing the right to patent the invention in the other countries—each patent application will be treated as if it were applied for at the same time as the first application, and thus as if it were applied for before public disclosure. Without this treaty, it would be nearly impossible for an inventor to patent an invention first in the United States and then in other countries because U.S. patent applications are made available to the public.

Another very significant international patent treaty is the *Patent Cooperation Treaty*, or PCT. This treaty facilitates the application for a patent in multiple countries. An inventor can apply for a patent to a single PCT governmental receiving office, and that application reserves the inventor's right to file for patent protection in 153 countries for up to two-and-half years. Once the inventor has filed the application, a PCT governmental searching office will perform the patent search for the application (this search verifies that the invention is not already subject to a prior claim). Once the search is completed, the inventor can choose to enter Chapter II of the process wherein the PCT governmental office assesses the patentability of the invention subject to the standards of the Patent Cooperation Treaty. Eventually, the inventor must have the PCT application filed in each of the national patent offices in which the inventor is seeking protection.

Filing a single PCT application offers numerous advantages. First, applying for the PCT patent buys the inventor the option to apply to multiple nations later without committing the inventor to the expense of those multiple applications. With a PCT application, the inventor can establish a date of application in multiple countries (protecting the inventor's priority over later claims), while paying only the single PCT application fee rather than the numerous national application fees. Though the inventor will eventually have to pay for national applications in the countries in which protection is sought, the inventor can delay those costs. Thus, the inventor has time to assess the likelihood

of the patent being granted and the potential profitability of the invention. If the PCT process suggests that the patent will not be granted or if it appears the invention has limited potential for earning returns, the inventor can forgo the expense of applying to the national offices.

Another advantage of the PCT process is that it helps make the results of patent applications more uniform. Though individual countries are not required to grant a patent to those inventions that are granted a patent by the PCT governing office, the granting of the patent by the PCT provides persuasive evidence in favor of granting the patent in the individual national offices.

Patent Strategies

It is typical to assume that an inventor seeks a patent because they desire to make and sell the invention themselves. However, inventors and firms may monetize patents in a range of different ways, including licensing the technology to others or selling the patent rights to another firm that can better utilize the technology.[5] Furthermore, whereas the conventional wisdom is that most inventors prefer to keep the details of their invention secret before the patent is granted (to prevent rivals from having access to their proprietary knowledge), this turns out not to be the case. A study by Stuart Graham and Deepak Hegde found that the vast majority of patentees prefer to disclose their patent applications *before* they are granted. Both large and small inventors, across all major technology fields exhibited this preference for early disclosure, presumably because it allows them to publicize their invention's quality and scope to competitors, external investors, and potential licensees. Disclosure via patent application also establishes the date from which patentees can enjoy provisional patent rights.[6]

Firms may also seek patents just to limit the options of competitors or to earn revenues through aggressive patent lawsuits. These actions are sometimes referred to as "**patent trolling**." A patent troll's primary purpose in owning patents is to extort money from other firms. For example, a patent troll might buy a patent from a bankrupt firm to sue another company that it claims is infringing on the purchased patent. Apple claims to be the number one target for patent trolls, having faced nearly 100 lawsuits between 2011 and 2014.[7] In the United States, Patent Assertion Entities (businesses that acquire patents from third parties and seek to generate revenue by targeting those who are using technologies that might be protected by the patents), filed 1656 cases in 2020 alone.[8] This type of predatory patenting has sparked an effort by the U.S. Federal Government to make patent granting stricter and to impose penalties against spurious patent lawsuits.[9]

patent trolling
A pejorative term for when an individual or firm misuses patents against other individuals or firms in attempt to extract money from them.

In industries with complex technologies such as computers, software, and telecommunications, a dense web of overlapping patents known as "**patent thickets**" can make it very difficult for firms to compete without falling prey to patent suits by other firms in that technology domain. This can seriously stifle innovation and has resulted in the rather peculiar strategy of firms buying bundles of patents to create war chests that they hope will deter the patent attacks of others. For example, in 2011, the bankrupt Nortel auctioned off its massive patent portfolio. The auction was won by a consortium called Rockstar Bidco that included Microsoft, Apple, RIM, Sony, and Ericsson, who paid $4.5 billion for the war chest, beating out Google which bid $4.4 billion. Google subsequently bought 1030 IBM patents that covered a range

patent thickets
A dense web of overlapping patents that can make it difficult for firms to compete or innovate.

of technologies, from the fabrication of microprocessing chips, object-oriented programming, and other business processes. These patents were not necessary for Google's business directly; rather they provided a retaliation threat to others that might attack them through patent suits.[10] Google also bought Motorola Mobility for $12.5 billion the same year, and it was widely believed that the purchase was almost solely for Motorola's patents, which would bolster Google's position in the lawsuit they expected would arise from the Nortel patents.[11] In October of 2013, confirming Google's fears, Rockstar Bidco filed suit against Google and seven companies that make phones for Google's Android operating system.[12] Google countersued based on its own patents, and in November 2014, it was reported that Google and Rockstar had reached a settlement.[13]

Trademarks and Service Marks

A trademark is a word, phrase, symbol, design, or other indicator that is used to distinguish the source of goods from one party from the goods of others. The "Intel Inside" logo on many computers is one example of a trademark, as is the familiar Nike "swoosh" symbol. A service mark is basically the same as a trademark, but distinguishes the provider of a service rather than a product. Often the term *trademark* is used to refer to both trademarks and service marks.

Trademarks and service marks can be embodied in any indicator that can be perceived through one of the five senses. Most marks are embodied in visual indicators, such as words, pictures, and slogans. However, marks are also registered that use other senses such as sound (e.g., tones that are associated with a particular company or brand) or smells (as in fragrance). Trademark rights may be used to prevent others from using a mark that is similar enough to be confusing, but they may not be used to prevent others from producing or selling the same goods or services under a clearly different mark.

In most countries, the rights to a trademark or service mark are established in the legitimate use of the mark and do not require registration; however, registration provides several advantages. First, registering the mark provides public notice of the registrant's claim of ownership over the mark. Second, marks must be registered before a suit can be brought in federal court against an infringement of the mark. Third, registration can be used to establish international rights over the trademark, as when the U.S. registration is used to establish registration in other countries, and to protect the mark against infringement from imported products. As of April 2021, the U.S. Patent and Trademark Office charged between $275 and $650 application fee to register a trademark. It normally took 10 to 16 months to receive certification from the U.S. Patent and Trademark Office, but the protection offered by the registration of the trademark begins from the date of filing. Unlike patents and copyrights, trademark protection can last as long as the trademark is in use, but the registration requires periodic renewal.

Major International Trademark Treaties

Nearly all countries offer some form of trademark registration and protection. National or regional offices maintain a Register of Trademarks that contains information on all trademark registrations and renewals. To eliminate the need to register separately in each country (or region), the World Intellectual Property Organization

administers a System of International Registration of Marks governed by two treaties: the Madrid Agreement Concerning the International Registration of Marks and the Madrid Protocol. Countries that adhere to either (or both) the Madrid Agreement or Madrid Protocol are part of the Madrid Union. Any individual that lives in, is a citizen of, or maintains an establishment in a Madrid Union country can register with the trademark office of that country and obtain an international registration that provides protection in as many other Madrid Union countries as the applicant chooses. As of June 2019, there were 104 member countries of the Madrid Union.

Copyright

Copyright is a form of protection granted to works of authorship. In the United States, the authors of original literary, dramatic, musical, artistic, and certain other intellectual works can obtain copyright protection.[14] Like trademarks, the rights of copyright protection are established by legitimate use of the work. This protection is available whether or not the work is published and prevents others from producing or distributing that work. Under section 106 of the 1976 Copyright Act, the owner of the copyright has the exclusive right to do (or authorize others to do) the following:

- Reproduce the work in copies or phonorecords.
- Prepare derivative works based upon the work.
- Distribute copies or phonorecords of the work to the public by sale or other transfer of ownership, or by rental, lease, or lending.
- Perform the work publicly, in the case of literary, musical, dramatic, and choreographic works, pantomimes, and motion pictures and other audiovisual works.
- Display the copyrighted work publicly, in the case of literary, musical, dramatic, and choreographic works, pantomimes, and pictorial, graphic, or sculptural works, including the individual images of a motion picture or other audiovisual work.
- Perform the work publicly by means of a digital audio transmission (in the case of sound recordings).

There are, however, limitations to these rights. In particular, in the United States, the doctrine of fair use stipulates that in most circumstances it is not a violation of copyright for others to use copyrighted material for purposes such as criticism, comment, news reporting, teaching, scholarship, or research. Furthermore, some types of work cannot be protected by copyright. For example, work that has not been fixed in a tangible form of expression (e.g., a choreographed dance or improvisational speech that was not notated or recorded) is not eligible for copyright protection. Titles, names, short phrases, slogans, familiar symbols, and lists of ingredients also cannot be copyrighted.

Unlike patent protection, copyright protection is secured automatically when an eligible work is created and fixed in a copy or phonorecord for the first time. No publication or registration with the Copyright Office is necessary to establish this copyright, though registering the copyright is advantageous in that it establishes a public record of the copyright claim and is required before filing an infringement suit in court. As of April 2021, basic online registration of copyright with the U.S. Copyright Office costs $35, and it took about 3 to 10 months to receive a certificate of registration.

Before 1978, copyright protection lasted only 28 years from the time it was secured (though during the last year the author could opt to renew the protection for an additional term). Revisions to U.S. copyright law, however, give copyright protection to works created after 1978 that lasts for the author's life plus an additional 70 years.

Copyright Protection around the World

As with patents and trademarks, no international copyright law automatically protects an author's work throughout the world. Copyright protection varies from country to country. However, most countries do offer copyright protection to both domestic and foreign works, and there are international copyright treaties for simplifying the process of securing such protection. One of the most significant is the Berne Union for the Protection of Literary and Artistic Property (known as the Berne Convention). The Berne Convention specifies a minimum level of copyright protection for all member countries, and it requires member countries to offer the same protection to both its own citizens and foreign nationals. Other treaties include the Universal Copyright Convention (UCC); the Rome Convention for the Protection of Performers, Producers of Phonograms and Broadcasting Organizations; the Brussels Convention Relating to the Distribution of Program-Carrying Signals Transmitted by Satellite; and the World Intellectual Property Organization Copyright Treaty.

TRADE SECRETS

trade secret Information that belongs to a business that is held private.

Rather than disclose detailed information about a proprietary product or process in exchange for the grant of a patent, inventors or firms often will choose to protect their intellectual property by holding it as a **trade secret**. A trade secret is information that belongs to a business that is generally unknown to others. Trade secrets need not meet many of the stringent requirements of patent law, enabling a broader class of assets and activities to be protectable. For example, while the formula for a beverage is not patentable, it can be considered a trade secret. Trade secret law traces its history back to Roman law punishing individuals who induced someone to reveal the details of their employer's commercial affairs.[15]

Information is typically considered to be a trade secret only if it (*a*) offers a distinctive advantage to the company in the form of economic rents, and (*b*) remains valuable only as long as the information remains private. Examples of trade secrets might include information about a firm's customers, its marketing strategies, or its manufacturing processes. Trade secret law protects such information from being wrongfully taken by another party. In the United States, trade secret law is implemented at the state level, but the Uniform Trade Secret Act attempts to make these laws consistent from state to state.

For information to qualify as a trade secret under the Uniform Trade Secret Act, the information must meet the following three criteria:

- The information must not be generally known or readily ascertainable through legitimate means.
- The information must have economic importance that is contingent upon its secrecy.
- The trade secret holder must exercise reasonable measures to protect the secrecy of the information.

If information meets these criteria, the Uniform Trade Secret Act stipulates that its owner can prevent others from benefiting from this information without the owner's permission. In particular, the act states that no individual or group can copy, use, or otherwise benefit from a trade secret without the owner's authorization if they meet *any* of the following conditions:

1. They are bound by a duty of confidentiality (e.g., employees and lawyers).
2. They have signed a nondisclosure agreement.
3. They acquire the secret through improper means such as theft or bribery.
4. They acquire the information from someone who did not have the right to disclose it.
5. They learn about the secret by mistake but have reason to know that the information was a protected trade secret.

In most regions, if owners of a trade secret believe that another party has stolen or improperly disclosed their trade secret, they can ask a court to issue an injunction against further use of the secrets, and they may also be able to collect damages for any economic harm suffered by the improper use of the trade secret. For example, in November 2002, Procter & Gamble sued Potlatch Corporation, claiming that Potlatch had stolen trade secret methods used to produce Bounty paper towels and Charmin bath tissue by hiring away two of Procter & Gamble's paper manufacturing experts. Potlatch is a large, private-label tissue manufacturer that produces toilet paper, facial tissues, napkins, and paper towels for grocery store chains such as Albertsons and Safeway. By March 2003, the two companies had reached an agreement to settle out of court, keeping the terms of the settlement confidential.[16]

THE EFFECTIVENESS AND USE OF PROTECTION MECHANISMS

The methods used to protect innovation—and their effectiveness—vary significantly both within and across industries.[17] In some industries, such as pharmaceuticals, legal protection mechanisms such as patents are very effective. In other industries, such as electronics, patents and copyright provide relatively little protection because other firms can often invent around the patent without infringing on it (as IBM discovered when Compaq was able to produce a computer nearly identical to its personal computer design). It is also notoriously difficult to enforce patents protecting industrial processes such as manufacturing techniques. If patents provide little protection, the firm may rely more heavily on trade secrets; however, the ability to protect trade secrets also varies with the nature of the technology and the industry context. To protect a trade secret, a firm must be able to expose its product to the public without revealing the underlying technology, but in many cases, revealing the product reveals all.

For some competitive situations, protecting a technology may not be as desirable as liberally diffusing it. In industries characterized by increasing returns, firms sometimes choose to liberally diffuse their technologies to increase their likelihood of rising to the position of dominant design. As discussed in Chapter Four, learning-curve effects and network externalities can cause some industries to demonstrate increasing returns to adoption: The more a technology is adopted, the more valuable it becomes.[18]

This dynamic can lead to winner-take-all markets that create natural monopolies. A firm that controls the standard can reap monopoly rents and can exert significant architectural control over both its own industry and related industries.[19]

This enviable position can be so lucrative that firms may be willing to lose money in the short term to improve their technology's chance of rising to the position of dominant design. Thus, firms may liberally diffuse their technologies (through, e.g., **open source software** or liberal licensing arrangements) to accelerate the technology's proliferation and thereby jump-start the self-reinforcing feedback effect that can lead to the technology's dominance. However, the firm often faces a dilemma: If it liberally diffuses the technology to would-be competitors, it relinquishes the opportunity to capture monopoly rents when and if the technology emerges as a dominant design. Furthermore, once control of a technology is relinquished, it can be very hard to regain; thus, such diffusion may result in the firm losing all hope of controlling the technology. Finally, liberal diffusion of the technology can result in the fragmentation of the technology platform: As different producers add improvements to the technology that make it better fit their needs, the "standard" may be split into many nonstandardized versions (as with UNIX, as described in more detail later in the chapter). To resolve these trade-offs, firms often adopt a strategy of partial protection for their innovations, falling somewhere on the continuum between wholly proprietary systems and wholly open systems.

open source software
Software whose code is made freely available to others for use, augmentation, and resale.

Wholly Proprietary Systems versus Wholly Open Systems

Wholly proprietary systems are those based on technology that is company-owned and protected through patents, copyrights, secrecy, or other mechanisms. Such technologies may be legally produced or augmented only by their developers. Wholly proprietary systems are often not compatible with the products offered by other manufacturers. Because their operation is based on protected technology, other manufacturers are often unable to develop components that may interact with the proprietary system. Proprietary systems typically provide their developers with the opportunity to appropriate rents from the technology. However, they might also be less likely to be adopted readily by customers as a result of their higher costs and the inability to mix and match components.

wholly proprietary systems
Goods based on technology that is owned and vigorously protected through patents, copyrights, secrecy, or other mechanisms. Wholly proprietary technologies may be legally produced and augmented only by their developers.

In **wholly open systems**, the technology used in a product or process is not protected by secrecy or patents; it may be based on available standards or it may be new technology that is openly diffused to other producers. Wholly open technologies may be freely accessed, augmented, and distributed by anyone. Such technologies are usually quickly commoditized and provide little appropriability of rents to their developers.

wholly open systems
Goods based on technology that is not protected and that is freely available for production or augmentation by other producers.

Many technologies are neither wholly proprietary nor wholly open—they are partially open, utilizing varying degrees of control mechanisms to protect their technologies. It is useful to think of a control continuum that stretches from wholly proprietary to wholly open (see Figure 9.2). For instance, most of the major video game console producers (Nintendo, Sony, and Microsoft) utilize a wholly proprietary strategy for their consoles, but a limited licensing policy for their games. The licensing policies are designed to encourage developers to produce games for the systems, while enabling the console producers to retain a great deal of control over the games produced. All games developed for the consoles must be approved by the console producer before

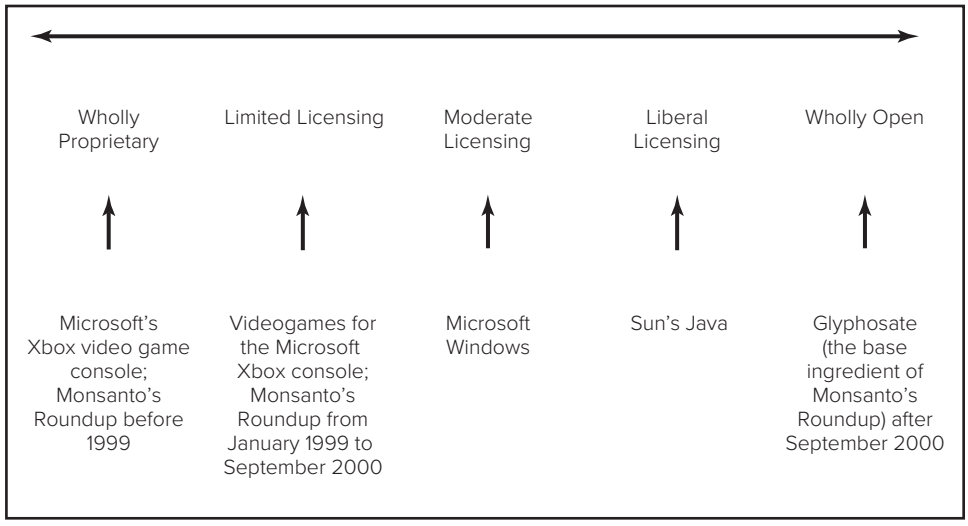


FIGURE 9.2
Examples on the Continuum from Wholly Proprietary to Wholly Open

they can be made commercially available. For example, in the case of Microsoft, would-be Xbox games developers must first apply to the Xbox Registered Developer Program (for established games developers) or the Xbox Incubator Program (for smaller or newer games developers). If accepted into one of these two programs, the developer will receive access to development tools, but this does not guarantee the approval of any resulting game titles. The games are subjected to a separate, rigorous approval process.

By contrast, the licensing policies for Microsoft's Windows are more open. Windows is protected by copyright, and Microsoft defends its exclusive right to augment the software; however, it also permits complementary goods providers to access portions of the source code to facilitate development of complementary goods, licenses the rights to such providers to produce complementary applications, and licenses **original equipment manufacturers (OEMs)** to distribute the software by bundling it with hardware. Those who purchase a license for the software can execute and bundle the software with other goods but may not augment the software. For example, software applications developers may produce and distribute value-added applications for use with Windows as long as those applications do not affect the functionality of the Windows program itself.

original equipment manufacturers (OEMs)
Firms that assemble goods using components made by other manufacturers, also called value-added resellers (VARs).

Sun's "community source" (as opposed to "open source") policy for Java is even more open. This policy grants anyone immediate access to the complete source code for Java and allows users to develop commercial applications based on the code, or to augment the code for their own implementations. These developers pay no license fee to Sun. However, any augmentation to the core structure of Java must be approved by the Java Community Process, which is managed by Sun. Sun's "community source" principle is meant to encourage the broader software community to improve Java and develop complementary applications, but it allows Sun to retain some control over the core platform to ensure that the platform does not become fragmented through unmanaged development by the software community.

Theory in Action IBM and the Attack of the Clones

In 1980, IBM was in a hurry to introduce a personal computer. When personal computers first began to emerge at the end of the 1970s, most of the major computer manufacturers considered it no more than a peculiar product for a hobbyist market. The idea that individuals would want personal computers on their desks seemed ludicrous. However, as total U.S. personal computer sales reached $1 billion, IBM began to worry that the personal computer market could actually turn out to be a significant computer market in which IBM had no share. To bring a personal computer to market quickly, IBM decided to use many off-the-shelf components from other vendors, including Intel's 8088 microprocessor and Microsoft's software. However, IBM was not worried about imitators because IBM's proprietary basic input/output system (BIOS), the computer code that linked the computer's hardware to its software, was protected by copyright. While other firms could copy the BIOS code, doing so would violate IBM's copyright and incur the legendary wrath of IBM's legal team.

However, getting around IBM's copyright turned out not to be difficult. Copyright protected the written lines of code, but not the functions those codes produced. Compaq was able to reverse-engineer the BIOS in a matter of months without violating IBM's copyright. First, a team of Compaq programmers documented every function the IBM computer would perform in response to a given command, without recording the code that performed the function. This list of functions was then given to another team of "virgin" programmers (programmers who were able to prove that they had never been exposed to IBM's BIOS code).[a] These programmers went through the list of functions and wrote code to create identical functions. The result was a new BIOS that acted just like an IBM BIOS but did not violate its copyright. Compaq sold a record-breaking 47,000 IBM-compatible computers in its first year, and other clones were quick to follow.

[a] Robert Cringely, *Accidental Empires* (New York: HarperCollins, 1992).

Many technologies that were once wholly proprietary or partially open become wholly open once their patents or copyrights expire. For instance, Monsanto's highly profitable Roundup herbicide is based on a patented chemical ingredient called glyphosate. This extremely potent herbicide was adopted by farmers in more than 100 countries and accounted for a substantial portion of Monsanto's sales.[20] However, facing impending expiration of its patents, Monsanto began to license the rights to glyphosate production to a few other companies (including Dow Agrosciences, DuPont, and Novartis) in 1999. In September 2000, the U.S. patent on glyphosate expired, and any chemical company was free to produce and sell glyphosate-based herbicides in the United States, making glyphosate a wholly open technology.

Advantages of Protection

Because proprietary systems offer greater rent appropriability, their developers often have more money and incentive to invest in technological development, promotion, and distribution. If a single firm is the primary beneficiary of its technology's success, it has much greater incentive to invest in further developing the technology. The profits from the technology may be directly reinvested in further improvements in the technology. The sponsor of a proprietary technology might also adopt a penetration pricing strategy (i.e., it may offer the technology at a low price or free) to rapidly build its installed base, it may spend aggressively on advertising to increase awareness of the technology, and it

may even subsidize the production of complementary goods to increase the desirability of its technology to customers. A firm may be willing to lose money in the short term to secure the technology's position as the standard, because once the technology has emerged as a standard, the payoff can be substantial and enduring. By contrast, when multiple firms can produce a technology, losing money on the technology in the short term to promote it as a standard is highly risky because the long-term distribution of the payoffs is uncertain. While the technology's developer may have borne the bulk of the cost in developing the technology, multiple firms may vie for the profits to be made on the technology.

architectural control
The ability of a firm (or group of firms) to determine the structure, operation, compatibility, and development of a technology.

Protecting the technology also gives the developing firm architectural control over the technology. **Architectural control** refers to the firm's ability to determine the structure and operation of the technology, and its compatibility with other goods and services. It also refers to the firm's ability to direct the future development path of the technology. Architectural control can be very valuable, especially for technologies in which compatibility with other goods and services is important. By controlling the technology's architecture, the firm can ensure that the technology is compatible with its own complements, while also restricting its compatibility with the complements produced by others.[21] The firm can also control the rate at which the technology is upgraded or refined, the path it follows in its evolution, and its compatibility with previous generations. If the technology is chosen as a dominant design, the firm with architectural control over the technology can have great influence over the entire industry. Through selective compatibility, it can influence which other firms do well and which do not, and it can ensure that it has a number of different avenues from which to profit from the platform.

Microsoft's Windows is the quintessential embodiment of this strategy. Because Windows is the dominant operating system in the personal computing market and because it serves as the interface between a computer's hardware and software, Microsoft has considerable market power and architectural control over the evolution of the personal computer system. Among other things, Microsoft has been able to incorporate ever more utility programs into the core program, thereby expanding and taking over the roles of many other software components. Once a user purchased an operating system, uninstaller programs, disk-compression programs, and memory management programs separately, but Windows 95 and 98 integrated all these products and more into the operating system. This "feature creep" had a major impact on competition in the industry; many utility producers such as Qualitas, Stac Electronics, Microhelp, Quarterdeck, and others were forced to abandon their once-profitable products.

Advantages of Diffusion

The primary argument for diffusing a technology instead of protecting it is that open technologies may accrue more rapid adoptions. If multiple firms are producing, distributing, and promoting the technology, the technology's installed base may accumulate much more rapidly than if one firm alone is responsible for such activities. Competition among producers may drive the price of the technology down, making it more attractive to customers. Both customers and complementary goods providers

may also perceive the technology as better (or its future more certain) if there are multiple companies backing the technology. This perception can lead to much more rapid adoption of the technology by customers and complementary goods providers, which further stimulates more companies to back the technology. Thus, a liberal diffusion strategy can stimulate the growth of the installed base and availability of complementary goods.[22]

Google used a liberal diffusion strategy to help its Android operating system for smartphones become widely adopted. Google's liberal licensing policy for Android attracted many handset manufacturers and applications developers to the system, enabling its ecosystem to grow rapidly. A larger ecosystem with more phone models, price points, and applications, in turn, attracted a wider range of customers.

Open technologies can also benefit from the collective development efforts of parties external to the sponsoring firm. For instance, Netscape Navigator, UNIX, and Linux are all technologies that have benefited significantly from external development. By making the source code freely available to the vast world of developers who could benefit from the technology, the technologies reaped the advantages of having a much larger pool of talent and resources directed at improving the technologies than could have been rallied by the original developers.

External development, however, poses some costs and risks. First, external development efforts typically lack the coordination of internal development. External developers may have very diverse objectives for the technology; rather than work together toward some unified vision of what the technology could achieve in the future, they might work in different, possibly even conflicting, directions.[23] Much of their effort may be redundant, as different external developers work on solving the same problems without communicating with each other. Finally, whether and how these improvements get incorporated into the technology and disseminated to other users of the technology can prove very problematic. UNIX provides a stark example of this.

UNIX was an operating system first developed by AT&T's Bell Laboratories in 1969. Though a Department of Justice injunction forbade AT&T from selling software commercially, it made the source code for the product available through licensing arrangements. Early licensees (notably, University of California–Berkeley) began using and adapting the software for their purposes, causing many incompatible versions of the software to emerge. Though the software community made several attempts to standardize the UNIX operating language, their efforts failed. AT&T also challenged the commercialization of several UNIX variants, but to no avail. Ultimately, AT&T sold the division responsible for UNIX to Novell, and Novell handed over the rights to the UNIX trademark to the X/Open standards-setting body.[24]

Given the range of advantages (and risks) of protecting versus diffusing a technology, a firm must carefully consider the following factors in deciding whether, and to what degree, it should protect its innovation.

Production Capabilities, Marketing Capabilities, and Capital

If the firm is unable to produce the technology at sufficient volume or quality levels (or market the technology with sufficient intensity), then protecting the technology so that the firm is its sole provider may significantly hinder its adoption. For example, when JVC was promoting its VHS standard for video recorders, its management knew JVC

was at a disadvantage in both manufacturing and marketing capabilities compared to Sony (which was promoting the Beta technology). JVC chose to vigorously pursue both licensing and OEM agreements, lining up Hitachi, Matsushita, Mitsubishi, and Sharp to boost the technology's production rate.

Similarly, if complementary goods influence the value of the technology to users, then the firm must (*a*) be able to produce the complements in sufficient range and quantity, (*b*) sponsor their production by other firms, or (*c*) encourage collective production of the complements through a more open technology strategy. The only firms that have been successful in the U.S. video game industry were those that were able to produce games in-house (ensuring that a wide range of games would be available at the console's launch) and that encouraged third-party development of games (to ensure that the number of game titles grew quickly). Both Nintendo and Sega had previous arcade experience, and thus possessed considerable game development expertise. Microsoft had long been a producer of PC-based video games, so it had some game developing experience, and it also acquired a few small game developers (e.g., Bungie) to expand its expertise in developing console-type games.[25] Sony had no prior game experience, but aggressively acquired in-house developers, licensed external developers, and set up a program with Metrowerks to provide developer tools that would make it easier for external developers to produce PlayStation games. If a firm lacks the production capability or expertise to produce a sufficient range of complementary goods, or the capital to acquire such capabilities quickly, it should encourage collective production of complements through a more open technology strategy and utilize forms of sponsorship.

Industry Opposition against Sole-Source Technology

Sometimes other industry members are able to exert strong pressure against the adoption of a technology that would give one (or a few) producer(s) undue control and power, causing a technology that is restricted to such production to be rejected or more hotly contested than a more open technology. This was the case with Sony and Philips' Super Audio CD (SACD) audio format. Sony and Philips had jointly created the original compact disc (CD) format and split the royalties on every CD player sold, totaling hundreds of millions of dollars. The rest of the world's leading consumer electronics producers (including Hitachi, JVC, Matsushita, Mitsubishi, and Toshiba) and record producers (including Time Warner and Seagram's Universal Music group) banded together to form the Digital Video Disk (DVD) Audio consortium. This consortium's purpose is to promote the DVD Audio standard that is intended to displace the CD and enable royalties to be split among the 10 companies that control the patents.[26] Industry observers note that a driving force underlying the formation of the consortium was to prevent Sony and Philips from controlling yet another generation of audio formats. The degree of industry opposition to a sole-source technology needs to be considered when the firm formulates its technology strategy. If the industry is able to pose significant opposition, the firm may need to consider a more open technology strategy to improve the technology's likelihood of being chosen as a dominant design.

Resources for Internal Development

If a firm does not have significant resources (capital, technological expertise) to invest in the technology's functionality, it may have difficulty producing a technology that has

an initial performance level, and rate of improvement, that the market finds attractive. In such instances, it can be valuable to tap the external development efforts of other firms (or individuals) through utilizing a more open technology strategy. For example, when Netscape found itself in a race to match browser capabilities with Microsoft, it was at a tremendous disadvantage in both human resources and capital. Microsoft had legions of internal developers and a lot of money to invest in Explorer; there was no way that Netscape could match those resources internally. Instead, Netscape tapped the external development community by giving them access to its source code and incorporating their improvements into the Navigator product.

Control over Fragmentation

For technologies in which standardization and compatibility are important, maintaining the integrity of the core product is absolutely essential, and external development can put it at risk. As the UNIX example illustrates, if the developing firm relinquishes all control over the development of the technology, the technology will have no shepherd with the ability and authority to direct its trajectory and ensure that a single standard remains intact. This suggests that the developer of any technology that requires standardization and compatibility should retain some degree of control over the technology, or find/establish another governing body with the authority to do so.

Incentives for Architectural Control

Architectural control over the evolution of a technology is always valuable; however, it becomes particularly valuable if a firm is a significant producer of complements to the technology in question. A firm with architectural control can typically design the technology to be compatible with its own complements and incompatible with those of competitors. If the technology is chosen as the dominant design, this architectural control allows the firm to ensure that it reaps the lion's share of the rewards in complements production. Furthermore, by making the technology selectively compatible with some competitors and not others, the firm can exert great influence over the competitive field.

Architectural control can also enable the firm to direct the development efforts put into the technology so that it exploits the firm's core competencies. Technology trajectories are path dependent; minor events in their evolution can set them careening off into unexpected directions. A firm that has a significant stake in a particular evolution path (because, e.g., it has technological competencies that are much more amenable to one path of evolution than other potential paths) may place a high value on architectural control, which can enable it to co-opt or destroy less favorable development paths by denying their progenitors access to the market.

Summary of Chapter

1. The degree to which a firm can capture the rents from its innovation efforts is largely determined by the degree to which competitors can quickly and easily imitate the innovation. Some innovations are inherently difficult to copy; others are difficult to copy because of the mechanisms the firm uses to protect its innovation.
2. The three primary legal mechanisms used to protect innovation in most countries are patents, trademarks, and copyrights. Each mechanism is designed to protect a different type of work or good.

3. International treaties have helped to harmonize patent, trademark, and copyright laws around the world. Most countries now have patent, trademark, and copyright laws of some form, and in some instances protection can be applied for in multiple countries simultaneously.
4. Trade secrets provide another mechanism of protecting innovation. Firms that protect their intellectual property as a trade secret often have legal recourse if another party wrongfully takes and uses such property.
5. Legal mechanisms for protecting innovation are more effective in some industries than others; in some industries, inventing around a patent or copyright is relatively easy. Similarly, in some industries, it is nearly impossible to protect an innovation by using trade secrets because commercializing the innovation reveals its underlying technologies.
6. Sometimes the choice between protecting versus diffusing a technology is not obvious. Both strategies offer potential advantages. Many firms use neither a wholly open nor a wholly proprietary strategy, but rather a partially open strategy.
7. Protecting an innovation helps ensure that the firm earns the lion's share of the returns from the innovation. These returns can then be reinvested in further developing the technology, promoting the technology, and producing complementary goods.
8. Protecting an innovation also preserves the firm's architectural control, enabling it to direct the technology's development, determine its compatibility with other goods, and prevent multiple incompatible versions of the technology from being produced by other firms.
9. Diffusing a technological innovation can encourage multiple firms to produce, distribute, and promote the technology, possibly accelerating its development and diffusion. Diffusion can be particularly useful in industries that accrue increasing returns to adoption. It is also useful when the firm has inadequate resources to be the sole developer, producer, distributor, and marketer of a good.

Discussion Questions

1. What are the differences between patents, copyrights, and trademarks?
2. What factors should a firm considering marketing its innovation in multiple countries use in formulating its protection strategy?
3. When are trade secrets more useful than patents, copyrights, or trademarks?
4. Identify a situation in which none of the legal protection mechanisms discussed (patents, copyrights, trademarks, and trade secrets) will prove useful.
5. Describe a technological innovation not discussed in the chapter, and identify where you think it lies on the control continuum between wholly proprietary and wholly open.
6. What factors do you believe influenced the choice of protection strategy used for the innovation identified above? Do you think the strategy was a good choice?

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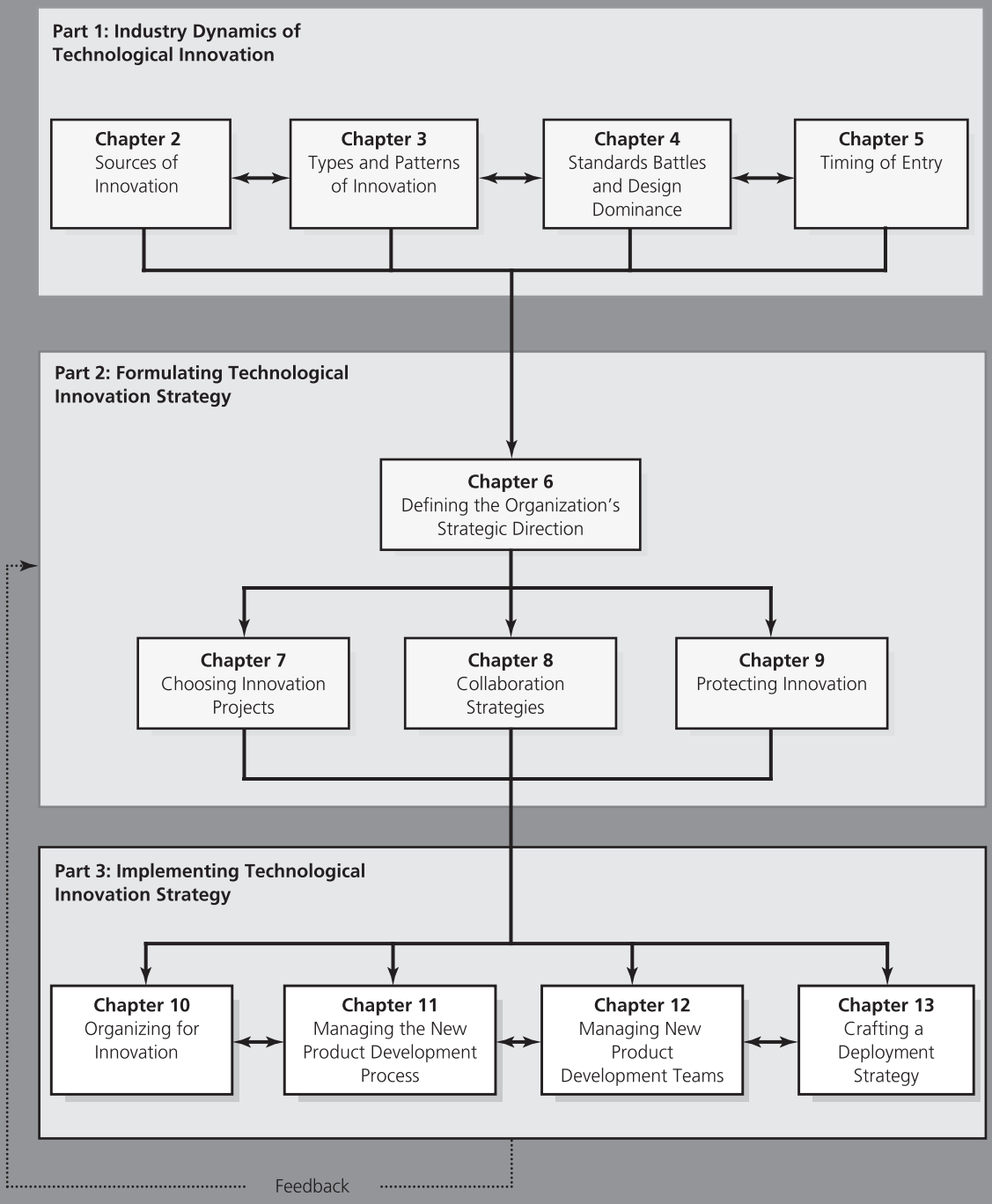
Part **Three**

Implementing Technological Innovation Strategy

In this section, we will cover the key aspects of implementing a technological innovation strategy, including:

- Structuring the firm to improve its likelihood of innovating, its effectiveness at new product development, and its speed of new product development.
- Managing new product development processes to maximize fit with customer needs, while simultaneously minimizing development cycle time and controlling development costs.
- Composing, structuring, and managing new product development teams to maximize new product development effectiveness.
- Crafting a strategy for effectively deploying the innovation into the marketplace, including timing, licensing strategies, pricing strategies, distribution, and marketing.

Implementing Technological Innovation Strategy



Chapter **Ten**

Organizing for Innovation

How Apple Organizes for Innovation

As of 2021, Apple Inc. was the world's largest technology company by revenue ($274.5 billion in 2020[a]), and the world's most valuable company by market capitalization (over $2 trillion as of March, 2021[b]). Apple produces a range of iconic products, including the Macintosh personal computers, iPhones, iPods, iPads, Apple Watches, Apple TV, iTunes, iMovie, GarageBand, and more. Apple typically designs both the hardware and software of its products, and even operates its own stores.

Given Apple's size and scope, one might expect it to have a multidivisional structure with business units based on products (smartphones, computers, music, software applications, etc.) and perhaps also with geographical divisions (like North America, Europe, Asia, etc.) While many firms are initially formed with a simple functional structure (i.e., one where departments are based on functions like manufacturing, marketing, and human resources), typically, as organizations grow and become more complex, they are reorganized into multidivisional structures where departments are based on products and/or markets. Leaders of these business units often have control over their key functions and accountability for their performance.[c] This enables a large organization to behave more like a federation of smaller organizations, simultaneously enabling them to focus their attention on their key markets and (hopefully) helping them stay agile.

Steve Jobs and Steve Wozniak founded Apple in 1976, but Steve Jobs resigned in 1985 after the Board of Directors and then CEO John Scully stripped him of his power and duties at the firm. The company, however, did not fare well without its intrepid leader and after over a decade of performance declines, Jobs agreed to come back. When Jobs returned to Apple in 1997, he found a multidivisional organization where General Managers ran divisions dedicated to products like the Macintosh, information appliances, server products and more.[d] However, the company was struggling to develop the next generation of its

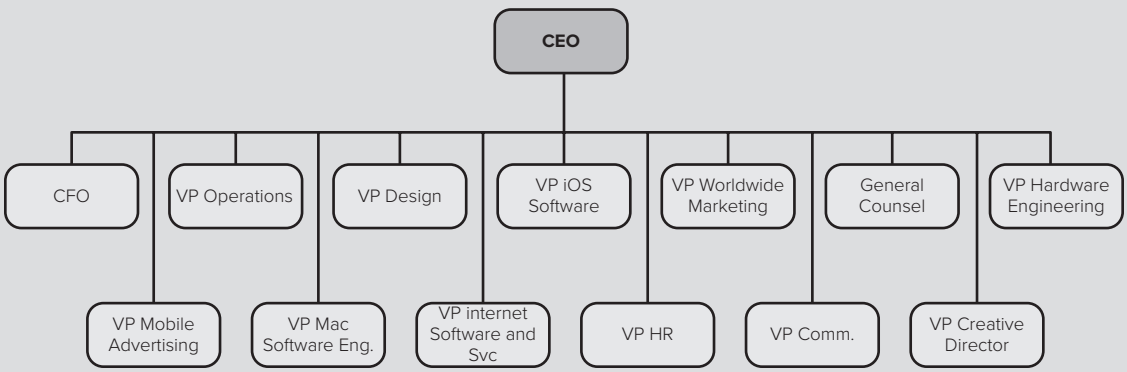
Mac operating system (it had invested $500 million but had experienced multiple delays and still did not have a market-ready product) and had incurred massive losses. Many were predicting that the firm would declare bankruptcy. The three CEOs that had sequentially reigned at Apple in Jobs' absence (John Scully, Mike Spindler, and Gilbert Amelio) had all been seasoned executives, but none had been capable of reviving Apple's magic.[e]

Jobs believed one of the problems with Apple was the multidivisional structure. The managers of the divisions tended to compete with each other rather than collaborate. They focused on the profit and loss statements ("P&Ls") of their individual divisions and were far less likely to create company-wide synergies. Furthermore, this meant that core areas of expertise, such as developing intuitive interfaces, for example, had to be spread across different business units, forfeiting depth and functional specialization in exchange for breadth. Jobs thus immediately laid off the general managers of all the business units and made the decision to organize Apple by functions. All the hardware engineers, for example, were consolidated into a single group that served the entire company. Likewise with software engineers. The entire company would have a single P&L. Not coincidentally, this structure would also give Jobs direct control over all operations at Apple, which was just the way Jobs liked it. Apple has grown significantly in both scale and scope since Jobs' 1998 reorganization, but it retains a (mostly) functional structure to this day (see Figure 1 below).[f]

Jobs believed that the functional structure offered several advantages for Apple. First, Jobs thought that only those people with deep technological expertise in their area should make decisions—"experts leading experts" rather than experts reporting to general managers.[g] These experts, he believed, would be more likely to hold strong views on optimal product design and to understand the products at a very deep level. Second, consolidating all the functional experts

FIGURE 1
Apple Organization Chart in 2021

Source: Adapted from information from Apple Annual Report



in an area into a single department gave Apple deep bench strength in that area and permitted more technical specialization. For example, Apple's computers, iPhones, iPads, and iPods all required highly intuitive interfaces. Having all the interface designers work together on all the products meant that some experts could focus on security, while others focused on reliability or responsiveness, etc. Each expert could leverage their best-in-class expertise across all the Apple product lines. Furthermore, having the functional experts together also helped to ensure that Apple's products work seamlessly together—a feature that has become core to the Apple experience. Third, because senior executives were rewarded based on overall company performance rather than the performance of a particular product, they were incentivized to create big innovations that could be leveraged company-wide.

Notably, Apple's organization structure puts a lot of responsibility on its CEO for coordinating across the functions. That can be an asset or a liability, depending on the CEO's talent, vision and expertise. This structure can also reinforce a focus on a relatively narrow range of products—a key tenet of strategy to which Jobs adhered. Whereas companies like Sony and Nike had become famous for using incremental innovation to rapidly spawn hundreds of different versions of their products—a phenomenon referred to as "mass customization"—Jobs preferred to invest all the company's money and energy on a few blockbuster products. As Jobs described, "People think focus means saying *yes* to the thing you've got to focus on. But that's not what it means at all. It means saying no to the hundred other good ideas that there are. You have to pick carefully. I'm actually as proud of the things we haven't done as the things we have done. Innovation is saying no to 1000 things."[h]

When Tim Cook took over from Steve Jobs (due to Jobs' untimely illness and death), he recognized the value of the organization Jobs had created and sought to preserve it, saying, "Apple has a culture of excellence that is, I think, so unique and special. I'm not going to witness or permit the change of it."[i] As of 2021, he had mostly preserved the functional structure Steve Jobs had put into place, but observers wondered just how long a behemoth like Apple could adhere to such a simple structure, and in particular whether it would work as well with someone other than Steve Jobs at its helm.

Discussion Questions

1. What are some of the advantages and disadvantages of having a functional structure at a company as big as Apple?
2. What kinds of companies do you think would benefit more (or less) from having the degree of centralized control that Apple uses?
3. Do you think Apple's functional structure influence the types of innovation it chooses to pursue? Do you think it influences Apple's success at innovation?
4. What do you think are the pros and cons of having a single P&L across the whole company?

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OVERVIEW

The structure of an organization and the degree to which it uses formalized and standardized procedures and controls can significantly influence its likelihood of innovating, the effectiveness of its innovation projects, and the speed of its new product development processes.[1] For example, it is often argued that small, flexible organizations with a minimum of rules and procedures will encourage creativity and experimentation, leading to more innovative ideas. At the same time, it is also frequently pointed out that well-developed procedures and standards can ensure that the organization makes better development investment decisions and is able to implement projects quickly and efficiently. How then do managers decide what structure and controls would make the most sense for their firm?

A vast majority of firms use some type of product team structure to organize their new product development process, and we will look closely at how teams are composed and structured in Chapter Twelve, Managing New Product Development Teams. This chapter focuses on the organization-wide structural dimensions that shape the firm's propensity and ability to innovate effectively and efficiently. We will review the research on how firm size and structural dimensions such as formalization, standardization, and centralization affect a firm's innovativeness. By focusing on these underlying structural dimensions, we will elucidate why some structures may be better for encouraging the creativity that leads to idea generation, while other structures may be better suited for efficient production of new products. We will also explore structural forms that attempt to achieve the best of both worlds—the free-flowing organic and entrepreneurial structures and controls that foster innovation, plus the formalized and standardized forms that maximize efficiency while ensuring coherence across all of the corporation's development activities. The chapter then turns to the challenge of managing innovation across borders. Multinational firms face particularly difficult

questions about where to locate—and how to manage—their development activities. We will review some of the work emerging on how multinational firms can balance the trade-offs inherent in these choices.

SIZE AND STRUCTURAL DIMENSIONS OF THE FIRM

Size: Is Bigger Better?

In the 1940s, Joseph Schumpeter challenged supporters of antitrust law by proposing that large firms would be more effective innovators.[2] Schumpeter pointed out that (1) capital markets are imperfect, and large firms are better able to obtain financing for R&D projects, and (2) firms with larger sales volume over which to spread the fixed costs of R&D would experience higher returns than firms with lower sales volume. Large firms are also likely to have better-developed complementary activities such as marketing or financial planning that enable them to be more effective innovators, and they are also likely to have greater global reach to obtain information or other resources.

Another advantage of size may arise in scale and learning effects. If large firms spend more on R&D in an absolute sense, they might also reap economies of scale and learning curve advantages in R&D—that is, they may get better and more efficient at it over time.[3] Through investing in R&D, the firm develops competencies in the new product development process and thus may improve its development process. It may accumulate better research equipment and personnel. Furthermore, as a large firm gains experience in choosing and developing innovation projects, it may learn to make better selections of projects that fit the firm's capabilities and have a higher likelihood of success.

Large firms are also in a better position to take on large or risky innovation projects than smaller firms.[4] For example, only a large company such as Boeing could develop and manufacture a 747, and only large pharmaceutical companies can plow millions of dollars into drug development in hopes that one or two drugs are successful.[5] This suggests that in industries that have large development scale (i.e., the average development project is very big and costly), large firms will tend to outperform small firms at innovation. In theory a coalition of small firms ought to achieve the same scale advantages, but in practice, coordinating a coalition of firms tends to be very difficult. While a single large firm can exert hierarchical authority over all of the development activities to ensure cooperation and coordination, coalitions often do not have such a well-defined system of authority and control.

On the other hand, as a firm grows, its R&D efficiency might decrease because of a loss of managerial control.[6] That is, the bigger a firm gets the more difficult it can become to effectively monitor and motivate employees. Furthermore, as a firm grows, it becomes increasingly difficult for individual scientists or entrepreneurs to appropriate the returns of their efforts; therefore their incentives diminish.[7] Thus, as the firm grows, the effectiveness of its governance systems may decrease.

Large firms may also be less innovative because their size can make them less nimble and responsive to change. Large firms typically have more bureaucratic inertia due to many layers of authority and well-developed policies and procedures.[8] For example, in the early 1980s, Xerox discovered that the administrative layers it had added to prevent errors in new product development had the unintended effect of blocking a project's

progress, making product development cycles unacceptably long and putting Xerox at a disadvantage to more nimble Japanese competitors.[9]

High numbers of employees, large fixed-asset bases, and a large base of existing customers or supplier contracts can also be sources of inertia, making it difficult for the firm to change course quickly. As the number of employees grows, communication and coordination may become more difficult and prone to decision-making delays. When large firms have large fixed-asset bases and/or significant fixed costs, they often prefer to stick with existing sources of cash flow rather than gambling on big changes. Strategic commitments to customers and suppliers can also tie the firm to its existing businesses and technologies, making it more difficult to respond to technological change. Strategic commitments can thus lead to an *Icarus Paradox*—a firm's prior success in the market can hinder its ability to respond to new technological generations.

Small firms are often considered more flexible and entrepreneurial than large firms. They are unencumbered by multiple layers of administration, large fixed-asset bases, or strategic commitments to large numbers of employees, customers, and suppliers. Small firms may also find it much simpler to monitor employees and reward them for their effort or success at innovation.[10] Because resources are less abundant, small firms may also be more motivated to choose projects more carefully, leading to higher rates of new product success.

A number of empirical studies have attempted to test whether large size improves or hampers innovation productivity. Several studies of patent counts, new drug introductions, and technological innovations that improve product performance have indicated that small firms often outperform large firms in innovation.[11] For example, a few studies of patenting output have concluded that small firms appear to spend their R&D dollars more carefully and are more efficient, receiving a larger number of patents per R&D dollar.[12] One study of 116 firms developing new business-to-business products also found that small firms (those with annual sales less than $100 million) had significantly shorter development cycles than large firms (those with $100 million and more in sales), even when considering the relative magnitude of the innovation.[13] However, a few studies have indicated that large firms may still outperform small firms in innovation in some industries.[14]

While the firm's overall size is not an easy-to-manipulate attribute of the firm, many firms have found ways of making even large firms feel small. One primary method is to break the overall firm into several smaller subunits, and then encourage an entrepreneurial culture within these subunits. Multiple studies have observed that in industries characterized by high-speed technological change, many large and hierarchical firms have been **disaggregated** (or "unbundled") into networks of smaller, often more specialized, autonomous divisions or independent firms.[15] In such industries, many firms have undergone large-scale downsizing, with many functions and layers of management eliminated. The giant multidivisional firms of the twentieth century were replaced by leaner firms that were more focused and flexible, loosely coupled in a network of alliances, supplier relationships, and distribution agreements.[16] This phenomenon led to the rise of terms such as *virtual organization,*[17] *network organization,*[18] and *modular organization.*[19]

disaggregated When something is separated into its constituent parts.

Since firms also use big company–small company hybrids to vary other structural dimensions of the firm (including formalization, standardization, and centralization), these ambidextrous approaches to organizing will be covered in more depth after the structural dimensions of the firm are reviewed.

STRUCTURAL DIMENSIONS OF THE FIRM

Firms vary on a number of structural dimensions that can influence the amount, type, and effectiveness of their innovation. Key structural dimensions include centralization, formalization, and standardization.

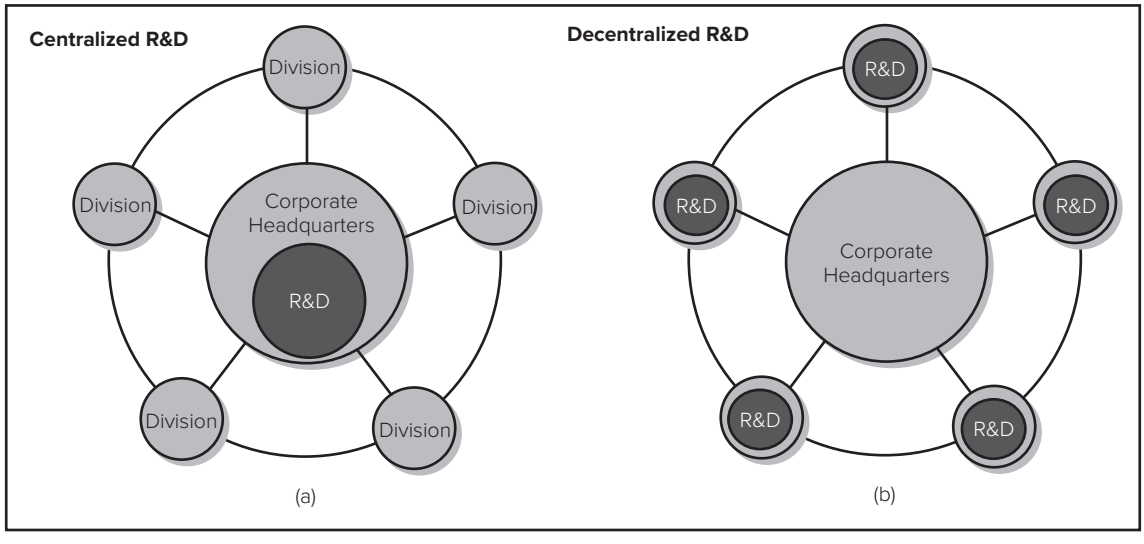
Centralization

centralization/ decentralization Centralization is the degree to which decision-making authority is kept at top levels of management. Decentralization is the degree to which decision-making authority is pushed down to lower levels of the firm.

Centralization is the degree to which decision-making authority is kept at top levels of the firm, while **decentralization** is the degree to which decision-making authority is pushed down to lower levels of the firm. Centralization can refer both to the geographical location of activities (that is, the degree to which activities are performed in a central location for the firm) and to where power and authority over activities are located. That is, activities might occur in locations far from the corporate headquarters, but the authority and decision making over those activities may be retained at headquarters—leading to greater centralization than their physical location would suggest.

For firms that have multiple R&D projects ongoing, whether to centralize or decentralize R&D activities is a complex issue. Decentralizing R&D activities to the divisions of the firm enables those divisions to develop new products or processes that closely meet their particular division's needs (see Figure 10.1). The solutions they develop are more likely to fit well within the operating structure of the division, and be closely matched to the requirements of the customers served by that division. The decentralization of development projects also enables the firm to take advantage of the diversity of knowledge and market contacts that may exist in different divisions. Consistent with this, studies by Felipe Csaszar show that when decision-making about

FIGURE 10.1
Centralized and Decentralized R&D Activities



new projects is pushed down to the lowest levels of the firm, the firm ends up taking on both a greater quantity and variety of projects. Though there will be more failed projects, the firm makes fewer "errors of omission."[20] However, there is much risk of reinventing the wheel when R&D activities are decentralized. Many redundant R&D activities may be performed in multiple divisions, and the full potential of the technology to create value in other parts of the firm may not be realized. Furthermore, having multiple R&D departments may cause each to forgo economies of scale and learning-curve effects.

By contrast, if the firm centralizes R&D in a single department, it may maximize economies of scale in R&D, enabling greater division of labor among the R&D specialists and maximizing the potential for learning-curve effects through the development of multiple projects. It also enables the central R&D department to manage the deployment of new technologies throughout the firm, improving the coherence of the firm's new product development efforts and avoiding the possibility that valuable new technologies are underutilized throughout the organization. For example, in the late 1980s, Intel realized that, as a result of the rising complexity and information processing demands in the semiconductor industry, its decentralized process development (which was scattered across diverse business groups) was resulting in serious delays and cost overruns. In the 1990s Intel thus centralized all process development, giving a single fabrication facility full responsibility for all new process generation. This development group would have maximum development resources (the highest in the industry). Once a new development process was completed and tested, it was replicated (in a process known in Intel as "copy exactly") in all of the company's other fabrication facilities.

The use of a centralized versus decentralized development process varies by type of firm and industry. For example, a study by Laura Cardinal and Tim Opler found that research-intensive firms that were highly diversified were more likely to establish separate research and development centers to facilitate communication and transfer of innovation across divisions.[21] A study by Peter Golder, on the other hand, found that consumer products companies tend to utilize more decentralized R&D, tailoring projects to local markets, while firms in electronics industries tend to centralize R&D in centers of excellence that are devoted to leveraging particular competencies.[22]

There is some disagreement about whether centralization enhances or impedes a firm's flexibility and responsiveness to technological change (or other environmental shifts). A highly centralized firm may be better able to make a bold change in its overall direction because its tight command-and-control structure enables it to impose such change on lower levels of the firm in a decisive manner. Decentralized firms may struggle to get the cooperation from all the divisions necessary to undergo a significant change. But decentralized firms may be better able to respond to some types of technological or environmental change because not all decisions need to be passed up the hierarchy to top management; employees at lower levels are empowered to make decisions and changes independently and thus may be able to act more quickly.

formalization
The degree to which the firm utilizes rules, procedures, and written documentation to structure the behavior of individuals or groups within the organization.

Formalization and Standardization

Formalization and standardization are closely related structural dimensions of organizations. The **formalization** of the firm is the degree to which the firm utilizes rules,

procedures, and written documentation to structure the behavior of individuals or groups within the organization. **Standardization** is the degree to which activities in a firm are performed in a uniform manner. The rules and procedures employed in formalization can facilitate the standardization of firm activities and help to regulate employee behavior by providing clear expectations of behavior and decision-making criteria. Formalization can substitute for some degree of managerial oversight, and thereby help large companies run smoothly with fewer managers. This is demonstrated in the accompanying Theory in Action about 3M, where both Lehr and Jacobson responded to the difficulty of managing the growing firm by imposing more discipline and rules. By creating formal processes for choosing and managing development projects, Lehr and Jacobsen hoped to improve the overall efficiency and coherence of the firm's many decentralized development activities. However, high degrees of formalization can also make a firm rigid.[23] If a firm codifies all of its activities with detailed procedures, it may stifle employee creativity. Employees may not feel empowered or motivated to implement new solutions. This is also noted in the 3M example, when employee resentment of the new planning methods led to morale and motivation problems.

standardization
The degree to which activities are performed in a uniform manner.

Similarly, while standardization can ensure that activities within the firm run smoothly and yield predictable outcomes, standardization can also stifle innovation. Standardization may be used to ensure quality levels are met and that customers and suppliers are responded to consistently and equitably. However, by minimizing variation, standardization can limit the creativity and experimentation that leads to innovative ideas.

Mechanistic versus Organic Structures

The combination of formalization and standardization results in what is often termed a **mechanistic** *structure.* Mechanistic structures are often associated with greater operational efficiency, particularly in large-volume production settings. The careful adherence to policies and procedures combined with standardization of most activities results in a well-oiled machine that operates with great consistency and reliability.[24] For example, Dell Computer achieves its operational excellence, delivering products cost-effectively and with minimal inconvenience, by being highly standardized, disciplined, and streamlined.[25] While mechanistic structures are often associated with high centralization, it is also possible to have a highly decentralized mechanistic structure by using formalization as a substitute for direct oversight. By establishing detailed rules, procedures, and standards, top management can push decision-making authority to lower levels of the firm while still ensuring that decisions are consistent with top management's objectives.

mechanistic
An organization structure characterized by a high degree of formalization and standardization, causing operations to be almost automatic or mechanical.

organic
An organization structure characterized by a low degree of formalization and standardization. Employees may not have well-defined job responsibilities and operations may be characterized by a high degree of variation.

Mechanistic structures, however, are often deemed unsuitable for fostering innovation. Mechanistic structures achieve efficiency by ensuring rigid adherence to standards and minimizing variation, potentially stifling creativity within the firm. **Organic** structures that are more free-flowing, and characterized by low levels of formalization and standardization, are often considered better for innovation and dynamic environments.[26] In the organic structure, employees are given far more latitude in their job responsibilities and operating procedures. Because much innovation arises from experimentation and improvisation, organic structures are often thought to be better for innovation despite their possible detriment to efficiency.[27]

Theory in Action Shifting Structures at 3M

In 1916, William McKnight, the then-general manager of sales and production for 3M, authorized the creation of the company's first research laboratory to improve 3M's sandpaper. McKnight had a strong belief in the power of individual entrepreneurship and innovation. He encouraged innovation through a combination of setting ambitious goals for new product development and giving individuals considerable freedom in how they pursued those goals. For example, McKnight established a companywide objective that 25 percent of sales should come from products created in the last five years. He also endorsed a "bootlegging" program whereby researchers could spend up to 15 percent of their time on whatever projects they were interested in pursuing.

As the firm grew, McKnight continued to support a centralized R&D lab while also encouraging divisions to pursue their own development initiatives in response to market needs they encountered. However, as 3M's product portfolio grew, it became increasingly difficult for 3M to manage functions such as production and sales. In 1944, McKnight began to experiment with an even more decentralized organizational form wherein divisions would have not only their own R&D labs, but also their own production operations and sales force. McKnight believed that small independent businesses would grow faster than a large company, leading to his "grow and divide" philosophy: Each division would be independent, and as its development projects grew into successful departments, they too would be spun off into new divisions.

By 1980, when Lou Lehr took the helm, 3M had grown to have 85 basic technologies and competed in about 40 major product markets. Lehr feared that 3M's greatest strength had become its weakness—the proliferation of independent businesses had led to a fragmentation of effort. Lehr worried that divisions might be wasting too much time on redundant activities and not taking advantage of the opportunity to leverage technologies across multiple divisions where they might be valuable. He wanted to ensure that divisions with related technologies would cooperate on development projects, and that new technologies would be diffused across the company. So he consolidated the company's 42 divisions and 10 groups into four business sectors based on their relatedness of technology. He also created a three-tiered R&D system: central research laboratories that concentrated on basic research with long-term potential, sector labs that would serve groups of related divisions and develop core technologies to drive medium-term (5 to 10 years) growth, and division labs that would continue to work on projects with immediate applications. Lehr also imposed much more formal planning on the development process—some of 3M's managers began to refer to it as "planning by the pound." He also eliminated many projects that had been struggling for years.

The arrival of "Jake" Jacobson in 1986 as the new CEO led 3M into an era of even greater discipline. Jacobson increased the target of sales from products developed within the past five years to 35 percent. He increased the R&D funding rate to about twice that of other U.S. companies, but also directed the company to become more focused in its project selection and to shorten development cycle times. He also implemented a companywide move toward using teams for development rather than encouraging individual entrepreneurs. Though Jacobson's initiatives improved efficiency, many researchers began to resent some of the changes. They believed that the move to manage all development projects with teams was destroying the individualistic culture of entrepreneurship at 3M and that the focus on discipline came at the expense of creativity and excitement. Motivation and morale problems began to emerge.

Thus, when "Desi" Desimone became the CEO in 1991, he eased the company back toward a slightly looser, more entrepreneurial focus. He believed his predecessors had established a good architecture for ensuring that innovation did not run away in an uncontrolled fashion, but he also believed the company needed more balance between freedom and control, as reflected in the following quote:

> Senior management's role is to create an internal environment in which people understand and value 3M's way of operating. It's a culture in which innovation and respect for the individual are still central. If you have senior management who have internalized the principles, you create a trust relationship in the company.

continued

concluded

The top knows it should trust the process of bottom-up innovation by leaving a crack open when someone is insistent that a blocked project has potential. And the lower levels have to trust the top when we intervene or control their activities.

Source: Adapted from C. Bartlett and A. Mohammed, "3M: Profile of an Innovating Company," Harvard Business School case no. 9-395-016, 1995.

Size versus Structure

Many of the advantages and disadvantages of firm size that were discussed at the beginning of the chapter are related to the structural dimensions of formalization, standardization, and centralization. Large firms often make greater use of formalization and standardization because as the firm grows it becomes more difficult to exercise direct managerial oversight. Formalization and standardization ease coordination costs, at the expense of making the firm more mechanistic. Many large firms attempt to overcome some of this rigidity and inertia by decentralizing authority, enabling divisions of the firm to behave more like small companies. For example, firms such as General Electric, Hewlett-Packard, Johnson and Johnson, and General Motors have attempted to take advantage of both bigness and smallness by organizing their companies into groups of small companies that can access the large corporation's resources and reach while retaining a small company's simplicity and flexibility.[28] The next section examines several methods by which firms can achieve some of the advantages of large size, and the efficiency and speed of implementation afforded by mechanistic structures, while simultaneously harnessing the creativity and entrepreneurial spirit of small firms and organic structures.

The Ambidextrous Organization: The Best of Both Worlds?

Most firms must simultaneously manage their existing product lines with efficiency, consistency, and incremental innovation, while still encouraging the development of new product lines and responding to technological change through more radical innovation. Tushman and O'Reilly argue that the solution is to create an ambidextrous organization.[29] An **ambidextrous organization** is a firm with a complex organizational form that is composed of multiple internally inconsistent architectures that can collectively achieve both short-term efficiency and long-term innovation.[30] Such firms might utilize mechanistic structures in some portions of the firm and organic structures in others. This is one of the rationales for setting up an R&D division that is highly distinct (either geographically or structurally) from the rest of the organization; a firm can use high levels of formalization and standardization in its manufacturing and distribution divisions, while using almost no formalization or standardization in its R&D division. Incentives in each of the divisions can be designed around different objectives, encouraging very different sets of behavior from employees. A firm can also centralize and tightly coordinate activities in divisions that reap great economies of scale such as manufacturing, while decentralizing activities such as R&D into many small units so that they behave like small, independent ventures. Whereas traditionally research emphasized the importance of diffusing information across the firm and ensuring cross-fertilization of ideas across new product development efforts, recent research suggests that some amount of isolation of teams, at least in early development stages, can

ambidextrous organization The ability of an organization to behave almost as two different kinds of companies at once. Different divisions of the firm may have different structures and control systems, enabling them to have different cultures and patterns of operations.

be valuable. When multiple teams interact closely, there is a risk that a solution that appears to have an advantage (at least at the outset) will be too rapidly adopted by other teams. This can cause all of the teams to converge on the same ideas, thwarting the development of other creative approaches that might have advantages in the long run.[31] Consistent with this, a significant body of research on "**skunk works**" has indicated that there can be significant gains from isolating new product development teams from the mainstream organization.[32] Separating the teams from the rest of the organization permits them to explore new alternatives, unfettered by the demands of the rest of the organization.

Skunk Works® Skunk Works® is a term that originated with a division of Lockheed Martin that was formed in June of 1943 to quickly develop a jet fighter for the U.S. Army. It has evolved as skunk works to refer more generally to new product development teams that operate nearly autonomously from the parent organization, with considerable decentralization of authority and little bureaucracy.

Similarly, firms that have multiple product divisions might find that one or more divisions need a more organic structure to encourage creativity and fluid responses to environmental change, while other divisions benefit from a more structured and standardized approach. For example, when *USA Today* decided to establish an online version of the popular newspaper, management discovered it would need more flexible procedures to respond to both rapid technological change and the real-time information updating requirements of the online paper. The paper would also require different incentive schemes to attract and retain technologically savvy employees. The company established the online paper as a separate division with a different reporting structure, less formalization, a different pay structure, and even different cultural norms about appropriate work attire and working hours.

Apple provides another example. In 1980, Apple was churning out Apple II personal computers at a fast clip. However, Steve Jobs was not content with the product design; he wanted a product that would revolutionize the world by dramatically changing the way people interact with computers. He wanted to develop a computer so user-friendly and self-contained that it would appeal even to people who had no interest in the technological features of computers—it would become an extension of their everyday lives. Jobs began working with a team of engineers on a new project called Macintosh (originally developed by another Apple engineer, Jef Raskin). Jobs did not believe that the growing corporate environment at Apple was conducive to nurturing a revolution, so he created a separate division for the Macintosh that would have its own unique culture. He tried to instill a free-spirited entrepreneurial atmosphere reminiscent of the company's early beginnings in a garage, where individualistic and often eccentric software developers would flourish. The small group of team members was handpicked and sheltered from normal corporate commitments and distractions. He encouraged the Macintosh team members to consider themselves renegades, and even hung a pirate's skull-and-crossbones flag over their building. Jobs would also take the team on regular retreats to isolated resorts and reaffirm the renegade culture with quotes like "It's more fun to be a pirate than to join the Navy."[33]

If big firms can have internal structures with the incentives and behavior of small firms, then much of the logic of the impact of firm size on technological innovation rates becomes moot. A single organization may have multiple cultures, structures, and processes within it; large firms may have entrepreneurial divisions that can tap the greater resources of the larger corporation, yet have the incentive structures of small firms that foster the more careful choice of projects or enhance the motivation of R&D scientists. Such entrepreneurial units may be capable of developing discontinuous innovations within the large, efficiency-driven organizations that tend to foster incremental innovations.

Firms can also achieve some of the advantages of mechanistic and organic structures by alternating through different structures over time.[34] Schoonhoven and Jelinek studied Intel, Hewlett-Packard, Motorola, Texas Instruments, and National Semiconductor and found that these firms maintained a "dynamic tension" between formal reporting structures, quasiformal structures, and informal structures.[35] While the organizations had very explicit reporting structures and formalized development processes, the organizations were also reorganized frequently to modify reporting relationships and responsibilities in response to a changing environment. Thus, while the organizations used seemingly mechanistic structures to ensure systematic and efficient production, frequent reorganizing enabled the firms to be flexible.

These firms also used what Schoonhoven and Jelinek term *quasiformal structures* in the form of teams, task forces, and dotted-line relationships (that is, reporting relationships that were not formally indicated on the organizational chart). These quasiformal structures were more problem-focused and could change faster than the rest of the company. They also provided a forum for interaction across divisions and thus played an important boundary-spanning role. One advantage of quasiformal structures is that they fostered interactions based on interests rather than on hierarchy. This can foster more employee motivation and cross-fertilization of ideas. As noted by one employee: "Sometimes [innovation] happens in the men's room. One guy's talking to another guy, and another guy's standing, eavesdropping on the conversation, scribbling on a napkin."[36] Some of the downsides to such quasiformal structures were that they required time to manage, and they could be hard to kill. Since the quasi structures were not part of the formal reporting structure, it could sometimes be difficult to establish who had the authority to disband them.

MODULARITY AND "LOOSELY COUPLED" ORGANIZATIONS

Another method firms use to strike a balance between efficiency and flexibility is to adopt standardized manufacturing platforms or components that can then be mixed and matched in a modular production system. This enables them to achieve standardization advantages (such as efficiency and reliability) at the component level, while achieving variety and flexibility at the end product level.

Modular Products

Modularity refers to the degree to which a system's components may be separated and recombined.[37] Making products modular can exponentially increase the number of possible configurations achievable from a given set of inputs.[38] For example, many of IKEA's shelving systems are designed so that users can mix and match a number of components to meet their needs. The shelves and supports come in a range of standardized sizes, and they can all be easily attached with standardized connectors. Similarly, some stoves now offer customers the ability to expand the range of the stove's functionality by removing the burners and plugging in other cooking devices such as barbecue grills and pancake griddles. Publishers have even embraced modularity by offering digital content that enables instructors to assemble their own textbooks from book chapters, articles, cases, or their own materials.

Many other products are produced in a modular way, even though the customer does not perceive the modularity. By standardizing a number of common components and using flexible manufacturing technologies that can quickly shift from one assembly

configuration to another, companies can produce a wide variety of product models just by changing which components are combined, while still achieving economies of scale and efficiency in the individual components. For example, Chrysler achieves one of the fastest new product development cycles in the automobile industry while also keeping new product development costs low through its practice of using a few standard platforms upon which all of its new car models are built. Tata Motors, the Indian company that introduced a $2500 car in 2008, used modularity even more dramatically. The Nano is built in components that can be sold and shipped in kits to be assembled and serviced by local entrepreneurs. This both enables the distribution of the Nano to be fast and streamlined and enables better penetration of remote rural markets.[39]

Modularity is achieved in product design through the specification of standard interfaces. For example, by designing all of its shelving components to work with its standardized connectors, IKEA ensures that components can be freely mixed and matched. Individual components can be changed without requiring any design changes in the other components. Because modularity enables a wider range of end configurations to be achieved from a given set of inputs, it provides a relatively cost-effective way for firms to meet heterogeneous customer demands. Furthermore, since modularity can enable one component to be upgraded without changing other components, modularity can enable firms and customers to upgrade their products without replacing their entire system. The personal computer is an excellent example of a modular system that enables upgrading. For example, if users want their personal computer to have more memory or a better monitor, they do not need to replace their entire computer system—they can simply purchase and install additional memory or a new monitor.

Modular products become more valuable when customers have heterogeneous demands and there are diverse options for meeting them. For example, suppose a car may be assembled from a range of components. The wider the range of components that may be recombined into a car, the wider is the range of possible car configurations achievable through modularity, and the greater is the potential opportunity cost of being "locked in" to a single configuration. Furthermore, the more heterogeneous customers are in their demand for car features, the less likely they are to agree on a single configuration. By employing modularity, heterogeneous customers can choose a car configuration that more closely meets their preferences.[40] By contrast, if customers all want the same thing, then there is little to be gained through offering a modular system—it will be a simple matter to determine the best combination of components to meet customer demands and integrate them into a nonmodular system.

When products are made more modular, it enables the entire production system to be made more modular. The standard interfaces reduce the amount of coordination that must take place between the developers of different components, freeing them to pursue more flexible arrangements than the typical organizational hierarchy.[41] Thus more modular products are often associated with more modular organizations with less centralization.[42] Such flexible arrangements are referred to as "loosely coupled organizational structures," as described in the next section.

Loosely Coupled Organizational Structures

Organizations can also be made modular through the adoption of structures that enable "loose coupling."[43] In a loosely coupled structure, development and production

activities are not tightly integrated but rather achieve coordination through their adherence to shared objectives and common standards. If, for example, each development group agrees to a development plan and standardized interfaces that enable the components they develop to connect and interact effectively, there may be little need for close coordination between the groups. The standard interface provides "embedded coordination" among all the development and production participants.[44] This can enable components of a product to be produced by highly autonomous divisions of the firm, or even by multiple independent firms.

Advances in information technology have also enabled loosely coupled organizational structures to become more common.[45] Information technology can enable a firm to access and process more information at a lower cost, vastly increasing the firm's options for development configurations.[46] For example, information technology lowers a firm's search costs for locating suitable development partners, as well as the costs of monitoring the partner's performance. This was clearly demonstrated in a study by Nick Argyres of the development of the B-2 "Stealth" bomber, a highly advanced military aircraft, developed jointly by Northrop, Boeing, Vaught, and General Electric.[47] Argyres found that enhanced information technology limited the need for coordination of activities through hierarchical control. By using information technology and developing a standard interface—a shared "technical grammar" that facilitated communication across firms—the firms involved in the development of the bomber could work autonomously, yet cooperatively.

Less need for integration frees firms to pursue more flexible research and development and production configurations. For instance, firms can become more specialized by focusing on a few key aspects of technological innovation that relate closely to the firm's core competencies, while obtaining other activities through outsourcing or alliances. By focusing on those activities in which the firm has a competitive advantage, the firm can improve its chance of developing a product that has a price-to-value ratio that attracts customers while reducing the overhead and administrative complexity of maintaining a wide scope of activities. This can cause whole industries to be transformed as large vertically integrated firms are displaced by nimbler, more specialized producers.[48] For example, when computer workstations displaced their more integrated predecessors, minicomputers (which were traditionally built using a proprietary central processor, combined with a proprietary system bus, and run with a proprietary operating system), the entire computer industry began to become more modular as integrated producers like Prime, Wang, and Data General were displaced by a network of producers (including Sun Microsystems, Silicon Graphics, and Motorola), whose components could be combined in numerous end product configurations. The platform ecosystems described in Chapter Four are another example of loosely coupled organizations, as are distributed innovation systems that use crowdsourcing like Linux and Wikipedia.[49]

There are, however, disadvantages of loose coupling. Many activities reap significant synergies by being integrated.[50] In particular, activities that require the frequent exchange of complex or tacit knowledge are likely to need closer integration than a loosely coupled development configuration can offer. For example, suppose the design of a delivery mechanism for a drug will require intensive coordination with the design of the drug itself. It may be that the strength and dosage of the drug must be carefully calibrated

Theory in Action

The Loosely Coupled Production of Boeing's 787 Dreamliner[a]

When Boeing launched its sales program for the yet-to-be-built 787 Dreamliner in late 2003, it rapidly became the fastest-selling commercial jet liner in history. By 2011, Boeing had received more than 800 advance orders for the aircraft—more than any other plane in history.[b] The Dreamliner marked an important turning point for the company. Boeing had not built an all-new airliner since 1994, when the 777 took to the sky. Since that time, Airbus had led the way in aerospace innovation, while Boeing had been content to stretch and refine its existing families of airplanes such as the 737 and 747. Many had begun to believe that Boeing no longer had what it took to build an entirely new aircraft.[c] The Dreamliner's success or failure would thus send strong signals to the market about the company's prospects for the future.

The Dreamliner was a super-efficient long-range mid-sized airliner. It would be the first commercial jet manufactured primarily from carbon fiber composites, enabling it to be significantly lighter and thus more fuel efficient than traditional commercial jets. Because the composite material could be more easily sculpted than aluminum, the wings of the jet would have graceful curves like a bird's wings. Furthermore, since the composite material was exceptionally strong and resistant to corrosion, the cabin could be both more pressurized and more humidified, making air travel more comfortable.[d] Composites also allowed Boeing to easily assemble the forward, center, and rear sections of the fuselage, the wings, the horizontal stabilizers, and the vertical fins as large individual modules that could be quickly snapped together to form the airplane, instead of constructing the aircraft piece by piece using aluminum sheets as prior aircraft had been constructed.[e]

The innovations of the Boeing 787 program extended well past the actual composition of the aircraft. For the 787 project, Boeing also revolutionized the structure of the production processes involved in building a commercial aircraft. The production of the 787 would be significantly more loosely coupled than any commercial aircraft to date. Dozens of partners from around the world built and preassembled large pieces of the plane which were then delivered to the Boeing plant for final assembly.[f] For example, Mitsubishi, Kawasaki, and Fuji, all of Japan, were contracted to produce the wings, forward fuselage, and center wing box, respectively. Saab makes the cargo doors, and Alenia Aeronautica of Italy produces a horizontal stabilizer and central fuselage. Dozens of companies from other countries contribute other parts.[g] Roughly 70 percent of the Dreamliner would be built outside of the United States. The dramatic increase in outsourcing was expected to provide a range of benefits that included spreading the risk of developing the aircraft, containing costs, and improving the prospects for foreign sales since purchasers and their governments often like to see work done on the aircraft in their countries.[h]

Though Boeing had outsourced a portion of the work on its planes for decades, the 787 ushered in a new era of outsourcing, Boeing's role would shift from being the traditional designer and manufacturer to becoming "an essential elements company, reserving for itself optimum design and integration tasks and relying on a select group of outsiders for everything else."[i] The revolutionary new production process was not without its challenges, however. The sheer complexity of the project and the large number of suppliers involved made coordination much more complicated. Breakdowns in this coordination had led to several delays. Though the first Dreamliner had been slated to take flight in August of 2007, customers did not actually take delivery of the first Dreamliners until late 2011. The challenges of coordinating suppliers around the globe had led to numerous production delays and design adjustments. Boeing's managers indicated that the company would make little profit on the first several dozen planes because even after they rolled off the assembly line, they would require corrective work, including the repair of parts and design changes.[j] Boeing's management acknowledged that some mistakes had been made in the supply chain, and Engineering Vice President Mike Denton indicated that the company was considering bringing more of the work back in-house. He noted, "We will probably do

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more of the design and even some of the major production for the next new airplanes ourselves as opposed to having it all out with the partners."[k]

[a] Adapted from "The Loosely Coupled Production of Boeing's 787 Dreamliner" by Jaspal Singh and Melissa A. Schilling, New York University teaching case.
[b] Christopher Drew, "Boeing Posts 20% Profit Gain But Cuts Forecast For 2012 As Jet Completion Slows," *New York Times,* January 25, 2012.
[c] Michael V. Copeland, "Boeing's Big Dream," *Fortune* 157, no. 9 (2008): 180–91.
[d] S. Holmes, "Better Living at 30,000 Feet," *BusinessWeek,* August 2, 2007.
[e] Roger Renstrom, "Boeing's Big Gamble: Half-Plastic Dreamliner," *Plastics News,* July 2, 2007.
[f] P. Hise, "How Many Small Businesses Does It Take to Build a Jet?" *Fortune Small Business* 17, no. 6 (2007): 42–5.
[g] J. Weber, "Boeing to Rein in Dreamliner Outsourcing," *BusinessWeek Online,* January 19, 2009: 10.
[h] Ibid., and M. Mecham, "The Flat-Earth Airplane," *Aviation Week & Space Technology,* July 3, 2006: 43.
[i] Mecham, "The Flat-Earth Airplane."
[j] Christopher Drew, "Boeing Posts 20% Profit Gain But Cuts Forecast For 2012 As Jet Completion Slows." *New York Times,* January 25, 2012.
[k] Weber, "Boeing to Rein in Dreamliner Outsourcing."

and adjusted in accordance to the speed at which the delivery mechanism releases the drug. Alternative materials considered for the delivery mechanism may also have to be evaluated for their risk of potential interaction with chemicals used in the drug solution. If ongoing intensive coordination is required, the development activities might be better carried out through close integration of all parties.

An integrated firm also has mechanisms for resolving conflict that may be more effective or less expensive than those available in the market.[51] For example, if a dispute should arise over the development of a new product among development groups that are within the same firm, top managers can decide what action to take and exercise their authority over the development groups. But if the development groups are in separate companies, developing a new product in a collaboration agreement, neither firm may possess the authority to resolve the dispute and enforce a particular outcome. If the firms are unable to resolve the dispute themselves, they may face going to court or arbitration to resolve the dispute, an expensive and time-consuming option.

USING CULTURE AND NORMS TO FOSTER INNOVATION

Significant research suggests that culture and norms can have a profound influence on both the quantity and nature of innovation in the firm. Some of the key ways managers can use culture and norms to foster innovation include:

1. Creating Norms of Meritocracy, Flexibility and Autonomy

The norms of an organization signal whether employees are empowered to challenge assumptions and come up with new ideas, or are expected to adhere to the current way of doing things. Giving people flexible roles and autonomy, for example, will both attract more creative employees and nurture the creative side of existing members. It is also important to make sure that every person feels empowered to contribute their ideas—not just those in positions of authority, or those in designated creative roles. At Pixar, for example, teams use "dailies," where artists have their ongoing work reviewed

by directors and peers. An informal audience of people from all levels in the organization provides direct feedback about both the creative and technical elements of the project. As Brad Bird, Oscar-winning director of films such as *The Incredibles* and *Ratatouille,* described, "As individual animators, we all have different strengths and weaknesses, but if we can interconnect all our strengths, we are collectively the greatest animator on earth. . . We're going to look at your scenes in front of everybody. Everyone will get humiliated and encouraged together. If there is a solution, I want everyone to hear the solution, so everyone adds it to their tool kit."[52] At first, people at Pixar were afraid to speak up, but after two-months of seeing artists benefit by hearing the blunt suggestions of Bird and others, people began to feel safe enough to contribute their own suggestions.[53]

Pixar was also able to attract and nurture creative people by giving them considerable autonomy. Teams choose their own hours, attire, office arrangements, project management routines, and meeting structure. As described by Ed Catmull, president of both Pixar and Walt Disney Animation Studios, "We believe the creative vision propelling each movie comes from one or two people and not from either corporate executives or a development department. Our philosophy is: You get creative people, you bet big on them, you give them enormous leeway and support, and you provide them with an environment in which they can get honest feedback from everyone."[54]

2. Eliminating Norms of Consensus

If leaders of organizations want employees to challenge the accepted wisdom, it is important to abolish practices and norms that require people to come to consensus prior to making decisions or moving forward with a project. Requiring consensus can force people to prematurely converge on mediocre ideas; if individuals believe that a consensus must be achieved to move forward, they will be more reluctant to dissent, and more readily pile onto the ideas put forward by others, particularly if they seem uncontroversial (and therefore more likely to achieve consensus). Making consensus the objective risks also making orthodox solutions the objective.[55] Consider the development of Apple's first iPod, launched in October of 2001. In the year 2000, the consumer portable audio device market was littered with failed devices. It was also a market in which Apple had no experience. If Steve Jobs had sought a consensus about where Apple's development efforts should focus it is unlikely that Apple's R&D engineers or product managers would have achieved consensus that Apple should enter the portable audio market. Yet it was precisely this move that led not only to iPod, but ultimately also iTunes and iPhone—products that were disproportionately responsible for making Apple the most valuable company in the world.

Research suggests that it can also be very useful to let teams that disagree about a solution pursue different—even competing—paths.[56] For example, at CERN (the European Organization for Nuclear Research that operates the Large Hadron Collider) teams of physicists and engineers accelerate and collide particles to simulate "the big bang," in hopes of advancing our understanding of the origins of the universe. Groups of scientists, however, often disagree about the best solution to particular problems. Management at CERN recognized that a solution that initially appears to be inferior to others could turn out to be better than others with further development. This means that if solutions have to "compete" to be accepted too early in their development, the

organization could end up committing major resources to a solution that appears better initially but might not be best in the long run. CERN thus encourages multiple teams to work on their own solutions separately, and only after the teams have had significant time to develop their solution, do the teams meet and compare alternatives. In this way, CERN helps teams achieve some of the advantages of separateness that the breakthrough innovators reap from their innate sense of detachment. The teams are encouraged to follow their own paths, without being constrained by the beliefs of other teams.

3. Solo Ideation versus Brainstorming Groups

There is considerable controversy over whether brainstorming groups foster or thwart creativity. In 1953, Alex Osborne wrote a highly influential book, Applied Imagination, in which he stated that "the average person can think up twice as many ideas when working with a group than when working alone." This argument found great popularity in both businesses and business schools, leading to a firmly entrenched belief that brainstorming groups were more creative than individuals. However, dozens of subsequent laboratory studies found results opposite to Osborne's claim: brainstorming groups produced fewer ideas, and ideas of less novelty, than the sum of the ideas created by the same number of individuals working alone.[57] These studies found that people in groups may be less creative due to both evaluation apprehension and production blocking. Evaluation apprehension refers to the fact that people tend to self-censor many of their ideas in group brainstorming sessions for fear of being judged negatively by others. Production blocking refers to the phenomenon whereby as people take turns to voice their ideas, those speaking later may forget their ideas before having a chance to voice them. Furthermore, the process of attending to another person's ideas redirects a listener's train of thought, essentially hijacking their own idea generation process. A "meta-analysis" of 20 brainstorming group studies found that productivity losses in both quantity and quality of ideas were highly significant, and large in magnitude. Losses increased with larger groups, and with supervision by an experimenter or other authority. It also found that production losses were greatest for evaluation apprehension, followed by production blocking.[58]

In both organizations and educational settings, working in teams has become a norm. Team work can be very valuable, but the research above suggests that to ensure individuals bring as much to the team as possible—especially when the objective is a creative solution—individuals need time to work alone before the group effort begins. They might be encouraged to commit their ideas to paper, and to flesh them out in at least a rudimentary way, before they are at risk of being extinguished by social processes. A creative idea can be fragile—easily swept away by the momentum of a group conversation. Almost every team suffers from some degree of group think; individuals who are more outspoken or who have forceful personalities can dominate the conversation and the decision making. They can herd a team onto a particular trajectory without even intending to do so. A little isolation and solitude can give other individuals a better chance to develop their creative ideas.[59]

The payoff of a person working alone on their own projects, tapping into their intrinsic motivation (as discussed in Chapter Two), has been the source of several of Google's most famous products. For example, Google research scientist Krishna Bharat created Google News in the aftermath of the September 11th, 2001, attack on

the World Trade Center, to meet his own needs for keeping up with the rapid emergence of news related to the event. The product was so useful, Google launched a beta version for use by the public one year later, in September of 2002. Gmail was also an independent project of an engineer at Google. Paul Buchheit, who had previously worked on Google Groups, began working on Gmail in August 2001. Because there was a large volume of internal email in Google, he wanted to develop a product that would both enable a user to keep her email indefinitely (requiring large storage capabilities) and enable rapid search (so that a particular email of interest could be quickly found). The result was Gmail, which was formally launched in 2004.

4. Lowering the Price of Failure

To foster a culture of innovation, many firms have found it useful to lower the price of failure, and even celebrate bold-but-intelligent failures. Former CEO of Procter & Gamble A. G. Lafley, for example, made fearlessness in the face of failure a core strategy of breakthrough innovation at the firm. "We learn," he said, "much more from failure than we do from success."[60] Similarly, at frog Design, failure is treated as a form of useful progressive experimentation. There is no finger pointing or admonishment; instead frog Design managers use an end-of-project retrospective framework to rapidly learn from the company's failures (as well as its successes). Theodore Forbath, Global Vice President of Innovation Strategy at frog Design, notes: "Company leaders can reinforce acceptance of failure by publicly celebrating projects that didn't quite meet expected results, but that were successful in providing new learning."[61] Pharmaceutical giant Eli Lilly even throws parties to celebrate its best failures (i.e., the ones that did not meet expected results but led to the greatest learning for the organization), and Tata Group gives an annual prize for the best failures in its organization.[62]

5. Cultivating Idealistic Goals

Many firms have found it valuable to organize the firm around a grand and noble mission that both fosters intrinsic motivation and helps to rally others to its cause. For example, Bristol-Myers Squibb's mission is "To discover, develop and deliver innovative medicines that help patients prevail over serious diseases." Google's well-known mission is "To organize the world's information and make it universally accessible and useful." These lofty goals, if well ingrained throughout the organization, can become an organizing principle that shapes employee behavior even without direct oversight or incentives. When those goals have a social component that employees embrace as being meaningful or intrinsically valuable (such as improving people's quality of life), that intrinsic motivation can inspire employees to invest more effort in coming up with better ways to achieve the mission and to bring those innovations to fruition.[63]

MANAGING INNOVATION ACROSS BORDERS

The organization of innovation activities becomes particularly interesting for multinational firms. Many of the same issues that shape the centralization-versus-decentralization decision discussed earlier become highly amplified in the multinational firm. Foreign markets offer highly diverse sources of information and other resources. They may also have highly diverse product needs and different operating norms. This prompts many

firms to consider decentralizing R&D to take advantage of local information and tailor innovation activities to the local market. However, innovations developed in this decentralized manner might never be diffused to the other divisions. The customization of products and processes to the local markets makes them particularly difficult to transfer to divisions serving different markets. Divisions that are accustomed to developing their own innovations may be reluctant to share them with others for fear of giving away their proprietary knowledge. They may also be reluctant to adopt other divisions' innovations because of the belief that innovations that are not developed locally will not suit their local market needs (a phenomenon known as *not-invented-here syndrome*). However, much of the value creation potential of a multinational is the opportunity to leverage technological innovation (and other core competencies) into multiple markets. Allowing innovation activities to become completely autonomous and disconnected risks forfeiting this opportunity. How does the multinational resolve this dilemma? A series of studies by Christopher Bartlett and Sumantra Ghoshal highlight some advantages and disadvantages of various approaches to the management of multinational innovation. They identify four primary strategies used by firms: center-for-global, local-for-local, locally leveraged, and globally linked.[64]

center-for-global strategy
When all innovation activities are conducted at a central hub and innovations are then diffused throughout the company.

The **center-for-global strategy** entails conducting all innovation activities at a centralized hub. These innovations are then deployed globally throughout the company. The centralization of innovation activities enables management to:

- Tightly coordinate all R&D activities (across both functions and projects).
- Achieve greater specialization and economies of scale in R&D activities while avoiding duplication of activities in multiple divisions.
- Develop and protect core competencies.
- Ensure that innovations are standardized and implemented throughout the company.

Managers are likely to choose a center-for-global approach to innovation when they have a strong desire to control the evolution of a technology, when they have strong concerns about the protection of proprietary technologies, when development activities require close coordination, or when there is a need to respond quickly to technological change and dispersed efforts are likely to create inefficiencies.[65] However, a center-for-global approach tends to not be very responsive to the diverse demands of different markets. Furthermore, the divisions that serve these markets might resist adopting or promoting centrally developed innovations. As a result, innovations developed centrally may not closely fit the needs of foreign markets and may also not be deployed quickly or effectively.

local-for-local strategy
When each division or subsidiary of the firm conducts its own R&D activities, tailored for the needs of the local market.

A **local-for-local strategy** is the opposite of the center-for-global strategy. Each national subsidiary uses its own resources to create innovations that respond to the needs of its local market. A local-for-local strategy takes advantage of access to diverse information and resources, and it customizes innovation for the needs and tastes of the local market. Managers are likely to choose a local-for-local strategy when divisions are very autonomous and when markets are highly differentiated.

There are several downsides to the local-for-local strategy, however. It can result in significant redundancy in activities as each division reinvents the wheel. Furthermore, each division may suffer from a lack of scale in R&D activities, and there is a risk that valuable innovations will not be diffused across the firm.

Over time, firms have developed variants of these strategies that attempt to reap advantages of both the center-for-global and local-for-local strategies. Bartlett and Ghoshal identify one such strategy as the **locally leveraged strategy**. A firm implementing a locally leveraged strategy attempts to take the most creative resources and innovative developments from the divisions and deploy them across the company. This strategy enables the firm to take advantage of the diverse ideas and resources created in local markets, while leveraging these innovations across the company. One way this strategy is employed in consumer markets is to assign an individual the role of international brand custodian. This person is responsible for ensuring that a successful brand is deployed into the firm's multiple markets while also maintaining consistency in the product's image and positioning.[66] Such a strategy can be very effective if different markets the company serves have similar needs.

locally leveraged strategy
When each division or subsidiary of the firm conducts its own R&D activities, but the firm attempts to leverage resulting innovations throughout the company.

Another approach, the **globally linked strategy**, entails creating a system of decentralized R&D divisions that are connected to each other. Each geographically decentralized division might be charged with a different innovation task that serves the global company's needs. For example, a multinational auto manufacturer may empower one of its European divisions with the responsibility for developing new subcompact models that most closely fit the European markets but that may ultimately also be sold in the United States, Canada, and South America. In the meantime, its American division might bear the bulk of the responsibility for collaborating with other manufacturers to develop more efficient manufacturing processes that will ultimately be deployed corporatewide. Thus, while innovation is decentralized to take advantage of resources and talent pools offered in different geographic markets, it is also globally coordinated to meet companywide objectives. This approach also attempts to enable the learning accrued through innovation activities to be diffused throughout the firm. This strategy can be quite powerful in its ability to tap and integrate global resources, but it is also expensive in both time and money as it requires intensive coordination.

globally linked strategy
Innovation activities are decentralized, but also centrally coordinated for the global needs of the corporation.

In both the locally leveraged and globally linked strategies, R&D divisions are decentralized and linked to each other. The difference lies in the missions of the R&D divisions. In the locally leveraged strategy, the decentralized R&D divisions are largely independent of each other and work on the full scope of development activities relevant to the regional business unit in which they operate. This means, for example, that if their regional business unit produces and sells health care items, beauty care products, and paper products, the R&D division is likely to work on development projects related to all of these products. However, to ensure that the best innovations are leveraged across the company, the company sets up integrating mechanisms (such as holding regular cross-regional meetings, or establishing a liaison such as an international brand custodian) to encourage the divisions to share their best developments with each other. By contrast, in the globally linked strategy, the R&D divisions are decentralized, but they each play a different role in the global R&D strategy. Instead of working on all development activities relevant to the region in which they operate, they specialize in a particular development activity. For example, an R&D division may be in a regional business unit that produces and sells health care, beauty care, and paper products, but its role may be to focus on developing paper innovations, while other R&D divisions in the firm work on health care items or beauty care products. Or it might focus on basic chemistry applications relevant to all of the products, while another division explores packaging innovations, and so on. The role of the division

should exploit some local market resource advantage (such as abundant timber or a cluster of chemical technology firms). This strategy attempts to take advantage of the diversity of resources and knowledge in foreign markets, while still linking each division through well-defined roles in the company's overall R&D strategy.

Bartlett and Ghoshal argue that, overall, the multinational firm's objective is to make centralized innovation activities more effective (that is, better able to serve the various local markets) while making decentralized innovation activities more efficient (that is, eliminating redundancies and exploiting synergies across divisions). Bartlett and Ghoshal propose that firms should take a *transnational* approach wherein resources and capabilities that exist anywhere within the firm can be leveraged and deployed to exploit any opportunity that arises in any geographic market. They argue that this can be achieved by:

- Encouraging reciprocal interdependence among the divisions of the firm (that is, each division must recognize its dependency on the other divisions of the firm).
- Utilizing integration mechanisms across the divisions, such as division-spanning teams, rotating personnel across divisions, and so on.
- Balancing the organization's identity between its national brands and its global image.

Ericsson provides an excellent example of this approach. Instead of using a strictly centralized or decentralized structure for its innovation activities, Ericsson's structure ebbs and flows between centralization and decentralization. Sometimes Ericsson increases levels of centralization and global integration for particular projects, while other times it decentralizes much more authority over innovation activities to its geographically dispersed divisions. Similar to the dynamic tension approach described by Jelinek and Schoonhoven, Ericsson regularly modifies its structure to adjust the balance between integration and autonomy. To encourage interunit integration, Ericsson also sends teams of 50 to 100 engineers to a different subsidiary for a year or two. Such member rotation programs facilitate the diffusion of knowledge throughout the firm.[67] Furthermore, encouraging engineers to become integrated into multiple areas of the company helped the engineers identify with both the global company and particular divisions.

Summary of Chapter

1. The impact of firm size on innovation has been debated for more than 50 years. Size is thought to confer advantages such as economies of scale in R&D, greater access to complementary resources (like capital and market access), and learning benefits. However, size may also be associated with disadvantages such as inertia and governance problems.
2. Many firms attempt to make big companies feel small by breaking them into networks of more specialized divisions. These divisions can behave like smaller, more entrepreneurial firms.
3. Structural dimensions of the firm, including formalization, standardization, and centralization, also affect the firm's propensity to innovate and its effectiveness at innovation. Formalization and standardization tend to improve efficiency, but can stifle experimentation and creativity. Centralization has a more ambiguous effect

on innovation; in some cases, centralization can enable significant innovation to occur more rapidly, and in other situations, decentralization fosters more innovation by enabling managers to respond quickly to local needs.

4. Traditionally, scholars have divided organization structures into two major types: mechanistic structures, which are highly formalized and standardized, and are good for efficient production, and organic structures, which are loose and free flowing and are good for creativity and experimentation.
5. Ambidextrous organizations attempt to achieve both the efficiency advantages of large mechanistic firms and the creativity and entrepreneurial spirit of small organic firms. These firms may have divisions with different structures and control schemes, or they may alternate between different structures.
6. Recently, many firms have begun forming loosely coupled networks both within and between firms to conduct development activities. Part of this transition is attributed to the rise in information technology and the resultant decrease in coordination costs.
7. The culture and norms of the organization can significantly influence the kinds of employees it attracts and their ability and willingness to innovate. Research suggests that cultivating norms of meritocracy and autonomy, eliminating norms of consensus, encouraging solo ideation, lowering the price of failure, and cultivating idealistic goals can all help to foster innovation.
8. Multinational firms face significant challenges in determining where and how to conduct their R&D activities. One primary challenge is to balance the need to tap the knowledge and resources of local markets while also achieving coherence across the corporation and ensure that technological innovations are diffused and leveraged throughout the organization.

Discussion Questions

1. Are there particular types of innovation activities for which large firms are likely to outperform small firms? Are there types for which small firms are likely to outperform large firms?
2. What are some advantages and disadvantages of having formalized procedures for improving the effectiveness or efficiency of innovation?
3. What factors should a firm consider when deciding how centralized its R&D activities should be? Should firms employ both centralized and decentralized R&D activities?
4. Why is the tension between centralization and decentralization of R&D activities likely to be even greater for multinational firms than for firms that compete in one national market?
5. What are some of the advantages and disadvantages of the transnational approach advocated by Bartlett and Ghoshal?

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Chapter **Eleven**

Managing the New Product Development Process

Scrums, Sprints, and Burnouts: Agile Development at Cisco Systems

Cisco Systems, founded in 1984 in San Francisco, had grown to become a global leader in networking technology such as routers, servers, switches, networking software, security software, and more. For most of the company's existence, it had used the "waterfall" method to develop software.[a] The typical waterfall method began with analysis that would result in documents specifying the business case, product requirements, marketing requirements, and so on. Different teams would then be formed to sequentially design, build, test, and deploy the product. A team could begin working only when the previous team had completed its stage of the process, and the entire process could take 18 months or more.[b]

This approach was similar to "stage-gate" methods and consistent with how the company developed hardware products. But managers in the software divisions were unsatisfied; they worried that their development cycles were too long compared to other software companies, making it hard to compete. In 2014, they decided to try a new approach called *agile development* to make software development faster and more flexible at the firm.

Agile development began as a set of principles laid out by a group of 17 software engineers at a three-day retreat at a ski resort in Snowbird Utah in 2001.[c] Their objective was to come to an agreement about how to make software development faster and leaner. They ended up developing a set of core values and practices that emphasize collaboration, self-organization, and cross-functional teams.[d] The key distinction between traditional waterfall methods and agile methods was that rather than designing a complete product upfront and moving through a sequential process that culminates in testing and release, the product is broken up into many smaller parts or features that are built and

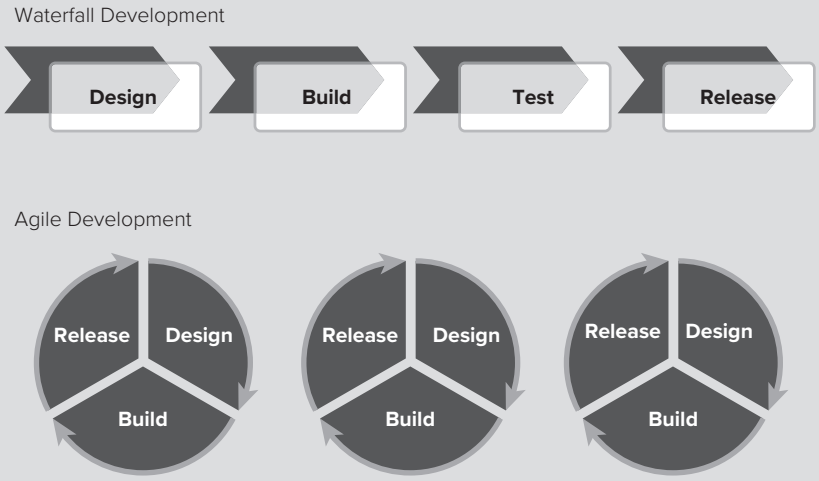


FIGURE 1
Waterfall versus Agile Development

released quickly, enabling the developers to get feedback and fix bugs early (see Figure 1). The process also gave developers considerably more autonomy.

The method rapidly grew in popularity, attracting companies such as Google, Spotify, Netflix, and Twitter. Furthermore, though it was designed for software development, it had also been adopted for managing other kinds of projects, and had even been adopted by companies such as Lockheed Martin, Walmart, and ExxonMobil.[e]

The Agile Development Process

In agile development, a manager deemed the *product owner* (a person at the organization who represents the customer's interests) assembles a complete list of functions to be developed for the product based on *user stories*—short descriptions of functions described by customers in their own words. This list is called the *product backlog*. Work on the product backlog is organized into a series of *sprints,* periods of roughly two weeks in which a small set of features from the product backlog are developed and tested. Work is conducted by *scrum teams,* small, self-organizing teams with no titles or team manager. There is no formal task assignment; each member just contributes in whatever way they can to complete the work and the team makes decisions as a whole. Sometimes agile development projects also have a *scrum master* who acts as a coach for multiple scrum teams. The scrum master does not provide day-to-day direction or impose any particular technical solution; rather, the scrum master's job is to help guide the scrum process itself.

The teams figure out what items they can commit to and create a *sprint backlog*—a list of tasks they will complete during the sprint. Each day of the sprint, all the team members and the product owner attend a quick scrum meeting of 15 minutes max where they share what they worked on the previous day, what they will work on that day, and any obstacles to their progress. During the sprint, the scrum team takes a small set of features from idea through coding and testing.

At the end of each sprint, there should be work that can be demonstrated to a client—a *minimum viable product* or MVP. This enables customers to provide

feedback early and often, helping the scrum teams weed out or refine their ideas. A *burndown chart* shows the amount of work remaining in a sprint or a product release, and is used to determine whether a sprint or release is on schedule.

"Release Early, Release Often"

In agile development, rather than having grand comprehensive product redesigns, the product is constantly, incrementally adapted. Introducing small changes one or a few at a time helps to reduce risk, and also improves transparency about what works and what does not work. By contrast, when many changes are introduced simultaneously, it is harder to tell why the overall product succeeds or fails.

For this approach to work, a product has to be fairly modular. That is, it must be possible for a large product to be broken down into many smaller, relatively independent problems that can be worked on separately (something that is not possible with all products).[f] Furthermore, a potential downside of the agile approach is that it may be more difficult (or less likely) to make large-scale systemic changes to a product.

In the right setting, however, agile development can accelerate product development, improve customer satisfaction, and even improve employee satisfaction because the process gives them much more autonomy and a sense of ownership in their jobs. As one team member at Cisco put it, "My boss used to come and tell me to get my team to do this or do that. Now, I tell him that I cannot tell my team to do this or that; I can suggest it to them, but they will discuss and decide if it's the right thing to do."[g]

Discussion Questions

1. What are some of the advantages and disadvantages of the agile development process?
2. How is agile development similar to or different from (a) the stage-gate process and (b) the parallel development process described in the chapter?
3. What are some of the likely changes agile development requires in managing development personnel?
4. What kinds of projects do you think agile development is appropriate for? What kinds of projects do you think it might be inappropriate for?

[a] https://www.scaledagileframework.com/cisco-case-study/

[b] Roger Chen, Ramya Ravichandar, and Don Proctor, "Managing the Transition to the New Agile Business and Product Development Model: Lessons from Cisco Systems," Indiana University Kelley School of Business Teaching Case (2016).

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OVERVIEW

In many industries, the ability to develop new products quickly, effectively, and efficiently is now the single most important factor driving firm success. In industries such as computer hardware and software, telecommunications, automobiles, and consumer electronics, firms often depend on products introduced within the past five years for more than 50 percent of their sales. Yet despite the avid attention paid to new product development, the failure rates for new product development projects are still agonizingly high. By many estimates, more than 95 percent of all new product development projects fail to result in an economic return.[1] Many projects are never completed, and of those that are, many flounder in the marketplace. Thus, a considerable amount of research has been focused on how to make the new product development process more effective and more efficient. This chapter discusses some strategic imperatives for new product development processes that have emerged from the study of best–and worst–practices in new product development.

We will begin by looking at the three key objectives of the new product development process: maximizing fit with customer requirements, minimizing cycle time, and controlling development costs. We then will turn to methods of achieving these objectives, including adopting parallel development processes, using project champions, and involving customers and suppliers in the development process. Next we will look at a number of tools firms can utilize to improve the effectiveness and efficiency of the development process, including creating go/kill decision points with stage-gate processes, defining design targets with quality function deployment, reducing costs and development time with design for manufacturing and CAD/CAM systems, and using metrics to assess the performance of the new product development process.

OBJECTIVES OF THE NEW PRODUCT DEVELOPMENT PROCESS

For new product development to be successful, it must simultaneously achieve three sometimes-conflicting goals: (1) maximizing the product's fit with customer requirements, (2) minimizing the development cycle time, and (3) controlling development costs.

Maximizing Fit with Customer Requirements

For a new product to be successful in the marketplace, it must offer more compelling features, greater quality, or more attractive pricing than competing products. Despite the obvious importance of this imperative, many new product development projects fail to achieve it. This may occur for a number of reasons. First, the firm may not have a clear sense of which features customers value the most, resulting in the firm's overinvesting in some features at the expense of features the customer values more. Firms may also overestimate the customer's willingness to pay for particular features, leading them to produce feature-packed products that are too expensive to gain significant market penetration. Firms may also have difficulty resolving heterogeneity in customer demands; if some customer groups desire different features from other groups, the firm may end up producing a product that makes compromises between these conflicting demands, and the resulting product may fail to be attractive to any of the customer groups.

Numerous new products have offered technologically advanced features compared to existing products but have failed to match customer requirements and were subsequently rejected by the market. For example, consider Apple's Newton MessagePad, a relatively early entrant into the personal digital assistant market. The Newton was exceptional on many dimensions. It had a highly advanced ARM610 RISC chip for superior processing performance. Its operating system was object oriented (a feature that software programmers had been clamoring for), and Apple openly licensed the architecture to encourage rapid and widespread adoption by other vendors. Also, its weight, size, and battery life were better than many of the other early competitors. However, the Newton MessagePad was still much too large to be kept in a pocket, limiting its usefulness as a handheld device. Many corporate users thought the screen was too small to make the product useful for their applications. Finally, early problems with the handwriting recognition software caused many people to believe the product was fatally flawed.

Another example is Philips' attempt to enter the video game industry. In 1989, Philips introduced its Compact Disc Interactive (CD-i). The CD-i was a 32-bit system (introduced well before Sega's 32-bit Saturn or Sony's 32-bit PlayStation), and in addition to being a game player, it offered a number of educational programs and played audio CDs. However, Philips had overestimated how much customers would value (and be willing to pay for) these features. The CD-i was priced at $799, more than double the cost of Nintendo or Sega video game systems. Furthermore, the product was very complex, requiring a half-hour demonstration by a skilled sales representative. Ultimately, the product failed to attract many customers and Philips abandoned the product.

Minimizing Development Cycle Time

Even products that achieve a very close fit with customer requirements can fail if they take too long to bring to market. As discussed in Chapter Five, bringing a product to market early can help a firm build brand loyalty, preemptively capture scarce assets, and build customer switching costs. A firm that brings a new product to market late may find that customers are already committed to other products. Also, a company that is able to bring its product to market early has more time to develop (or encourage others to develop) complementary goods that enhance the value and attractiveness of the product.[2] Other things being equal, products that are introduced to the market earlier are likely to have an installed base and availability of complementary goods advantage over later offerings.

development cycle time
The time elapsed from project initiation to product launch, usually measured in months or years.

Another important consideration regarding **development cycle time** relates to the cost of development and the decreasing length of product life cycles. First, many development costs are directly related to time. Both the expense of paying employees involved in development and the firm's cost of capital increase as the development cycle lengthens. Second, a company that is slow to market with a particular generation of technology is unlikely to be able to fully amortize the fixed costs of development before that generation becomes obsolete. This phenomenon is particularly vivid in dynamic industries such as electronics where life cycles can be as short as 12 months (e.g., personal computers and semiconductors). Companies that are slow to market may find that by the time they have introduced their products, market demand has already shifted to the products of a subsequent technological generation.

Finally, a company with a short development cycle can quickly revise or upgrade its offering as design flaws are revealed or technology advances. A firm with a short development cycle can take advantage of both first-mover *and* second-mover advantages.

Some researchers have pointed out the costs of shortening the development cycle and rushing new products to market. For example, Dhebar points out that rapid product introductions may cause adverse consumer reactions; consumers may regret past purchases and be wary of new purchases for fear they should rapidly become obsolete.[3] Other researchers have suggested that speed of new product development may come at the expense of quality or result in sloppy market introductions.[4] Compressing development cycle time can result in overburdening the development team, leading to problems being overlooked in the product design or manufacturing process. Adequate product testing may also be sacrificed to meet development schedules.[5] However, despite these risks, most studies have found a strong positive relationship between speed and the commercial success of new products.[6]

Controlling Development Costs

Sometimes a firm engages in an intense effort to develop a product that exceeds customer expectations and brings it to market early, only to find that its development costs have ballooned so much that it is impossible to recoup the development expenses even if the product is enthusiastically received by the market. This highlights the fact that development efforts must be not only *effective,* but also *efficient.* Later in the chapter, ways to monitor and control development costs are discussed.

SEQUENTIAL VERSUS PARTLY PARALLEL DEVELOPMENT PROCESSES

partly parallel development process
A development process in which some (or all) of the development activities at least partially overlap. That is, if activity A would precede activity B in a partly parallel development process, activity B might commence before activity A is completed.

Before the mid-1990s, most U.S. companies proceeded from one development stage to another in a sequential fashion (see Figure 11.1a). The process included a number of gates at which managers would decide whether to proceed to the next stage, send the project back to a previous stage for revision, or kill the project. Typically, R&D and marketing provided the bulk of the input in the opportunity identification and concept development stages, R&D took the lead in product design, and manufacturing took the lead in process design. According to critics, one problem with such a system emerges at the product design stage when R&D engineers fail to communicate directly with manufacturing engineers. As a result, product design proceeds without manufacturing requirements in mind. A sequential process has no early warning system to indicate that planned features are not manufacturable. Consequently, cycle time can lengthen as the project iterates back and forth between the product design and process design stages.[7]

To shorten the development process and avoid time-consuming and costly iterations between stages of the development cycle, many firms have adopted a **partly parallel development process**, as shown in Figure 11.1b.[8] Product design is initiated before concept development is complete, and process design is begun long before product

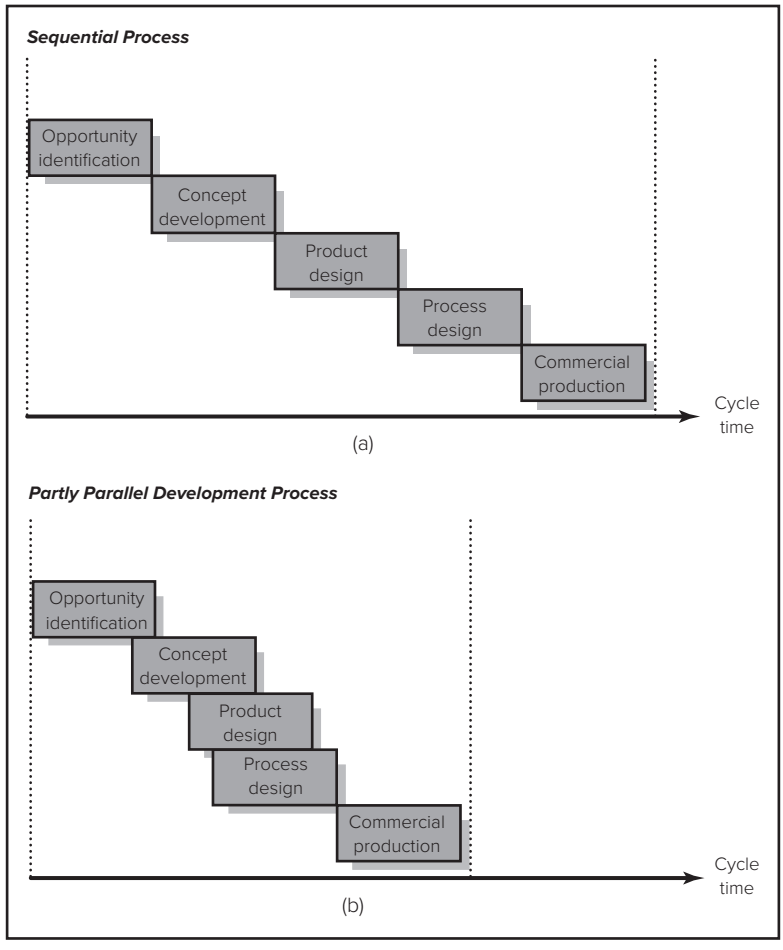


FIGURE 11.1
Sequential versus Partly Parallel Development Processes

concurrent engineering
A design method in which stages of product development (e.g., concept development, product design, and process design) and planning for later stages of the product lifecycle (e.g., maintenance, disposal, and recycling) occur simultaneously.

design is finalized, enabling much closer coordination between the different stages and minimizing the chance that R&D will design products that are difficult or costly to manufacture. This should eliminate the need for time-consuming iterations between design stages and shorten overall cycle time. One type of parallel development process, **concurrent engineering**, involves not only conducting the typical product development stages simultaneously but also takes into account downstream stages of a product's lifecycle such as maintenance and disposal.

Parallel development processes are not universally endorsed, however. In some situations, using a parallel development process can substantially increase the risks or costs of the development process. If, for example, variations in product design require significant changes to the process design, beginning process design before product design is finalized can result in costly rework of the production process. Such risks are especially high in markets characterized by rapid change and uncertainty.[9]

Theory in Action The Development of Zantac

In the 1970s, Glaxo Holdings PLC of Great Britain was one of the larger health care conglomerates in the world, known principally for its baby food, but it needed a new hit product to stimulate sales. While contemplating research possibilities, the head of Glaxo's research laboratory, David Jack, attended a lecture by James Black, a Nobel Prize–winning scientist and researcher for U.S.-based SmithKline Beecham. During the lecture, Black described a new possibility for treating ulcers that involved compounds called H_2 blockers that would inhibit gastrointestinal cells from secreting acid. Jack was intrigued. Ulcers were a common problem, and thus represented a large market opportunity for an effective solution. Jack began experimenting with different compounds in pursuit of a formula that would be safe and effective. Unfortunately, researchers at SmithKline Beecham beat Glaxo to the finish line, introducing Tagamet in 1977. Tagamet revolutionized ulcer treatment, and sales grew phenomenally.[a]

Discouraged but not thwarted, Jack's team kept working. Other companies (including Merck and Eli Lilly) were also developing their own ulcer treatments, and Jack believed that beating them to market might still give the company a shot at a significant share. In that same year, the team came up with a compound based on ranitidine (Tagamet was based on a compound called cimetidine) that achieved the desired objectives. However, Jack realized that if Glaxo was going to beat Merck and Eli Lilly to market, it would need to radically shorten the typical 10-year testing period required to secure regulatory approval and bring the product to market. To achieve this, Jack proposed the first parallel development process used in the pharmaceutical industry. Instead of following the typical sequence of testing (e.g., from rats to monkeys, and from short-term toxicity to long-term toxicity), Jack proposed doing all of the tests concurrently.[b] This intensified development process could potentially cut the cycle time in half—to five years—however, it would also be expensive and risky. If the development efforts increased the research costs substantially, it would be much harder to recoup those expenses through sales of the drug.

Fortunately for Jack's team, Paul Girolami, Glaxo's director of finance, chose to champion the project. Girolami argued that the company should be willing to risk its range of decently profitable products for one potentially sensational drug, stating, "Having all your eggs in one basket concentrates the mind because you had better make sure it is a good basket."[c] Not only was he able to convince the company that it was worth investing in the shortened development process, but also insisted that the product be modified so that it could be taken once a day (Tagamet required twice-a-day use) and so that the product would have fewer side effects than Tagamet. These features would help differentiate Zantac as a superior product, and it was hoped they would enable Glaxo to take share away from SmithKline Beecham. The development process was successful, and the product was ready for launch in 1982. To recoup its development costs, Girolami chose a premium pricing strategy for the product (one-third higher than Tagamet), arguing that its advantages would warrant its additional cost. He also insisted that the product be launched globally in all major markets, and he set up a distribution alliance with Hoffman-LaRoche to speed up the product's penetration of the U.S. market.

Girolami's strategies were successful, and by the end of the year, Zantac was stealing about 100,000 patients a month from Tagamet. By 1987, Zantac sales had exceeded Tagamet's, and by 1991, Zantac became the world's No. 1 selling prescription drug and the first drug ever to achieve $1 billion in U.S. sales.[d] Both David Jack and Paul Girolami were knighted, and Sir Paul Girolami was appointed chairman of Glaxo.[e]

[a] A. Corsig, T. Soloway, and R. Stanaro, "Glaxo Holdings PLC: Zantac," in *New Product Success Stories,* ed. R. Thomas (New York: John Wiley & Sons, Inc., 1995), 242–52.

[b] Ibid.

[c] C. Kennedy, "Medicine Man to the World," *Director* 46, no. 4 (1992): 106–10.

[d] "Anti-Ulcer Drugs: Too Much Acid," *The Economist* 318, no. 7700 (1991): 82–4.

[e] Corsig, Soloway, and Stanaro, "Glaxo Holdings PLC: Zantac."

Furthermore, once process design has commenced, managers may be reluctant to alter the product design even if market testing reveals that the product design is suboptimal. It is precisely these risks that the stage-gate* process (discussed later in the chapter) attempts to minimize.

PROJECT CHAMPIONS

A number of studies on new product development have suggested that firms should assign (or encourage) a senior member of the company to champion a new product development project.[10] Senior executives have the power and authority to support and fight for a project. They can facilitate the allocation of human and capital resources to the development effort, ensuring that cycle time is not extended by resource constraints, and help ensure that the project can sustain the necessary momentum to surmount the hurdles that inevitably will arise.[11] A senior project champion also can stimulate communication and cooperation between the different functional groups involved in the development process. Given that interfunctional communication and cooperation are necessary both to compress cycle time and to achieve a good fit between product attributes and customer requirements, the use of executive sponsors can improve the effectiveness of the development process. As of 2001, 68 percent of North American firms, 58 percent of European firms, and 48 percent of Japanese firms reported using senior managers to champion new product development projects.[12] An example of a successful use of project championing is described in the accompanying Theory in Action: The Development of Zantac.

Risks of Championing

Vigorous project championing, however, also has its risks. A manager's role as champion may cloud judgment about the true value of the project. Optimism is the norm in product development—surveys indicate a systematic upward bias in estimates of future cash flows from a project.[13] In the role of champion, this optimism is often taken to extreme levels. Managers may fall victim to escalating commitment and be unable (or unwilling) to admit that a project should be killed even when it is clear to many others in the organization that the project has gone sour, or the factors driving the project's original value are no longer relevant. While it is common to read stories about projects that succeed against all odds because of the almost fanatical zeal and persistence of their champions, bankruptcy courts are full of companies that should have been less zealous in pursuing some projects. Managers who have invested their reputations and years of their lives in development projects may find it very difficult to cut their losses, in much the same way that individuals tend to hold losing stocks much longer than they should due to the temptation to try to recoup what they have lost. Though the champion's seniority is an asset in gaining access to resources and facilitating coordination, this same seniority may also make others in the firm unwilling to challenge the project champion even if it has become apparent that the project's expected value has turned negative.[14]

Firms may benefit from also developing "antichampions" who can play the role of devil's advocate. Firms should also encourage a corporate culture open to the expression

*Note: Stage-Gate® is a registered trademark of Stage-Gate International Inc.

Research Brief Five Myths about Product Champions

Stephen Markham and Lynda Aiman-Smith argue that a number of myths have become widely accepted about new product champions. While Markham and Aiman-Smith believe that product champions are critical to new product development, they also argue that for product champions to be effective, their role in the development process must be completely understood. Markham and Aiman-Smith conducted a systematic review of the theoretical and empirical literature on product champions and identified five popular myths:

Myth 1: *Projects with champions are more likely to be successful in the market.* Markham and Aiman-Smith's review of the empirical data on use of project champions found that projects with champions were just as likely to be market failures as market successes. Markham and Aiman-Smith point out that while champions may improve the likelihood of a project being completed, the factors determining its market success are often beyond the champion's control.[a]

Myth 2: *Champions get involved because they are excited about the project, rather than from self-interest.* Markham and Aiman-Smith report that empirical evidence suggests champions are more likely to support projects that will benefit the champion's own department.[b]

Myth 3: *Champions are more likely to be involved with radical innovation projects.* Empirical evidence from multiple large sample studies indicates that champions were equally likely to be involved with radical versus incremental innovation projects.

Myth 4: *Champions are more likely to be from high (or low) levels in the organization.* Markham and Aiman-Smith argue that there are myths about both high-level and low-level managers being more likely to be product champions. Though stories abound featuring prominent senior managers supporting projects, as do stories featuring low-level champions fighting vigorously for a project's success, empirical evidence suggests that champions may arise from any level in the organization. (Note that this research does not indicate champions from all levels of the firm are equally effective.)

Myth 5: *Champions are more likely to be from marketing.* Markham and Aiman-Smith argue that while anecdotal evidence may more often emphasize champions who have marketing backgrounds, an empirical study of 190 champions found that champions arose from many functions of the firm. Specifically, the study found that 15 percent of champions were from R&D, 14 percent were from marketing, 7 percent were from production and operations, and 6 percent were general managers. Interestingly, 8 percent of champions were potential users of the innovations.[c]

[a] Stephen Markham, Stephen Green, and Raja Basu, "Champions and Antagonists: Relationships with R&D Project Characteristics and Management," *Journal of Engineering and Technology Management* 8 (1991): 217–42; Stephen Markham and Abbie Griffin, "The Breakfast of Champions: Associations between Champions and Product Development Environments, Practices, and Performance," *The Journal of Product Innovation Management* 15 (1998): 436–54; and Stephen Markham, "Corporate Championing and Antagonism as Forms of Political Behavior: An R&D Perspective," *Organization Science* 11 (2000): 429–47.

[b] Stephen Markham, "Corporate Championing and Antagonism as Forms of Political Behavior."

[c] Diana Day, "Raising Radicals: Different Processes for Championing Innovative Corporate Ventures," *Organization Science* 5 (1994): 148–72.

of dissenting opinion, and champions should be encouraged to justify their projects on the basis of objective criteria, without resorting to force of personality.[15] The accompanying Research Brief describes five myths that have become widely accepted about project champions.

INVOLVING CUSTOMERS AND SUPPLIERS IN THE DEVELOPMENT PROCESS

As mentioned previously, many products fail to produce an economic return because they do not fulfill customer requirements for performance and price, or because they take too long to bring to market. Both of these problems can be reduced by involving customers and suppliers in the development process.

Involving Customers

Firms often make decisions about projects on the basis of financial considerations and level of production and technical synergy achieved by the new product proposal rather than on marketing criteria. This can lead to an overemphasis on incremental product updates that closely fit existing business activities.[16] The screening decision should focus instead on the new product's advantage and superiority to the consumer, and the growth of its target market.[17] The end customer is often the one most able to identify the maximum performance capabilities and minimum service requirements of a new product. Including the end customer in the actual development team or designing initial product versions and encouraging user extensions can help the firm focus its development efforts on projects that better fit customer needs.[18] Distributors can also be valuable partners in the new product development process. These organizations will often be the first to know who is buying the product, how they are using it, and be the first to hear of problems with the product or suggestions for how it might be improved.[19]

agile development
A process commonly used in software whereby the overall product is broken down into smaller independent pieces that are worked on by autonomous, self-organizing teams. Features are developed and presented to customers quickly so that the overall product can be rapidly and continuously adapted.

Customers may be involved in the new product development process as an information source, or as actual co-developers of a new product.[20] Many firms use *beta testing* to get customer input early in the development process. A *beta version* of a product is an early working prototype of a product released to users for testing and feedback. Beta versions also enable a firm to signal the market about its product features before the product reaches the commercial production stage. **Agile development** processes (that are now often used in software development) take this approach even further. In agile development, the product is divided into many smaller features or functionalities, and these are rapidly developed into *minimum viable products* and presented to the customer for feedback, enabling rapid incremental adaptation. Other firms involve customers in the new product development process in even more extensive ways, such as enabling customers to *co-create* the end product (this is discussed more in the section below on crowdsourcing).

lead users
Customers who face the same general needs of the marketplace but are likely to experience them months or years earlier than the rest of the market and stand to benefit disproportionately from solutions to those needs.

Some studies suggest that firms should focus on the input of lead users in their development efforts rather than a large sample of customers. **Lead users** are those who face the same needs of the general marketplace but face them months or years earlier than the bulk of the market, and expect to benefit significantly from a solution to those needs.[21] According to a survey by the Product Development & Management Association, on average, firms report using the lead user method to obtain input into 38 percent of the projects they undertake. Not surprisingly, when customers help co-create an innovation, the resulting innovations tend to better fit their needs or expectations.[22] More detail on how firms use lead users is provided in the accompanying Theory in Action section: The Lead User Method of Product Concept Development.

Theory in Action — The Lead User Method of Product Concept Development

Hilti AG, a European manufacturer of construction components and equipment, turned to the lead user method in its development of a pipe hanger (a steel support that fastens pipes to walls or ceilings of buildings). The firm first used telephone interviews to identify customers who had lead user characteristics (were ahead of market trends and stood to benefit disproportionately from the new solution). The lead users were invited to participate in a three-day product concept generation workshop to develop a pipe hanging system that would meet their needs. At the end of the workshop, a single pipe hanger design was selected as the one that best met all the lead users' objectives. The company then presented this design to 12 routine users (customers who were not lead users but who had a long, close relationship with Hilti). Ten of the 12 routine users preferred the new design to previously available solutions, and all but one of the 10 indicated they would be willing to pay a 20 percent higher price for the product. Not only was the project successful, but the lead user method was also faster and cheaper than the conventional market research methods the firm had used in the past to develop its product concepts. Hilti's typical process took 16 months and cost $100,000, but the lead user method took 9 months and cost $51,000.

Source: Cornelius Herstatt and Eric von Hippel, "Developing New Product Concepts via the Lead User Method: A Case Study in a Low-Tech Field," *Journal of Product Innovation Management* 9 (1992): 213–21.

Involving Suppliers

Much of the same logic behind involving customers in the new product development process also applies to involving suppliers. By tapping into the knowledge base of its suppliers, a firm expands its information resources. Suppliers may be actual members of the product team or consulted as an alliance partner. In either case, they can contribute ideas for product improvement or increased development efficiency. For instance, a supplier may be able to suggest an alternative input (or configuration of inputs) that would achieve the same functionality but at a lower cost. Additionally, by coordinating with suppliers, managers can help to ensure that inputs arrive on time and that necessary changes can be made quickly to minimize development time.[23] Consistent with this argument, research has shown that many firms produce new products in less time, at a lower cost, and with higher quality by incorporating suppliers in integrated product development efforts.[24]

Boeing's development of the 777 involved both customers and suppliers on the new product development team; United employees (including engineers, pilots, and flight attendants) worked closely with Boeing's engineers to ensure that the airplane was designed for maximum functionality and comfort. Boeing also included General Electric and other parts suppliers on the project team, so that the engines and the body of the airplane could be simultaneously designed for maximum compatibility.

crowdsourcing A distributed problem-solving model whereby a design problem or production task is presented to a group of people who voluntarily contribute their ideas and effort in exchange for compensation, intrinsic rewards, or a combination thereof.

Crowdsourcing

Firms can also open up an innovation task by directing an *innovation challenge* to third parties such as the general public, or specific, targeted groups of innovators from different networks. Sometimes firms work with third parties directly, and other times they

use a professional crowdsourcing service provider with their own network of innovators. For example, one professional crowdsourcing service provider, NineSigma, manages innovation challenges using a network of more than two million scientists and engineers globally. Similarly, Topcoder helps firms access a community of more than one million software coders. Some of these service providers (such as InnoCentive) operate their service as a closed network where the firm seeking the innovation does not know the details of the solution provider. Others (such as NineSigma) operate open networks where the firm seeking the innovation solution can see all the solution proposals submitted, as well as all contact details of all solution providers that submitted a response to the challenge.

Crowdsourcing challenges typically go through a four step process:

1. **Need translation.** A clear, concise, and compelling need statement is articulated that reduces industry jargon to a minimum, and that brings the challenge down to its most basic science. For example, NineSigma helped a client seeking ways to reduce wrinkles in shirts coming out of a dryer by producing a statement that read: "Our client is seeking ways to reduce surface tension of an organic material." The advantage of such a statement is that the specific application is removed, which invites interest from solution providers from seemingly unrelated industries. In this example, a professor doing integrated circuit research had developed a special polymer that was the solution most favored by the client. The need statement is usually a short one- to two-page document, often called a *Request for Proposal.* Andy Zynga of NineSigma notes, "It is very important to have a very clean and concise need statement to trigger interest. There may be a temptation to put two needs into one statement, but that is highly discouraging to solution providers and will reduce the chances of success."[25]
2. **Connecting.** The innovation challenge must be broadcast to the network of potential solution providers that have been selected as most suitable to respond.
3. **Evaluation/Selection.** Submitted proposals get an in-depth review, and the most interesting solution proposals get selected and collated in the form of a report.
4. **Acquisition.** The firm engages with the solution provider and negotiates an agreement to transfer knowledge, a license, patent, and so on. This usually involves a monetary or other compensation scheme. It may also be necessary to adapt the incoming solution to the specific needs of the firm.

People participate in crowdsourcing for a variety of reasons that often do not include monetary rewards. For example, Ben & Jerry's asked its customers to invent new varieties of ice cream flavors—the submitters of the best flavors were given a trip to the Dominican Republic to see a sustainable fair trade cocoa farm. However, individuals also often participate for the sheer excitement and challenge of solving the problem,[26] or for social or reputational benefits.[27] For example, Fiat Brazil used crowdsourcing to develop a new concept car called the *Fiat Mio* ("My Fiat"). Fiat created a Web site inviting people to create the car of the future. More than 17,000 people from around the world submitted over 11,000 ideas—and not just in the design. Participants were invited to contribute solutions at every stage of the development process, including solving problems related to fuel efficiency and production. Participants received no rewards from their participation other than the pleasure they derived from interacting with Fiat and with each other, and the satisfaction they felt at having their ideas incorporated

into the car. Hundreds of Fiat Mio's co-creators turned up at the unveiling of the car at a Sao Paulo motor show.

Thousands of companies and many public bodies have used crowdsourcing to solve challenges that seemed almost impossible to solve. For example, reducing plastics in our oceans, addressing the opioid crisis, and recapturing carbon are "Grand Challenges" that are being tackled with crowdsourcing approaches right now. Crowdsourcing is not appropriate for all types of projects, however, and many high profile crowdsourcing attempts have failed because moving a project from the idea stage to the implementation stage often requires a vastly different set of skills and organizational resources.

TOOLS FOR IMPROVING THE NEW PRODUCT DEVELOPMENT PROCESS

Some of the most prominent tools used to improve the development process include stage-gate processes, quality function deployment ("house of quality"), design for manufacturing, failure modes and effects analysis, and computer-aided design/computer-aided manufacturing. Using the available tools can greatly expedite the new product development process and maximize the product's fit with customer requirements.

Stage-Gate Processes

As discussed in a previous section, escalating commitment can lead managers to support projects long after their expected value has turned negative, and the cost of pushing bad projects forward can be very high. To help avoid this, many managers and researchers suggest implementing tough **go/kill decision points** in the product development process. The most widely known development model incorporating such go/kill points is the stage-gate process developed by Robert G. Cooper.[28] The stage-gate process provides a blueprint for moving projects through different stages of development. Figure 11.2 shows a typical stage-gate process.

go/kill decision points
Gates established in the development process where managers must evaluate whether or not to kill the project or allow it to proceed.

At each stage, a cross-functional team of people (led by a project team leader) undertakes parallel activities designed to drive down the risk of a development project. At each stage of the process, the team is required to gather vital technical, market, and financial information to use in the decision to move the project forward (go), abandon the project (kill), hold, or recycle the project.

In Stage 1, the team does a quick investigation and conceptualization of the project. In Stage 2, the team builds a business case that includes a defined product, its business justification, and a detailed plan of action for the next stages. In Stage 3, the team begins the actual design and development of the product, including mapping out the manufacturing process, the market launch, and operating plans. In this stage, the team also defines the test plans utilized in the next stage. In Stage 4, the team conducts the verification and validation process for the proposed new product, and its marketing and production. At Stage 5, the product is ready for launch, and full commercial production and selling commence.[29]

Preceding each stage is a go/kill gate. These gates are designed to control the quality of the project and to ensure that the project is being executed in an effective and efficient manner. Gates act as the funnels that cull mediocre projects. Each gate has three components: *deliverables* (these are the results of the previous stage and are the inputs

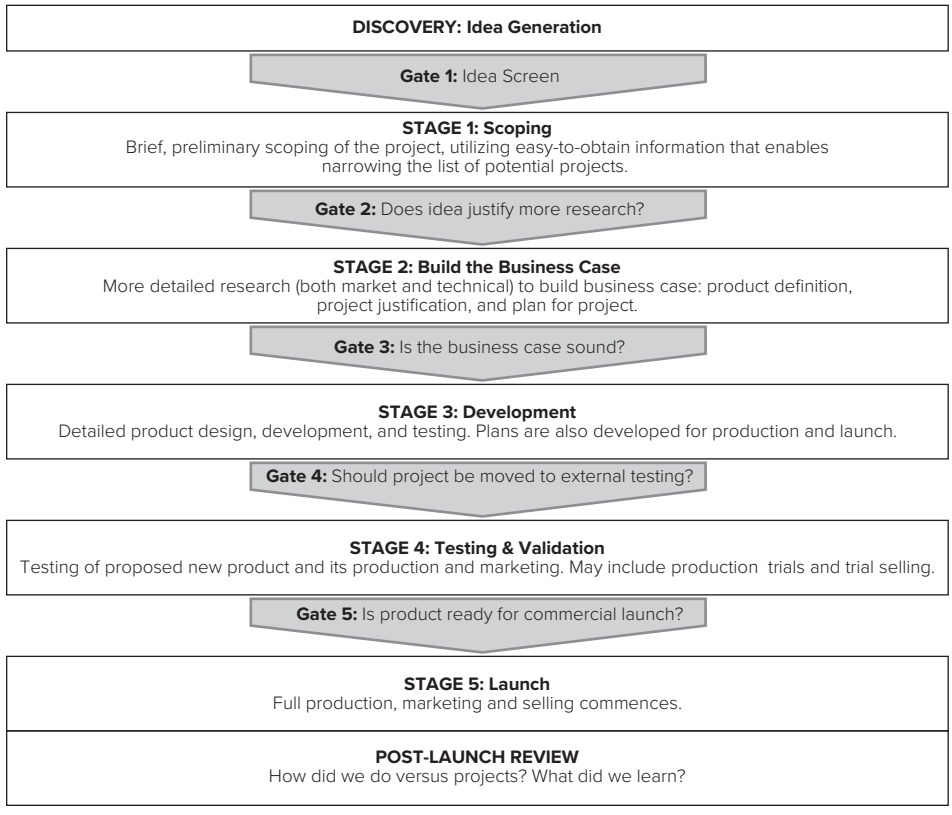


FIGURE 11.2
Typical Stage-Gate Process, from Idea to Launch

Source: Robert G. Cooper, "Stage-Gate Idea to Launch System," *Wiley International Encyclopedia of Marketing: Product Innovation & Management* 5, B. L. Bayus (ed.), (West Sussex UK: Wiley, 2011).

for the gate review), *criteria* (these are the questions or metrics used to make the go/kill decision), and *outputs* (these are the results of the gate review process and may include a decision such as go, kill, hold, or recycle; outputs should also include an action plan for the dates and deliverables of the next gate).

Because each stage of a development project typically costs more than the stage preceding it, breaking down the process into stages deconstructs the development investment into a series of incremental commitments. Expenditures increase only as uncertainty decreases. Figure 11.3 shows the escalation costs and cycle time for each stage of a typical development process in a manufacturing industry.

Many companies have adapted the stage-gate process to more specifically meet the needs of their firm or industry. For example, while managers at Exxon were strong advocates of using a stage-gate process to track and manage development projects, they also felt that the standard five-stage system did not adequately address the needs of a company in which basic research was a primary component in generating innovations. Exxon managers created their own extended stage-gate system to include directed basic research. The resulting stage-gate system included two basic research stages (Stages A and B in Figure 11.4) and five applied research and development stages. In Stage A, the company identifies the potential business incentives and competitive advantages of an envisioned technology. The company then develops a basic research plan that

FIGURE 11.3
Escalation of Development Time and Costs by Stage

Source: Adapted from Frederick Buggie, "Set the 'Fuzzy Front End' in Concrete," *Research Technology Management* 45, no. 4 (2002): 11–4.

Stage	Time	Cost
0. "Here's an idea!"		
1. Formulate–describe and sketch	1 week	$100
2. Conduct preliminary investigations	2 weeks	$1,000
3. Design and define specifications	1 month	$10,000
4A. Develop prototype and test		
4B. Market research		
4C. Strategic fit evaluation and NPV risk analysis	2 months	$100,000
5A. Scale up, build pilot plant		
5B. Market test	8 months	$1 million
6A. Build plant		
6B. Promote, launch, market	16 months	$10 million

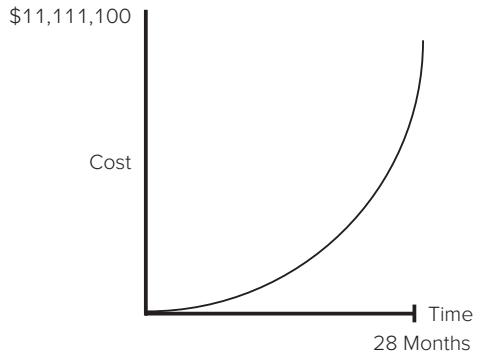
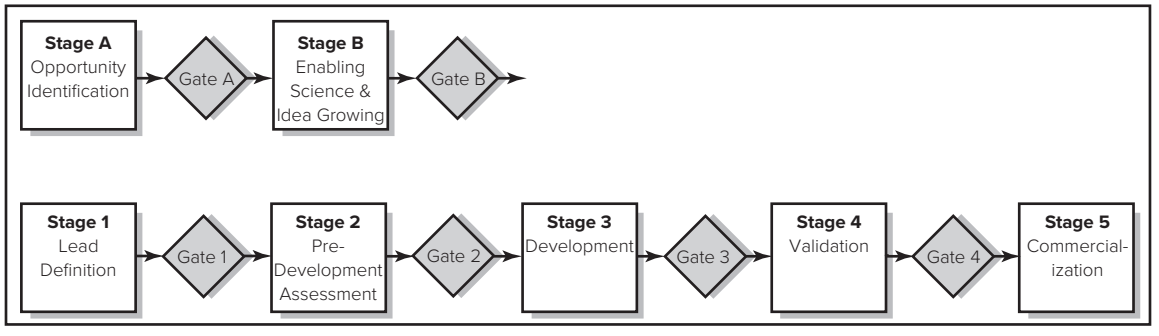


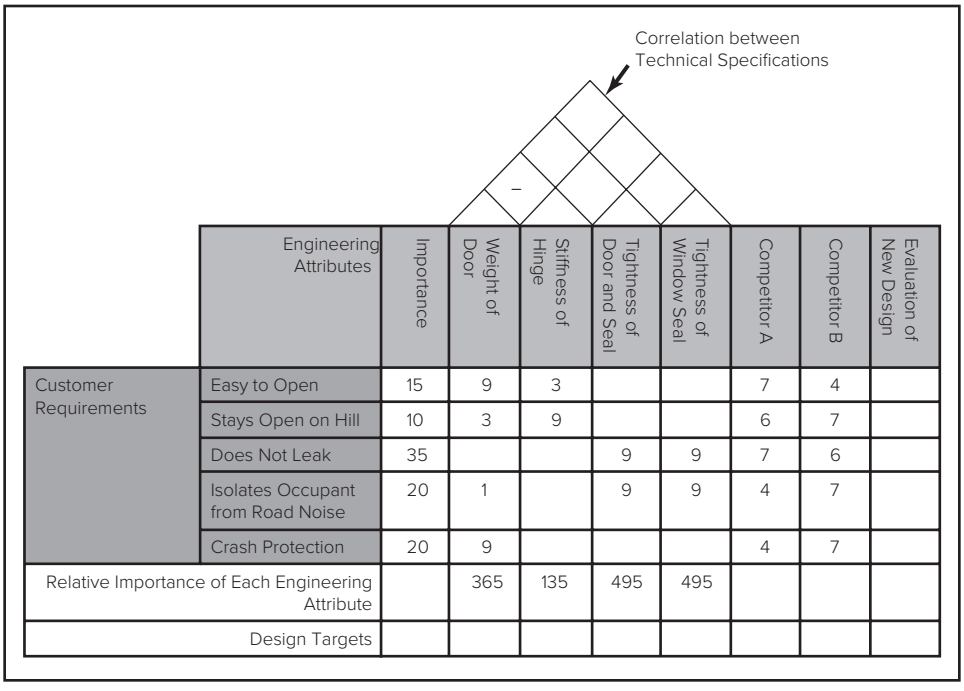
FIGURE 11.4
Exxon Research and Engineering's Stage-Gate System



establishes specific scientific deliverables, the methods of achieving these deliverables, and the required resources. In Stage B, Exxon's research division begins to execute the plan developed in Stage A, using scientific methods to generate leads for addressing the business opportunity. Stage 1 then identifies the best leads, using "proof-of-principle" assessments to establish whether the leads are feasible.[30] Stages 2 through 5 proceed according to a typical stage-gate process.

According to studies by the Product Development and Management Association, nearly 60 percent of firms (including IBM, Procter & Gamble, 3M, General Motors, and Corning) use some type of stage-gate process to manage their new product development process. Corning has made the process mandatory for all information system development projects, and Corning managers believe that the process enables them to better estimate the potential payback of any project under consideration. They also report that the stage-gate process has reduced development time, allows them to identify projects that should be killed, and increases the ratio of internally developed products that result in commercial projects.[31]

FIGURE 11.5
Quality Function Deployment House of Quality for a Car Door



	Engineering Attributes	Importance	Weight of Door	Stiffness of Hinge	Tightness of Door and Seal	Tightness of Window Seal	Competitor A	Competitor B	Evaluation of New Design
Customer Requirements	Easy to Open	15	9	3			7	4	
	Stays Open on Hill	10	3	9			6	7	
	Does Not Leak	35			9	9	7	6	
	Isolates Occupant from Road Noise	20	1		9	9	4	7	
	Crash Protection	20	9				4	7	
Relative Importance of Each Engineering Attribute			365	135	495	495			
Design Targets									

Quality Function Deployment (QFD)—The House of Quality

QFD was developed in Japan as a comprehensive process for improving the communication and coordination among engineering, marketing, and manufacturing personnel.[32] It achieves this by taking managers through a problem-solving process in a very structured fashion. The organizing framework for QFD is the "house of quality" (see Figure 11.5). The house of quality is a matrix that maps customer requirements against product attributes. This matrix is completed in a series of steps.

1. The team must first identify customer requirements. In Figure 11.5, market research has identified five attributes that customers value most in a car door: it is easy to open and close, it stays open on a hill, it does not leak in the rain, it isolates the occupant from road noise, and it protects the passengers in the event of crashes.
2. The team weights the customer requirements in terms of their relative importance from a customer's perspective. This information might be obtained from focus group sessions or direct interaction with the customers. The weights are typically entered as percentages, so that the complete list totals 100 percent.
3. The team identifies the engineering attributes that drive the performance of the product–in this case, the car door. In Figure 11.5, four attributes are highlighted: the weight of the door, the stiffness of the door hinge (a stiff hinge helps the door stay open on a hill), the tightness of the door seal, and the tightness of the window seal.
4. The team enters the correlations between the different engineering attributes to assess the degree to which one characteristic may positively or negatively affect another. The correlations are entered into the matrix that creates the peaked roof

of the house. In this case, the negative sign between door weight and hinge stiffness indicates that a heavy door reduces the stiffness of the hinge.

5. The team fills in the body of the central matrix. Each cell in the matrix indicates the relationship between an engineering attribute and a customer requirement. A number (in this example, one, three, or nine) is placed in the cell located at the intersection of each row (customer requirements) with each column (engineering attributes), which represents the strength of relationship between them. A value of one indicates a weak relationship, a three indicates a moderate relationship and a nine indicates a strong relationship. The cell is left blank if there is no relationship. The ease of opening the door, for example, is strongly related to the weight of the door and moderately related to the stiffness of the door hinge, but is not related to the tightness of the door seal or window seal.
6. The team multiplies the customer importance rating of a feature by its relationship to an engineering attribute (one, three, or nine). These numbers are then summed for each column, yielding a total for the relative importance of each engineering attribute. For example, the stiffness of the hinge influences how easy the door is to open, and whether the door stays open on a hill. Thus to calculate the relative importance of the stiffness of the hinge, the team multiplies the customer importance rating of how easy the door is to open by its relationship to the stiffness of the hinge ($15 \times 3 = 45$), then multiplies the customer importance rating of the door staying open on a hill by its relationship to the stiffness of the hinge ($10 \times 9 = 90$), and then adds these together for the total relative importance of the hinge stiffness ($45 + 90 = 135$). These scores indicate that the tightness of the door and window seals is the most important engineering attribute, followed by the weight of the door.
7. The team evaluates the competition. A scale of one to seven is used (one indicating a requirement is not addressed, and seven indicating a requirement is completely satisfied) to evaluate the competing products (in this case A and B) on each of the customer requirements. These scores go in the right-hand "room" of the house of quality.
8. Using the relative importance ratings established for each engineering attribute and the scores for competing products (from step 7), the team determines target values for each of the design requirements (e.g., the door's optimal weight in pounds).
9. A product design is then created based on the design targets from step 8. The team then evaluates the new design that was created. The team assesses the degree to which each of the customer requirements has been met, entering a one to seven in the far right column of the house of quality, permitting it to compare the new design with the scores of the competing products.

The great strength of the house of quality is that it provides a common language and framework within which the members of a project team may interact. The house of quality makes the relationship between product attributes and customer requirements very clear, it focuses on design trade-offs, it highlights the competitive shortcomings of the company's existing products, and it helps identify what steps need to be taken to improve them. The house of quality is used in settings as diverse as manufacturing, construction,

FIGURE 11.6
Design Rules for Fabricated Assembly Products

Source: Adapted from Melissa A. Schilling and Charles W. L. Hill, "Managing the New Product Development Process," *Academy of Management Executive* 12, no. 3: 67–81.

Design Rule	Impact on Performance
Minimize the number of parts	Simplifies assembly; reduces direct labor; reduces material handling and inventory costs; boosts product quality
Minimize the number of part numbers (use common parts across product family)	Reduces material handling and inventory costs; improves economies of scale (increases volume through commonalty)
Eliminate adjustments	Reduces assembly errors (increases quality); allows for automation; increases capacity and throughput
Eliminate fasteners	Simplifies assembly (increases quality); reduces direct labor costs; reduces squeaks and rattles; improves durability; allows for automation
Eliminate jigs and fixtures	Reduces line changeover costs; lowers required investment

police service, and educational curriculum design.[33] Advocates of QFD maintain that one of its most valuable characteristics is its positive effect upon cross-functional communication and, through that, upon cycle time and the product/customer fit.[34]

Design for Manufacturing

Another method of facilitating integration between engineering and manufacturing, and of bringing issues of manufacturability into the design process as early as possible, is the use of design for manufacturing methods (DFM). Like QFD, DFM is simply a way of structuring the new product development process. Often this involves articulating a series of design rules. Figure 11.6 summarizes a set of commonly used design rules, along with their expected impact on performance.

As shown in Figure 11.6, the purpose of such design rules is typically to reduce costs and boost product quality by ensuring that product designs are easy to manufacture. The easier products are to manufacture, the fewer the assembly steps required, the higher labor productivity will be, resulting in lower unit costs. DEKA Research makes a point of bringing manufacturing into the design process early, because as founder Dean Kamen points out, "It doesn't make sense to invent things that ultimately are made of unobtanium or expensium."[35] In addition, designing products to be easy to manufacture decreases the likelihood of making mistakes in the assembly process, resulting in higher product quality.

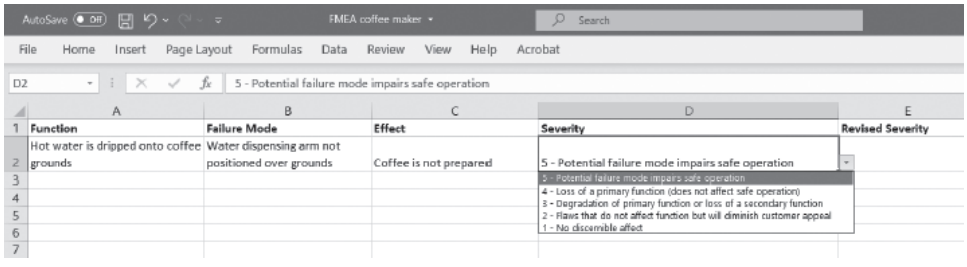
The benefits of adopting DFM rules can be dramatic. Considering manufacturing at an early stage of the design process can shorten development cycle time. In addition, by lowering costs and increasing product quality, DFM can increase the product's fit with customer requirements. For example, when NCR used DFM techniques to redesign one of its electronic cash registers, it reduced assembly time by 75 percent, reduced the parts required by 85 percent, utilized 65 percent fewer suppliers, and reduced direct labor time by 75 percent.[36]

Failure Modes and Effects Analysis

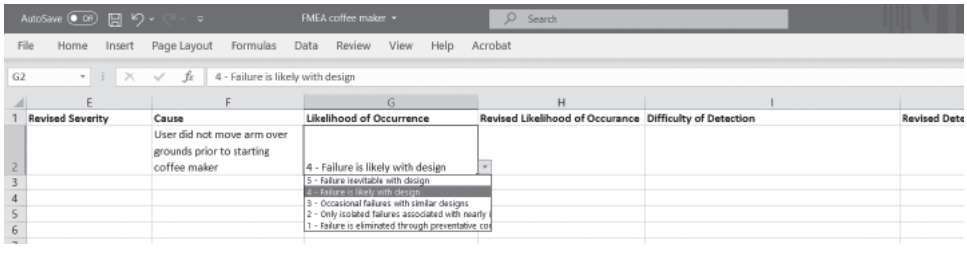
Failure modes and effects analysis (FMEA) is a method by which firms identify potential failures in a system, classify them according to their severity, likelihood, and

detectability, and put a plan into place to address them. First, potential failure modes are identified. For example, a firm developing a commercial aircraft might consider failure modes such as "landing gear does not descend," or "communication system experiences interference"; a firm developing a new line of luxury hotels might consider failure modes such as "a reservation cannot be found," or "guest experiences poor service by room service staff." Potential failure modes are then evaluated on three criteria of the risk they pose: severity, likelihood of occurrence, and difficulty of detection.

For example, suppose we wanted to apply FMEA to improve the design of a drip coffee maker.[37] In our example, users sometimes do not position the water dispensing arm correctly over the coffee grounds before closing the top of the coffee maker and starting the brew cycle. As a result, hot water (rather than coffee) fills the coffee pot and could also potentially spill out of the coffee maker. For this example, we could use the spreadsheet below, where we first identify the function that should occur, the failure mode, its effect, and score the severity of this effect. In this example, we are using scores from one to five, but a different range of scores (e.g., one to ten) may also be used. In this example, the severity of the outcome is the maximum score, five, as not only does the problem cause the device to fail to perform its primary function (brewing coffee) but it also presents a potential safety hazard as hot water might injure the user. We will leave the "revised severity" cell empty until after we have addressed the problem.

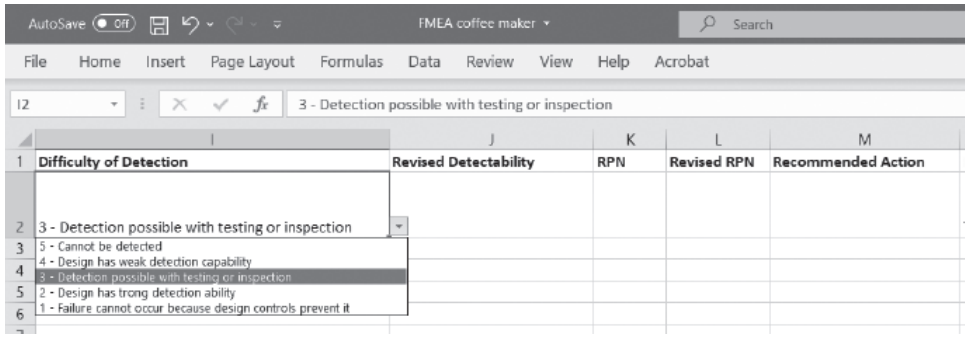


We then identify the cause, and we score the likelihood of occurrence, which in this example we have assigned a four as the failure mode is likely but not inevitable. We will fill the "revised likelihood of occurrence" after the problem is addressed.

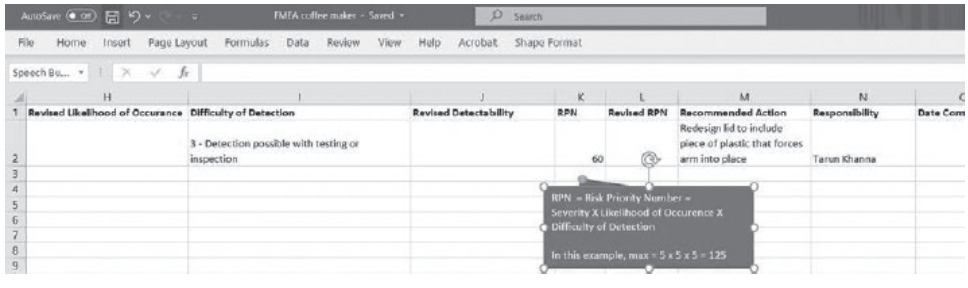


Next, we score the difficulty of detection, that is, how easy is it to determine the failure will occur. In this example we have scored the failure mode with a three, that is, detection possible with testing or inspection, as the user can easily ascertain that the

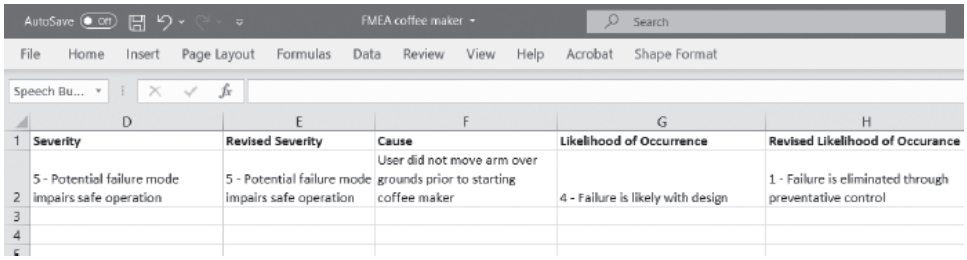
arm is in the wrong position if they open the top of the coffee maker and visually inspect its alignment over the coffee grounds. We will leave the "revised detectability" cell empty until after we have addressed the problem.



We can now multiple the scores for severity, likelihood of occurrence, and difficulty of detection to create a risk priority number. Because we have used scales from one to five for the three categories of severity, likelihood of occurrence and difficulty of detection, the maximum risk priority number is 125 and the minimum is one. In our example, the risk priority number is 60 (i.e., $5 \times 4 \times 3$). We might compare the risk priority number for this failure mode to other potential failure modes of the coffee maker to prioritize our efforts, or we might have an absolute cutoff (e.g., "no greater than 15") that we might decide is allowable and we thus must address any failure mode that has a higher risk priority number than this cutoff. In our example, we have determined that the recommended action is to redesign the lid to include a piece of plastic that forces the arm into alignment over the grounds when the user closes the lid, and we have assigned Tarun Khanna responsibility for ensuring this action is completed.

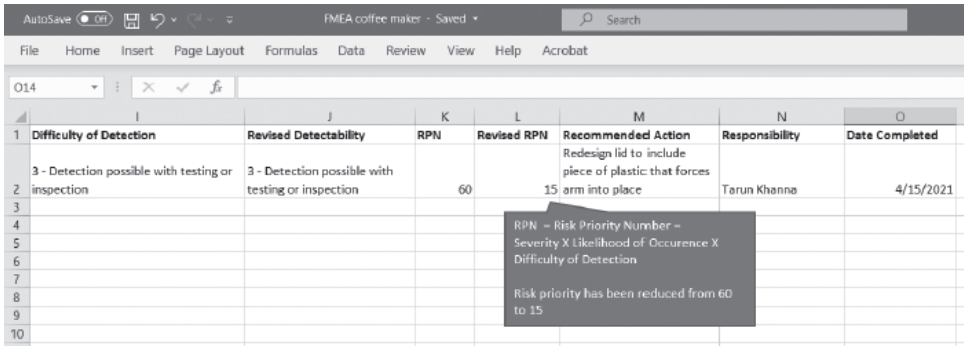


After the recommended action is complete, we can reevaluate the severity, likelihood of occurrence and difficulty of detection. In our example, the redesigned lid does not change the severity of failure should the failure occur, but it lowers the likelihood of occurrence to one because it prevents the occurrence from happening almost completely.



	D	E	F	G	H
1	**Severity**	**Revised Severity**	**Cause**	**Likelihood of Occurrence**	**Revised Likelihood of Occurance**
2	5 - Potential failure mode impairs safe operation	5 - Potential failure mode impairs safe operation	User did not move arm over grounds prior to starting coffee maker	4 - Failure is likely with design	1 - Failure is eliminated through preventative control
3					
4					

The redesign does not change the difficulty of detection of the failure. However, by lowering the likelihood of occurrence, this modest change lowers the risk priority number to fifteen and the failure mode is no longer deemed high risk.



	I	J	K	L	M	N	O
1	**Difficulty of Detection**	**Revised Detectability**	**RPN**	**Revised RPN**	**Recommended Action**	**Responsibility**	**Date Completed**
2	3 - Detection possible with testing or inspection	3 - Detection possible with testing or inspection	60	15	Redesign lid to include piece of plastic that forces arm into place	Tarun Khanna	4/15/2021
3							
4							
5							
6							
7							
8							
9							
10							

RPN – Risk Priority Number – Severity X Likelihood of Occurence X Difficulty of Detection

Risk priority has been reduced from 60 to 15

FMEA was originally introduced in the 1940s by the U.S. Armed Forces and was initially adopted primarily for development projects in which the risks posed by failure were potentially very severe. For example, FMEA was widely used in the Apollo Space Program in its mission to put a man on the moon, and was adopted by Ford after its extremely costly experience with its Pinto model (the location of the gas tank in the Pinto made it exceptionally vulnerable to collisions, leading to fire-related deaths; Ford was forced to recall the Pintos to modify the fuel tanks, and was forced to pay out record-breaking sums in lawsuits that resulted from accidents).[38] Soon, however, FMEA was adopted by firms in a wide range of industries, including many types of manufacturing industries, service industries, and health care. A recent PDMA study found that firms report using FMEA in 40 percent of the projects they undertake.[39]

Computer-Aided Design/Computer-Aided Engineering/Computer-Aided Manufacturing

Computer-aided design (CAD) and computer-aided engineering (CAE) are the use of computers to build and test product designs. Rapid advances in computer technology have enabled the development of low-priced and high-powered graphics-based workstations. With these workstations, it is now possible to achieve what could previously be done only on a supercomputer: construct a three-dimensional "working" image of a product or subassembly. CAD enables the creation of a three-dimensional model; CAE makes it possible to virtually test the characteristics (e.g., strength, fatigue, and reliability) of this model. The combination enables product prototypes to be developed and

Theory in Action Computer-Aided Design of an America's Cup Yacht

Team New Zealand discovered the advantages of using sophisticated computer-aided-design techniques in designing the team's 1995 America's Cup yacht. The team had traditionally relied on developing smaller-scale prototypes of the yacht and testing the models in a water tank. However, such prototypes took months to fabricate and test and cost about $50,000 per prototype. This greatly limited the number of design options the team could consider. However, by using computer-aided-design technologies, the team could consider many more design specifications more quickly and inexpensively. Once the basic design is programmed, variations on that design can be run in a matter of hours, at little cost, enabling more insight into design trade-offs. Computer-aided design also avoided some of the problems inherent in scaling up prototypes (some features of the scaled-down prototype boats would affect the flow of water differently from full-scale boats, resulting in inaccurate results in prototype testing). The team would still build prototypes, but only after considering a much wider range of design alternatives using computer-aided-design methods. As noted by design team member Dave Egan,

> Instead of relying on a few big leaps, we had the ability to continually design, test, and refine our ideas. The team would often hold informal discussions on design issues, sketch some schematics on the back of a beer mat, and ask me to run the numbers. Using traditional design methods would have meant waiting months for results, and by that time, our thinking would have evolved so much that the reason for the experiment would long since have been forgotten.

Source: M. Iansiti and A. MacCormack, "Team New Zealand," Harvard Business School case no. 9-697-040, 1997.

tested in virtual reality. Engineers can quickly adjust prototype attributes by manipulating the three-dimensional model, allowing them to compare the characteristics of different product designs. Eliminating the need to build physical prototypes can reduce cycle time and lower costs as illustrated in the accompanying Theory in Action: Computer-Aided Design of an America's Cup Yacht. Visualization tools and 3-D software are even being used to allow nonengineering customers to see and make minor alterations to the design and materials.

Computer-aided manufacturing (CAM) is the implementation of machine-controlled processes in manufacturing. CAM is faster and more flexible than traditional manufacturing.[40] Computers can automate the change between different product variations and allow for more variety and customization in the manufacturing process.

three-dimensional printing
A method whereby a design developed in a computer-aided design program is printed in three dimensions by laying down thin strips of material until the model is complete.

A recent incarnation of computer-aided manufacturing is **three-dimensional printing** (also known as additive manufacturing), whereby a design developed in a computer-aided design program is literally printed by laying down thin horizontal cross sections of material until the model is complete. Unlike traditional methods of constructing a model, which typically involve machining a mold that can take several days to complete, three-dimensional printing can generate a model in a few hours. By 2021, three-dimensional printing was being used to create products as diverse as food, clothing, jewelry, solid-state batteries, and even titanium landing gear brackets for supersonic jets.[41] Biotechnology firms were even using three-dimensional printing for use in creating organs by depositing layers of living cells onto a gel medium.[42] This method has recently begun rapidly replacing injection molding for products that are produced in relatively small quantities.

Theory in Action Postmortems at Microsoft

At Microsoft, almost all projects receive either a postmortem discussion or a written postmortem report to ensure that the company learns from each of its development experiences. These postmortems tend to be extremely candid and can be quite critical. As noted by one Microsoft manager, "The purpose of the document is to beat yourself up." Another Microsoft manager notes that part of the Microsoft culture is to be very self-critical and never be satisfied at getting things "halfway right." A team will spend three to six months putting together a postmortem document that may number anywhere from less than 10 pages to more than 100. These postmortem reports describe the development activities and team, provide data on the product size (e.g., lines of code) and quality (e.g., number of bugs), and evaluate what worked well, what did not work well, and what the group should do to improve on the next project. These reports are then distributed to the team members and to senior executives throughout the organization.

Source: Michael A. Cusumano and R. W. Selby, *Microsoft Secrets* (New York: Free Press, 1995).

TOOLS FOR MEASURING NEW PRODUCT DEVELOPMENT PERFORMANCE

Many companies use a variety of metrics to measure the performance of their new product development process. In addition to providing feedback about a particular new product, such performance assessments help the company improve its innovation strategy and development processes. For example, evaluating the performance of its new product development process may provide insight into which core competencies the firm should focus on, how projects should be selected, whether or not it should seek collaboration partners, how it should manage its development teams, and so on.

Both the metrics used by firms and the timing of their use vary substantially across firms. In a survey by Goldense and Gilmore, 45 percent of companies reported using periodic reviews at calendar periods (e.g., monthly or weekly) and at predetermined milestones (e.g., after product definition, after process design, post launch, etc.).[43] Microsoft, for example, uses postmortems to measure new product development performance, as described in the accompanying Theory in Action: Postmortems at Microsoft. Measures of the success of the new product development process can help management to:

- Identify which projects met their goals and why.
- Benchmark the organization's performance compared to that of competitors or to the organization's own prior performance.
- Improve resource allocation and employee compensation.
- Refine future innovation strategies.[44]

Multiple measures are important because any measure used singly may not give a fair representation of the effectiveness of the firm's development process or its overall innovation performance. Also, the firm's development strategy, industry, and other environmental circumstances must be considered when formulating measures and interpreting results. For example, a firm whose capabilities or objectives favor development of breakthrough projects may experience long intervals between product introductions and receive a low score on measures such as cycle time or percent of sales

earned on projects launched within the past five years, despite its success at its strategy. Conversely, a firm that rapidly produces new generations of products may receive a high score on such measures even if it finds its resources are overtaxed and its projects are overbudget. Additionally, the success rate of new product development can vary significantly by industry and project type. Some authors argue that even firms with excellent new product development processes should not expect to have a greater than 65 percent success rate for all new products launched.[45]

New Product Development Process Metrics

Many firms use a number of methods to gauge the effectiveness and efficiency of the development process. These measures capture different dimensions of the firm's ability to successfully shepherd projects through the development process. To use such methods, it is important to first define a finite period in which the measure is to be applied in order to get an accurate view of the company's current performance; this also makes it easier for the manager to calculate a response. The following questions can then be asked:

1. What was the average cycle time (time to market) for development projects? How did this cycle time vary for projects characterized as breakthrough, platform, or derivative?
2. What percentage of development projects undertaken within the past five years met all or most of the deadlines set for the project?
3. What percentage of development projects undertaken within the past five years stayed within budget?
4. What percentage of development projects undertaken within the past five years resulted in a completed product?

Overall Innovation Performance

Firms also use a variety of methods to assess their overall performance at innovation. These measures give an overall view of the bang for the buck the organization is achieving with its new product development processes. Such measures include:

1. What is the firm's return on innovation? (This measure assesses the ratio of the firm's total profits from new products to its total expenditures, including research and development costs, the costs of retooling and staffing production facilities, and initial commercialization and marketing costs.)
2. What percentage of projects achieve their sales goals?
3. What percentage of revenues are generated by products developed within the past five years?
4. What is the firm's ratio of successful projects to its total project portfolio?

Summary of Chapter

1. Successful new product development requires achieving three simultaneous objectives: maximizing fit with customer requirements, minimizing time to market, and controlling development costs.
2. Many firms have adopted parallel development processes to shorten the development cycle time and to increase coordination among functions such as R&D, marketing, and manufacturing.

3. Many firms have also begun using project champions to help ensure a project's momentum and improve its access to key resources. Use of champions also has its risks, however, including escalating commitment and unwillingness of others in the organization to challenge the project.
4. Involving customers in the development process can help a firm ensure that its new products match customer expectations. In particular, research indicates that involving lead users can help the firm understand what needs are most important to customers, helping the firm to identify its development priorities. Involving lead users in the development process can also be faster and cheaper than involving a random sample of customers in the development process.
5. Many firms use beta testing to get customer feedback, exploit external development of the product, and signal the market about the firm's upcoming products.
6. Firms can also involve suppliers in the development process, helping to minimize the input cost of a new product design and improving the likelihood that inputs are of appropriate quality and arrive on time.
7. Stage-gate processes offer a blueprint for guiding firms through the new product development process, providing a series of go/kill gates where the firm must decide if the project should be continued and how its activities should be prioritized.
8. Quality function deployment can be used to improve the development team's understanding of the relationship between customer requirements and engineering attributes. It can also be a tool for improving communication between the various functions involved in the development process.
9. Failure Modes and Effects Analysis can be used to help firms prioritize their development efforts in order to reduce the likelihood of failures that will have the greatest impact on the quality, reliability, and safety of a product or process.
10. Design for manufacturing and CAD/CAM are additional tools development teams can use to reduce cycle time, improve product quality, and control development costs.
11. Firms should use a variety of measures of their new product development effectiveness and overall innovation performance to identify opportunities for improving the new product development process and improving the allocation of resources.

Discussion Questions

1. What are some of the advantages and disadvantages of a parallel development process? What obstacles might a firm face in attempting to adopt a parallel process?
2. Consider a group project you have worked on at work or school. Did your group use mostly sequential or parallel processes?
3. Name some industries in which a parallel process would not be possible or effective.
4. What kinds of people make good project champions? How can a firm ensure that it gets the benefits of championing while minimizing the risks?
5. Is the stage-gate process consistent with suggestions that firms adopt parallel processes? What impact do you think using stage-gate processes would have on development cycle time and development costs?
6. What are the benefits and costs of involving customers and suppliers in the development process?

Suggested Further Reading

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Chapter **Twelve**

Managing New Product Development Teams

Magna International's Carbon Fiber "Lightweighting" Project

Magna International is a Canada-based company that produces components and technology for automakers such as General Motors, BMW, Tesla, Volkswagen, and others. Its 2020 sales of $32.6 billion made it the largest North American supplier to the auto manufacturing industry.

Magna prided itself on being an innovation-focused company. Prior to 2004 all of Magna's R&D work was decentralized, with each product division conducting its own R&D work (and as Magna management noted, that was mostly "D," a.k.a., development work, rather than basic research). However, in 2004, Magna's top management team decided to put together a corporate R&D group with the mandate to understand "megatrends"—large scale technological changes that would be likely to affect a wide range of Magna's customers. Some of these megatrends would be translated into corporate-wide initiatives to develop new technologies and capabilities.

One such megatrend that Magna execs identified shortly after forming the corporate R&D group was carbon fiber composites for "lightweighting" (i.e., making products that are typically made of steel significantly lighter). Magna had been approached by several Chinese automakers about projects involving the creation of ultralight cars, and Magna had begun conducting studies of what the best options might be. Then in 2007, BMW announced a plan dubbed "Project I" whereby the company would develop a new sub-brand to tackle fuel efficiency issues. Project I would involve the development of at least two new car models—the i3 and i8. These cars would be all electric, and it was hoped that they would make extensive use of lightweight carbon-fiber reinforced plastic to help offset the weight of the electric drive system (BMW estimated that the batteries, motors and electronics of an electric-drive system was about 440 pounds more than a comparable gasoline drivetrain).[a]

Carbon fiber composites were anticipated to be expensive, but they offered a number of benefits. In addition to saving weight, carbon fiber would never rust,

and could be glued and bolted together (a process far safer than the stamping and welding used for metal auto bodies). A carbon fiber body also had the unique property of not deforming in a crash; if part of it were damaged that part could simply be cut out and a new section of carbon fiber glued in its place. This was expected to significantly lower the cost of repair and consequently, would lead to lower insurance prices.

BMW issued a Request for Proposals (RFP) to its key suppliers asking them to identify how they might contribute to the project. Tom Pilette, then VP of Product and Process Development at Magna saw this as the perfect opportunity to advance Magna's carbon fiber ambitions. He proposed to the top management team that the company develop a scalable manufacturing method for carbon fiber composites. Pilette's proposal specified that the company would commit approximately $4 million over 36 months, and it would include sub projects like creating a mechanical material library, evaluating crash performance, developing repair procedures, and more. The ultimate objective of the project would be to establish Magna as a global volume supplier of carbon fiber composites for a wide range of manufacturers—not just BMW.

Pilette's proposal was greeted with enthusiasm from the top management team at Magna. Though it would take several months to be formally approved, Pilette was given almost $100K to begin laying the groundwork for the project immediately, signaling that the project would be approved. Pilette set about composing the core project team. First, he chose a project lead—Will Harney, a plastics engineer with significant seniority in the firm who knew everybody. Pilette knew that Harney's deep and wide connections in the firm would help to ensure the team obtained access to the expertise and resources it needed. Harney was also known for challenging the status quo, which was exactly what Pilette was looking for. As Pilette noted, "I wanted somebody who was not afraid to wear that title. He had a lot of responsibility to manage a big visible project at Magna and he needed to think independently."[b] Harney was responsible for ensuring the project was on time, within budget and met expectations.

Next, Pilette and his senior staff laid out the project objectives and time frame to become BMW's lead supplier within 36 months. Pilette and Harney then worked together to identify the other core team members they needed to help complete the project. Harney was based in Toronto at Magna's Canada headquarters, but he would need to lead a team that was spread across three different "Centers of Excellence" around the world. The core team was composed of seven people who were dedicated to the project 100 percent of the time. Five were collocated in Canada, one was in the Central Europe, and one was in Detroit. These seven people were all engineers but came from different engineering specialties, including advanced automation, manufacturing, process and product development, and materials engineering. Each of the members of this core team also had the authority to draw from the talent in their local Center of Excellence, but those people were more likely to work on the project only part time. On average, there were about three people working on the project in Central Europe and three in Detroit at any point in time, in addition to the five people who were in Toronto.

Each of the Centers of Excellence had a different focus. Toronto was known for its composites laboratory and is where much of the material development took place. The Central European center worked on other aspects of manufacturing development, and Detroit was focused on the business side of the project and developed business models. Each Center also managed several sub projects and had its own resource team, budget and timing plan. The core team had weekly project reviews via conference calls and also met in person once a month to update their plans, meet with potential customers, and create reports that would go to Magna's top management team. Furthermore, every quarter the core team compiled an overall progress report.

Team member compensation was a combination of salary plus bonus, where the bonus was based on successful implementation of the project as a team (60 percent) and individual contribution (40 percent). Furthermore, all employees had stock in the firm so to some degree they were all motivated by Magna's overall success. When asked whether working across three different cultures posed challenges with team cohesion or coordination, Pilette responded, "The culture is engineering. Everyone just wants to solve the problem."[c] All team members were encouraged to express their views and there was frequent debate within the team over the right course of action. Often the decision would come down to a vote rather than achieving a unanimous agreement. As Pilette noted, "We cannot always wait for everyone to agree. We can never recover lost time from indecision. We will act even if there isn't consensus."[d]

One such example of action without consensus occurred early in the team's process when a disagreement broke out over the best way to configure the molding technique to produce the carbon forms. One group within the team, the more senior members of Magna, wanted to use a conventional compression molding tooling technique. This technique had traditionally been used with polyester and glass, which would flow, fill the mold, and combine together. However, more junior members of the team argued that this process would not work with carbon fiber because it was more difficult to combine the epoxy resin and fiber in the composite; the epoxy resin was likely to just flow past the carbon fibers without becoming integrated with the fibers. This group argued that the mold should be inverted to help trap the resin and seal it to the fiber. The team knew that if they chose incorrectly, it could cost them a lot of time and money to recover. After heated debate, the team held a vote and decided to go with the conventional technique. Almost immediately it was apparent that it was a disaster. The mold trials of prototype parts had poor surface quality because some of the fibers had remained dry. The team had to go back and recreate the mold according to the method the more junior group had proposed. This new method worked beautifully, and Magna ultimately patented that process.[e]

While the experience with the mold could have bred dissensus and animosity in the team, instead it had the opposite effect. As Pilette described, "The whole process brought the whole team together; the old guys started listening to the young guys more and respected them more. It also empowered the young guys to speak up more and to really stretch themselves and innovate more. That early win under their belt made them say 'wow that worked, let's do more!'"[f]

Ultimately BMW never awarded the contract to any of the would-be contenders; they decided to produce a carbon fiber manufacturing process themselves in house. However, in the process of pursuing the contract with BMW, Magna invented many new innovative product and process technologies that subsequently put Magna in a leadership position and helped it to become known as an innovator. In 2016, Magna International was awarded a JEC Americas Innovation Award for its molding process used in making carbon-fiber hoods for Cadillac that were 72 percent lighter than steel hoods.[g] Then in 2018 Magna won first place awards at the Society of Plastics Engineers Automotive Awards for both its lightweight composite liftgates (used by Jeep), and for its carbon fiber subframe (jointly created with Ford Motor Company) that achieved a 34 percent mass savings compared to stamped steel.[h] Magna had become known as one of the most innovative lightweighting companies in the world.

By 2021, carbon fiber still did not account for a large portion of auto manufacturing due to its cost, but it was widely believed that it would play a much more significant role as the industry transitioned to electric vehicles. As noted by Graham Burrow, president of the Magna Exteriors group, "The trend is toward more plastic or plastic derivatives in a car and less metals . . . that's why we want to stick with the technology, to develop it and bring down the cost by reducing cycle time and working on different blends of materials so it's more affordable. I think we're getting a lot closer."[i]

Discussion Questions

1. How would you characterize the structure of the core team used by Magna for the lightweighting project? What do you think are the pros or cons of this type of team for this project?
2. What do you think the benefits and costs were of Magna's team working across three different geographic locations? Do you think those benefits and costs would be different today given the advances that have been made in communication technologies?
3. What are the pros and cons of Pilette's approach of having the teams vote on decisions? What other alternatives are there, and what are their pros and cons?

[a] Jake Holmes, "Ten things we learned about the BMW i3 and i8," *Automobile Magazine,* August 1, 2011.

[b] Interview with Tom Pilette, Magna International, May 19, 2021.

[c] Interview with Tom Pilette, Magna International, May 26, 2021.

[d] Interview with Tom Pilette, Magna International, May 19, 2021.

[e] Interview with Tom Pilette, Magna International, May 19, 2021.

[f] Interview with Tom Pilette, Magna International, May 26, 2021.

[g] 2016. Magna Press Release, "Magna wins 2016 JEC Americas Innovation Award," May 16th.

[h] Evan Milberg, "Magna wins big at SPE for automotive composite innovation," *Composites Manufacturing,* July 23, 2018.

[i] S. Hernandez McGavin, "Magna: The time for carbon fiber is now; supplier sees cost improvement," *Automotive News,* April 17, 2017.

OVERVIEW

New product development often requires activities that are the responsibility of different departments within the organization. To facilitate coordination and cooperation across division boundaries, many organizations create cross-functional new product development teams to lead and manage the development process for the project. There is considerable variation, however, in how teams are formed and managed. In this chapter, we will look at several factors that affect the new product development team's performance, including its size, composition, structure, administration, and leadership.

CONSTRUCTING NEW PRODUCT DEVELOPMENT TEAMS

In constructing new product development teams, the organization must consider how the team's size and composition will affect its mix of skills, its access to resources, and its effectiveness in providing communication and coordination across the divisions.

Team Size

New product development teams may range from a few members to hundreds of members. For example, the development team that created the IBM personal computer had 19 members, but the average team size for development projects at IBM is close to 200.[1] The Yahoo! Internet portal was developed by 13 software developers, split into several small teams of one to three members.[2] Though there is considerable research suggesting that individuals may be better off working alone during early ideation phases (see the Research Brief on "Why Brainstorming Teams Kill Breakthrough Innovation"), teams are often extremely valuable for refining and executing on those ideas. By combining the efforts and expertise of multiple individuals, groups can often outperform individuals on many problem-solving tasks, implying that the size of the development team might be related to its potential for success.[3]

Bigger, however, is not always better. Large teams can create more administrative costs and communication problems, leading to costly delays. Additionally, the larger the team, the harder it can be to foster a shared sense of identity among team members. Further, as the size of the team increases, the potential for **social loafing** also increases. Social loafing occurs when, as the size of the team increases, individuals perceive that they will not receive full credit (or blame) for their contribution to the group effort and so their effort and commitment decrease.[4] The average team size used by U.S. organizations is 11 members,[5] but there is considerable variance in the size of teams used by organizations, and each team may vary in size over the course of a new product development project.

social loafing When an individual in a team does not exert the expected amount of effort and relies instead on the work of other team members.

Team Composition

A lack of communication among the marketing, R&D, and manufacturing functions of a company can be extremely detrimental to new product development. A lack of cross-functional communication can lead to a poor fit between product attributes and customer requirements. R&D cannot design products that fit customer requirements

Research Brief Why Brainstorming Teams Kill Breakthrough Ideas[a]

Over a half a century ago, Alex Osborne wrote an influential book called *Applied Imagination* that opined, "the average person can think up twice as many ideas when working with a group than when working alone." Managers must have been convinced because brainstorming teams took off in popularity and are still used widely to this day. In fact, in business schools it is almost heretical to argue that teams are not more creative than individuals.

The only problem is that Osborne was wrong. Dozens of laboratory studies tried to confirm Osborne's arguments, but found the opposite: brainstorming groups produced fewer ideas, and ideas of less novelty, than the sum of the ideas created by the same number of individuals working alone.[b]

How could this be? Aren't ideas supposed to cross-fertilize, generating new and unusual hybrids?[c] It turns out group idea cross-fertilization is more likely to result in ideas that are mediocre compromises; more exciting ideas come from going solo. There are three main reasons that groups are less creative than individuals working on their own.

1. FEAR OF JUDGMENT

A series of studies by Professors Michael Diehl, Wolfgang Stroebe, Bernard Nijstad, Paul Pauhus, and others found that people self-censor many of their most creative ideas in group brainstorming sessions for fear of being judged negatively by others.[d] When the scientists told groups that their ideas would be judged by their peers, they came up with significantly fewer and less novel ideas than groups that were told they would be evaluated by anonymous judges.

As Isaac Asimov, one of the most famous science fiction writers of all time (and also a biochemistry professor at Boston University) put it, "My feeling is that as far as creativity is concerned, isolation is required. . . . The presence of others can only inhibit this process, since creation is embarrassing. For every new good idea you have, there are a hundred, ten thousand foolish ones, which you naturally do not care to display."[e]

2. PRODUCTION BLOCKING

When people take turns to voice their ideas, those bringing up the rear may forget their ideas before having a chance to voice them. Worse still, the process of attending to another person's ideas redirects a listener's train of thought, essentially hijacking their own idea generation process. Scientists were able to demonstrate this by separating individuals into rooms where they would speak their ideas into a microphone when lights indicated it was their turn. In some of the rooms the individuals could hear the contributions of others, and in some they could not. This study resulted in big creativity losses: being required to wait to give ideas caused people to submit far fewer ideas, and even fewer ideas if they could hear the contributions of others.

Now imagine what happens when people do not have to take turns but instead volunteer ideas at will: the most outgoing people in the group will dominate the idea submission while the quieter people, or those more worried about social pressure, do not submit many (or any) of their ideas. Furthermore, if they do submit their ideas, they may submit only those ideas that build upon the ideas that were already contributed—a sure way to drive out novelty.

3. FEASIBILITY TRUMPS ORIGINALITY

Another series of studies by Professor Eric Rietzschel and colleagues shows that teams are not just bad for idea *generation*; they even impair idea *selection*.[f] If you let people work alone to generate ideas but then let the group select the best ideas to pursue, they will tend to make decisions that reduce novelty. Studies showed that when groups interactively ranked their "best" ideas, they chose ideas that were less original than the average of the ideas produced, and more feasible than the average of the ideas produced. In other words, people tended to weight "feasible" more highly than "original." If a brainstorming group is intended to elicit novel ideas, asking groups to select and submit their best ideas is not the way to achieve that outcome.

continued

concluded

THE BENEFITS OF SPENDING TIME ALONE

Solitude is extremely valuable for creativity; it affords a person the time to think and pursue those things they find intrinsically interesting. It can help them to develop their own beliefs about how the world works, and to develop a self-concept that is less structured by the opinions of others.[g]

Research on serial breakthrough innovators show that many spent significant time alone, investing heavily in self education.[h] Most were voracious readers—Elon Musk, for example, often read for 10 hours a day, and has his brother Kimbal recalled, "If it was the weekend he could go through two books in a day."[i] Elon Musk himself noted in a *Rolling Stone* interview, "I was raised by books. Books and then my parents."[j]

The preceding suggests that when managers want employees to come up with breakthroughs, they need to give them some time alone to ponder their craziest of ideas and follow their paths of association into unknown terrain. They should be urged to come up with ideas freely, without fear of judgment. They should be encouraged to commit their ideas to paper, and to flesh each of them out before exposing them to others. Managers can also follow Google and 3M's lead and give employees in creative roles a significant percentage of their time (e.g., 3M uses 15 percent of work hours) to pursue products of their own creation and choosing. Google's Gmail and Google News, 3M's Post-it Notes, and many other products, were developed this way.

A creative idea can be fragile—easily swept away by the momentum of a group conversation. Almost every team suffers from some degree of groupthink; individuals who are more outspoken or who have forceful personalities can dominate the conversation and the decision making. They can herd a team onto a particular trajectory without even intending to do so, or worse, they can bring everyone to a mediocre compromise. A little isolation and solitude can give other individuals a better chance to develop their breakthrough ideas. These individuals should have time to elaborate their ideas and engage in some initial planning and prototyping before their idea are subjected to review or moved into team development.

[a] Adapted from Melissa A. Schilling, "Why brainstorming groups kill breakthrough ideas (and what to do instead)," Inc., February 9, 2018.

[b] Michael Diehl and Wolfgang Stroebe, "Productivity Loss in Brainstorming Groups: Toward the Solution of a Riddle," *Journal of Personality and Social Psychology* 53 (1987): 497–509; Brian Mullen, Craig Johnson, and Eduardo Salas, "Productivity Loss in Brainstorming Groups: A Meta-Analytic Integration," *Basic and Applied Social Psychology* 12, no. 1 (1991): 3–23; and W. Stroebe, B. A. Nijstad, and E. F. Rietzshel, "Productivity Loss in Brainstorming Groups: The Evolution of a Question," in *Advances in Experimental Social Psychology,* eds. M. P. Zanna, and J. M. Olson (San Diego, CA: Academic Press, 2010), 43: 157–203.

[c] Larry Alton, "5 Strategies for team brainstorming to use in your next meeting," Inc., October 30, 2017.

[d] Michael Diehl and Wolfgang Stroebe, "Productivity Loss in Brainstorming Groups: Toward the Solution of a Riddle," *Journal of Personality and Social Psychology* 53 (1987): 497–509; Paul B. Pauhus et al., "Perception of Performance in Group Brainstorming: The Illusion of Group Productivity," *Personality and Social Psychology Bulletin* 19 (1993): 78–89; and Brian Mullen, Craig Johnson, and Eduardo Salas, "Productivity Loss in Brainstorming Groups: A Meta-Analytic Integration," *Basic and Applied Social Psychology* 12, no. 1 (1991): 3–23.

[e] Isaac Asimov, "Isaac Asimov Asks, 'How Do People Get New Ideas?'" *MIT Technology Review* (1959 essay, reproduced in October 20, 2014).

[f] Eric F. Rietzschel, Bernard Nijstad, and Wolfgang Stroebe, "Productivity Is Not Enough: A Comparison of Interactive and Nominal Brainstorming Groups on Idea Generation and Selection," *Journal of Experimental Social Psychology* 42, no. 2 (2006): 244–51.

[g] Christopher R. Long and James R. Averill, "Solitude: An Exploration of Benefits of Being Alone," *Journal for the Theory of Social Behavior* 33 (2003): 21–44.

[h] Melissa A. Schilling, *Quirky: The Remarkable Story of the Traits, Foibles, and Genius of Breakthrough Innovators Who Changed the World* (New York: Public Affairs, 2018).

[i] Ashlee Vance, *Elon Musk: Tesla, SpaceX, and the Quest for a Fantastic Future* (New York: Harper Collins, 2015).

[j] Neil Strauss, "Elon Musk: The Architect of Tomorrow," Rolling Stone (November 15, 2017).

unless it receives *and* attends to input from marketing regarding those requirements. The manufacturing/R&D interface is also of critical importance because of manufacturing's role in determining two key attributes of a product—*quality* and *price.* By working closely with R&D, manufacturing can ensure that R&D designs products that

are relatively easy to manufacture. Designing for ease of manufacturing can lower both unit costs and product defects, which translates into a lower final price and higher quality. Similarly, a lack of cross-functional communication between functions can lead to longer cycle times as a product iterates back and forth between different stages in the process.

cross-functional teams
Teams whose members are drawn from multiple functional areas in the firm such as R&D, marketing, manufacturing, distribution, and so on.

One of the ways that firms address this problem is by building cross-functional product development teams.[6] **Cross-functional teams** include members drawn from more than one functional area, such as engineering, manufacturing, or marketing.[7] For instance, in Chrysler's "vehicle deployment platform teams," team members are drawn from design, engineering, purchasing, manufacturing, product planning, finance, and marketing. Firms around the world rely heavily on cross-functional teams for their new product development efforts. In 2000, 77 percent of U.S. firms, 67 percent of European firms, and 54 percent of Japanese firms reported using cross-functional teams.[8]

Teams that are composed of people from diverse backgrounds have several advantages over teams that are drawn from only one or a few functional areas.[9] A greater variety of specialists provides a broader knowledge base and increases the cross-fertilization of ideas.[10] Having specialists from different areas also allows the project to draw on a wider mix of information sources in the environment through scanning activities.[11] Functional experts often actively read journals and are involved in associations that directly affect their trade. These activities can lead to the creation and improvement of innovative ideas, as well as provide solutions to product development problems.[12] By combining members of different functional areas into one project team, a wide variety of information sources can be ensured.

A number of arguments also support other types of diversity. Individuals who enter the organization at different times (organizational tenure diversity) are likely to have different contacts outside of the team, enabling the team to draw from a wider mix of resources. Teams that incorporate cultural diversity should show better problem solving by incorporating multiple viewpoints, and teams composed of members who are diverse in terms of education, gender, or age can help ensure a variety of viewpoints are considered and external resources are tapped.[13] Studies have demonstrated that demographic diversity in teams can increase innovative outcomes and overall performance.[14]

Diversity of team members, however, can also raise coordination and communication costs. Individuals tend to interact more frequently and more intensely with other individuals whom they perceive as being similar to them on one or more dimensions.[15] This phenomenon is known as **homophily**. Research on homophily suggests that individuals prefer to communicate with others they perceive as similar to them because it is easier and more comfortable to communicate with those who have similar dialects, mental models, and belief systems.[16] The perception of similarity can also be self-reinforcing—as individuals interact with frequency and intensity, they can develop a common dialect, greater trust, and greater familiarity with the knowledge each possesses. The common dialect, trust, and familiarity, in turn, make the individuals both more willing and more able to exchange information effectively in future interactions. When individuals perceive others as being very different from them, they may be less willing

homophily
The tendency for individuals to like other people whom they perceive as being similar to themselves.

to interact frequently or intensely, and it may be more difficult for them to develop a shared understanding. Heterogeneous teams often have greater difficulty integrating objectives and views, leading to conflict and lower group cohesion.[17] Research has also indicated, however, that the communication and coordination differences between heterogeneous or homogeneous teams diminish if the groups maintain long-term contact. Presumably, through extensive interaction, heterogeneous teams learn to manage their group processes better.[18]

In sum, heterogeneous teams should possess more information, on average, than homogeneous groups. The heterogeneity of a team can also increase the creativity and variance in decision making, leading to more innovative outcomes and higher overall performance.[19] However, to realize this potential performance advantage, heterogeneous teams may require long-term contact and incentives to foster communication and cooperation.

The ability of team members to communicate and cooperate effectively is also a function of the personalities of the individuals on the team. A study by Susan Kichuk and Willi Wiesner explored whether five personality factors (conscientiousness, extroversion, neuroticism, agreeableness, and openness to experience) influenced the likelihood of success in new product development teams. Kichuk and Wiesner found that the personality characteristics that enhanced the success of a new product development team were high extroversion, high agreeableness, and low neuroticism.[20]

THE STRUCTURE OF NEW PRODUCT DEVELOPMENT TEAMS

Teams can be structured in a number of ways. One well-known typology classifies teams into four types: functional, lightweight, heavyweight, and autonomous.[21] Figure 12.1 depicts each of these types graphically.

Functional Teams

In functional teams, members remain in their functional departments (e.g., R&D, marketing, manufacturing, etc.), and report to their regular functional manager (see Figure 12.1, panel a); however, they may meet periodically to discuss the project. Such teams are usually temporary, and individuals may spend less than 10 percent of their time working on team-related activities. Functional teams also typically do not have a project manager or dedicated liaison personnel. While this team structure is straightforward to implement because it requires little (if any) deviation from the firm's normal operations, this structure provides little opportunity for cross-functional coordination.[22] Further, since individuals are still evaluated and rewarded almost exclusively based on their functional performance, the team members may have little commitment to the development project. Functional teams are more likely to be appropriate for derivative projects that primarily affect only a single function of the firm.

Lightweight Teams

In lightweight teams, members still reside in their functional departments, and functional supervisors retain authority over evaluation and rewards (see Figure 12.1, panel b).

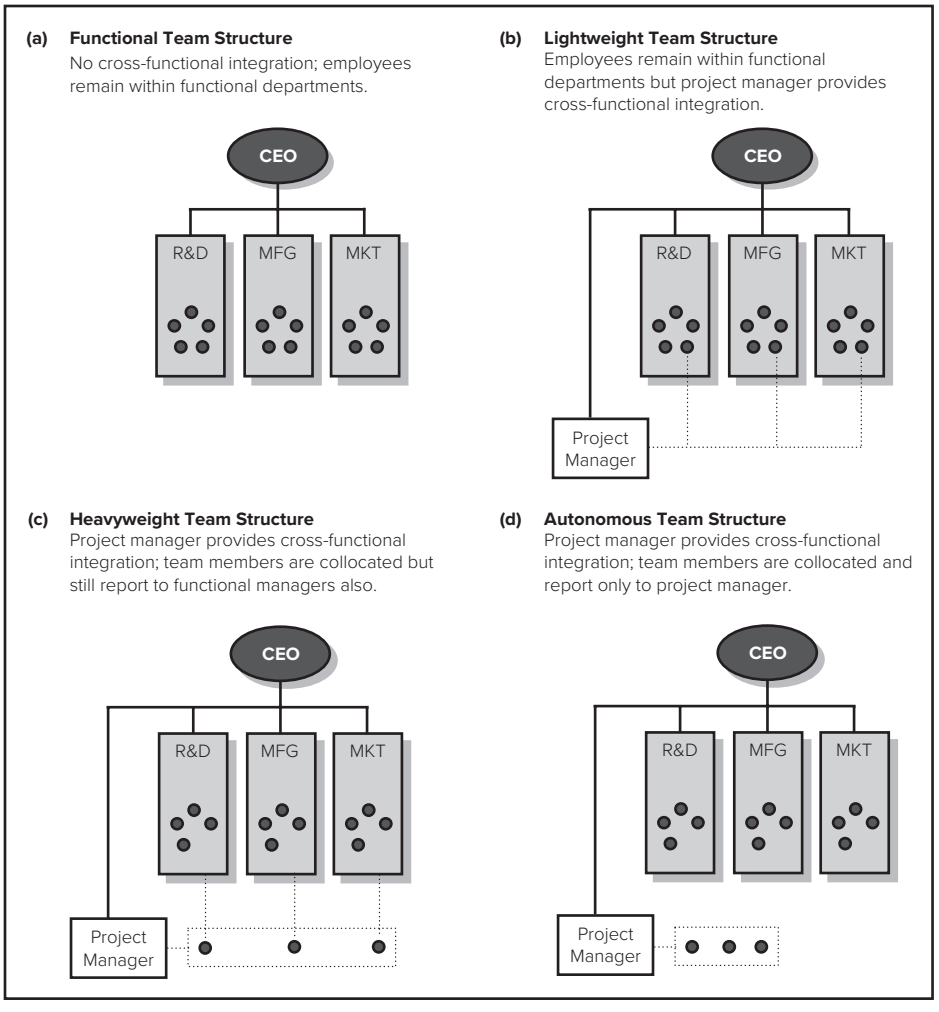


FIGURE 12.1
Types of Development Teams

Like functional teams, lightweight teams are typically temporary, and members spend the majority of their time on their normal functional responsibilities (up to 25 percent of their time might be spent on team-related activities). However, lightweight teams have a project manager and dedicated liaison personnel who facilitate communication and coordination among functions. Managers of lightweight teams are normally junior or middle management employees, who are not able to exert significant influence or authority over team members. As a result of these factors, lightweight teams offer a small improvement in team coordination and likelihood of success over functional teams. Such a team structure might be appropriate for derivative projects where high levels of coordination and communication are not required.

Heavyweight Teams

In heavyweight teams, members are removed from their functional departments so that they may be *collocated* with the project manager (see Figure 12.1, panel c). Project managers of heavyweight teams are typically senior managers who outrank functional managers, and have significant authority to command resources, and evaluate and reward team members.[23] The core group of team members in the heavyweight team is often dedicated full-time to the project. This combination of factors helps ensure that the team has strong cross-functional coordination and communication, and that team members are significantly committed to the development project. However, heavyweight teams are still often temporary; thus, the long-term career development of individual members continues to rest with their functional managers rather than the project manager. This type of team structure offers a significant improvement in communication and coordination over functional teams, and it is typically considered appropriate for platform projects.

Autonomous Teams

In autonomous teams, members are removed from their functional departments and dedicated full-time (and often permanently) to the development team (see Figure 12.1, panel d). Team members are collocated with the project manager, who is a very senior person in the organization. The project manager of an autonomous team is given full control over resources contributed from different functional departments, and the project manager has exclusive authority over the evaluation and reward of team members. Autonomous teams often do not conform to the operating procedures of the rest of the organization; instead they are permitted to create their own policies, procedures, and reward systems.[24] Autonomous teams are also held fully accountable for the success of the project; in many ways, autonomous teams act like independent divisions of the firm. Autonomous teams typically excel at rapid and efficient new product development, particularly when such development requires breaking away from the organization's existing technologies and routines. Thus, autonomous teams are typically considered to be appropriate for breakthrough projects and some major platform projects. They can be the birthplace of new business units.[25] However, the independence of the autonomous teams can cause them to underutilize the resources of the parent organization. Furthermore, autonomous teams are often hard to fold back into the organization if the project is completed or terminated. Many autonomous teams thus go on to become separate divisions of the firm, or may even be spun off of the firm as a subsidiary.

Figure 12.2 summarizes key dimensions across which the four teams vary, including a number of points that have not yet been dealt with in the text. The potential for conflict between the functions and the team, particularly the project manager, rises with the move from functional teams to autonomous teams. The independence of heavyweight and autonomous teams may prompt them to pursue goals that run counter to the interests of the functions. Senior managers should keep such conflict in check.

FIGURE 12.2
Summary of Characteristics of Team Types

Characteristics	Functional Team	Lightweight Team	Heavyweight Team	Autonomous Team
Project manager	None	Junior or middle manager	Senior manager	Senior manager
Power of project manager	NA	Low	High	Very high
Time spent on team activities	Up to 10%	Up to 25%	100%	100%
Location of team members	Functions	Functions	Collocated with project manager	Collocated with project manager
Length of commitment to team	Temporary	Temporary	Long-term but ultimately temporary	Permanent
Evaluation of team members	Functional heads	Functional heads	Project manager and functional heads	Project manager
Potential for conflict between team and functions	Low	Low	Moderate	High
Degree of cross-functional integration	Low	Moderate	High	High
Degree of fit with existing organizational practices	High	High	Moderate	Moderate-low
Appropriate for:	Some derivative projects	Derivative projects	Platform projects/ breakthrough projects	Platform projects/ breakthrough projects

THE MANAGEMENT OF NEW PRODUCT DEVELOPMENT TEAMS

For a new product development team to be effective, its leadership and administrative policies should be matched to the team's structure and needs.

Team Leadership

The team leader is responsible for directing the team's activities, maintaining the team's alignment with project goals, and serving as a communicator between the team and senior management. In heavyweight and autonomous teams, the team leader may also be the person who is primarily responsible for the evaluation, compensation, and promotion of individual team members. Effective team leaders are often much more directly related to the team's success than senior management or project champions. This may be because team leaders interact much more frequently with the team and more directly influence the team's behavior.[26]

As described in the team type and structure section above, different types of teams have different leadership needs. For instance, while lightweight teams might have a junior or middle-management leader who provides basic coordination between the functional groups, heavyweight and autonomous teams require senior managers with significant experience and organizational influence. In heavyweight and autonomous teams, the project manager must be someone who can lead and evaluate the team members, champion the development project both within the team and to the wider organization, and act as a translator between the various functions.[27] In particular, project managers in heavyweight and autonomous teams must have high status within the organization, act as a concept champion for the team within the organization, be good at conflict resolution, have multilingual skills (i.e., they must be able to talk the language of marketing, engineering, and manufacturing), and be able to exert influence upon the engineering, manufacturing, and marketing functions.[28] Other things being equal, teams whose project managers are deficient in one or more of these dimensions will have a lower probability of success.[29]

Team Administration

To ensure that members have a clear focus and commitment to the development project, many organizations now have heavyweight and autonomous teams develop a project charter and contract book. The *project charter* encapsulates the project's mission and articulates exact and measurable goals for the project. It might include a vision statement for the project (e.g., "Dell laptops will be the market standard for performance and value") and a background statement for why this project is important for the organization. The charter may describe who is on the team, the length of time members will spend on the team, and the percentage of their time that will be spent on team activities.[30] It may also stipulate the team's budget, its reporting timeline, and the key success criteria of the project (e.g., meeting a particular time-to-market goal, exceeding customer satisfaction criteria established for the project, capturing a target amount of market share within a defined period of time, etc.). Establishing an explicit set of goals for the project helps ensure that the team members have a common understanding of the project's overall purpose and priorities. Goals also help to structure the new product development process and can facilitate cooperation by keeping team members oriented toward a common outcome.[31]

Once the team charter is established, core team members and senior managers must negotiate a *contract book*. The contract book defines in detail the basic plan to achieve the goal laid out in the project charter. Typically, the contract book will estimate the resources required, the development time schedule, and the results that will be achieved. The contract book provides a tool for monitoring and evaluating the team's performance in meeting objectives by providing a set of performance benchmarks and deadlines to which the team's performance can be compared. More important, however, the contract book is an important mechanism for establishing team commitment to the project and a sense of ownership over the project. After negotiation and acceptance of this contract, all parties often sign the contract book as an indication of their intention to honor the plan and achieve the results. Team members who sign the contract book typically feel a greater sense of duty to work toward the project's goals. Furthermore, signing the contract book can give team members a

sense of ownership over the project and empowerment to make decisions about the project. This ownership and empowerment can help team members identify with a project's outcome and can encourage them to exert extra effort to ensure its success.[32]

Managing Virtual Teams

virtual teams Teams in which members may be a great distance from each other, but are still able to collaborate intensively via advanced information technologies such as videoconferencing, groupware, and e-mail or Internet chat programs.

Recent advances in information technology have enabled companies to make greater use of virtual teams. **Virtual teams** are teams in which members may be a great distance from each other, but are still able to collaborate intensively via advanced information technologies such as videoconferencing, groupware, and e-mail or Internet chat programs. Virtual teaming can enable individuals with unique skills to work on a project, regardless of their location. By meeting virtually, individuals who live at great distances can collaborate without incurring travel costs or disruption to their lives.[33] This is especially valuable for a company whose operations are highly global. For example, SAP Aktiengesellschaft is headquartered in Walldorf, Germany, but has large R&D centers in India, China, Israel, and the United States. Each location has deep expertise in particular areas but lacks functional breadth due to specialization. SAP's managers choose employees from different locations to assemble virtual teams that optimally integrate the expertise needed for a given project.[34] Similarly, when IBM began deploying more of its products globally, it increased its use of virtual teams. About one-third of IBM's employees will participate in virtual teams at some point in their career. When IBM needs to staff a project, it gives a list of the needed skills to the human resource division, which identifies an appropriate pool of people. If the skills and talent of the people are more important than their ability to meet face-to-face, a virtual team is formed.[35]

Virtual teams pose a distinct set of management challenges, however. As described earlier in the chapter, much of the work on the structure of new product development teams has emphasized the importance of collocation. Collocation facilitates communication and collaboration by giving team members opportunities for rich face-to-face communication and informal interaction.[36] Proximity and frequent interaction help teams to develop shared norms and a dialect for communicating about the project. Virtual teams, by contrast, must often rely on communication channels that are much less rich than face-to-face contact and face significant hurdles in establishing norms and dialects and may suffer from greater conflict. They may also have trouble negotiating multiple time zones, which can lead to frustration.[37]

In the forming of virtual teams, it is important to select personnel who are both comfortable with the technologies being used to facilitate collaboration and who have strong interpersonal skills.[38] Team members must be able to work independently and have a strong work ethic. Since distance makes it easy for team members to deflect opportunities for interaction, it is important to choose individuals who tend to seek interaction rather than avoid it. It is important that members of the team establish standards for how quickly they will respond to messages, and how often they will be available for synchronous communications (communications where the involved parties must participate at the same time, such as telephone calls, videoconferencing, and instant messaging).[39] Furthermore, because many of the opportunities for informal interaction may be lost in a virtual environment, more types of interaction may have

Research Brief Virtual International R&D Teams

Gassman and von Zedtwitz build on the transnational corporation model discussed in Chapter Ten by examining how such firms coordinate their international innovation efforts via virtual teams. As in some of the arguments made in Chapter Ten about loosely coupled R&D activities, virtual international R&D teams may jointly work on a single development project, utilizing information technologies (rather than geographical proximity) to achieve coordination. However, while information technology decreases the need to collocate R&D activities, it does not readily solve problems related to building trust and transferring tacit knowledge. The type of innovation project being undertaken and the type of knowledge that must be shared should influence the degree to which firms rely on decentralized, virtual coordination processes.

Gassman and von Zedtwitz studied 37 technology-intensive multinationals and identified four patterns of teams: (1) decentralized self-coordination, (2) system integrator as coordinator, (3) core team as system architect, and (4) centralized venture team. In the *decentralized self-coordinating* teams, there was no single source of power or authority over the teams. Teams communicated primarily through telephone, the Internet, shared databases, and groupware. Coordination was relatively weak and relied largely on a strong corporate culture. Decentralized self-coordination was more likely to arise if there were well-developed standard interfaces between components being developed in different locales; thus, it tended to be suited to modular innovation as opposed to architectural innovation (see Chapter Three).

In teams with a *system integrator as R&D coordinator*, a single individual or office takes responsibility for helping different divisions coordinate. The system integrator helps to build a common understanding of the project among each of the divisions, translates knowledge from one division to another, and tracks progress and contributions. While the overall project is decentralized, the system integrator enables some centralized coordination.

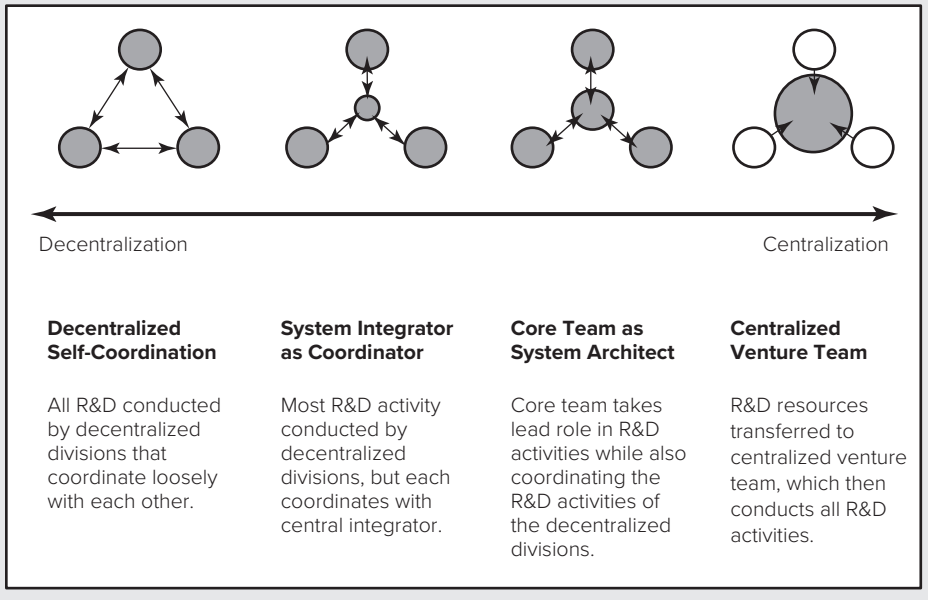
In the *core team as system architect* model, a core team of key decision makers from all of the decentralized R&D groups meets regularly to coordinate the otherwise decentralized groups. The core team often includes a strong project manager, leaders from each of the decentralized groups, and occasionally external customers or consultants. The core team constructs the overall architecture of the development project and maintains its coherence throughout its development. Because the core team has more direct authority over the individual divisions than the system integrator described above, the core team is better able to resolve conflict and enforce standards across the divisions. Because core teams can provide a significant level of integration across the divisions, core teams are often able to conduct architectural innovation. In the *centralized venture team,* R&D personnel and resources are relocated to one central location to enable maximum integration and coordination. The team is likely to have a very powerful senior project manager with significant authority to allocate resources and define the responsibilities of individual team members. Gassman and von Zedtwitz describe two examples of centralized venture teams—Asea Brown's "High Impact Projects" and Sharp's "Golden Badge" projects. Because of their high expense, such teams are likely to be used only for strategic innovations of the utmost importance.

Gassman and von Zedtwitz's model is summarized in Figure 12.3. Overall, Gassman and von Zedtwitz argue that innovations that are radical, are architectural, or require the intensive transfer of complex or tacit knowledge will require greater centralization. Innovations that are incremental, are modular, and do not require the frequent transfer of complex or tacit knowledge can be more decentralized.

continued

concluded

FIGURE 12.3
Gassman and von Zedtwitz's Typology of International Virtual Teams



Source: Adapted from O. Gassman and M. von Zedtwitz, "Trends and Determinants of Managing Virtual R&D Teams," *R&D Management* 33, no. 3 (2003): 243–62.

to be incorporated into the ground rules of the team.[40] For example, the team leader might schedule daily or weekly unstructured "chat" times where team members are required to participate in a group conference call or online conference to share ideas that may not be uncovered in the team's more formal interactions.

Virtual teams also face challenges in developing trust, resolving conflict, and exchanging tacit knowledge, as discussed in the accompanying Research Brief about virtual international R&D teams.

Summary of Chapter

1. Bringing multiple people together into a team enables multiple bases of expertise to be collectively directed toward problem solving; thus, teams are powerful mechanisms for problem solving. However, if teams become too big, administrative costs and communication problems can become significant.
2. Diversity of team members ensures that the team can draw on different perspectives and bases of expertise. In particular, functional diversity is often sought in new product development teams. Cross-functional teams enable design, manufacturing, and marketing objectives to be integrated in the new product development process.
3. Diversity of team members ensures that the individuals in the team not only possess different knowledge or viewpoints, but also have different sources of extra-team resources upon which to draw through boundary-spanning activities.

4. Diversity can also make it more difficult for teams to develop a common understanding of the new product development project and can result in lower group cohesion. Teams may need long-term contact and incentives for cooperation to overcome these challenges.
5. The way in which a team is structured (collocation, permanence, supervisory relationships, etc.) significantly influences how team members interact and the likely outcomes of a development project. Different types of teams are appropriate for different types of development projects.
6. Attributes of the team leader (seniority, authority, multilingual skills) must match the team type for teams to be most effective.
7. Many firms have teams develop and sign a project charter and contract book to ensure that all team members have a common understanding of the project's goals and possess a sense of ownership and commitment to the project's success.
8. When a company wishes to form a team with individuals who have unique skills but live great distances from each other, it might opt to form a virtual team. Virtual teams use information technologies to achieve communication and coordination. Virtual teams face a distinct set of challenges in promoting participation, cooperation, and trust. As a result, they require special consideration of the selection of team members and the team administration processes.

Discussion Questions

1. What are the trade-offs in choosing a team's size and level of diversity?
2. How can managers ensure that a team reaps the advantages of diversity while not being thwarted by some of the challenges team diversity raises?
3. Identify an example of a development project, and what type of team you believe they used. Do you think this was the appropriate type of team given the nature of the project?
4. What are some advantages and disadvantages of collocation? For what types of projects are virtual teams inappropriate?

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Chapter **Thirteen**

Crafting a Deployment Strategy

Deployment Tactics in the Global Video Game Industry

In the global video game industry, the introduction of each generation of console has ushered in a new battle for market dominance. New entrants have made startling entrances and toppled seemingly invincible incumbents. Game developers, distributors, and customers have had to watch the battle closely in order to place their bets about which console would emerge as the generation's victor. Each generation has also revealed which deployment strategies have paid off—or proven fatal—for the contenders.[a]

Pong: The Beginning of an Era

In the fall of 1972, Nolan Bushnell founded a company called Atari and introduced Pong, a Ping-Pong-like game that was played on a user's television set with the aid of the Atari console. Though considered to be the "second generation" of home video game consoles (a console called the Odyssey, released by Magnavox early 1972 is considered the first generation), Pong was the first home console to become known around the world. In its first year, Pong earned more than $1 million in revenues. Pong and over 60 similar knockoffs soon flooded the market. In these early years of the video game industry, swift advances in integrated circuits enabled a rapid proliferation of new consoles and games. By 1984, video game console and games sales had reached $3 billion in the United States alone. However, console makers in this era did not utilize strict security measures to ensure that only authorized games could be played on their consoles, leading to explosive growth in the production of unauthorized games (games produced for a console without authorization of that console's producer). As a result, the market was soon saturated with games of dubious quality, and many unhappy retailers were stuck with video game inventories they were unable to move. Profits began to spiral downward, and by 1985, many industry observers were declaring the video game industry dead.

The Emergence of 8-Bit Systems

Much to everyone's surprise, however, two new companies from Japan entered the U.S. video game market, initiating the home video game console's third generation: Nintendo, with its 8-bit Nintendo Entertainment System (NES), introduced in 1985, and Sega, which launched its 8-bit Master System in the United States in 1986. Unlike Atari, which had spent only a few hundred thousand dollars on advertising, Sega and Nintendo spent $15 million promoting their systems. Both systems offered technological advantages over the previous generation of video game consoles. Both companies also agreed to sell to many retailers on consignment, helping to eliminate risk to the retailer of getting stuck with unsold merchandise. Sega's Master System appeared to be slightly superior to Nintendo's, but Nintendo spent much more on the development of quality games and characters and had more game titles available than Sega. The NES sold over 1 million units in the first year, sold 19 million units by 1990, and could be found in more than one-third of the households in America and Japan.[b]

From 1985 to 1989, Nintendo held a near monopoly of the U.S. video game industry. The company sold its consoles for a price very close to production costs, while earning the bulk of its profits from games. Nintendo both made games for its system in-house and licensed third-party developers to produce games through very strict licensing policies that (*a*) limited the number of titles a developer could produce each year, (*b*) required the developer to preorder a minimum number of cartridges from Nintendo (which had its own contract manufacturers produce the games), and (*c*) restricted the developers from making similar games for other consoles. Nintendo also restricted the volume and pricing of consoles sold through distributors, ensuring that no single distributor acquired significant bargaining power.[c] Nintendo's restrictive policies were very profitable; however, they also incurred sanctions by the Federal Trade Commission, and they alienated distributors and developers, potentially leaving the company more vulnerable to a competitor.

The 16-Bit Video Game Systems

In September 1989, Sega launched the fourth generation of video game console competition when it introduced its 16-bit Genesis to the U.S. video game market. The Genesis offered dramatic performance enhancement over 8-bit systems. Further, Sega leveraged its popular arcade games to the Genesis and made it backward compatible with its 8-bit Master System games. There were 20 Genesis game titles available by December 1989. NEC also introduced a 16-bit system, the TurboGrafx-16, in the fall of 1989 and had 12 game titles by December 1989. Though Nintendo had its own 16-bit system in the works, it delayed introducing it to the United States for fear of cannibalizing its 8-bit system sales.

By the end of 1989, Sega had already sold 600,000 consoles in the United States, and NEC had sold 200,000. In 1990 and 1991, both Sega and NEC added game titles to their lists, bringing their totals to 130 and 80, respectively. By the end of 1991, Sega had sold 2 million consoles in the United States, and NEC had sold 1 million. Unlike Sega, which produced a major portion of its games in-house, NEC relied completely on external games developers, who found the

system to have only a small technological advantage over 8-bit systems.[d] Developers began to abandon the NEC platform, and NEC exited the market in 1991. Nintendo finally introduced its own 16-bit Super Nintendo Entertainment System (SNES) in 1991, but it was too late to quell Sega's momentum. In 1992, Nintendo controlled 80 percent of the video game market (based on combined 8-bit and 16-bit sales), but by 1994, Sega was the market leader.

Like Nintendo, Sega made little profit on the consoles and focused instead on increasing unit sales to drive game sales and software developer royalties. Sega, however, used less restrictive licensing arrangements than Nintendo and rapidly lured a large number of developers to make Sega game titles. Further, though Nintendo could have made its 16-bit system backward compatible, thus linking the value consumers possessed in their 8-bit game libraries to the new system, Nintendo chose to make the system incompatible with the 8-bit games. By the end of 1991, the SNES had 25 game titles compared to the 130 available for Genesis. Nintendo had given Sega two years of installed base lead on a system that offered a significant technological advantage and then entered the market at a ground-zero position with respect to the availability of complementary goods. The consequence of Nintendo's late move is aptly captured in a review of video game players published in *Fortune*: "To tell the truth, Nintendo just isn't cool anymore. This one is 16 bits, so it's better than the original Nintendo. But the company only made it to compete with Sega, and most kids already have that. So they don't need Super Nintendo, unless they're jerks and have to have everything. That's just idiotic."[e] Over time, sales of the Nintendo SNES accelerated, and it would ultimately prove to be one of the more successful game systems ever introduced, but Nintendo's near-monopoly position had been broken; Sega had successfully leapfrogged Nintendo.

32/64-Bit Systems

In the late 1980s and early 1990s, new competitors were attracted to the video game market, with fifth generation machines that had either 32- or 64-bit systems. In 1989, Philips announced its 32-bit Compact Disc Interactive (CD-i), an interactive multimedia compact disc system that would serve as a game player, teaching tool, and music system. However, the CD-i was very complex, requiring a 30-minute demonstration. Furthermore, it was expensive—introduced at $799 and later reduced to a below-cost $500 (more than twice the cost of Nintendo or Sega systems).[f] While the product was actually much more than a video game console, customers compared it to the popular Nintendo and Sega systems and were dismayed by its price and complexity. Making matters worse, Philips was reluctant to disclose the technical specifications, greatly limiting the software development for the system. The Philips CD-i never attained more than a 2 percent market share.[g] Other companies also introduced 32-bit systems, including Turbo Technologies' Duo and 3DO's Interactive Multiplayer, but the cost of the systems ($600 to $700) was prohibitive. Turbo Tech's Duo was very short-lived and received little attention. But 3DO's system received considerable attention. The company was founded in October 1993 by Trip Hawkins, formerly of video games developer Electronic Arts. However, 3DO's unique strategy of licensing out all game and

hardware production made it impossible to achieve the low console prices of Sega and Nintendo by subsidizing console production with game royalties. The hardware producers (Matsushita and Panasonic) for 3DO did not sell games and were unwilling to sell the consoles without a margin. Sales of the machine never took off, and 3DO exited the market. Atari also made a surprising return to the video game market in 1993 with the technologically advanced Jaguar. However, Atari's long struggle had not inspired great confidence in either developers or distributors, and several of the large retail chains chose not to carry the product.[h]

In 1995, two 32-bit systems arrived on the scene that would survive: Sega's Saturn and Sony's PlayStation. Both systems were introduced with great fanfare and considerable developer support. Although only Sega had experience and brand image in the video game market, Sony entered with tremendous brand image in consumer electronics and access to (and leverage in) extensive distribution channels in electronics and media. To rapidly gain insight into the toy industry, Sony hired experienced toy industry veteran Bruce Stein to head the video game unit. Sony's size and previous success in several electronics markets (including the development and control of the compact disc format) also enabled it to persuade several games developers (including Electronic Arts, the largest game developer in the United States at that time) to produce only PlayStation titles for the first six months after its introduction. Sony's console was designed to play games on compact discs (CDs) rather than the cartridge style games Nintendo and Sega used, and while many skeptics at first argued that the CDs would have slower game play and higher piracy rates, moving to the CD would ultimately prove to be a huge advantage. Games could be developed and manufactured at a much lower cost, which in turn, enabled Sony to allow developers to produce games in smaller batches, lowering their risk. By the end of 1995, there were already 50 PlayStation titles and this number had grown to 800 by the end of 2000.

Though Sega's Saturn had beaten Sony's PlayStation to market by several months, it was shipped to only four retailers due to limited supply: Toys "R" Us, Babbage's, Software Etc., and Electronics Boutique. This aggravated retailers such as Best Buy and Walmart, which had long supported Sega.[i] Developers also felt that it was easier to program for the PlayStation than the Saturn, causing it to lose crucial developer support.[j] By the end of 1996, the installed base of Sony PlayStation in the United States (2.9 million units) was more than double that of the Sega Saturn (1.2 million units).

In 1996, after more than two years of announcements, Nintendo finally introduced its 64-bit game system called Nintendo 64. Though only two software titles were available at the console's release (one being Super Mario), the game units were sold out within weeks of their release. Though Nintendo's 64-bit system gained rapid consumer acceptance, neither Nintendo nor Sega was able to reclaim dominance over the video game industry. Though several new entrants (and one returning entrant, Atari) had tried to break into the video game industry through technological leapfrogging, only Sony had the successful combination of a product with a technological advantage, strategies, and resources that enabled it to rapidly build installed base and availability of complementary goods, and a reputation that signaled the market that this was a fight it could win.

128-Bit Systems

In September 1999, Sega launched the first sixth-generation console, its 128-bit Dreamcast, a $199 gaming system that enabled access to the Internet. Before the Dreamcast release, Sega was suffering from its lowest market share in years at 12 percent. The Dreamcast was the first 128-bit system to market, and 514,000 units were sold in the first two weeks. An installed base of 5 million was achieved by October 2000. Sega's success turned out to be short-lived, however. In March 2000, Sony launched its 128-bit PlayStation 2 (PS2) in Japan and introduced the system to the United States in October. Despite price cuts on the Dreamcast and a promotion rebate that would make the console essentially free (in exchange for a two-year contract for Sega's SegaNet Internet service), the Dreamcast was crushed in the holiday sales season. In early 2001, Sega announced it would cease making consoles and transform itself into a third-party developer of games for other consoles.

Sony's PS2 was an unprecedented success. Not only did it offer a significant technological advantage over the 32-bit systems, but it was also backward compatible, enabling gamers to play their PlayStation games on the console until they amassed new game libraries.[k] During the opening sales weekend of March 4, 2000, PS2 sales reached about 1 million units, a figure that eclipsed by 10 times the amount of original PlayStation units sold during the three-day release period in 1994. Demand for the new unit was so high that on the opening day of pre-orders on Sony's Web site, more than 100,000 hits in one minute were received, and Sony was forced to briefly shut down the Web site.

At the time of the PS2 release, Nintendo had just postponed the launch of its new 128-bit system, the GameCube, to a release date in the first half of 2001. Unlike the PS2, the GameCube did not offer backward compatibility with Nintendo 64 games. The GameCube was also targeted toward a younger market (8- to 18-year-olds) than Sony's 16- to 24-year-old demographic. The real threat to Sony's PS2 came in the form of a new entrant to the video console industry: Microsoft's Xbox. The Xbox, launched in November 2001, was targeted at the 18- to 34-year-old male, positioning it directly against the PS2.

Microsoft had previously produced PC-based computer games (such as Flight Simulator and the Age of Empires series) and operated an online gaming service (Microsoft Gaming Zone) and thus had some familiarity with the industry. However, it did not have either the arcade experience of Sega or Nintendo or the consumer electronics experience of Sony. By the time the Xbox hit the market, PS2 already had a significant lead in installed base and availability of games (more than 300 PS2 game titles were available at the end of 2001), but Microsoft was counting on the technological advantages offered by the Xbox to tip consumer preferences. The Xbox offered a faster processor and more memory than the PS2. Furthermore, customers did not have to trade-off technological advantages against price: The Xbox launched at a retail price of $299, significantly less than its production costs (it is estimated that Microsoft lost between $100 and $125 per unit).[l]

To rapidly deploy the console and build installed base, Microsoft leveraged its existing relationships with distributors that carried its software, though it was now forced to seek much greater penetration into distributors such as Toys "R" Us, Babbages, and Circuit City. Microsoft also faced the challenge of cultivating

a radically different brand image in the game console market than the one that it had achieved in the software market, and to make much greater use of marketing channels such as television advertising and gaming magazines. To that end, Microsoft budgeted $500 million to be spent over 18 months to market the Xbox—more than any other marketing campaign in the company's history.[m] Microsoft planned to produce 30 to 40 percent of its games in-house and gave away $10,000 game development kits to attract third-party games developers.

Both the Xbox and Nintendo's GameCube were launched just in time for the extremely important 2001 Christmas season and sold briskly. By the year's end, it was estimated that 1.3 million GameCube units had been sold, and 1.5 million Xbox units had been sold.[n] However, both of the new consoles were outrun by PS2, which sold 2 million units in the month of December alone. This market share pattern remained remarkably consistent over the next few years. By the end of fiscal year 2005, Microsoft reported it had shipped a total of 22 million Xbox consoles, which was slightly more than Nintendo's 20.6 million GameCube shipments, and far behind the Sony PS2's 100 million consoles shipped.[o]

The Seventh Generation: A Second Round of Competition in 128-bit Systems

In late 2005, Microsoft was first to introduce a seventh-generation console: Xbox 360. Though severe manufacturing shortages resulted in only 600,000 units being sold in the 2005 Christmas season, Microsoft was hoping its head start over Sony and Nintendo (both of which would not debut their next generation consoles until late 2006) would enable it to attain a dominant position.

The console was designed around a custom IBM processor that had three Power PC processors on a single chip, and a custom graphics processor from ATI. The result was a powerful console that generated high-definition video with stunning graphics. The Xbox 360 was also backward compatible with a portion of the Xbox game library (though not with all Xbox games). At launch, buyers could get a basic model for $299 or a premium model for $399.[p] More important (from Microsoft's perspective at least), the console was supposed to be for more than just games—it was Microsoft's next attempt to secure control over the digital living room. Users could download music, movies, TV shows, and purchase premium content. With an optional high-definition DVD drive users could also watch high-definition movies (or at least they could have if Microsoft's bet on Toshiba's HD-DVD standard had not gone awry—in early 2008, Toshiba conceded defeat to Sony in the high-definition DVD format war, and announced that it would stop making the drives). By early 2006, Microsoft had sold 3.2 million Xbox 360s. The number would have been higher but the company could not keep up with demand.[q]

Sony launched its PlayStation 3 (PS3) in November of 2006. The console had a powerful IBM cell processor; it included Sony's Blu-ray disc player (for playing high-definition DVDs) and launched with a hefty price tag of $599 for a 60-gigabyte model, or $499 for a 20-gigabyte model. Estimates put the cost of the components used to produce the consoles at $840 and $805, respectively, meaning that Sony lost more than $200 on every unit.[r] Sony claimed that the PS3 was backward compatible with all games written for the Playstation and PS2,

but it turned out that not all of the older games would play on the new system. Though the console sold out within minutes of its launch, by early 2007, Sony had sold only 3.5 million PS3s worldwide—significantly less than it had forecasted.

Instead of joining Sony and Microsoft in a technological arms race, Nintendo changed the rules of the game with its Wii console in time for the Christmas season of 2006. Instead of a controller with buttons or a joystick that players had to vigorously manipulate, it offered an innovative wireless motion-sensing remote that enabled users to simulate real play, such as swinging a tennis racket in a tennis match or punching an opponent in a boxing match. The console was also launched at a price of $250—significantly cheaper than the Xbox 360 or PS3. It was fully compatible with GameCube games, and because it was much cheaper to develop a Wii game (e.g., as little as $5 million compared to the $20 million for a PS3 game), it attracted third-party developers in droves. The net result was dramatic—the console attracted casual gamers in unprecedented numbers, and from a remarkably wide range of demographics. Wiis were being used in nursing homes, for Wii bowling leagues, and on cruise ships.[s] Instead of simply being purchased by soccer moms for their children, they were being played by the soccer moms themselves. By mid-2007, the Wii was selling twice as fast as the Xbox 360 and four times as fast as the PS3. While Microsoft and Sony lost money on every console in hopes of profiting on future game sales, Nintendo was estimated to make $50 on every unit of the Wii sold.

In fall of 2008, Microsoft slashed the price of the Xbox 360 to $199, making it the cheapest console of the generation. However, even Don Mattrick, senior vice president of Microsoft's Interactive Entertainment Business admitted that the Xbox 360 was unlikely to overtake the Wii, noting: "I'm not at a point where I can say we're going to beat Nintendo . . . we will sell more consoles this generation than Sony."[t] Sony followed suit by lowering the price of its starter model to $399. The years 2009 and 2010 were worse for all of the console companies—the recession lead to a significant drop in sales and profits for all three companies and they responded by dropping the price of their consoles. However, the pattern of sales remained the same. By December of 2010, over 75 million Wii's had been sold worldwide, compared to 45 million Xbox 360's, and 42 million PS3s.

In 2010, both Sony and Microsoft launched their own motion-based controllers. The Playstation Move was a handheld motion sensor wand, similar to the Wii Remote. Microsoft's Kinect, on the other hand, did not require holding anything—it was a webcam-style peripheral that detected user motion within a range of play in front of the device. Both devices were sold as add-on units to the companies' consoles and were priced at over $150 (nearly as much as the entire Wii console). However, despite the high price, by March 2011, Microsoft reported that it had sold more than ten million Kinect sensors worldwide, making it the fastest selling consumer device ever, according to the Guinness World Records Committee.

The Eighth Generation: Increasing Competition from Mobile Devices

The eighth generation saw the video game console makers focusing on further integration with other media and connectivity. In late 2012, Nintendo launched

its Wii U, a tablet/controller hybrid with an embedded touch screen that would enable users to get a different perspective than the one on the main display. The Wii U was backward compatible with most Wii games, but the initial lineup of games written specifically for the device was weak at the consoles launch. Furthermore, because Nintendo had beaten Sony and Microsoft to market by a year, many customers decided to wait until they could compare all three consoles, leading to sluggish sales for the Wii U.

Sony launched its PlayStation 4 (PS4) in November of 2013, a device which would emphasize connectivity with other devices (such as the PlayStation Vita or Apple iOS or Android powered mobile devices), social game play, and cloud-based subscription gaming. The device launched at a price of $399, signaling that Sony had learned a painful-but-important lesson with PS3. Microsoft released its Xbox One the same month, billing the device as an "all-in-one entertainment system." Like the PS4, the device enabled game streaming from the cloud and included a Blu-Ray player. However, it also gave users the ability to connect to their television set-top box and overlay live television with gaming in a split-screen and to record and share highlights of game play. The console was launched at a price of $499.

The eight generation also brought increasing competition from new players. Mobile device platforms such as Android and Apple's iOS began to account for significantly more gaming. Though the devices were small and the games lacked the high fidelity and motion-based play offered by the home consoles, they made up for it—at least in part—by a rapid proliferation of free or low-cost games and the fact that their owners tended to have them available at all times. Smartphones had much lower game development costs and offered access to a very wide range of consumers. As a result, smartphone platforms not only attracted existing game developers but also helped spur the rise of many new game developers. By 2018, there were more than 800,000 mobile games available (a number unimaginable on home consoles) and mobile gaming accounted for more than half of the $138 billion in global gaming revenues.[u] Increased competition and digital distribution had also given rise to new revenue models such as "freemium" where the initial game was provided for free but consumers spent money through in-game purchases.[v]

The Ninth Generation: Repeating Old Patterns?

In November 2020, Microsoft released its Series S and Series X, and Sony released its PlayStation 5 (PS5), officially starting the ninth generation of video game console competition. Notably this represented a later start for a new generation; while previous generations of video game consoles had tended to be about five years long, the eighth generation had been about eight years long, with both Microsoft and Sony opting to introduce mid-generation updates to their eighth-generation consoles.[w] Many industry analysts believed this signaled that the companies were hoping to move away from the very stark generational cycles that had long characterized the industry.

Not surprisingly, the ninth-generation consoles offered faster computation and graphics processors, 4K (or even 8K) resolution, and high-performance solid-state memory drives. Notably, both consoles were designed to enable fast

loading times for games that were streamed online, in a nod to the shift to online multiplayer gaming.[x] Both consoles were also both mostly backward compatible with games developed for the eighth generation. The lowest price version of the PS5 launched with a $399 price; the Xbox Series S launched at $299 and the Series X launched at $499. All three consoles used similar central processing units and offered the same networking options, but they differed in their graphics processors, memory, and storage. Though exact tear-down costs of the consoles were not available, a Microsoft Executive questioned in the 2021 lawsuit between Epic and Apple confirmed what industry observed had speculated for years: Microsoft does not earn any profit on sales of Xbox hardware. Asked about the profit Microsoft makes on consoles, Lori Wright, Microsoft's head of Xbox business development replied, "We don't; we sell the consoles at a loss." Given the similar parts and construction of the PlayStation and its price, it was very likely that the PlayStations were also sold at a loss—both companies needed their consoles to drive heavy games sales in order to turn a profit.

The Nintendo Switch had been launched in March of 2017, to staunch the losses Nintendo was suffering from the failed Wii U[y], and rapidly became the became the fastest selling game console of all time. However, its position vis-à-vis Xbox and PlayStation ninth generation consoles was unclear. It was priced at $299 (the same as the Xbox Series S) but had significantly lower processing power. That combined with its launch timing made many view it as a late eighth generation console rather than an early ninth generation console. The switch also did double duty as a home console and a portable device, making it even harder to define.

In their first month of launch PS5 and Xbox models X and S (combined) sold 4.5 million units and 2.4 million units respectively, suggesting that Sony had a clear lead over Microsoft.[z] Nintendo Switch, however, outsold them both, selling about 5 million units in December of 2020, and bringing the Switch installed base to 79 million—about 60 percent higher than Xbox One's installed base (49.6 million units) and almost 70 percent of Sony PS4's installed base (114.9 million units).[aa]

The market seemed to be clearly stabilized into an oligopoly of three heavyweight competitors. The only real threat appeared to be PC gaming, which was on the rise again. According to a study by DFC Intelligence, of the 3.1 billion people who played video games worldwide in 2020, about half played mostly or entirely on a PC. Was the supremacy of consoles approaching its end?[ab]

Discussion Questions

1. What factors do you think enabled Sega to break Nintendo's near monopoly of the U.S. video game console market in the late 1980s?
2. Why did Nintendo choose not to make its early video game consoles backward compatible? What were the advantages and disadvantages of this strategy?
3. What strengths and weaknesses did Sony have when it entered the video game market in 1995? What strengths and weaknesses did Microsoft have when it entered the video game market in 2001?

4. In what ways did Nintendo's Wii and Switch break with the norms of competition in the video game industry? How defensible was Nintendo's position?
5. Comparing the deployment strategies used by the firms in each of the generations, can you identify any timing, licensing, pricing, marketing, or distribution strategies that appear to have consistently influenced firms' success and failure in the video game industry?

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OVERVIEW

The value of any technological innovation is only partly determined by what the technology can *do*. A large part of the value of an innovation is determined by the degree to which people can understand it, access it, and integrate it within their lives. Deployment is not just a way for the firm to earn revenues from its innovations; deployment is a core part of the innovation process itself.

Deployment strategies can influence the receptivity of customers, distributors, and complementary goods providers. Effective deployment strategies can reduce uncertainty about the product, lower resistance to switching from competing or substitute goods, and accelerate adoption. Ineffective deployment strategies can cause even brilliant technological innovations to fail. As shown in the opening vignette, 3DO's Interactive Multiplayer and Philips' Compact Disk Interactive were two of the first 32-bit video game systems introduced to the market and offered significant technological advantages over previous generations of consoles. However, both were priced so high and introduced with so few games that neither was able to attract a significant share of the market. When Sega introduced its 32-bit system to the market—beating Sony's PlayStation by several months—the price was low and some popular games were available, but weak distribution ultimately hobbled the console's deployment. On the other hand, despite being a newcomer to the video game industry, Sony's exceptionally executed deployment strategies for the PlayStation made the console a nearly overnight success. Sony used a combination of intense marketing, low prices, strong games availability, and aggressive distribution to ensure that the product launched with unmistakable impact.

We will cover five key elements of the deployment process in this chapter: launch timing, licensing and compatibility, pricing, distribution, and marketing. Several of these topics warrant entire courses and textbooks in their own right; only an introduction to the issues most central to the deployment of a new technological innovation will be covered here.

LAUNCH TIMING

As illustrated by the video game industry, the timing of the product launch can be a significant part of a company's deployment strategy. For example, even though Nintendo had a 16-bit video game system in development when Sega's 16-bit Genesis was introduced, Nintendo delayed introducing a 16-bit system for fear of cannibalizing its 8-bit system sales. The advantages and disadvantages of being a first, early-but-not-first, or late mover were discussed in Chapter Five; the focus here is on how a firm can use timing as a deployment strategy.

Strategic Launch Timing

Generally, firms try to decrease their development cycles in order to decrease their costs and to increase their timing of entry options, but this does not imply that firms should always be racing to launch their products as early as possible. A firm can strategically use launch timing to take advantage of business cycle or seasonal effects, to position its product with respect to previous generations of related technologies, and

to ensure that production capacity and complementary goods or services are in place. The role of each of these tactics is illustrated in the video game industry.

Nintendo, Sony, and Microsoft all took advantage of seasonal effects by introducing their consoles shortly before Christmas so that the hype of the consoles' launch would coincide with the Christmas buying season. The majority of video game consoles are sold in December. By launching their consoles close to December, firms could target their advertising for this time and leverage the free publicity that surrounded a console's launch such as press releases announcing the introduction and external product reviews.

Because the video game industry is characterized by distinct generations of technology, the timing of a console's launch also plays a key role in its positioning within a technological generation and with respect to competing consoles. If a console is introduced too early, it may receive a tepid welcome because customers want to wait to compare the consoles with others that will compete in the generation. Furthermore, by launching well ahead of competitors, a console maker may forfeit the opportunity to incorporate more advanced technology or may create customer confusion about which generation the product belongs to. For example, though the Xbox offered a processor that was double the speed of the PS2, its introductory timing positioned it as being in the same generation as the PS2. Many customers saw it as a competitor to a product they already had, rather than as a next generation technology. If the console is introduced too late, the company can lose its image as a technological leader and may have already conceded a considerable installed base lead to earlier entrants. This is aptly illustrated in the quote about Nintendo's late introduction of the SNES in the opening vignette: "To tell the truth, Nintendo just isn't cool anymore. This one is 16 bits, so it's better than the original Nintendo. But the company only made it to compete with Sega, and most kids already have that. So they don't need Super Nintendo. . . ."

Finally, timing the introduction of a console to coincide with production capacity and games availability has proven very important in the video game console industry. For example, in Sega's rush to ensure that the Saturn beat Sony's PlayStation to market, it introduced the product before it had built adequate production capacity. Sega was subsequently unable to stock many important distributors, and it alienated companies that had supported Sega in previous generations. Similarly, the importance of having games available at the time of launch is also clearly demonstrated: Every video game console producer that has been successful in at least one generation (e.g., Atari, Nintendo, Sega, Sony, Microsoft) has ensured that games would be available at the console's launch, even if that meant buying games development companies to force them to produce compatible games! Games availability was also encouraged through licensing strategies, as discussed later in the chapter.

cannibalization When a firm's sales of one product (or at one location) diminish its sales of another of its products (or at another of its locations).

Optimizing Cash Flow Versus Embracing Cannibalization

A second key point about timing of entry is also illustrated in the video game industry. For firms introducing a next generation technology into a market in which they already compete, entry timing can become a decision about whether and to what degree to embrace **cannibalization**. Traditionally, research on product life cycles has emphasized the importance of timing new product introduction so as to optimize cash flows or profits from each generation and minimize cannibalization. If a firm's current product is very profitable, the firm will often delay introduction of a next generation

product until profits have begun to significantly decrease for the current product. This strategy is intended to maximize the firm's return on investment in developing each generation of the product. However, in industries driven by technological innovation, delaying the introduction of a next generation product can enable competitors to achieve a significant technological gap. If competitors introduce products that have a large technological advantage over the firm's current products, customers might begin abandoning the firm's technology.

Instead, if the firm invests in continuous innovation and willingly cannibalizes its existing products with more advanced products, the firm can make it very difficult for other firms to achieve a technological lead large enough to prove persuasive to customers. By providing incentives for existing customers to upgrade to its newest models, the firm can further remove any incentive customers have to switch to another company's products when they purchase next generation technology. Many would argue that this is where Nintendo made a key mistake. In the late 1980s, Nintendo was deriving significant profits from its 8-bit system and thus was reluctant to cannibalize those sales with a 16-bit system. However, by not embracing cannibalization, Nintendo enabled Sega to steal customers away by offering a product with a significant technological advantage.

LICENSING AND COMPATIBILITY

Chapter Nine revealed how making a technology more open (i.e., not protecting it vigorously or partially opening the technology through licensing) could speed its adoption by enabling more producers to improve and promote the technology and allowing complementary goods developers to more easily support the technology. However, the chapter also pointed out that making a technology completely open poses several risks. First, if a firm completely opens its technology, other producers may drive the price of the technology down to a point at which the firm is unable to recoup its development expense. If competition drives the price down so no producer earns significant margins on the technology, no producer will have much incentive to further develop the technology. Finally, opening a technology completely may cause its underlying platform to become fragmented as different producers alter it to their needs, resulting in loss of compatibility across producers and the possible erosion of product quality.

Each of these effects was demonstrated in the opening vignette. By not protecting their technologies enough, video game console producers in the first generation relinquished their ability to control game production quantity and quality. The resulting market glut of poor-quality games decimated the video game industry. But Nintendo's highly restrictive licensing policies for its 8-bit system made games developers eager to give their support to the first rival that appeared viable. In the 16-bit, 32/64-bit, and 128-bit generations, the console makers sought to achieve a delicate balance of making licensing open enough to attract developer support while making licensing strict enough to control game quantities and quality.

In deploying a technological innovation, often a firm must decide how compatible (or incompatible) to make its technology with that provided by others or with previous generations of its own technology. If there is an existing technology with a large installed base or availability of complementary goods, the firm can sometimes leverage

the value of that installed base and complementary goods by making its technology compatible with current products. For instance, producers of IBM-compatible computers (as detailed in Chapter Nine) were able to tap IBM's installed base and complementary goods advantages by offering computers that operated identically to those made by IBM. Users of IBM compatibles reaped the same installed base advantages and had access to all the same software as they would have with an IBM computer.

If the firm wishes to avoid giving away its own installed base or complementary goods advantages to others, it may protect them by ensuring its products are incompatible with those of future entrants. Most competitors in the U.S. video game industry (with the exception of Atari) have been fairly successful at this strategy. Nintendo, for example, uses a security chip to ensure that only licensed Nintendo games may be played in its consoles, and only Nintendo consoles may be used to play Nintendo games.

backward compatible When products of a technological generation can work with products of a previous generation. For example, a computer is backward compatible if it can run the same software as a previous generation of the computer.

Firms must also decide whether or not to make their products **backward compatible** with their own previous generations of technology. Nintendo repeatedly opted not to make its consoles backward compatible, believing it would be more profitable to require customers to purchase new games. This is understandable given that the consoles were sold at cost and profits were made through game sales; however, it also meant that Nintendo forfeited a significant potential source of advantage over Sega. In contrast, Sega made its 16-bit Genesis compatible with its 8-bit Master System games—though this may not have proven terribly persuasive to customers given the limited success of the Master System. More significantly, Sony made its PS2 console backward compatible with PlayStation games, thereby not only ensuring a tremendous existing library of compatible games at its launch but also providing a significant incentive to PlayStation owners who were considering upgrading to a 128-bit system to choose the PS2 as opposed to Sega's Dreamcast, or waiting for the Xbox or GameCube.

Many products use backward compatibility to ease customer adoption of upgraded technology. Blu-ray video players, for example, will play standard DVDs in addition to Blu-ray DVDs, and most audio devices are designed to work with standard 3.5-millimeter headphone connectors even if they enable other more advanced connection types. If these products were not backward compatible, it might slow customer adoption of the new technology.

Sometimes backward compatibility is at odds with the firm's technological objectives of the upgrade, creating a difficult strategic decision about whether or not to offer backward compatibility. When Apple introduced the iPhone 7, for example, it abandoned the analog headphone jack standard to most other audio devices, stating that it was time to move on to wireless headphone solutions. As Phil Schiller, Apple's marketing chief noted, "Some people have asked why we would remove the analog headphone jack from the iPhone. . . . It really comes down to one word: courage. The courage to move on to do something new that betters all of us."[1] Apple's Lightning port could be used with standard headphones with a special adapter, but this proved to be a source of irritation for many consumers who neither wanted to move to wireless headphones (because of their expense and their need to be charged) nor to carry the special adapter. Moving to a nonbackward compatible headphone jack was, on the other hand, very effective at boosting sales of Apple and Beats (which is owned by Apple) headphones.

Some firms use a particularly powerful strategy that combines continuous innovation with backward compatibility. A firm that both innovates to prevent a competitor from

creating a technological gap and utilizes backward compatibility so that its new platform or models are compatible with previous generations of complementary goods can leverage the existing value yielded by a large range of complementary goods to its new platforms. While such a strategy may cause the firm to forfeit some sales of complementary goods for the new platform (at least initially), it can also effectively link the generations through time and can successfully transition customers through product generations while preventing competitors from having a window to enter the market. Microsoft has utilized this strategy deftly with Windows—though the operating system is regularly updated, each successive generation provides backward compatibility with most of the major software applications developed for previous generations. Thus, customers can upgrade without having to replace their entire libraries of software applications.

PRICING

Pricing is a crucial element in the firm's deployment strategy. Price simultaneously influences the product's positioning in the marketplace, its rate of adoption, and the firm's cash flow. Before a firm can determine its pricing strategy, it must determine the objectives it has for its pricing model. For example, if a firm is in an industry plagued with overcapacity or intense price competition, the firm's objective may be simply *survival.* A **survival price** strategy prices goods to cover variable costs and some fixed costs. It is a short-run strategy, however; in the long run, the firm will want to find a way to create additional value. One common pricing objective is to *maximize current profits.* Under this pricing strategy, the firm first estimates costs and demand and then sets the price to maximize cash flow or rate of return on investment. This strategy emphasizes current performance, but may sacrifice long-term performance.

survival pricing
When the price of a good is set to cover variable costs and part of fixed costs. This is a short-run strategy that does not create long-term profits for the firm.

For new technological innovations, firms often emphasize either a *maximum market skimming* objective or a *maximum market share* objective. To skim the market, firms will initially set prices high on new products. The high price may signal the market that the new product is a significant innovation that offers a substantial performance improvement over previously available products. The high price can also help the firm recoup initial development expenses, assuming there is also high initial demand. However, high initial prices may also attract competitors to the market and can slow adoption of the product. If costs are expected to decline rapidly with the volume of units produced, a skimming strategy can actually prove less profitable than a pricing strategy that stimulates more rapid customer adoption.

penetration pricing
When the price of a good is set very low (or free) to maximize the good's market share.

When achieving high volume is important, firms will often emphasize a maximum market share objective. To maximize market share, firms often use **penetration pricing**. The firm will set the lowest price possible hoping to rapidly attract customers, driving volume up and production costs down. Effective utilization of penetration pricing often requires that the firm builds large production capacity in advance of demand. In the short run, the firm may bear significant risk from this capital investment, and it may lose money on each unit if the price is less than its initial variable costs. However, if its volume increases and drives its production costs down, the firm can achieve a very powerful position: It can have a low-cost position that enables it to earn profits despite a low price, and it can have a substantial share of the market.

Firms in industries characterized by increasing returns (strong learning-curve effects and/or network externalities) will often use the objective of maximizing market share and a penetration pricing strategy. In such industries, there is strong pressure for the industry to adopt a single dominant design (as discussed in Chapter Four). It is in the firm's best interest to accelerate adoption of its technology, building its installed base, attracting developers of complementary goods, and riding down the learning curve for its production costs.

For example, Honda's first hybrid electric vehicle, the Insight, was introduced at a price ($20,000) that actually caused Honda to lose money on each Insight it sold. However, Honda believed the hybrid technology would become profitable in the long term, and that the experience it would gain by working with hybrid technology and the continuance of its "green" car company image were strong enough motivations to sell the Insight at a loss for the first few years.[2]

Sometimes firms price below cost because the losses are expected to be recouped through profits on complementary goods or services. In the video game industry, this has proven to be a very important strategy. Nintendo, Sega, Sony, and Microsoft have each sold their video game consoles at a price very close to (or below) production costs while profiting from subsequent game sales and licensing royalties. Similarly, printers are often sold below cost with the intention of making profits on compatible ink cartridges. This is sometimes called the "razor and razor blade strategy" because companies such as Gillette and Schick famously were early users of this strategy, offering razors at a very low price but charging high margins on replacement cartridges. The key to making this strategy pay off is to be able to control compatibility with complements, and to be able to offer complements at a sufficient range of price and quality to meet customer needs. Printers do not recoup their costs, for example, if customers are able to seek out and use cheaper ink cartridges that do not pay a license fee to the printer company.[3]

Firms can also influence cash flow and the customers' perception of costs through manipulating the timing of when the price of a good is paid. For instance, while the most typical pricing model requires the customer to pay the full price before taking ownership, other pricing models enable the customer to delay paying the purchase price by offering a free trial for a fixed time. This permits the customer to become familiar with the benefits of the product before paying the price, and it can be very useful when customers face great uncertainty about a new product or service. Another pricing model enables customers to pay as they go, such as through leasing programs, or a pricing model whereby the initial product is free (or available at a low price) but the customer pays for service. For example, when cable television subscribers order cable service, they typically pay little or no fee for the equipment and instead pay a significant amount (often between $20 and $90 depending on the package) for monthly service that may include some portion for the equipment expense.

freemium
A pricing model where a base product or service is offered for free, but a premium is charged for additional features or service.

Another way to manipulate consumers' perception of the costs is through a **"freemium"** model, where the base product is free, but additional features or capacity have a price. For example, when Drew Houston and Arash Ferdowsi founded Dropbox, a popular cloud storage and file synchronization service, they quickly realized they could not afford to use advertising programs such as Google's AdWords to promote the service—it was simply too expensive. They thus combined a freemium model with a very successful referral program. First, users could get an initial storage allotment for free but would have to pay to get additional storage. Since users' reliance upon the

service and storage needs both tend to increase over time, ultimately they often end up paying for additional capacity. By early 2018, Dropbox had more than 500 million registered users, of which more than 11 million were paying. In some situations, however, a freemium pricing model may signal the consumer that a product is less valuable. Research by professor Joost Rietveld using experiments and the Steam gaming platform, for example, found that consumers spent more time using a game, and spent more on in-game purchases when they had to pay for the base game.[4]

When it is unclear how customers will respond to a particular price point, firms often use introductory pricing that indicates the pricing is for a stipulated time. This allows the company to test the market's response to a product without committing to a long-term pricing structure.

DISTRIBUTION

Selling Direct versus Using Intermediaries

manufacturers' representatives
Independent agents that promote and sell the product lines of one or a few manufacturers. They are often used when direct selling is appropriate but the manufacturer does not have a sufficiently large direct sales force to reach all appropriate market segments.

wholesalers
Companies that buy manufacturer's products in bulk, and then resell them (often in smaller or more diverse bundles) to other supply channel members such as retailers.

retailers
Companies that sell goods to the public.

Firms can sell their products directly to users through their direct sales force or an online ordering system or mail-order catalog. Alternatively, firms can use intermediaries such as **manufacturers' representatives**, **wholesalers**, and **retailers**. Selling direct gives the firm more control over the selling process, pricing, and service. For example, when Tesla Motors launched its Model S electric vehicle, its managers suspected that the dealer networks used by other automakers were not the best distribution method: electric vehicles require more explanation to consumers about estimating fuel savings, how often and where they will charge their automobiles, battery life and resale value, and so on. Salespeople at dealerships would require extensive training, and additional incentives to motivate them to invest that amount of time in customer education. Management at Tesla Motors thus decided to utilize a direct-to-customer model whereby it operates its own boutique stores in high-traffic urban locations.

In many situations, however, selling direct can be impractical or overly expensive. Intermediaries provide a number of important services that can make distribution more efficient. First, wholesalers and retailers *break bulk.* In general, manufacturers prefer to sell large quantities of a limited number of items, while customers prefer to buy limited quantities of a large number of items. Wholesalers and retailers can pool large orders from a large number of manufacturers and sell a wider range of goods in small quantities to customers.

For example, a typical book publisher produces a limited range of book titles, but desires to sell them in high volume. The average final consumer may wish to purchase only one copy of a particular book title, but often wants a wide range of book titles to choose from. Both wholesalers and retailers provide valuable bulk-breaking services in this channel. A wholesaler such as Ingram will purchase pallets of books from many different publishers such as McGraw-Hill, Simon & Schuster, and Hachette Group. It then breaks apart the pallets and reassembles bundles of books that include titles from multiple publishers but have fewer copies of any particular book title. These bundles are then sold to retailers such as Barnes & Noble, which offers a wide range of titles sold on an individual basis. Though publishers could sell directly to final consumers using the Internet or a mail-order catalog, customers would have to examine

the offerings of many different publishers to be able to consider the same range of books offered by a retailer.

Intermediaries also provide a number of other services such as transporting goods, carrying inventory, providing selling services, and handling transactions with customers. Many intermediaries also offer greater convenience for customers by offering geographically dispersed retail sites. Location convenience can be particularly important if customers are geographically dispersed and they are likely to want to examine or try different product options or to need on-site service. By contrast, if the product is primarily sold to a few industrial customers or if the product can be routinely ordered without close examination, trial, or service, geographic dispersion may be less important.

original equipment manufacturer (or value-added reseller)
A company that buys products (or components of products) from other manufacturers and assembles them or customizes them into a product that is then sold under the OEM's own name.

Original equipment manufacturers (OEMs) (also called **value-added resellers**, or VARs) provide an even more crucial role in the distribution process. An OEM buys products (or components of products) from other manufacturers and assembles them into a product that is customized to meet user needs. The OEM then sells this customized product under its own name and often provides marketing and service support for the product. OEMs are very common in the computer and electronics industries where manufacturers are often specialized in the production of individual components but users prefer to purchase whole, assembled products. Dell Computer, for example, is a very successful OEM in the computer industry. OEMs can provide a very valuable coordinating function in an industry by aggregating components and providing a single contact point for the customer.

disintermediation
When the number of intermediaries in a supply channel is reduced; for example, when manufacturers bypass wholesalers and/or retailers to sell directly to end users.

In some industries, advances in information technology (such as the Internet) have enabled **disintermediation** or a reconfiguration in the types of intermediaries used. For example, online investing services such as E-trade or Ameritrade caused some disintermediation in the investment market by enabling customers to bypass brokers and place their own stock or bond orders online. In industries where the product is information that can be conveyed digitally, such as newspapers, software, and music, the Internet can deliver the product from the manufacturer straight to the consumer. In most industries, however, information technology has simply shifted the roles of intermediaries or expanded the services they provide. For example, online stores such as Dell.com or Amazon.com enable customers to bypass traditional retail outlets such as computer stores or bookstores. However, in most cases this has not shortened the supply chain by which goods are delivered to customers—it has just rerouted it. In other instances, moving commerce online has required creating additional intermediaries (such as companies specialized in delivering the goods of others) or enhanced the services that intermediaries provide. For example, while grocers traditionally required customers to provide their own distribution for "the last mile" (the distance between the store and the customer's home), online grocery shopping shifts the responsibility of moving goods "the last mile" to the stores, requiring them either to develop their own delivery services or to purchase delivery services from other providers. Barnes & Noble uses online sales to complement its bricks-and-mortar retail outlets: Customers can come into the stores to see and physically handle books (an option many book shoppers express a strong preference for), but they can also order books online—from home or from within a Barnes & Noble store—if they are looking for a book that is not stocked at a convenient location.

To determine whether to use intermediaries and what type of intermediaries would be appropriate, the firm should answer the following questions:

1. *How does the new product fit with the distribution requirements of the firm's existing product lines?* Whether the firm already has an existing sales channel that would suit the product will be a primary consideration in how the product should be distributed. For example, if the firm already has a large direct sales force and the new product would fit well with this direct sales system, there may be no need to consider other distribution options. On the other hand, if the firm does not have an existing direct sales force, it will have to determine whether the new product warrants the cost and time of building a direct sales force.
2. *How numerous and dispersed are customers, and how much product education or service will customers require? Is prepurchase trial necessary or desirable? Is installation or customization required?* If customers are dispersed but require little product education or service, mail order or online ordering may suffice. On the other hand, if customers are dispersed and require moderate amounts of education, service, or prepurchase trial, using intermediaries is often a good option because they can provide some on-site education and service and/or trial. If customers are not dispersed, or will require *extensive* education and service, it may be necessary for the firm to provide the education and service directly. Furthermore, if the product will require installation or customization, the firm will often need to employ either a direct sales force or an intermediary capable of providing extensive service.
3. *How are competing products or substitutes sold?* The firm must consider how competing or substitute products are sold, because this both determines the nature of the existing distribution channel options and shapes customer expectations about how products will be purchased. For example, if customers are used to purchasing the product in a retail environment where the product can be viewed and handled and where customers can receive personal sales assistance, they may be reluctant to switch to a sales channel with less contact, such as online purchasing or mail order. How the product is sold may also affect the product's positioning from the perspective of the customer. For example, if competing products are primarily sold in a high-contact mode such as specialty stores or via a direct sales force, selling the new product in a lower-contact channel such as mass discounters or through mail order might cause the customer to perceive the product as being of lower quality or more economical. Market research can assess how the sales channel influences the customer's perception of the product.

Strategies for Accelerating Distribution

When the industry is likely to select a single technology as the dominant design, it can be very important to deploy the technology rapidly. Rapid deployment enables the technology to build a large installed base and encourages the developers of complementary goods to support the technology platform. As the technology is adopted, producer and user experience can be used to improve the technology, and producer costs should also decrease due to learning effects and economies of scale. The firm can use a variety of strategies to accelerate distribution, such as forging alliances with distributors, creating bundling relationships, sponsoring or contracting with large customer groups, and providing sales guarantees.[5]

Alliances with Distributors

Firms introducing a technological innovation can use strategic alliances or exclusivity contracts to encourage distributors to carry and promote their goods. By providing a distributor a stake in the success of the new technology, the firm may be able to persuade the distributor to carry and promote the new technology aggressively. Firms that already have relationships with distributors for other goods are at an advantage in pursuing this strategy; firms without such relationships may need to cultivate them, or even consider forward vertical integration to ensure that their product is widely available.

Lack of distribution may have contributed significantly to the failure of the Sega Saturn to gain installed base. Sega had very limited distribution for its Saturn launch, which may have slowed the building of its installed base both directly (because customers had limited access to the product) and indirectly (because distributors that were initially denied product may have been reluctant to promote the product after the limitations were lifted). Nintendo, by contrast, had unlimited distribution for its Nintendo 64 launch, and Sony not only had unlimited distribution, but also had extensive experience negotiating with retailing giants such as Walmart for its consumer electronics products. Consequently, Sony PlayStation had better distribution on its first day of business than the Sega Saturn, despite Sega's decade of experience in the market.[6]

Bundling Relationships

Firms can also accelerate distribution of a new technology by bundling it with another product that is already in wide use.[7] Bundling enables the new technology to piggyback on the success of another product that already has a large installed base. Once customers acquire the new product in tandem with something else that they already use, switching costs may prevent customers from changing to a different product, even if the different product might have initially been preferred. As customers become familiar with the product, their ties to the technology (for instance, through the cost of training) increase and their likelihood of choosing this technology in future purchase decisions may also increase. Bundling arrangements have proven to be a very successful way for firms to build their installed base and ensure provision of complementary goods. Consider, for example, SiriusXM radio, whose success is overwhelmingly attributed to its bundling arrangements with car companies. Sirius spends more than a billion dollars a year to subsidize the installation of SiriusXM equipment in new cars. Once people have used a free trial of Sirius radio in their car, almost 40 percent convert to paid accounts.[8]

Contracts and Sponsorship

Firms can also set up contractual arrangements with distributors, complementary goods providers, and even large end users (such as universities or government agencies) to ensure that the technology is used in exchange for price discounts, special service contracts, advertising assistance, or other inducements. For example, when medical equipment manufacturers introduce significantly new medical devices such as new ultrasound equipment or magnetic resonance imaging machines, they will often donate or lend a number of these machines to large teaching hospitals. As the new equipment's benefits become clear to the doctors and hospital administration, their likelihood of purchasing additional machines increases. Because large teaching hospitals train medical staff that

may ultimately work for other hospitals and are often influential leaders in the medical community, providing these hospitals with free equipment can be an effective way of encouraging the rest of the medical community to adopt the product.

Guarantees and Consignment

If there is considerable market uncertainty about the new product or service, the firm can encourage distributors to carry the product by offering them guarantees (such as promising to take back unsold stock) or agreeing to sell the product on consignment. For example, when Nintendo introduced the Nintendo Entertainment System to the U.S. market, distributors were reluctant to carry the console or games because many had been stuck with worthless inventory after the crash of the video game market in the mid-1980s. Nintendo agreed to sell the Nintendo Entertainment System to distributors on consignment: Nintendo would be paid only for consoles that were sold, rather than requiring distributors to buy consoles up front. Retailers bore little risk in distributing the good because unsold units could be returned to Nintendo, and the video game industry was reborn.

A similar argument can be made for offering guarantees to complementary goods producers. If complementary goods producers are reluctant to support the technology, the firm can guarantee particular quantities of complementary goods will be purchased, or it can provide the capital for production, thus bearing the bulk of the risk of producing complementary goods for the technology. The complementary goods producer may still have forfeited time or effort in producing goods that may not have a long-term market, but its direct costs will be less at risk.

MARKETING

The marketing strategy for a technological innovation must consider both the nature of the target market and the nature of the innovation. For example, is the target market composed primarily of large industrial users or individual consumers? Is the innovation likely to appeal only to technophiles or to the mass market? Are the benefits of the technology readily apparent, or will they require considerable customer education? Will customers respond more to detailed technical content or eye-catching brand images? Can the marketer alleviate customer uncertainty about the innovation? Major marketing methods are briefly reviewed next, along with how marketing can be tailored to particular adopter categories. Also explored is how marketing can shape perceptions and expectations about the innovation's installed base and availability of complementary goods.

Major Marketing Methods

The three most commonly used marketing methods include advertising, promotions, and publicity/public relations.

Advertising

Many firms use advertising to build public awareness of their technological innovation. Doing so requires that the firm craft an effective advertising message and choose advertising media that can convey this message to the appropriate target market.

In crafting an advertising message, firms often attempt to strike a balance between achieving an entertaining and memorable message versus providing a significant quantity of informative content. Too much focus on one or the other can result in advertisements that are memorable but convey little about the product, or advertisements that are informative but quickly lose the audience's attention. Many firms hire an advertising agency to develop and test an advertising message.

The media used are generally chosen based on their match to the target audience, the richness of information or sensory detail they can convey, their reach (the number of people exposed), and their cost per exposure. Some of the advantages and disadvantages of various advertising media are provided in Figure 13.1.

Promotions

Firms can also use promotions at the distributor or customer level to stimulate purchase or trial. Promotions are usually temporary selling tactics that might include:

- Offering samples or free trial.
- Offering cash rebates after purchase.
- Including an additional product (a "premium") with purchase.
- Offering incentives for repeat purchase.
- Offering sales bonuses to distributor or retailer sales representatives.
- Using cross-promotions between two or more noncompeting products to increase pulling power.
- Using point-of-purchase displays to demonstrate the product's features.

Publicity and Public Relations

Many firms use free publicity (such as articles that appear in a newspaper or magazine about the company or its product) to effectively generate word of mouth. For example, Pfizer's drug Viagra got an enormous amount of free exposure from unofficial celebrity endorsements and humorous coverage on TV shows such as *The Tonight Show* and *Late Show with David Letterman.* Other firms rely on internally generated publications (e.g., annual reports, press releases, articles written by employees for trade magazines or other media) to reach and influence target markets. **Viral marketing** is an attempt to capitalize on the social networks of individuals to stimulate word-of-mouth advertising. Information is sent directly to a set of targeted consumers (a process called "seeding") that are well-positioned in their social networks in some way (e.g., they may be "hubs" in that they have many more friends than others, or may have high potential for opinion leadership). The objective is to spark rapid spreading of the information through social networks, akin to a viral epidemic. Such strategies leverage the fact that people may be more receptive to, or have greater faith in, information that comes through personal contacts.[9] Firms may also sponsor special events (e.g., sporting events, competitions, conferences), contribute to good causes (e.g., charities), exhibit at trade associations, or encourage online consumer reviews to generate public awareness and goodwill.[10] Farmos even involved potential customers in the testing process of its drug Domosedan to generate awareness, as described in the accompanying Theory in Action.

viral marketing Sending information directly to targeted individuals in effort to stimulate word-of-mouth advertising. Individuals are typically chosen on the basis of their position or role in particular social networks.

FIGURE 13.1
Advantages and Disadvantages of Major Advertising Media

Media	Advantages	Disadvantages
Online Advertising: Pay-per-click (search engines)	Can be highly targeted to a particular audience; pay only for results (clicks); fast to deploy—can gain immediate visibility, and can be adjusted or deleted just as quickly; enables rapid and efficient tracking of responses for analyzing effectiveness of the ad	Vulnerable to click-through fraud (e.g., clicks by a competitor or an unhappy customer or employee), which could result in wasted advertising spend
Online Advertising: Social Media	Can connect with customers in a rich way; potential for broad reach and viral marketing; can be highly targeted to a particular audience; relatively inexpensive; can be quickly deployed and adjusted; can track visitors in real-time	Conversion of visitors to customers is often low; can be difficult to build awareness and traffic to social media site
Television	High sensory richness that combines sight, sound, and motion; high geographic and demographic reach; independent stations offer new opportunities to more directly target-specific audiences	Increasingly fragmented audience due to proliferation of stations; increasing use of DVR's enables viewers to skip the advertising; high absolute cost; fleeting exposure
Radio	High geographic and demographic selectivity; medium reach; relatively low cost	Audio presentation only; advertisers may need to buy ads with multiple stations to achieve desired audience reach; fleeting exposure
Newspaper	Timeliness; good local market coverage; broad acceptance; high believability; audience can keep or revisit the advertisement; wide price ranges available	Newspaper audiences are decreasing; easy for audience to skip over ad; relatively poor production quality; high advertising clutter; may be difficult to selectively target a particular audience
Magazine	High geographic and demographic selectivity; high quality visual production; long life; can enable significant technical content; good pass-along readership	Slow deployment (long ad purchase lead times); some waste circulation; may require advertising in multiple magazines to achieve desired reach
Direct Mail	High audience selectivity; no ad competition within the same medium; personalization; enables communication of significant technical content; may be passed along to others; responses can usually be efficiently tracked	Relatively high cost; "junk mail" image; requires access to good mailing lists; requires relatively long lead times for printing and mailing
Outdoor (e.g., billboards, banners)	High repeat exposure; low cost; low competition	Limited audience selectivity; very limited technical content
Telephone	High audience selectivity; can give personalized message	Relatively high cost; can be perceived as an annoyance

Tailoring the Marketing Plan to Intended Adopters

As described in Chapter Three, innovations tend to diffuse through the population in an s-shape pattern whereby adoption is initially slow because the technology is unfamiliar; it then accelerates as the technology becomes better understood and utilized by the mass market, and eventually the market is saturated so the rate of new

Theory in Action Generating Awareness for Domosedan

When the Finnish company Farmos Group Limited introduced its veterinary drug Domosedan, executives knew that building awareness of the drug among known opinion leaders would be crucial. Domosedan represented a disruptive innovation in painkillers for horses and cattle; it would significantly alter the way veterinarians performed their examinations and treatments. Unlike previous sedatives and painkillers used in the treatment of large animals, Domosedan enabled veterinarians to conduct clinical and surgical examinations without tying up or anesthetizing their patients. Animals could be treated while standing, and in most instances they would not have to be transported to the veterinarian's clinic.

Farmos knew that university professors and advanced practitioners were important opinion leaders in veterinary medicine. To educate this group and encourage them to support the product, Farmos asked them to help with the testing process required for the drug's approval and sales permit. University professors were involved in the preclinical testing, and visionary practitioners were utilized for clinical testing. By proactively involving these influential potential adopters, the testing simultaneously acted as a premarketing tool while establishing the drug's efficacy and safety. This enabled opinion leaders to acquire advanced knowledge of and experience with the product before it was released. By the time the drug was launched, many of these influential users were already enthusiastic supporters of the product.

Because the drug represented a scientific breakthrough, it was featured in presentations at scientific conferences and was investigated in numerous dissertations, generating further awareness and excitement about the drug. When it was launched in Finland, the company hosted a large dinner party for all practicing veterinarians to attend, creating a celebratory atmosphere for the drug's introduction. Farmos' tactics were successful—Domosedan was adopted rapidly, spreading quickly around the world, and became a significant commercial success.

Source: Adapted from Birgitta Sandberg, "Creating the Market for Disruptive Innovation: Market Proactiveness at the Launch Stage," *Journal of Targeting, Measurement and Analysis for Marketing* 11, no. 2 (2002): 184–96.

adoptions declines. These stages of adoption have been related to the adopter categories of *innovators* (in the very early stages); followed by *early adopters,* which cause adoption to accelerate; then the *early majority* and *late majority* as the innovation penetrates the mass market; and finally the *laggards* as the innovation approaches saturation.[11] The characteristics of these groups make them responsive to different marketing strategies.

Innovators and early adopters are typically looking for very advanced technologies that offer a significant advantage over previous generations. They are willing to take risks and to pay high prices, and they will accept some incompleteness in the product, but they may also demand considerable customization and technical support.[12] They are more likely to respond to marketing that offers a significant amount of technical content and that emphasizes the leading-edge nature of the innovation. Marketing channels that enable high content and selective reach are appropriate for this market. To market to the early majority, on the other hand, requires that the company communicate the product's completeness, its ease of use, its consistency with the customer's way of life, and its legitimacy. For this market segment, detailed technical information is not as important as using market channels with high reach and high credibility.

Firms often find it is difficult to make the transition between successfully selling to early adopters versus the early majority. While early adopters may be enthusiastic about the innovation's technological features, the early majority may find the product

too complex, expensive, or uncertain. This can result in a chasm in the product's diffusion curve: Sales drop off because the early adopter market is saturated and the early majority market is not yet ready to buy (see Figure 13.2).[13] The company must simultaneously weather a period of diminished sales while scaling up its production capacity and improving efficiency to target the mass market.

To target the late majority and laggards, firms will often use similar channels as those used to target the early majority, although emphasizing reducing the cost per exposure. The marketing message at this stage must stress reliability, simplicity, and cost-effectiveness. The marketing channel need not enable high content, but it must have high credibility and not be so expensive as to drive product costs up significantly.

Recently, marketers have begun to tap the contagion-like spread of information by targeting individuals most likely to rapidly spread information. This is described in detail in the accompanying Research Brief.

Using Marketing to Shape Perceptions and Expectations

As described in Chapter Four, when distributors and customers are assessing the value of a technological innovation, they are swayed not only by evidence of the innovation's actual value, but also by their perception of the innovation's value and their expectations for its value in the future. Advertising, promotions, and publicity can play a key role in influencing the market's perceptions and expectations about the size of the installed base and the availability of complementary goods. Preannouncements can generate excitement about a product before its release, while press releases extolling forecasted sales can convince customers and distributors that the product's installed base will increase rapidly. The firm can also shape expectations about the future of the technology by signaling the market (including distributors, end users, manufacturers of complementary goods, and perhaps even other potential contenders for the new standard) that this is a battle it intends to win and is capable of winning. The firm's reputation may create a signal about its likelihood of success. Firms may also use credible commitments such as major fixed capital investments and guarantees to convince stakeholders that the firm has what it takes to challenge the incumbents.

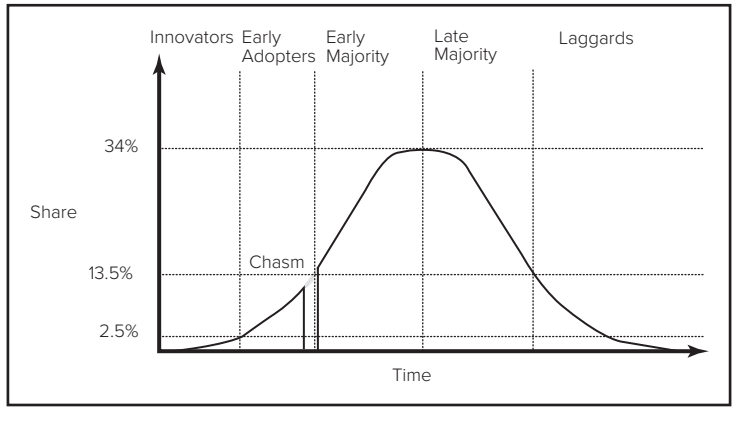


FIGURE 13.2
The Chasm between Early Adopters and Early Majority Customers

Preannouncements and Press Releases

A firm that aggressively promotes its products can increase both its actual installed base and its perceived installed base. Even products that have relatively small installed bases can obtain relatively large mindshares through heavy advertising. Since perceived installed base may drive subsequent adoptions, a large perceived installed base can lead to a large actual installed base. Such a tactic underlies the use of *vaporware*—preadvertised products that are not actually on the market yet and may not even exist—by many software vendors. By building the impression among customers that a product is ubiquitous, firms can prompt rapid adoption of the product when it actually is available. Vaporware may also buy a firm valuable time in bringing its product to market. If other vendors beat the firm to market and the firm fears that customers may select a dominant design before its offering is introduced, it can use vaporware to attempt to persuade customers to delay purchase until the firm's product is available.

The Nintendo 64 provides an excellent example. In an effort to forestall consumer purchases of 32-bit systems, Nintendo began aggressively promoting its development of a 64-bit system (originally named Project Reality) in 1994, though the product would not actually reach the market until September 1996. The project underwent so many delays that some industry observers dubbed it "Project Unreality." Another interesting vaporware example was Nintendo's rewritable 64M disk drive. Though the product was much hyped, it was never introduced.

Major video game producers also go to great lengths to manage impressions of their installed base and market share, often to the point of exaggeration or deception. For example, at the end of 1991, Nintendo claimed it had sold 2 million units of the SNES to the U.S. market, while Sega disagreed, arguing that Nintendo had sold 1 million units at most. Nintendo also forecast that it would sell an additional 6 million units by the end of 1992 (actual installed base of Super Nintendo systems in the United States reached just over 4 million units in 1992). By May 1992, Nintendo was claiming a 60 percent share of the 16-bit market, and Sega was claiming a 63 percent share. Similar tactics were deployed in the battle for the 32/64-bit market. For example, in October 1995, Sony announced to the press that it had presold 100,000 consoles in the United States, to which Mike Ribero, Sega's executive vice president for marketing and sales, countered that Sony's figures were deceptive, arguing that many preorders would never materialize into actual purchases.[14]

Reputationa

When a firm is poised to introduce a new technological innovation, its reputation for both technological and commercial competence will critically influence the market's expectation about its likelihood of success.[15] Customers, distributors, and complementary goods producers will use the firm's track record for technological innovation as an indicator of the new product's functionality and value. The firm's prior commercial success acts as an indicator of the firm's ability to build and manage the necessary support network around the new technology (distribution, advertising, alliances) to create the necessary momentum in the installed base–complementary goods cycle.

When Sega entered the video game market, it had the benefit of having several highly successful arcade games to its credit (both Atari and Nintendo had also been arcade game producers before developing home video games). The company had a reputation

Research Brief Creating an Information Epidemic

Some individuals, by virtue of their natural proclivities and talents, can initiate a cascade of information that travels with startling momentum through a population. Such individuals can have a remarkable effect on marketplace behavior. Gladwell identifies three distinct types of individuals who have such a disproportionate amount of influence: connectors, mavens, and salespersons.[a]

Connectors are individuals who tend to form an exceptionally large circle of acquaintances. Sociologists have found that if a random sample of people is asked to identify the individuals they know on a first-name basis, connectors will identify many times the number of people an average person identifies.[b] These people may have an exceptionally high social drive; they also tend to have a knack for remembering people's names and keeping track of social details such as birthdays. However, it is not just the quantity of acquaintances that distinguishes connectors. Connectors also tend to have a diverse array of affiliations. They may belong to a number of different kinds of clubs, associations, or other social institutions. They belong to multiple social worlds. Thus, connectors can bring together people who would otherwise be unlikely to meet.

Mavens are individuals who are driven to obtain and disseminate knowledge about one or more of their interests. Economists have widely studied "market mavens," otherwise known as "price vigilantes." These individuals will closely track the prices charged by various retailers (or other market outlets) and will vociferously complain if they find something inappropriate, such as a promotion that is misleading.[c] Other mavens may take great pride in always knowing the best restaurants or hotels, or they may be avid readers of *Consumer Reports*. Mavens not only collect information, but they are also keenly interested in educating others. They will frequently volunteer information and derive great pleasure out of helping other consumers.

Finally, *salespersons* are those individuals who are naturally talented persuaders. Such individuals are gifted at providing verbal responses that their listener is likely to find compelling. They may also have an acute ability to send and respond to nonverbal cues, enabling them to influence other people's emotional response to something. These individuals can infect others with their mood![d]

Any of these individuals is capable of sparking an information epidemic. While a connector with a valuable piece of information is likely to expose a great number and diversity of people, the maven is likely to convey the information to fewer people but in more detail, making it more convincing. The salesperson may not expose as many people as the connector and may not be driven to acquire and disseminate the volumes of information that the maven transmits, but the people the salesperson does transmit information to are likely to find it irresistible. Some individuals possess more than one of these traits simultaneously, making them a veritable typhoon of influence in the marketplace.

[a] Adapted from Malcolm Gladwell, *The Tipping Point* (Boston: Little, Brown and Company, 2000).

[b] Albert L. Barabasi, *Linked: The New Science of Networks* (Cambridge, MA: Perseus Books, 2002).

[c] Lawrence F. Feick and Linda L. Price, "The Market Maven: A Diffuser of Marketplace Information," *Journal of Marketing* 51 (1987): 83–97.

[d] Elaine Hatfield, John T. Cacioppo, and Richard. L. Rapson, *Emotional Contagion* (Cambridge: Cambridge University Press, 1994); and H. Friedman et al., "Understanding and Assessing Nonverbal Expressiveness: The Affective Communication Test," *Journal of Personality and Social Psychology* 39, no. 2 (1980): 333–51.

for developing exciting games, and this reputation may have facilitated customer acceptance of its 16-bit challenge to Nintendo's 8-bit dominance. By contrast, when Sony entered the video game market, it did not have the arcade background that underscored the other primary competitors. However, it did have a wealth of technological expertise as a consumer electronics manufacturer and exceptional brand equity in electronic products. Furthermore, Sony had demonstrated its ability to win a format war through

its successful introduction of the CD format (with Philips) that supplanted vinyl records and analog cassettes.

Similarly, reputation was probably Microsoft's greatest strength in the battle for dominance over 128-bit video game systems. Microsoft's near monopoly in the personal computer operating system market was achieved through its unrivaled skill in using network externalities to its advantage. Microsoft had skillfully leveraged its controlling share in PC operating systems into domination over many categories of the software market, obliterating many would-be competitors. Microsoft's reputation sent a strong signal to distributors, developers, and customers that would shape their expectations for its future installed base and availability of complementary goods. Microsoft's success was not assured, but it was a powerful force to be reckoned with.

Credible Commitments

A firm can also signal its commitment to an industry by making substantial investments that would be difficult to reverse. For example, it was well publicized that Sony spent more than $500 million developing the PlayStation, in addition to manufacturing the system and establishing an in-house games development unit. By contrast, 3DO's cumulative research and development costs at the launch of its multiplayer were less than $37 million, and the company utilized a strategy whereby all console and game production was performed by third parties. Thus, 3DO may not have signaled the market that it had enough confidence in the platform to bear the brunt of the capital risk.

Summary of Chapter

1. A firm can use its launch timing strategy to take advantage of business cycle or seasonal effects, to influence its positioning vis-à-vis competitors, and to ensure that production capacity and complementary goods are sufficiently available at time of launch.
2. The launch timing decision must also consider the need to harvest cash flows from existing product generations versus the advantages of willingly cannibalizing existing products to preempt competitors.
3. Successful deployment requires striking a careful balance between making a system open enough to attract complementary goods providers (and/or other producers if that is desirable) and protected enough to ensure that product quality, margins, and compatibility can be sustained.
4. Common pricing strategies for technological innovations include market skimming and penetration pricing. While the first attempts to maximize margins earned on early sales of the product, the second attempts to maximize market share. Pricing strategies should consider the firm's ability to earn profits from sales of complementary goods or services—if profits from complements are expected to be high, lower prices on the platform technology may be warranted.
5. Firms can manipulate the customer's perception of the product's price (and the timing of cash flows) through the timing of when the price is paid.
6. Intermediaries provide a number of valuable roles in the supply chain, including breaking bulk, transporting, carrying inventory, providing selling services, and managing customer transactions.

7. Sometimes a firm can accelerate distribution of its innovation by forging relationships with distributors, bundling the good with others that have a wider installed base, sponsoring large customer groups, or providing sales guarantees to distributors or complements producers.
8. Marketing methods vary in attributes such as cost, reach, information content, duration of exposure, flexibility of message, and ability to target particular segments of the market. When designing the marketing plan, the firm must take into account both the nature of the innovation (e.g., Is it complex? Are benefits easy to observe?) and the nature of the customer (e.g., Does the customer require in-depth technical detail? Is the customer likely to be influenced by brand images and/or reputation? How much uncertainty is the customer likely to tolerate?).
9. Marketing strategies can influence the market's perception of how widely used the product is or will be, and thus can influence the behavior of customers, distributors, and complementary goods producers. Preannouncements, the firm's reputation, and credible commitments can all influence the market's assessment of the product's likelihood of success.

Discussion Questions

1. Identify one or more circumstances when a company might wish to delay introducing its product.
2. What factors will (or should) influence a firm's pricing strategy?
3. Pick a product you know well. What intermediaries do you think are used in bringing this product to market? What valuable services do you think these intermediaries provide?
4. What marketing strategies are used by the producers of the product you identified for Question 3? What are the advantages and disadvantages of these marketing strategies?

Suggested Further Reading

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